

## **Solutions Manual for**

Heat and Mass Transfer: Fundamentals & Applications

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## **Chapter 9**

# **NATURAL CONVECTION**

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## Physical Mechanisms of Natural Convection

**9-1C** The convection heat transfer coefficient is usually higher in forced convection because of the higher fluid velocities involved.

**9-2C** The hot boiled egg in a spacecraft will cool faster when the spacecraft is on the ground since there is no gravity in space, and thus there will be no natural convection currents which is due to the buoyancy force.

**9-3C** The buoyancy force is proportional to the density of the medium, and thus is larger in sea water than it is in fresh water. Therefore, the hull of a ship will sink deeper in fresh water because of the smaller buoyancy force acting upwards.

**9-4C** The upward force exerted by a fluid on a body completely or partially immersed in it is called the buoyancy or “lifting” force. The buoyancy force is proportional to the density of the medium. Therefore, the buoyancy force is the largest in mercury, followed by in water, air, and the evacuated chamber. Note that in an evacuated chamber there will be no buoyancy force because of absence of any fluid in the medium.

**9-5C** There cannot be any natural convection heat transfer in a medium that experiences no change in volume with temperature.

**9-6C** The greater the volume expansion coefficient, the greater the change in density with temperature, the greater the buoyancy force, and thus the greater the natural convection currents.

**9-7C** The Grashof number represents the ratio of the buoyancy force to the viscous force acting on a fluid. The inertial forces in Reynolds number is replaced by the buoyancy forces in Grashof number.

**9-8** The volume expansion coefficient is defined as  $\beta = \frac{1}{\rho} \left( \frac{\partial \rho}{\partial T} \right)_P$ . For an ideal gas,  $P = \rho RT$  or  $\rho = \frac{P}{RT}$ , and thus

$$\beta = -\frac{1}{\rho} \left( \frac{\partial (P/RT)}{\partial T} \right)_P = \frac{1}{\rho} \left( \frac{-P}{RT^2} \right) = \frac{1}{\rho T} \left( \frac{P}{RT} \right) = \frac{1}{\rho T} (\rho) = \frac{1}{T}$$

**9-9** The volume expansion coefficient of saturated liquid water at 70°C is to be determined using its definition and the values tabulated in Table A-9.

**Assumptions** Density depends on temperature only and not pressure.

**Properties** The properties of sat. liq. water are listed in the following table:

| T, °C | $\rho$ , kg/m <sup>3</sup> | $\beta$ , K <sup>-1</sup> |
|-------|----------------------------|---------------------------|
| 65    | 980.4                      |                           |
| 70    | 977.5                      | $0.578 \times 10^{-3}$    |
| 75    | 974.7                      |                           |

**Analysis** The volume expansion coefficient is defined as

$$\beta = -\frac{1}{\rho} \left( \frac{\partial \rho}{\partial T} \right)_p$$

For density varying with temperature at constant pressure, we can approximate

$$\beta \approx -\frac{1}{\rho} \frac{\Delta \rho}{\Delta T} = -\frac{1}{\rho} \frac{\rho_1 - \rho_2}{T_1 - T_2} \quad \text{where} \quad T_1 = 65^\circ\text{C}, \quad T_2 = 75^\circ\text{C}, \quad \text{and} \quad \rho = 977.5 \text{ kg/m}^3$$

Hence, the volume expansion coefficient is calculated to be

$$\beta \approx -\frac{1}{977.5 \text{ kg/m}^3} \frac{(980.4 - 974.7) \text{ kg/m}^3}{(65 - 75) \text{ K}} = 5.83 \times 10^{-4} \text{ K}^{-1}$$

**Discussion** The calculated volume expansion coefficient is about 1% higher than the value listed in Table A-9 ( $5.78 \times 10^{-4} \text{ K}^{-1}$ ).

**9-10** Using the given  $\rho(T)$  correlation, the volume expansion coefficient of liquid water at 70°C is to be determined.

**Assumptions** Density depends on temperature only and not pressure.

**Properties** The volume expansion coefficient of liquid water at 70°C is  $5.78 \times 10^{-4} \text{ K}^{-1}$  (Table A-9).

**Analysis** The volume expansion coefficient is defined as

$$\beta = -\frac{1}{\rho} \left( \frac{\partial \rho}{\partial T} \right)_p = -\frac{1}{\rho} \left( \frac{d\rho}{dT} \right) = -\frac{1}{\rho} (-0.0736 - 0.0071T)$$

Hence, at  $T = 70^\circ\text{C}$  the volume expansion coefficient is

$$\begin{aligned} \beta &= -\frac{(-0.0736 - 0.0071T) \text{ kg/m}^3 \cdot \text{K}}{(1000 - 0.0736T - 0.00355T^2) \text{ kg/m}^3} \\ &= -\frac{[-0.0736 - 0.0071(70)] \text{ kg/m}^3 \cdot \text{K}}{[1000 - 0.0736(70) - 0.00355(70)^2] \text{ kg/m}^3} \\ &= 5.84 \times 10^{-4} \text{ K}^{-1} \end{aligned}$$

**Discussion** The calculated volume expansion coefficient is about 1% higher than the value listed in Table A-9 ( $5.78 \times 10^{-4} \text{ K}^{-1}$ ).

**9-11** The Grashof numbers for a plate placed in various fluids are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Properties are constant. 3 Air behaves as an ideal gas.

**Properties** The properties of air, liq. water, and engine oil are listed in the following table:

| Fluid                               | $T_f, ^\circ\text{C}$ | $\rho, \text{kg/m}^3$ | $\mu, \text{kg/m}\cdot\text{s}$ | $\beta, \text{K}^{-1}$ |
|-------------------------------------|-----------------------|-----------------------|---------------------------------|------------------------|
| Air (Table A-15)                    | 90                    | 0.9718                | $2.139 \times 10^{-5}$          | $2.755 \times 10^{-3}$ |
| Liq. water (Table A-9)              | 90                    | 965.3                 | $0.315 \times 10^{-3}$          | $0.702 \times 10^{-3}$ |
| Engine oil (Table A-13)             | 80                    | 852.0                 | $3.232 \times 10^{-2}$          | $0.700 \times 10^{-3}$ |
| For air (ideal gas) $\beta = 1/T_f$ |                       |                       |                                 |                        |

**Analysis** The Grashof number is given as

$$\text{Gr}_L = \frac{g\beta(T_s - T_\infty)L_c^3}{\nu^2} = \frac{g\beta(T_s - T_\infty)L_c^3}{(\mu/\rho)^2}$$

For air,

$$\text{Gr}_{L, \text{air}} = \frac{(9.81 \text{ m/s}^2)(2.755 \times 10^{-3} \text{ K}^{-1})(150 - 30) \text{ K}(0.1 \text{ m})^3}{(2.139 \times 10^{-5} / 0.9718)^2 \text{ m}^4/\text{s}^2} = \mathbf{6.69 \times 10^6}$$

The Grashof number for liquid water and engine oil are calculated similar to the calculation done for air above

$$\text{Gr}_{L, \text{water}} = \mathbf{7.76 \times 10^9}$$

$$\text{Gr}_{L, \text{oil}} = \mathbf{6.68 \times 10^5}$$

**Discussion** Higher value of the Grashof number implies increase in buoyancy force over the viscous force, which means increase in natural convection flow.

**Natural Convection over Surfaces**

**9-12C** Rayleigh number is the product of the Grashof and Prandtl numbers.

**9-13C** No, a hot surface will cool slower when facing down since the warmer air in this position cannot rise and escape easily.

**9-14C** The heat flux will be higher at the bottom of the plate since the thickness of the boundary layer which is a measure of thermal resistance is the lowest there.

**9-15C** A vertical cylinder can be treated as a vertical plate when  $D \geq \frac{35L}{Gr^{1/4}}$ .

**9-16** A vertical plate separates the hot water from the cold water. The temperature of the plate surface on the cold water side is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Temperature of each surface is constant. 3 The plate thermal conductivity is constant. 4 Radiation heat transfer is negligible.

**Properties** Assuming the surface temperature on the cold water side is  $T_{s,c} = (100 + 7)^\circ\text{C}/2 = 53.5^\circ\text{C}$ , thus  $T_{f,c} = (T_{s,c} + T_{\infty,c})/2 = (53.5 + 7)^\circ\text{C}/2 = 30.25^\circ\text{C}$ . Then, the properties of water at  $T_{f,c}$  are  $k = 0.6033 \text{ W/m}\cdot\text{K}$ ,  $\rho = 995.5 \text{ kg/m}^3$ ,  $\mu = 0.0007935 \text{ kg/m}\cdot\text{s}$ ,  $\nu = \mu/\rho = 7.971 \times 10^{-7} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 5.502$ ,  $\beta = 0.0003072 \text{ K}^{-1}$  (Table A-9).

The thermal conductivity of the plate is given as  $k_{\text{plate}} = 15 \text{ W/m}\cdot\text{K}$ .

**Analysis** The Rayleigh number is

$$\text{Ra} = \frac{g\beta(T_{s,c} - T_{\infty,c})L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.0003072 \text{ K}^{-1})(53.5 - 7) \text{ K} (0.2 \text{ m})^3}{(7.971 \times 10^{-7} \text{ m}^2/\text{s})^2} (5.502) = 9.708 \times 10^9$$

The Nusselt number relation for vertical plate is

$$\text{Nu} = \left\{ 0.825 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.492}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(9.705 \times 10^9)^{1/6}}{\left[ 1 + \left( \frac{0.492}{5.502} \right)^{9/16} \right]^{8/27}} \right\}^2 = 307.2$$

$$h = \frac{k}{L} \text{Nu} = \frac{0.6033 \text{ W/m}\cdot\text{K}}{0.2 \text{ m}} (307.2) = 926.7 \text{ W/m}^2 \cdot \text{K}$$

Thus, the rate of heat transfer balance for conduction through the plate thickness and natural convection is

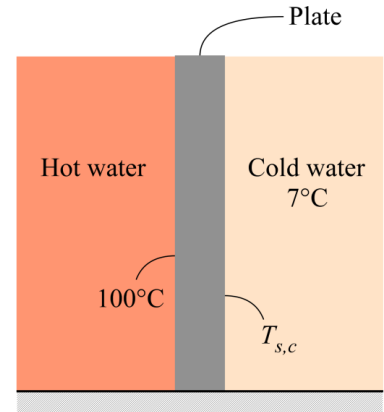
$$\frac{k_{\text{plate}}}{l} (T_{s,h} - T_{s,c}) = h(T_{s,c} - T_{\infty,c}) \rightarrow T_{s,c} = 43.5^\circ\text{C} \quad (\text{first iteration})$$

The above solution is repeated iteratively until  $T_{s,c}$  converges to  $T_{s,c} = 46^\circ\text{C}$ .

**Discussion** The results from the iterative solution are listed in the following table:

| Iter | $T_{s,c} [^\circ\text{C}]$ | Ra                  | Nu    |
|------|----------------------------|---------------------|-------|
| 1    | 53.5                       | $9.708 \times 10^9$ | 307.2 |
| 2    | 43.5                       | $5.915 \times 10^9$ | 264.6 |
| 3    | 47.2                       | $7.194 \times 10^9$ | 280.7 |
| 4    | 45.8                       | $6.694 \times 10^9$ | 274.7 |
| 5    | 46.3                       | $6.870 \times 10^9$ | 276.8 |
| 6    | <b>46.1</b>                | $6.799 \times 10^9$ | 276.0 |

As  $T_{s,c}$  changes through the iterations, so does the film temperature used for evaluating the properties.





**9-17** Reconsider Prob. 9-16. A vertical plate separates the hot water from the cold water. The effect of  $k_{\text{plate}}$  on  $T_{s,c}$  is to be investigated.

**Analysis** The problem is solved using EES, and the solution is given below.

**"GIVEN"**

$L=0.2$  [m]  
thickness=0.025 [m]  
 $T_{\text{infinity}_c}=7$  [C]  
 $T_{s_h}=100$  [C]

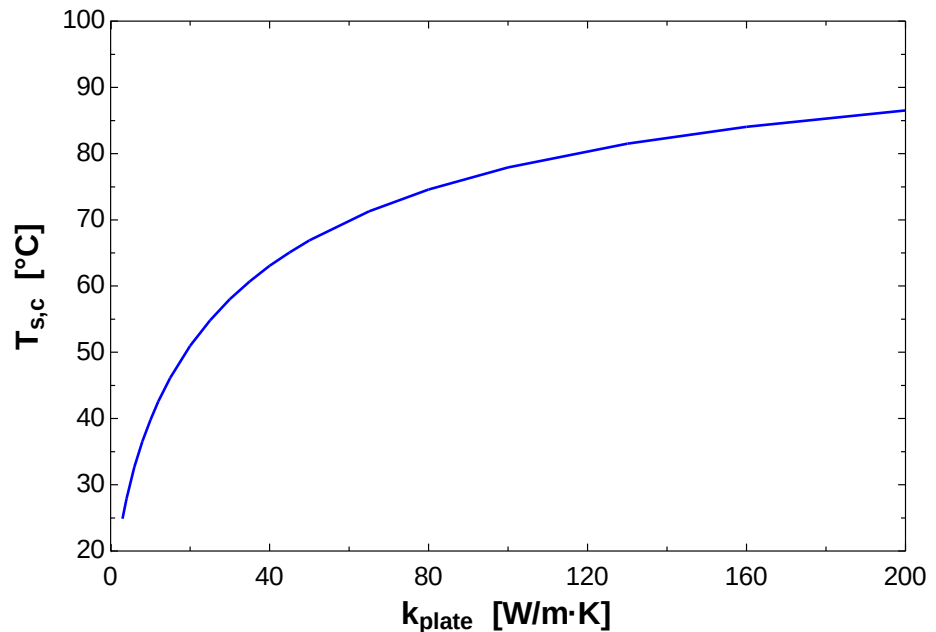
**"PROPERTIES"**

$g=9.81$  [m/s^2]  
Fluid\$='water'  
 $k=\text{Conductivity}(\text{Fluid}\$, T=T_{\text{film}_c}, x=0)$   
 $\text{Pr}=\text{Prandtl}(\text{Fluid}\$, T=T_{\text{film}_c}, x=0)$   
 $\rho=\text{Density}(\text{Fluid}\$, T=T_{\text{film}_c}, x=0)$   
 $\mu=\text{Viscosity}(\text{Fluid}\$, T=T_{\text{film}_c}, x=0)$   
 $\nu=\mu/\rho$   
 $\beta=\text{Volexpcoef}(\text{Fluid}\$, T=T_{\text{film}_c}, x=0)$   
 $T_{\text{film}_c}=1/2*(T_{s_c}+T_{\text{infinity}_c})$

**"ANALYSIS"**

$\text{Ra}=(g*\beta*(T_{s_c}-T_{\text{infinity}_c})*L^3)/\nu^2*\text{Pr}$   
 $\text{Nusselt}=(0.825+0.387*\text{Ra}^{(1/6)})/((1+(0.492/\text{Pr})^{(9/16)})^{(8/27)))^2$   
 $h=k/L*\text{Nusselt}$   
 $q_{\text{dot}}=k_{\text{plate}}/\text{thickness}*(T_{s_h}-T_{s_c})$   
 $q_{\text{dot}}=h*(T_{s_c}-T_{\text{infinity}_c})$

| $k_{\text{plate}}$ [W/m·K] | $T_{s,c}$ [°C] |
|----------------------------|----------------|
| 3                          | 24.89          |
| 4                          | 27.88          |
| 6                          | 32.71          |
| 8                          | 36.57          |
| 10                         | 39.80          |
| 12                         | 42.59          |
| 15                         | 46.16          |
| 20                         | 50.97          |
| 25                         | 54.82          |
| 30                         | 58.01          |
| 35                         | 60.72          |
| 40                         | 63.05          |
| 45                         | 65.09          |
| 50                         | 66.90          |
| 65                         | 71.27          |
| 80                         | 74.57          |
| 100                        | 77.90          |
| 130                        | 81.49          |
| 160                        | 84.06          |
| 200                        | 86.53          |



**Discussion** As  $k_{\text{plate}}$  increases, the thermal resistance of the plate reduces, therefore  $T_{s,c}$  increases with increasing value of  $k_{\text{plate}}$ .

**9-18**  A vertical ASTM A203 B steel plate with one surface subjected to natural convection with cold air at  $-70^{\circ}\text{C}$ . The rate of heat addition to the plate needed to keep the surface from cooling below  $-30^{\circ}\text{C}$  is to be determined.

**Assumptions** **1** Steady state conditions. **2** Air is an ideal gas. **3** The local atmospheric pressure is 1 atm. **4** Thermal radiation is negligible.

**Properties** The properties of air at the film temperature of  $T_f = (T_{\infty} + T_s)/2 = (-70 - 30)/2 = -50^{\circ}\text{C}$  are (Table A-15)  $\text{Pr} = 0.7440$ ,  $k = 0.01979 \text{ W/m}\cdot\text{K}$ ,  $\nu = 9.319 \times 10^{-6} \text{ m}^2/\text{s}$ ,  $\beta = 1/T_f = 1/(-50 + 273 \text{ K}) = 0.004484 \text{ K}^{-1}$

**Analysis** The characteristic length of the plate is  $L_c = L = 0.5 \text{ m}$ , and the Rayleigh number is

$$\text{Ra}_L = \frac{g\beta(T_s - T_{\infty})L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.004484 \text{ K}^{-1})(-30 + 70 \text{ K})(0.5 \text{ m})^3}{(9.319 \times 10^{-6} \text{ m}^2/\text{s})^2} (0.7440) = 1.8843 \times 10^9$$

The Nusselt number for natural convection in this case is determined from

$$\text{Nu} = \frac{hL}{k} = \left\{ 0.825 + \frac{0.387 \text{ Ra}_L^{1/6}}{[1 + (0.492/\text{Pr})^{9/16}]^{8/27}} \right\}^2$$

So, the natural convection heat transfer coefficient is

$$h = \frac{k}{L} \left\{ 0.825 + \frac{0.387 \text{ Ra}_L^{1/6}}{[1 + (0.492/\text{Pr})^{9/16}]^{8/27}} \right\}^2 = \frac{0.01917 \text{ W/m}\cdot\text{K}}{0.5 \text{ m}} \left\{ 0.825 + \frac{0.387(1.8843 \times 10^9)^{1/6}}{[1 + (0.492/0.7440)^{9/16}]^{8/27}} \right\}^2$$

$$h = 5.955 \text{ W/m}^2 \cdot \text{K}$$

The heat transfer rate required to keep the plate surface at  $-30^{\circ}\text{C}$  can be determined from the Newton's law of cooling:

$$\dot{Q} = hA_s(T_s - T_{\infty}) = hL^2(T_s - T_{\infty}) = (5.955 \text{ W/m}^2 \cdot \text{K})(0.5 \text{ m})^2(-30 + 70 \text{ K}) = \mathbf{59.6 \text{ W}}$$

**Discussion** With a heat addition rate of about 60 W or higher, the plate surface exposed to natural convection with the cold air can be maintained from being cooled below  $-30^{\circ}\text{C}$ .

**9-19**  A vertical ASTM B152 copper plate with one surface subjected to natural convection with hot air. The rate of heat removal from the plate is 90 W, and the maximum temperature that the air can reach without causing the surface temperature of the copper plate to increase above 260°C is to be determined.

**Assumptions** 1 Steady state conditions. 2 Air is an ideal gas. 3 The local atmospheric pressure is 1 atm. 4 Thermal radiation is negligible.

**Properties** The properties of air at the film temperature of  $T_f = 300^\circ\text{C}$  are (Table A-15)  $\text{Pr} = 0.6935$ ,  $k = 0.04418 \text{ W/m}\cdot\text{K}$ ,  $\nu = 4.765 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\beta = 1/T_f = 1/(300 + 273 \text{ K}) = 0.001745 \text{ K}^{-1}$

**Analysis** The characteristic length of the plate is  $L_c = L = 0.5 \text{ m}$ , and the Rayleigh number is

$$\text{Ra}_L = \frac{g\beta(T_\infty - T_s)L^3}{\nu^2} \text{Pr}$$

The Nusselt number for natural convection in this case is determined from

$$\text{Nu} = \frac{hL}{k} = \left\{ 0.825 + \frac{0.387 \text{Ra}_L^{1/6}}{[1 + (0.492/\text{Pr})^{9/16}]^{8/27}} \right\}^2$$

So, the natural convection heat transfer coefficient is

$$h = \frac{k}{L} \left\{ 0.825 + \frac{0.387 \text{Ra}_L^{1/6}}{[1 + (0.492/\text{Pr})^{9/16}]^{8/27}} \right\}^2$$

From energy balance, the rate of heat removal from the plate is equal to the heat transfer rate to the plate by natural convection from the hot air:

$$\dot{Q} = hA_s(T_\infty - T_s) = hL^2(T_\infty - T_s) = 90 \text{ W}$$

By substituting the equation for  $\text{Ra}_L$  into the equation for  $h$  and then substituting the resulting equation in the above Newton's law of cooling equation yields

$$\frac{k}{L} \left\{ 0.825 + \frac{0.387 [g\beta(T_\infty - T_s)L^3\text{Pr}/\nu^2]^{1/6}}{[1 + (0.492/\text{Pr})^{9/16}]^{8/27}} \right\}^2 L^2(T_\infty - T_s) = 90 \text{ W}$$

or

$$\frac{0.04418 \text{ W/m}\cdot\text{K}}{0.5 \text{ m}} \left\{ 0.825 + \frac{0.387 [(9.81 \text{ m/s}^2)(0.001745 \text{ K}^{-1})(T_\infty - T_s)(0.5 \text{ m})^3(0.6935)/(4.765 \times 10^{-5} \text{ m}^2/\text{s})^2]^{1/6}}{[1 + (0.492/0.6935)^{9/16}]^{8/27}} \right\}^2 (0.5 \text{ m})^2(T_\infty - T_s) = 90 \text{ W}$$

The temperature of the hot air can be solved implicitly by trial-and-error with  $T_s = 260^\circ\text{C}$ , which yields

$$T_\infty = \mathbf{340.9^\circ\text{C}}$$

**Discussion** With a heat removal rate from the copper plate at 90 W, the plate surface exposed to natural convection with the hot air would reach 260°C if the air temperature is at 340.9°C. Thus, the maximum temperature that the air can reach without causing the surface temperature of the copper plate to increase above its maximum use temperature is about 340°C.

With  $T_\infty = 340.9^\circ\text{C}$  and  $T_s = 260^\circ\text{C}$ , the film temperature becomes 300.5°C. Thus, 300°C is an appropriate temperature to evaluate the properties of air.

**9-20** A vertical plate separates the hot water from the cold air. The surface exposed to the cold air is subjected to radiation heat transfer also. The temperature of the plate surface exposed to the cold air is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Temperature on each surface is constant. 3 The plate thermal conductivity is constant. 4 Local atmospheric pressure is 1 atm. 5 The  $T_{\text{surr}}$  is the same as the cold air temperature.

**Properties** Assuming the surface temperature on the cold air side is  $T_{s,c} = (100 + 2)^\circ\text{C}/2 = 51^\circ\text{C}$ , thus  $T_{f,c} = (T_{s,c} + T_{\infty,c})/2 = (51 + 2)^\circ\text{C}/2 = 26.5^\circ\text{C}$ . Then, the properties of air at  $T_{f,c}$  and 1 atm pressure are  $k = 0.02562 \text{ W/m}\cdot\text{K}$ ,  $\nu = 1.576 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 0.7277$  (Table A-15), and  $\beta = 1/T_{f,c} = 1/299.5 \text{ K}$ .

The thermal conductivity and the emissivity of the plate are given as  $k_{\text{plate}} = 1.5 \text{ W/m}\cdot\text{K}$  and  $\varepsilon_{\text{plate}} = 0.73$ , respectively.

**Analysis** The Rayleigh number is

$$\text{Ra} = \frac{g\beta(T_{s,c} - T_{\infty,c})L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(26.5 + 273 \text{ K})^{-1}(51 - 2) \text{ K} (0.2 \text{ m})^3}{(1.576 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7277) = 3.762 \times 10^7$$

The Nusselt number relation for vertical plate is

$$\text{Nu} = \left\{ 0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.492}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(3.762 \times 10^7)^{1/6}}{\left[ 1 + \left( \frac{0.492}{0.7277} \right)^{9/16} \right]^{8/27}} \right\}^2$$

$$= 45.89$$

$$h = \frac{k}{L} \text{Nu} = \frac{0.02562 \text{ W/m}\cdot\text{K}}{0.2 \text{ m}} (45.89) = 5.879 \text{ W/m}^2 \cdot \text{K}$$

Thus, the rate of heat transfer balance for conduction through the plate thickness  $l$ , natural convection and radiation is

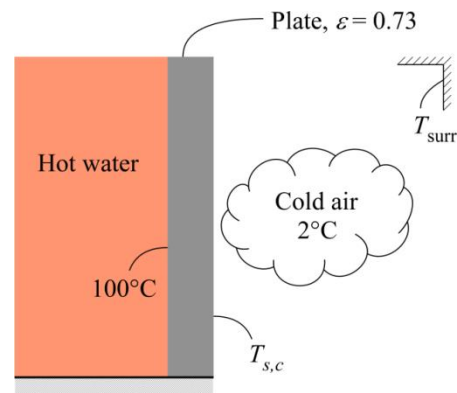
$$\frac{k_{\text{plate}}}{l} (T_{s,h} - T_{s,c}) = h(T_{s,c} - T_{\infty,c}) + \varepsilon_{\text{plate}} \sigma (T_{s,c}^4 - T_{\text{surr}}^4) \rightarrow T_{s,c} = 84.6^\circ\text{C} \quad (\text{first iteration})$$

The above solution is repeated iteratively until  $T_{s,c}$  converges to  $T_{s,c} = 83.7^\circ\text{C}$ .

**Discussion** The results from the iterative solution are listed in the following table:

| Iter | $T_{s,c} [^\circ\text{C}]$ | Ra                  | Nu    | $h [\text{W/m}^2 \cdot \text{K}]$ |
|------|----------------------------|---------------------|-------|-----------------------------------|
| 1    | 51                         | $3.762 \times 10^7$ | 45.89 | 5.879                             |
| 2    | 84.6                       | $4.934 \times 10^7$ | 49.66 | 6.670                             |
| 3    | 83.7                       | $4.912 \times 10^7$ | 49.60 | 6.653                             |
| 4    | 83.72                      | $4.913 \times 10^7$ | 49.60 | 6.654                             |
| 5    | <b>83.71</b>               | $4.912 \times 10^7$ | 49.60 | 6.653                             |

As  $T_{s,c}$  changes through the iterations, so does the film temperature used for evaluating the properties.





**9-21** Reconsider Prob. 9-20. A vertical plate separates the hot water from the cold air. The surface exposed to the cold air is subjected to radiation heat transfer also. The effect of the plate thickness on  $T_{s,c}$  is to be evaluated.

**Analysis** The problem is solved using EES, and the solution is given below.

**"GIVEN"**

$L=0.2$  [m]  
 $T_{\infty,c}=2$  [C]  
 $T_{s,h}=100$  [C]

**"PROPERTIES"**

$g=9.81$  [m/s<sup>2</sup>] "gravitational acceleration"  
 Fluid\$='air'

**"Cold air"**

$k=\text{Conductivity}(\text{Fluid}\$, T=T_{\text{film}_c})$   
 $\text{Pr}=\text{Prandtl}(\text{Fluid}\$, T=T_{\text{film}_c})$   
 $\rho=\text{Density}(\text{Fluid}\$, T=T_{\text{film}_c}, P=101.3)$   
 $\mu=\text{Viscosity}(\text{Fluid}\$, T=T_{\text{film}_c})$   
 $\nu=\mu/\rho$   
 $\beta=\text{Volexpcoef}(\text{Fluid}\$, T=T_{\text{film}_c})$   
 $T_{\text{film}_c}=1/2*(T_{s,c}+T_{\infty,c})$

**"Plate"**

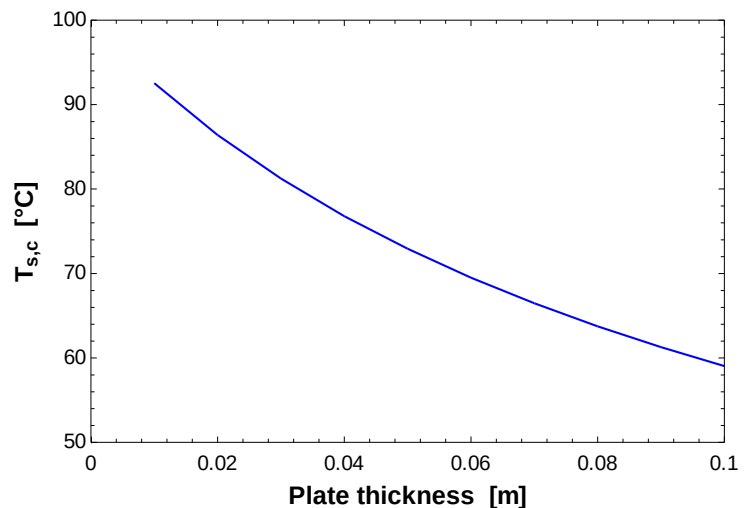
$k_{\text{plate}}=1.5$  [W/m-K]  
 $\epsilon_{\text{plate}}=0.73$

**"ANALYSIS"**

$\text{Ra}=(g*\beta*(T_{s,c}-T_{\infty,c})*L^3)/\nu^2*\text{Pr}$   
 $\text{Nusselt}=(0.825+0.387*\text{Ra}^{(1/6)})/((1+(0.492/\text{Pr})^{(9/16))}^{(8/27)))^2$   
 $h=k/L*\text{Nusselt}$   
 $\dot{q}=k_{\text{plate}}/\text{thickness}*(T_{s,h}-T_{s,c})$   
 $\dot{q}=h*(T_{s,c}-T_{\infty,c})+\sigma*\epsilon_{\text{plate}}*((T_{s,c}+273)^4-(T_{\infty,c}+273)^4)$

| Thickness [m] | $T_{s,c}$ [°C] |
|---------------|----------------|
| 0.01          | 92.54          |
| 0.02          | 86.40          |
| 0.03          | 81.23          |
| 0.04          | 76.79          |
| 0.05          | 72.92          |
| 0.06          | 69.51          |
| 0.07          | 66.47          |
| 0.08          | 63.74          |
| 0.09          | 61.27          |
| 0.10          | 59.03          |

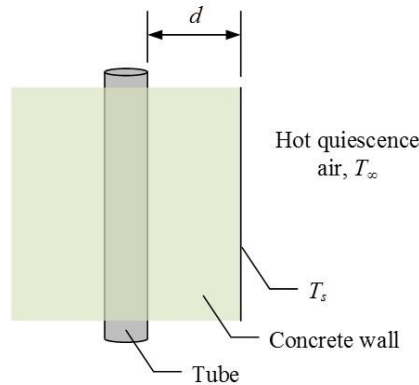
**Discussion** As the plate thickness increases, the thermal resistance of the plate increases, thus reducing the surface temperature on the cold air side.



**9-22**  A CPVC tube is embedded in a vertical concrete wall, where the tube surface is 3 cm from the wall surface. The surface of the wall is subjected to convection with hot quiescence air at 140°C. Would the tube comply with the ASME Code for Process Piping?

**Assumptions** 1 Steady state conditions. 2 Air is an ideal gas. 3 The local atmospheric pressure is 1 atm. 4 Thermal radiation is negligible. 5 No heat generation in the wall. 6 One-dimensional conduction. 7 Constant properties.

**Properties** The properties of air at the film temperature of  $T_f = (T_\infty + T_s)/2 = (140 + 100)/2 = 120^\circ\text{C}$  are (Table A-15)  $\text{Pr} = 0.7073$ ,  $k = 0.03235 \text{ W/m}\cdot\text{K}$ ,  $\nu = 2.522 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\beta = 1/T_f = 1/(120 + 273 \text{ K}) = 0.002545 \text{ K}^{-1}$



**Analysis** The characteristic length of the vertical wall is  $L_c = L = 1 \text{ m}$ , and the Rayleigh number is

$$\text{Ra}_L = \frac{g\beta(T_\infty - T_s)L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.002545 \text{ K}^{-1})(140 - 100 \text{ K})(1 \text{ m})^3}{(2.522 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7073) = 1.1105 \times 10^9$$

The Nusselt number for natural convection in this case is determined from

$$\text{Nu} = \frac{hL}{k} = \left\{ 0.825 + \frac{0.387 \text{ Ra}_L^{1/6}}{[1 + (0.492/\text{Pr})^{9/16}]^{8/27}} \right\}^2$$

So, the natural convection heat transfer coefficient is

$$h = \frac{k}{L} \left\{ 0.825 + \frac{0.387 \text{ Ra}_L^{1/6}}{[1 + (0.492/\text{Pr})^{9/16}]^{8/27}} \right\}^2 = \frac{0.03235 \text{ W/m}\cdot\text{K}}{1 \text{ m}} \left\{ 0.825 + \frac{0.387(1.1105 \times 10^9)^{1/6}}{[1 + (0.492/0.7073)^{9/16}]^{8/27}} \right\}^2$$

$$h = 4.103 \text{ W/m}^2 \cdot \text{K}$$

Applying energy balance, with natural convection at the concrete surface and conduction through the concrete wall,

$$\dot{q}_{\text{cond}} = \dot{q}_{\text{conv}}$$

$$k_{\text{wall}} \frac{T_s - T_d}{d} = h(T_\infty - T_s)$$

Solving for the temperature  $T_d$  in the concrete wall at  $d = 3 \text{ cm}$ , we get

$$T_d = T_s - \frac{hd}{k_{\text{wall}}} (T_\infty - T_s) = 100^\circ\text{C} - \frac{(4.103 \text{ W/m}^2 \cdot \text{K})(0.03 \text{ m})}{1.4 \text{ W/m}\cdot\text{K}} (140 - 100^\circ\text{C}) = 96.48^\circ\text{C} > 93.3^\circ\text{C}$$

**Discussion** With the tube surface located at the depth of 3 cm from the concrete surface, it will be in contact with the concrete at 96.5°C, which is over 3°C higher than the recommended maximum temperature for CPVC (ASME B31.3-2014, Table B-1). Therefore, the tube would not be in compliance.

If the depth is increased to  $d \geq 5.72 \text{ cm}$ , the temperature in the concrete wall would be 93.3°C or lower.

**9-23** C&S A vertical ASTM A240 410S stainless steel plate with one surface subjected to natural convection with cold gas at  $-70^\circ\text{C}$ . The type of gas alternates between carbon dioxide and hydrogen. The rate of heat addition to the plate needed to keep the surface from cooling below  $-30^\circ\text{C}$ , such that it is applicable to both carbon dioxide and hydrogen, is to be determined.

**Assumptions** 1 Steady state conditions. 2  $\text{CO}_2$  and  $\text{H}_2$  are ideal gases. 3 The local atmospheric pressure is 1 atm. 4 Thermal radiation is negligible.

**Properties** The properties of  $\text{CO}_2$  at the film temperature of  $T_f = (T_\infty + T_s)/2 = (-70 - 30)/2 = -50^\circ\text{C}$  are (Table A-16)  $\text{Pr} = 0.8019$ ,  $k = 0.01051 \text{ W/m}\cdot\text{K}$ ,  $\nu = 4.699 \times 10^{-6} \text{ m}^2/\text{s}$ ,  $\beta = 1/T_f = 1/(-50 + 273 \text{ K}) = 0.004484 \text{ K}^{-1}$

The properties of  $\text{H}_2$  gas at the film temperature of  $T_f = -50^\circ\text{C}$  are (Table A-16)  $\text{Pr} = 0.6562$ ,  $k = 0.1404 \text{ W/m}\cdot\text{K}$ ,  $\nu = 6.624 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\beta = 1/T_f = 1/(-50 + 273 \text{ K}) = 0.004484 \text{ K}^{-1}$

### Analysis

**For  $\text{CO}_2$  gas:** The characteristic length of the plate is  $L_c = L = 0.5 \text{ m}$ , and the Rayleigh number is

$$\text{Ra}_L = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.004484 \text{ K}^{-1})(-30 + 70 \text{ K})(0.5 \text{ m})^3}{(4.699 \times 10^{-6} \text{ m}^2/\text{s})^2} (0.8019) = 7.9876 \times 10^9$$

The Nusselt number for natural convection in this case is determined from

$$\text{Nu} = \frac{hL}{k} = \left\{ 0.825 + \frac{0.387 \text{ Ra}_L^{1/6}}{[1 + (0.492/\text{Pr})^{9/16}]^{8/27}} \right\}^2$$

So, the natural convection heat transfer coefficient is

$$h = \frac{k}{L} \left\{ 0.825 + \frac{0.387 \text{ Ra}_L^{1/6}}{[1 + (0.492/\text{Pr})^{9/16}]^{8/27}} \right\}^2 = \frac{0.01051 \text{ W/m}\cdot\text{K}}{0.5 \text{ m}} \left\{ 0.825 + \frac{0.387(7.9876 \times 10^9)^{1/6}}{[1 + (0.492/0.8019)^{9/16}]^{8/27}} \right\}^2$$

$$h = 5.024 \text{ W/m}^2 \cdot \text{K}$$

The heat transfer rate required to keep the plate surface at  $-30^\circ\text{C}$ , when exposed to the cold  $\text{CO}_2$  gas at  $-70^\circ\text{C}$  is

$$\dot{Q} = hA_s(T_s - T_\infty) = hL^2(T_s - T_\infty) = (5.024 \text{ W/m}^2 \cdot \text{K})(0.5 \text{ m})^2(-30 + 70 \text{ K}) = \mathbf{50.24 \text{ W}}$$

**For  $\text{H}_2$  gas:** The characteristic length of the plate is  $L_c = L = 0.5 \text{ m}$ , and the Rayleigh number is

$$\text{Ra}_L = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.004484 \text{ K}^{-1})(-30 + 70 \text{ K})(0.5 \text{ m})^3}{(6.624 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.6562) = 3.2893 \times 10^7$$

The Nusselt number for natural convection in this case is determined from

$$\text{Nu} = \frac{hL}{k} = \left\{ 0.825 + \frac{0.387 \text{ Ra}_L^{1/6}}{[1 + (0.492/\text{Pr})^{9/16}]^{8/27}} \right\}^2$$

So, the natural convection heat transfer coefficient is

$$h = \frac{k}{L} \left\{ 0.825 + \frac{0.387 \text{ Ra}_L^{1/6}}{[1 + (0.492/\text{Pr})^{9/16}]^{8/27}} \right\}^2 = \frac{0.1404 \text{ W/m}\cdot\text{K}}{0.5 \text{ m}} \left\{ 0.825 + \frac{0.387(3.2893 \times 10^7)^{1/6}}{[1 + (0.492/0.6562)^{9/16}]^{8/27}} \right\}^2$$

$$h = 12.22 \text{ W/m}^2 \cdot \text{K}$$

The heat transfer rate required to keep the plate surface at  $-30^\circ\text{C}$ , when exposed to the cold  $\text{H}_2$  gas at  $-70^\circ\text{C}$  is

$$\dot{Q} = hA_s(T_s - T_\infty) = hL^2(T_s - T_\infty) = (12.22 \text{ W/m}^2 \cdot \text{K})(0.5 \text{ m})^2(-30 + 70 \text{ K}) = \mathbf{122.2 \text{ W}}$$

**Discussion** Comparing the results for the  $\text{CO}_2$  and  $\text{H}_2$  gas, a heat addition rate of higher than 122 W is needed to keep the plate surface from cooling below  $-30^\circ\text{C}$ , such that it is applicable for both gases. The natural convection heat transfer coefficient for the  $\text{H}_2$  gas is about 2.4 times higher than that for the  $\text{CO}_2$  gas. The  $\text{H}_2$  gas has higher thermal conductivity (about 13 times higher than  $\text{CO}_2$ ), while the  $\text{CO}_2$  gas is a better thermal insulator.

**9-24** A thin vertical plate is subjected to uniform heat flux on one side and exposed to cool air on the other side. The heat flux on the plate is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Constant heat flux on the plate surface. 3 Thermal resistance in the plate is negligible. 4 Local atmospheric pressure is 1 atm. 5 The  $T_{\text{surr}}$  is the same as the cool air temperature.

**Properties** The film temperature is determined with the plate midpoint temperature,  $T_f = (T_{L/2} + T_\infty)/2 = (55 + 5)^\circ\text{C}/2 = 30^\circ\text{C}$ . Then, the properties of air at  $T_f = 30^\circ\text{C}$  are  $k = 0.02588 \text{ W/m}\cdot\text{K}$ ,  $\nu = 1.608 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 0.7282$  (Table A-15), and  $\beta = 1/T_f = 1/303 \text{ K} = 0.0033 \text{ K}^{-1}$ .

**Analysis** The Rayleigh number is

$$\text{Ra} = \frac{g\beta(T_{L/2} - T_\infty)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(30 + 273 \text{ K})^{-1}(55 - 5) \text{ K}(0.5 \text{ m})^3}{(1.608 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7282) = 5.699 \times 10^8$$

The Nusselt number relation for vertical plate is

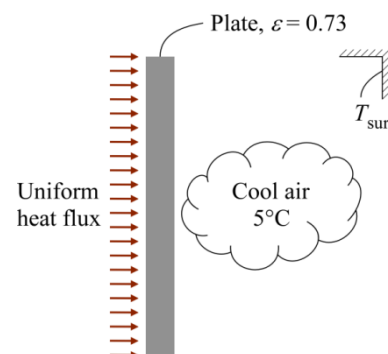
$$\text{Nu} = \left\{ 0.825 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.492}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(5.699 \times 10^8)^{1/6}}{\left[ 1 + \left( \frac{0.492}{0.7282} \right)^{9/16} \right]^{8/27}} \right\}^2 = 103.7$$

$$h = \frac{k}{L} \text{Nu} = \frac{0.02588 \text{ W/m}\cdot\text{K}}{0.5 \text{ m}} (103.7) = 5.368 \text{ W/m}^2 \cdot \text{K}$$

Thus, the heat flux on the plate surface can be determined from the heat loss by natural convection and radiation on the other side of the plate:

$$\begin{aligned} \dot{q}_s &= h(T_{L/2} - T_\infty) + \varepsilon\sigma(T_{L/2}^4 - T_{\text{surr}}^4) \\ &= (5.368 \text{ W/m}^2 \cdot \text{K})(55 - 5) \text{ K} + (5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(0.73)(328^4 - 278^4) \text{ K}^4 \\ &= \mathbf{500 \text{ W/m}^2} \end{aligned}$$

**Discussion** Natural convection contributes to about 54% of the  $500 \text{ W/m}^2$  heat flux. For constant surface heat flux, the plate midpoint temperature is used instead of the surface temperature in the evaluation of the fluid properties.



**9-25** A thin vertical plate is subjected to uniform heat flux on one side and exposed to hydrogen gas on the other side. The plate midpoint temperature is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Constant heat flux on the plate surface. 3 Thermal resistance in the plate is negligible. 4 Local atmospheric pressure is 1 atm. 5 The film temperature is 50°C (this assumption will be verified).

**Properties** The properties of H<sub>2</sub> gas at  $T_f = 50^\circ\text{C}$  are  $k = 0.1881 \text{ W/m}\cdot\text{K}$ ,  $\nu = 1.240 \times 10^{-4} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 0.7191$  (Table A-16), and  $\beta = 1/T_f = 1/323 \text{ K} = 0.003096 \text{ K}^{-1}$  (Table A-16).

**Analysis** With the assumption that  $T_f = 50^\circ\text{C}$ , the plate midpoint temperature is estimated as

$$T_{L/2} = 2T_f - T_\infty = 95^\circ\text{C}$$

The Rayleigh number is

$$\text{Ra} = \frac{g\beta(T_{L/2} - T_\infty)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(50 + 273 \text{ K})^{-1}(95 - 5) \text{ K}(0.5 \text{ m})^3}{(1.240 \times 10^{-4} \text{ m}^2/\text{s})^2} (0.7191) = 1.598 \times 10^7$$

The Nusselt number relation for vertical plate is

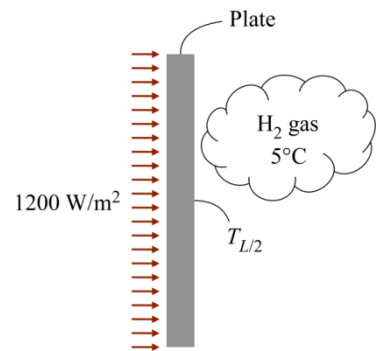
$$\text{Nu} = \left\{ 0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.492}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(1.598 \times 10^7)^{1/6}}{\left[ 1 + \left( \frac{0.492}{0.7191} \right)^{9/16} \right]^{8/27}} \right\}^2 = 35.74$$

$$h = \frac{k}{L} \text{Nu} = \frac{0.1881 \text{ W/m}\cdot\text{K}}{0.5 \text{ m}} (35.74) = 13.45 \text{ W/m}^2 \cdot \text{K}$$

Thus, the plate midpoint temperature from the heat flux as

$$\dot{q}_s = h(T_{L/2} - T_\infty) \rightarrow T_{L/2} = \frac{\dot{q}_s}{h} + T_\infty$$

$$T_{L/2} = \frac{1200 \text{ W/m}^2}{13.45 \text{ W/m}^2 \cdot \text{K}} + 5^\circ\text{C} = 94.2^\circ\text{C}$$



**Discussion** The assumed film temperature  $T_f = 50^\circ\text{C}$  is an appropriate assumption, since the determined  $T_{L/2} = 94.2^\circ\text{C}$  would give a film temperature of  $T_f = 49.6^\circ\text{C}$ . Otherwise,  $T_{L/2}$  would have to be solved iteratively.

For constant surface heat flux, the plate midpoint temperature is used instead of the surface temperature in the evaluation of the fluid properties.

**9-26** A thin vertical plate is subjected to uniform heat flux on one side and exposed to air on the other side. The plate midpoint temperatures for (a) a highly polished copper surface and (b) a black oxidized copper surface are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Constant heat flux on the plate surface. 3 Thermal resistance in the plate is negligible. 4 Local atmospheric pressure is 1 atm. 5 The  $T_{\text{surr}}$  is the same as the air temperature.

**Properties** We first assume the film temperature is  $T_f = 30^\circ\text{C}$ . Then, the properties of air at  $T_f$  are  $k = 0.02588 \text{ W/m}\cdot\text{K}$ ,  $\nu = 1.608 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 0.7282$  (Table A-15), and  $\beta = 1/T_f = 1/303 \text{ K} = 0.0033 \text{ K}^{-1}$ .

The emissivity of highly polished copper is  $\varepsilon = 0.02$  and of black oxidized copper is  $\varepsilon = 0.78$  (Table A-18).

**Analysis** With the assumption that  $T_f = 30^\circ\text{C}$ , the plate midpoint temperature is estimated as

$$T_{L/2} = 2T_f - T_\infty = 55^\circ\text{C}$$

The Rayleigh number is

$$\text{Ra} = \frac{g\beta(T_{L/2} - T_\infty)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(30 + 273 \text{ K})^{-1}(55 - 5) \text{ K}(0.5 \text{ m})^3}{(1.608 \times 10^{-5} \text{ m}^2/\text{s})^2}(0.7282) = 5.699 \times 10^8$$

The Nusselt number relation for vertical plate is

$$\text{Nu} = \left\{ 0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.492}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(5.699 \times 10^8)^{1/6}}{\left[ 1 + \left( \frac{0.492}{0.7282} \right)^{9/16} \right]^{8/27}} \right\}^2 = 103.7$$

$$h = \frac{k}{L} \text{Nu} = \frac{0.02588 \text{ W/m}\cdot\text{K}}{0.5 \text{ m}}(103.7) = 5.368 \text{ W/m}^2 \cdot \text{K}$$

(a) With the known surface heat flux of  $1000 \text{ W/m}^2$  and  $\varepsilon = 0.02$  (highly polished copper), the plate midpoint temperature can be determined as

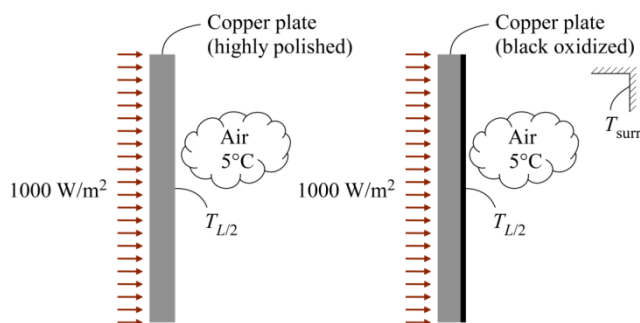
$$\dot{q}_s = h(T_{L/2} - T_\infty) + \varepsilon\sigma(T_{L/2}^4 - T_{\text{surr}}^4)$$

$$1000 \text{ W/m}^2 = (5.368 \text{ W/m}^2 \cdot \text{K})(T_{L/2} - 5) \text{ K} + (5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(0.02)[(T_{L/2} + 273)^4 - (5 + 273)^4] \text{ K}^4$$

Solving for the plate midpoint temperature yields  $T_{L/2} = 183.4^\circ\text{C}$  (first iteration)

The above solution is repeated iteratively until  $T_{L/2}$  converges to  $T_{L/2} = 147.2^\circ\text{C}$ . The results from the iterations are as follows:

| Iter | $T_{L/2}$ [ $^\circ\text{C}$ ] | Ra                  | Nu    | $h$ [ $\text{W/m}^2 \cdot \text{K}$ ] |
|------|--------------------------------|---------------------|-------|---------------------------------------|
| 1    | 55                             | $5.699 \times 10^8$ | 103.7 | 5.368                                 |
| 2    | 183.4                          | $8.427 \times 10^8$ | 116.6 | 7.122                                 |
| 3    | 141.6                          | $8.423 \times 10^8$ | 116.7 | 6.778                                 |
| 4    | 148.3                          | $8.457 \times 10^8$ | 116.8 | 6.842                                 |
| 5    | 147.0                          | $8.452 \times 10^8$ | 116.8 | 6.830                                 |
| 6    | <b>147.2</b>                   | $8.453 \times 10^8$ | 116.8 | 6.832                                 |



(b) With the known surface heat flux of  $1000 \text{ W/m}^2$  and  $\varepsilon = 0.78$  (black oxidized copper), the plate midpoint temperature can be determined as

$$\dot{q}_s = h(T_{L/2} - T_\infty) + \varepsilon\sigma(T_{L/2}^4 - T_{\text{surr}}^4)$$

$$1000 \text{ W/m}^2 = (5.368 \text{ W/m}^2 \cdot \text{K})(T_{L/2} - 5) \text{ K} + (5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(0.78)[(T_{L/2} + 273)^4 - (5 + 273)^4] \text{ K}^4$$

Solving for the plate midpoint temperature yields  $T_{L/2} = 92.9^\circ\text{C}$  (first iteration)

The above solution is repeated iteratively until  $T_{L/2}$  converges to  $T_{L/2} = 88.5^\circ\text{C}$ . The results from the iterations are as follows:

| Iter | $T_{L/2}$ [ $^\circ\text{C}$ ] | Ra                  | Nu    | $h$ [ $\text{W/m}^2 \cdot \text{K}$ ] |
|------|--------------------------------|---------------------|-------|---------------------------------------|
| 1    | 55                             | $5.699 \times 10^8$ | 103.7 | 5.368                                 |
| 2    | 92.9                           | $7.564 \times 10^8$ | 113.0 | 6.165                                 |
| 3    | 88.1                           | $7.401 \times 10^8$ | 112.3 | 6.085                                 |
| 4    | 88.6                           | $7.419 \times 10^8$ | 112.3 | 6.093                                 |
| 5    | <b>88.5</b>                    | $7.415 \times 10^8$ | 112.3 | 6.092                                 |

**Discussion** For part (a), the highly polished copper surface has a low emissivity, which limits the heat loss on the surface by radiation. For part (b), the black oxidized copper surface has a higher emissivity, which increases the heat loss on the surface by radiation. Therefore the plate midpoint temperature for part (a) is higher than that of part (b).

Note that as  $T_{L/2}$  changes through the iterations, so does the film temperature used for evaluating the properties.



**9-27** Reconsider Prob. 9-26. A thin vertical copper plate is subjected to uniform heat flux on one side and exposed to air on the other side. The effect of the heat flux on the plate midpoint temperature for (a) a highly polished copper surface and (b) a black oxidized copper surface is to be determined.

**Analysis** The problem is solved using EES, and the solution is given below.

**"GIVEN"**

L=0.5 [m]  
 T\_infinity=5 [C]  
 epsilon=0.02 "0.02 for (a) and 0.78 for (b)"

**"PROPERTIES"**

g=9.81 [m/s^2] "gravitational acceleration"  
 Fluid\$='air'

**"Air"**

k=Conductivity(Fluid\$, T=T\_film)  
 Pr=Prandtl(Fluid\$, T=T\_film)  
 rho=Density(Fluid\$, T=T\_film, P=101.3)  
 mu=Viscosity(Fluid\$, T=T\_film)  
 nu=mu/rho  
 beta=Volexpcoef(Fluid\$, T=T\_film)  
 T\_film=1/2\*(T\_0.5L+T\_infinity)

**"ANALYSIS"**

Ra=(g\*beta\*(T\_0.5L-T\_infinity)\*L^3)/nu^2\*Pr  
 Nusselt=(0.825+0.387\*Ra^(1/6))/((1+(0.492/Pr)^(9/16))^(8/27))^2  
 h=k/L\*Nusselt  
 q\_dot=h\*(T\_0.5L-T\_infinity)+sigma\*epsilon\*((T\_0.5L+273)^4-(T\_infinity+273)^4)

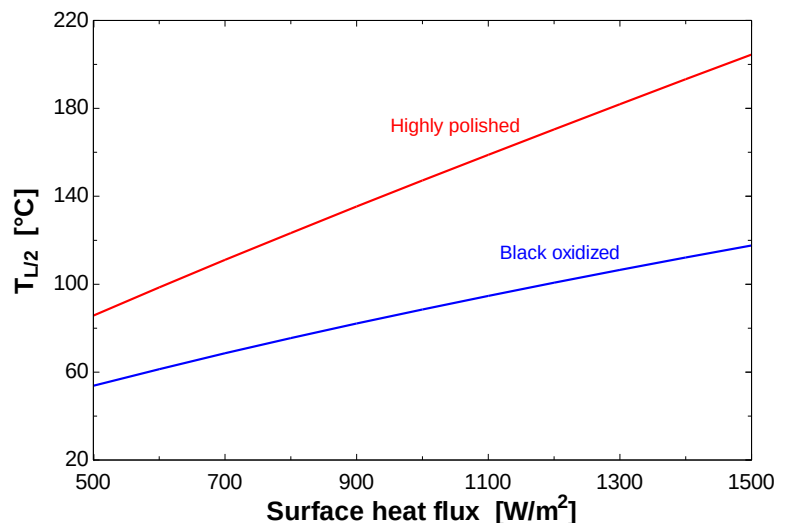
(a) Highly polished surface ( $\epsilon = 0.02$ )

| $\dot{q}_s$ [W/m <sup>2</sup> ] | $T_{s,c}$ [°C] | Ra        |
|---------------------------------|----------------|-----------|
| 500                             | 85.74          | 7.314E+08 |
| 600                             | 98.57          | 7.735E+08 |
| 700                             | 111.1          | 8.036E+08 |
| 800                             | 123.3          | 8.243E+08 |
| 900                             | 135.3          | 8.377E+08 |
| 1000                            | 147.2          | 8.453E+08 |
| 1100                            | 158.9          | 8.481E+08 |
| 1200                            | 170.5          | 8.473E+08 |
| 1300                            | 181.9          | 8.434E+08 |
| 1400                            | 193.3          | 8.371E+08 |
| 1500                            | 204.5          | 8.288E+08 |

(b) Black oxidized surface ( $\epsilon = 0.78$ )

| $\dot{q}_s$ [W/m <sup>2</sup> ] | $T_{s,c}$ [°C] | Ra        |
|---------------------------------|----------------|-----------|
| 500                             | 53.77          | 5.598E+08 |
| 600                             | 61.34          | 6.106E+08 |
| 700                             | 68.55          | 6.527E+08 |
| 800                             | 75.46          | 6.877E+08 |
| 900                             | 82.11          | 7.170E+08 |
| 1000                            | 88.52          | 7.416E+08 |
| 1100                            | 94.71          | 7.621E+08 |
| 1200                            | 100.7          | 7.794E+08 |
| 1300                            | 106.5          | 7.938E+08 |
| 1400                            | 112.2          | 8.058E+08 |
| 1500                            | 117.7          | 8.157E+08 |

**Discussion** The plate midpoint temperature for the highly polished copper surface is higher than that of the black oxidized copper surface. The highly polished copper surface has very low emissivity, thus the heat loss from the surface is mainly by natural convection. The black oxidized copper surface has significant heat loss by radiation in addition to natural convection, which causes the plate midpoint temperature to be lower.



**9-28** A street sign surface is subjected to radiation, the surface temperature of the street sign is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Properties are constant. 3 The surface temperature is constant. 4 The street sign is treated as a vertical plate. 5 Air is an ideal gas.

**Properties** The properties of air (1 atm) at 30°C are given in Table A-15:  $k = 0.02588 \text{ W/m}\cdot\text{K}$ ,  $\nu = 1.608 \times 10^{-5} \text{ m}^2/\text{s}$ , and  $\text{Pr} = 0.7282$ . Also,  $\beta = 1/T_f = 0.0033 \text{ K}^{-1}$ .

**Analysis** The Rayleigh number is

$$\text{Ra}_L = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.0033 \text{ K}^{-1})(T_s - 293 \text{ K})(0.2 \text{ m})^3}{(1.608 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7282) \dots (1)$$

Assuming the Rayleigh number is within  $10^4 < \text{Ra}_L < 10^9$ , the Nusselt number for vertical plate is

$$\text{Nu} = 0.59 \text{Ra}_L^{1/4}$$

or

$$h = \left( \frac{0.02588 \text{ W/m}\cdot\text{K}}{0.2 \text{ m}} \right) 0.59 \text{Ra}_L^{1/4} \dots (2)$$

From energy balance, we obtain

$$\alpha_s \dot{q}_{\text{solar}} = h[T_s - T_\infty] + \varepsilon \sigma [T_s^4 - T_{\text{surr}}^4] \dots (3)$$

Equations (1), (2), and (3) can be solved simultaneously to get the surface temperature. Copy the following lines and paste on a blank EES screen to solve the above equation:

```
g=9.81
k=0.02588
L=0.2
Pr=0.7282
q_incident=200
T_inf=25+273
T_surr=25+273
alpha=0.6
beta=1/(273+30)
epsilon=0.7
nu=1.608e-5
sigma=5.670e-8
Ra_L=g*beta*(T_s-T_inf)*L^3/nu^2*Pr
(h*L/(0.59*k))^4=Ra_L
alpha*q_incident=h*(T_s-T_inf)+epsilon*sigma*(T_s^4-T_surr^4)
```

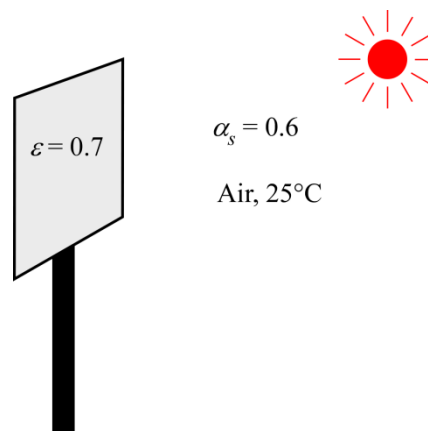
Solving by EES software, we get

$$\text{Ra}_L = 9.961 \times 10^6, \quad h = 4.289 \text{ W/m}^2 \cdot \text{K}, \quad \text{and} \quad T_s = 311.7 \text{ K}$$

Therefore, the surface temperature of the street sign is

$$T_s = 38.7^\circ\text{C}$$

**Discussion** The assumption that the Rayleigh number is within  $10^4 < \text{Ra}_L < 10^9$  turned out to be appropriate. Note that absolute temperatures must be used in calculations involving the radiation heat transfer equation.



**9-29** A glass window is considered. The convection heat transfer coefficient on the inner side of the window, the rate of total heat transfer through the window, and the combined natural convection and radiation heat transfer coefficient on the outer surface of the window are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm.

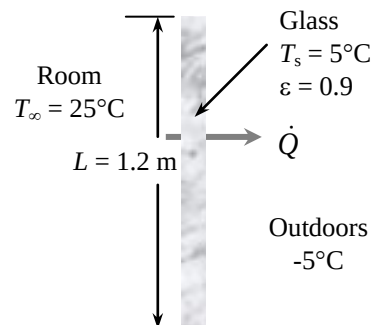
**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (5 + 25)/2 = 15^\circ\text{C}$  are (Table A-15)

$$k = 0.02476 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.470 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7323$$

$$\beta = \frac{1}{T_f} = \frac{1}{(15 + 273)\text{K}} = 0.003472 \text{ K}^{-1}$$



**Analysis** (a) The characteristic length in this case is the height of the window,  $L_c = L = 1.2 \text{ m}$ . Then,

$$\text{Ra} = \frac{g\beta(T_\infty - T_s)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003472 \text{ K}^{-1})(25 - 5 \text{ K})(1.2 \text{ m})^3}{(1.470 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7323) = 3.989 \times 10^9$$

$$\text{Nu} = \left\{ 0.825 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.492}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(3.989 \times 10^9)^{1/6}}{\left[ 1 + \left( \frac{0.492}{0.7323} \right)^{9/16} \right]^{8/27}} \right\}^2 = 189.7$$

$$h = \frac{k}{L} \text{Nu} = \frac{0.02476 \text{ W/m}\cdot^\circ\text{C}}{1.2 \text{ m}} (189.7) = \mathbf{3.915 \text{ W/m}^2\cdot^\circ\text{C}}$$

$$A_s = (1.2 \text{ m})(2 \text{ m}) = 2.4 \text{ m}^2$$

(b) The sum of the natural convection and radiation heat transfer from the room to the window is

$$\dot{Q}_{\text{convection}} = hA_s(T_\infty - T_s) = (3.915 \text{ W/m}^2\cdot^\circ\text{C})(2.4 \text{ m}^2)(25 - 5)^\circ\text{C} = 187.9 \text{ W}$$

$$\begin{aligned} \dot{Q}_{\text{radiation}} &= \varepsilon A_s \sigma (T_{\text{surr}}^4 - T_s^4) \\ &= (0.9)(2.4 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[(25 + 273 \text{ K})^4 - (5 + 273 \text{ K})^4] = 234.3 \text{ W} \end{aligned}$$

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{convection}} + \dot{Q}_{\text{radiation}} = 187.9 + 234.3 = \mathbf{422.2 \text{ W}}$$

(c) The outer surface temperature of the window can be determined from

$$\dot{Q}_{\text{total}} = \frac{kA_s}{t} (T_{s,i} - T_{s,o}) \longrightarrow T_{s,o} = T_{s,i} - \frac{\dot{Q}_{\text{total}} t}{kA_s} = 5^\circ\text{C} - \frac{(422.2 \text{ W})(0.006 \text{ m})}{(0.78 \text{ W/m}\cdot^\circ\text{C})(2.4 \text{ m}^2)} = 3.65^\circ\text{C}$$

Then the combined natural convection and radiation heat transfer coefficient on the outer window surface becomes

$$\dot{Q}_{\text{total}} = h_{\text{combined}} A_s (T_{s,o} - T_{\infty,o})$$

$$\text{or } h_{\text{combined}} = \frac{\dot{Q}_{\text{total}}}{A_s (T_{s,o} - T_{\infty,o})} = \frac{422.2 \text{ W}}{(2.4 \text{ m}^2)[3.65 - (-5)]^\circ\text{C}} = \mathbf{20.35 \text{ W/m}^2\cdot^\circ\text{C}}$$

Note that  $\Delta T = \dot{Q}R$  and thus the thermal resistance  $R$  of a layer is proportional to the temperature drop across that layer.

Therefore, the fraction of thermal resistance of the glass is equal to the ratio of the temperature drop across the glass to the overall temperature difference,

$$\frac{R_{\text{glass}}}{R_{\text{total}}} = \frac{\Delta T_{\text{glass}}}{\Delta T_{\text{total}}} = \frac{5 - 3.65}{25 - (-5)} = 0.045 \quad (\text{or } 4.5\%)$$

which is low. Thus it is reasonable to neglect the thermal resistance of the glass.

**9-30** A hot plate with an insulated back is considered. The rate of heat loss by natural convection is to be determined for different orientations.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm.

**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (55 + 25)/2 = 40^\circ\text{C}$  are (Table A-15)

$$k = 0.02662 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.702 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7255$$

$$\beta = \frac{1}{T_f} = \frac{1}{(40 + 273)\text{K}} = 0.003195 \text{ K}^{-1}$$

**Analysis** (a) When the plate is vertical, the characteristic length is the height of the plate.  $L_c = L = 0.6 \text{ m}$ . Then,

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003195 \text{ K}^{-1})(55 - 25 \text{ K})(0.6 \text{ m})^3}{(1.702 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7255) = 5.087 \times 10^8$$

$$\text{Nu} = \left\{ 0.825 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.492}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(5.087 \times 10^8)^{1/6}}{\left[ 1 + \left( \frac{0.492}{0.7255} \right)^{9/16} \right]^{8/27}} \right\}^2 = 100.1$$

$$h = \frac{k}{L} \text{Nu} = \frac{0.02662 \text{ W/m}\cdot^\circ\text{C}}{0.6 \text{ m}} (100.1) = 4.441 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = L^2 = (0.6 \text{ m})^2 = 0.36 \text{ m}^2$$

and  $\dot{Q} = hA_s(T_s - T_\infty) = (4.441 \text{ W/m}^2\cdot^\circ\text{C})(0.36 \text{ m}^2)(55 - 25)^\circ\text{C} = \mathbf{48.0 \text{ W}}$

(b) When the plate is horizontal with hot surface facing up, the characteristic length is determined from

$$L_s = \frac{A_s}{P} = \frac{L^2}{4L} = \frac{L}{4} = \frac{0.6 \text{ m}}{4} = 0.15 \text{ m}$$

Then,

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003195 \text{ K}^{-1})(55 - 25 \text{ K})(0.15 \text{ m})^3}{(1.702 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7255) = 7.948 \times 10^6$$

$$\text{Nu} = 0.59 \text{Ra}^{1/4} = 0.59(7.948 \times 10^6)^{1/4} = 31.32$$

$$h = \frac{k}{L_c} \text{Nu} = \frac{0.02662 \text{ W/m}\cdot^\circ\text{C}}{0.15 \text{ m}} (31.32) = 5.559 \text{ W/m}^2\cdot^\circ\text{C}$$

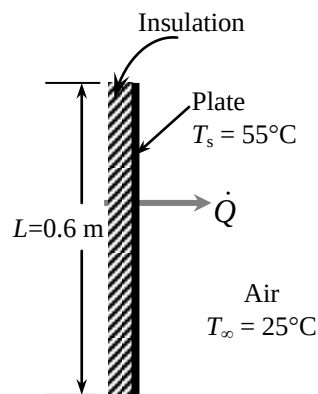
and  $\dot{Q} = hA_s(T_s - T_\infty) = (5.559 \text{ W/m}^2\cdot^\circ\text{C})(0.36 \text{ m}^2)(55 - 25)^\circ\text{C} = \mathbf{60.0 \text{ W}}$

(c) When the plate is horizontal with hot surface facing down, the characteristic length is again  $L_c = 0.15 \text{ m}$  and the Rayleigh number is  $\text{Ra} = 7.948 \times 10^6$ . Then,

$$\text{Nu} = 0.27 \text{Ra}^{1/4} = 0.27(7.948 \times 10^6)^{1/4} = 14.34$$

$$h = \frac{k}{L_c} \text{Nu} = \frac{0.02662 \text{ W/m}\cdot^\circ\text{C}}{0.15 \text{ m}} (14.34) = 2.544 \text{ W/m}^2\cdot^\circ\text{C}$$

and  $\dot{Q} = hA_s(T_s - T_\infty) = (2.544 \text{ W/m}^2\cdot^\circ\text{C})(0.36 \text{ m}^2)(55 - 25)^\circ\text{C} = \mathbf{27.5 \text{ W}}$





**9-31** Prob. 9-30 is reconsidered. The rate of natural convection heat transfer for different orientations of the plate as a function of the plate temperature is to be plotted.

**Analysis** The problem is solved using EES, and the solution is given below.

**"GIVEN"**

L=0.6 [m]  
 T\_infinity=25 [C]  
 T\_s=55 [C]

**"PROPERTIES"**

Fluid\$='air'  
 k=Conductivity(Fluid\$, T=T\_film)  
 Pr=Prandtl(Fluid\$, T=T\_film)  
 rho=Density(Fluid\$, T=T\_film, P=101.325)  
 mu=Viscosity(Fluid\$, T=T\_film)  
 nu=mu/rho  
 beta=1/(T\_film+273)  
 T\_film=1/2\*(T\_s+T\_infinity)  
 g=9.81 [m/s^2]

**"ANALYSIS"**

**"(a), plate is vertical"**

delta\_a=L  
 Ra\_a=(g\*beta\*(T\_s-T\_infinity)\*delta\_a^3)/nu^2\*Pr  
 Nusselt\_a=0.59\*Ra\_a^0.25  
 h\_a=k/delta\_a\*Nusselt\_a  
 A=L^2  
 Q\_dot\_a=h\_a\*A\*(T\_s-T\_infinity)

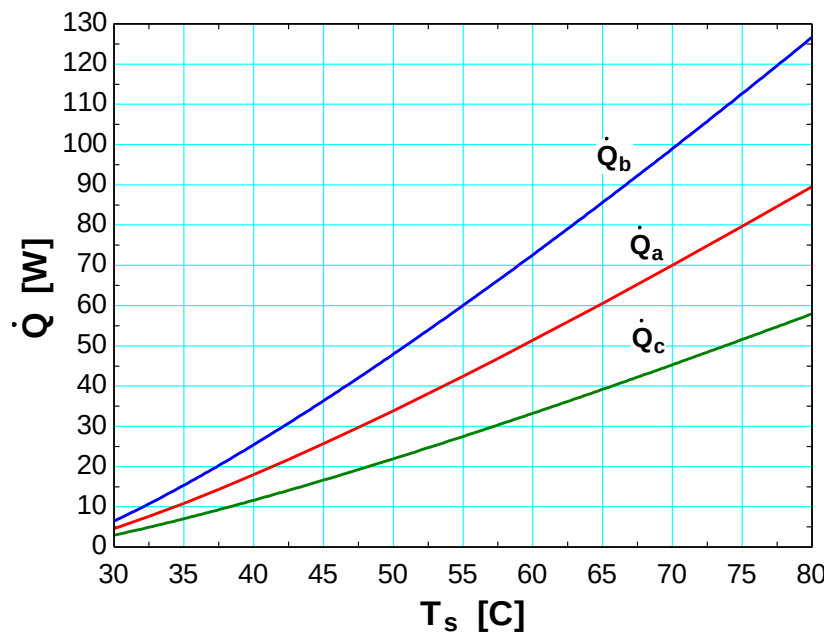
**"(b), plate is horizontal with hot surface facing up"**

delta\_b=A/p  
 p=4\*L  
 Ra\_b=(g\*beta\*(T\_s-T\_infinity)\*delta\_b^3)/nu^2\*Pr  
 Nusselt\_b=0.59\*Ra\_b^0.25  
 h\_b=k/delta\_b\*Nusselt\_b  
 Q\_dot\_b=h\_b\*A\*(T\_s-T\_infinity)

**"(c), plate is horizontal with hot surface facing down"**

delta\_c=delta\_b  
 Ra\_c=Ra\_b  
 Nusselt\_c=0.27\*Ra\_c^0.25  
 h\_c=k/delta\_c\*Nusselt\_c  
 Q\_dot\_c=h\_c\*A\*(T\_s-T\_infinity)

| $T_s$<br>[C] | $\dot{Q}_a$<br>[W] | $\dot{Q}_b$<br>[W] | $\dot{Q}_c$<br>[W] |
|--------------|--------------------|--------------------|--------------------|
| 30           | 4.573              | 6.467              | 2.959              |
| 32.5         | 7.582              | 10.72              | 4.907              |
| 35           | 10.85              | 15.34              | 7.022              |
| 37.5         | 14.32              | 20.26              | 9.27               |
| 40           | 17.97              | 25.41              | 11.63              |
| 42.5         | 21.76              | 30.78              | 14.08              |
| 45           | 25.69              | 36.33              | 16.62              |
| 47.5         | 29.73              | 42.04              | 19.24              |
| 50           | 33.87              | 47.9               | 21.92              |
| 52.5         | 38.12              | 53.9               | 24.67              |
| 55           | 42.45              | 60.03              | 27.47              |
| 57.5         | 46.86              | 66.27              | 30.33              |
| 60           | 51.35              | 72.62              | 33.23              |
| 62.5         | 55.91              | 79.07              | 36.19              |
| 65           | 60.54              | 85.62              | 39.18              |
| 67.5         | 65.24              | 92.26              | 42.22              |
| 70           | 69.99              | 98.98              | 45.3               |
| 72.5         | 74.8               | 105.8              | 48.41              |
| 75           | 79.67              | 112.7              | 51.56              |
| 77.5         | 84.59              | 119.6              | 54.74              |
| 80           | 89.56              | 126.7              | 57.96              |



**9-32** It is proposed that the side surfaces of a cubic industrial furnace be insulated for \$550 in order to reduce the heat loss by 90 percent. The thickness of the insulation and the payback period of the insulation to pay for itself from the energy it saves are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm.

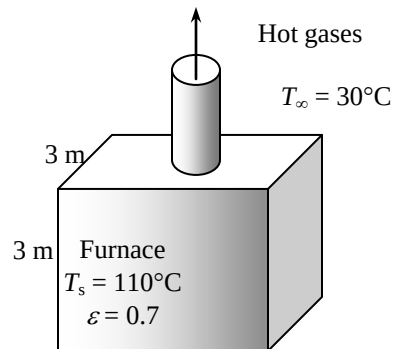
**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (110 + 30)/2 = 70^\circ\text{C}$  are (Table A-15)

$$k = 0.02881 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.995 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7177$$

$$\beta = \frac{1}{T_f} = \frac{1}{(70 + 273)\text{K}} = 0.002915 \text{ K}^{-1}$$



**Analysis** The characteristic length in this case is the height of the furnace,  $L_c = L = 3 \text{ m}$ . Then,

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.002915 \text{ K}^{-1})(110 - 30 \text{ K})(3 \text{ m})^3}{(1.995 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7177) = 1.114 \times 10^{11}$$

$$\text{Nu} = \left\{ 0.825 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.492}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(1.114 \times 10^{11})^{1/6}}{\left[ 1 + \left( \frac{0.492}{0.7177} \right)^{9/16} \right]^{8/27}} \right\}^2 = 545.1$$

$$h = \frac{k}{L_c} \text{Nu} = \frac{0.02881 \text{ W/m}\cdot^\circ\text{C}}{3 \text{ m}} (545.1) = 5.235 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = 4(3 \text{ m})^2 = 36 \text{ m}^2$$

Then the heat loss by combined natural convection and radiation becomes

$$\begin{aligned} \dot{Q} &= hA_s(T_s - T_\infty) + \varepsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4) \\ &= (5.235 \text{ W/m}^2\cdot^\circ\text{C})(36 \text{ m}^2)(110 - 30)^\circ\text{C} \\ &\quad + (0.7)(36 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[(110 + 273 \text{ K})^4 - (30 + 273 \text{ K})^4] \\ &= 33,779 \text{ W} \end{aligned}$$

Noting that insulation will reduce the heat losses by 90%, the rate of heat loss after insulation will be

$$\dot{Q}_{\text{saved}} = 0.9 \dot{Q}_{\text{no insulation}} = 0.9 \times 33,779 \text{ W} = 30,401 \text{ W}$$

$$\dot{Q}_{\text{loss}} = (1 - 0.9) \dot{Q}_{\text{no insulation}} = 0.1 \times 33,779 \text{ W} = 3378 \text{ W}$$

The furnace operates continuously and thus 8760 h. Then the amount of energy and money the insulation will save becomes

$$\text{Energy saved} = \dot{Q}_{\text{saved}} \Delta t = \frac{30,401 \text{ kJ/s}}{0.78} \left( \frac{1 \text{ therm}}{105,500 \text{ kJ}} \right) (8760 \times 3600 \text{ s/yr}) = 11,651 \text{ therms/yr}$$

$$\text{Money saved} = (\text{Energy saved})(\text{Unit cost of energy}) = (11,651 \text{ therms})(\$1.10/\text{therm}) = \$12,816$$

Therefore, the money saved by insulation will pay for the cost of \$550 in

$$550/(\$12,816/\text{yr}) = 0.04292 \text{ yr} = \mathbf{16 \text{ days}}.$$

Insulation will lower the outer surface temperature, the Rayleigh and Nusselt numbers, and thus the convection heat transfer coefficient. For the evaluation of the heat transfer coefficient, we assume the surface temperature in this case to be  $50^\circ\text{C}$ . The properties of air at the film temperature of  $(T_s + T_\infty)/2 = (50 + 30)/2 = 40^\circ\text{C}$  are (Table A-15)

$$k = 0.02662 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.702 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7255$$

$$\beta = \frac{1}{T_f} = \frac{1}{(40 + 273)\text{K}} = 0.003195 \text{ K}^{-1}$$

Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} Pr = \frac{(9.81 \text{ m/s}^2)(0.003195 \text{ K}^{-1})(50 - 30 \text{ K})(3 \text{ m})^3}{(1.702 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7255) = 4.239 \times 10^{10}$$

$$Nu = \left\{ 0.825 + \frac{0.387 Ra^{1/6}}{\left[ 1 + \left( \frac{0.492}{Pr} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387 (4.239 \times 10^{10})^{1/6}}{\left[ 1 + \left( \frac{0.492}{0.7255} \right)^{9/16} \right]^{8/27}} \right\}^2 = 400.5$$

$$h = \frac{k}{L} Nu = \frac{0.02662 \text{ W/m} \cdot ^\circ\text{C}}{3 \text{ m}} (400.5) = 3.554 \text{ W/m}^2 \cdot ^\circ\text{C}$$

$$A_s = 4 \times (3 \text{ m})(3 + 2t_{insul}) \text{ m}$$

The total rate of heat loss from the outer surface of the insulated furnace by convection and radiation becomes

$$\begin{aligned} \dot{Q} &= \dot{Q}_{conv} + \dot{Q}_{rad} = hA_s(T_s - T_\infty) + \varepsilon A_s \sigma (T_s^4 - T_{surr}^4) \\ 3378 \text{ W} &= (3.554 \text{ W/m}^2 \cdot ^\circ\text{C}) A(T_s - 30)^\circ\text{C} \\ &\quad + (0.7) A (5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4) [(T_s + 273 \text{ K})^4 - (30 + 273 \text{ K})^4] \end{aligned}$$

In steady operation, the heat lost by the side surfaces of the pipe must be equal to the heat lost from the exposed surface of the insulation by convection and radiation, which must be equal to the heat conducted through the insulation. Therefore,

$$\dot{Q} = \dot{Q}_{insulation} = kA_s \frac{(T_{furnace} - T_s)}{t_{ins}} \rightarrow 3378 \text{ W} = (0.038 \text{ W/m} \cdot ^\circ\text{C}) A_s \frac{(110 - T_s)^\circ\text{C}}{t_{insul}}$$

Solving the two equations above by trial-and-error (or better yet, an equation solver) gives

$$T_s = 41.2^\circ\text{C} \text{ and } t_{insul} = 0.0284 \text{ m} = \mathbf{2.84 \text{ cm}}$$

**9-33** A vertical plate with length  $L$  is placed in a quiescent air, and the expressions, having the form  $Nu = C Ra_L^n$ , for the average heat transfer coefficient are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties.

**Properties** The properties of air at  $T_f = 20^\circ\text{C}$  are  $k = 0.02514 \text{ W/m}\cdot\text{K}$ ,  $\nu = 1.516 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $Pr = 0.7309$  (from Table A-15). Also,  $\beta = 1/T_f = 0.003413 \text{ K}^{-1}$ .

**Analysis** The Rayleigh number ( $L_c = L$ ) is

$$Ra_L = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} Pr = \frac{(9.81 \text{ m/s}^2)(0.003413 \text{ K}^{-1})\Delta T L^3}{(1.516 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7309) = 1.065 \times 10^8 \Delta T L^3$$

For  $10^4 < Ra_L < 10^9$ , we have

$$Nu = \frac{hL}{k} = 0.59 Ra_L^{1/4} \quad \rightarrow \quad h = 0.59 \frac{k}{L} Ra_L^{1/4}$$

Substituting the  $Ra_L$  yields

$$h = 0.59 \left( \frac{0.02514}{L} \right) (1.065 \times 10^8 \Delta T L^3)^{1/4} = 1.51 (\Delta T / L)^{1/4} \quad 10^4 < Ra_L < 10^9$$

For  $10^{10} < Ra_L < 10^{13}$ , we have

$$Nu = \frac{hL}{k} = 0.1 Ra_L^{1/3} \quad \rightarrow \quad h = 0.1 \frac{k}{L} Ra_L^{1/3}$$

Substituting the  $Ra_L$  yields

$$h = 0.1 \left( \frac{0.02514}{L} \right) (1.065 \times 10^8 \Delta T L^3)^{1/3} = 1.19 \Delta T^{1/3} \quad 10^{10} < Ra_L < 10^{13}$$

**Discussion** The average heat transfer coefficient for laminar conditions ( $10^4 < Ra_L < 10^9$ ) is dependent on  $\Delta T$  and  $L$ . In turbulent conditions ( $10^{10} < Ra_L < 10^{13}$ ), the average heat transfer coefficient is not influenced by  $L$ .

**9-34** A circuit board containing square chips is mounted on a vertical wall in a room. The surface temperature of the chips is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm. 4 The heat transfer from the back side of the circuit board is negligible.

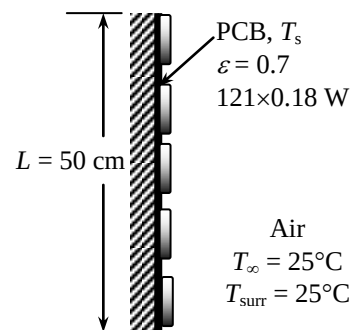
**Properties** Based on the problem statement, the properties of air at 1 atm and the anticipated film temperature of  $(T_s + T_\infty)/2 = (35 + 25)/2 = 30^\circ\text{C}$  are (Table A-15)

$$k = 0.02588 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.608 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7282$$

$$\beta = \frac{1}{T_f} = \frac{1}{(30 + 273)\text{K}} = 0.0033 \text{ K}^{-1}$$



**Analysis** The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown. We start the solution process by “guessing” the surface temperature to be  $35^\circ\text{C}$  for the evaluation of the properties and  $h$ . We will check the accuracy of this guess later and repeat the calculations if necessary. The characteristic length in this case is the height of the board,  $L_c = L = 0.5 \text{ m}$ . Then,

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.0033 \text{ K}^{-1})(35 - 25 \text{ K})(0.5 \text{ m})^3}{(1.608 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7282) = 1.140 \times 10^8$$

$$\text{Nu} = \left\{ 0.825 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.492}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(1.140 \times 10^8)^{1/6}}{\left[ 1 + \left( \frac{0.492}{0.7282} \right)^{9/16} \right]^{8/27}} \right\}^2 = 63.72$$

$$h = \frac{k}{L} \text{Nu} = \frac{0.02588 \text{ W/m}\cdot^\circ\text{C}}{0.5 \text{ m}} (63.72) = 3.30 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = (0.5 \text{ m})^2 = 0.25 \text{ m}^2$$

Considering both natural convection and radiation, the total rate of heat loss can be expressed as

$$\begin{aligned} \dot{Q} &= hA_s(T_s - T_\infty) + \varepsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4) \\ (121 \times 0.18) \text{ W} &= (3.30 \text{ W/m}^2\cdot^\circ\text{C})(0.25 \text{ m}^2)(T_s - 25)^\circ\text{C} \\ &\quad + (0.7)(0.25 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[(T_s + 273 \text{ K})^4 - (25 + 273 \text{ K})^4] \end{aligned}$$

Its solution is

$$T_s = 36.2^\circ\text{C}$$

which is sufficiently close to the assumed value in the evaluation of properties and  $h$ . Therefore, there is no need to repeat calculations by reevaluating the properties and  $h$  at the new film temperature.

**Discussion** The assumed film temperature of  $T_f = 30^\circ\text{C}$  is an appropriate assumption, since the determined  $T_s = 36.2^\circ\text{C}$  would give a film temperature of  $T_f = 30.6^\circ\text{C}$ . Otherwise,  $T_s$  would have to be solved iteratively.

**9-35** A circuit board containing square chips is positioned horizontally in a room. The surface temperature of the chips is to be determined for two orientations.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm. 4 The heat transfer from the back side of the circuit board is negligible.

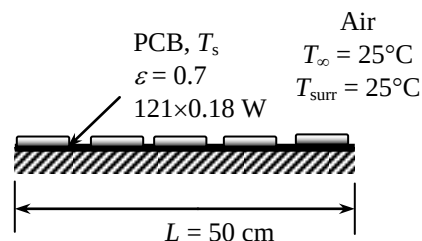
**Properties** Based on the problem statement, the properties of air at 1 atm and the anticipated film temperature of  $(T_s + T_\infty)/2 = (35 + 25)/2 = 30^\circ\text{C}$  are (Table A-15)

$$k = 0.02588 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.608 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7282$$

$$\beta = \frac{1}{T_f} = \frac{1}{(30 + 273)\text{K}} = 0.0033 \text{ K}^{-1}$$



**Analysis** The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown. We start the solution process by “guessing” the surface temperature to be  $35^\circ\text{C}$  for the evaluation of the properties and  $h$ . The characteristic length for both cases is determined from

$$L_c = \frac{A_s}{p} = \frac{(0.5 \text{ m})^2}{2[(0.5 \text{ m}) + (0.5 \text{ m})]} = 0.125 \text{ m}.$$

$$\text{Then, } Ra = \frac{g\beta(T_s - T_\infty)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.0033 \text{ K}^{-1})(35 - 25 \text{ K})(0.125 \text{ m})^3}{(1.608 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7282) = 1.781 \times 10^6$$

(a) Chips (hot surface) facing up:

$$Nu = 0.54 Ra^{1/4} = 0.54 (1.781 \times 10^6)^{1/4} = 19.73$$

$$h = \frac{k}{L_c} Nu = \frac{0.02588 \text{ W/m}\cdot^\circ\text{C}}{0.125 \text{ m}} (19.73) = 4.08 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = (0.5 \text{ m})^2 = 0.25 \text{ m}^2$$

Considering both natural convection and radiation, the total rate of heat loss can be expressed as

$$\begin{aligned} \dot{Q} &= hA_s(T_s - T_\infty) + \varepsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4) \\ (121 \times 0.18) \text{ W} &= (4.08 \text{ W/m}^2\cdot^\circ\text{C})(0.25 \text{ m}^2)(T_s - 25)^\circ\text{C} \\ &\quad + (0.7)(0.25 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[(T_s + 273 \text{ K})^4 - (25 + 273 \text{ K})^4] \end{aligned}$$

Its solution is  $T_s = 35.2^\circ\text{C}$

which is sufficiently close to the assumed value. Therefore, there is no need to repeat calculations.

(b) Chips (hot surface) facing down:

$$Nu = 0.27 Ra^{1/4} = 0.27 (1.781 \times 10^6)^{1/4} = 9.863$$

$$h = \frac{k}{L_c} Nu = \frac{0.02588 \text{ W/m}\cdot^\circ\text{C}}{0.125 \text{ m}} (9.863) = 2.04 \text{ W/m}^2\cdot^\circ\text{C}$$

Considering both natural convection and radiation, the total rate of heat loss can be expressed as

$$\begin{aligned} \dot{Q} &= hA_s(T_s - T_\infty) + \varepsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4) \\ (121 \times 0.18) \text{ W} &= (2.04 \text{ W/m}^2\cdot^\circ\text{C})(0.25 \text{ m}^2)(T_s - 25)^\circ\text{C} \\ &\quad + (0.7)(0.25 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[(T_s + 273 \text{ K})^4 - (25 + 273 \text{ K})^4] \end{aligned}$$

Its solution is  $T_s = 38.3^\circ\text{C}$

which is sufficiently close to the assumed value of  $35^\circ\text{C}$  in the evaluation of properties and  $h$ . Therefore, there is no need to repeat calculations.

**Discussion** The assumed film temperature of  $T_f = 30^\circ\text{C}$  is an appropriate assumption, since the determined  $T_s = 38.3^\circ\text{C}$  would give a film temperature of  $T_f = 31.7^\circ\text{C}$ . Otherwise,  $T_s$  would have to be solved iteratively.

**9-36** A printed circuit board (PCB) is placed in a room. The average temperature of the hot surface of the board is to be determined for different orientations.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm. 3 The heat loss from the back surface of the board is negligible.

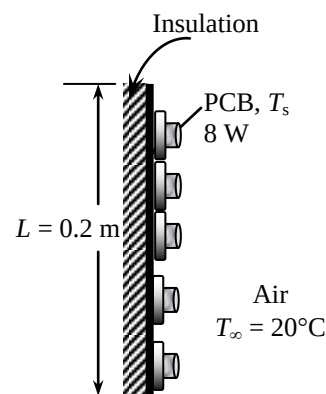
**Properties** We evaluate air properties at a film temperature of  $(T_s + T_\infty)/2 = 32.5^\circ\text{C}$  and 1 atm based on the problem statement. Then, for an air temperature of  $T_\infty = 20^\circ\text{C}$ , the corresponding surface temperature is  $T_s = 45^\circ\text{C}$ . The properties of air at 1 atm and  $32.5^\circ\text{C}$  are (Table A-15)

$$k = 0.02607 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.631 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7275$$

$$\beta = \frac{1}{T_f} = \frac{1}{(32.5 + 273)\text{K}} = 0.003273 \text{ K}^{-1}$$



**Analysis** The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown.

(a) **Vertical PCB**. We start the solution process by “guessing” the surface temperature to be  $45^\circ\text{C}$  for the evaluation of the properties and  $h$ . We will check the accuracy of this guess later and repeat the calculations if necessary. The characteristic length in this case is the height of the PCB,  $L_c = L = 0.2 \text{ m}$ . Then,

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003273 \text{ K}^{-1})(45 - 20 \text{ K})(0.2 \text{ m})^3}{(1.631 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7275) = 1.756 \times 10^7$$

$$\text{Nu} = \left\{ 0.825 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.492}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(1.756 \times 10^7)^{1/6}}{\left[ 1 + \left( \frac{0.492}{0.7275} \right)^{9/16} \right]^{8/27}} \right\}^2 = 36.78$$

$$h = \frac{k}{L} \text{Nu} = \frac{0.02607 \text{ W/m}\cdot^\circ\text{C}}{0.2 \text{ m}} (36.78) = 4.794 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = (0.15 \text{ m})(0.2 \text{ m}) = 0.03 \text{ m}^2$$

Heat loss by both natural convection and radiation heat can be expressed as

$$\dot{Q} = hA_s(T_s - T_\infty) + \varepsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4)$$

$$8 \text{ W} = (4.794 \text{ W/m}^2\cdot^\circ\text{C})(0.03 \text{ m}^2)(T_s - 20)^\circ\text{C} + (0.8)(0.03 \text{ m}^2)(5.67 \times 10^{-8})[(T_s + 273)^4 - (20 + 273 \text{ K})^4]$$

Its solution is

$$T_s = 46.6^\circ\text{C}$$

which is sufficiently close to the assumed value of  $45^\circ\text{C}$  for the evaluation of the properties and  $h$ .

(b) **Horizontal, hot surface facing up** Again we assume the surface temperature to be  $45^\circ\text{C}$  and use the properties evaluated above. The characteristic length in this case is

$$L_c = \frac{A_s}{p} = \frac{(0.20 \text{ m})(0.15 \text{ m})}{2(0.2 \text{ m} + 0.15 \text{ m})} = 0.0429 \text{ m}$$

Then

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003273 \text{ K}^{-1})(45 - 20 \text{ K})(0.0429 \text{ m})^3}{(1.631 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7275) = 1.728 \times 10^5$$

$$\text{Nu} = 0.54 \text{Ra}^{1/4} = 0.54(1.728 \times 10^5)^{1/4} = 11.01$$

$$h = \frac{k}{L_c} \text{Nu} = \frac{0.02607 \text{ W/m}\cdot^\circ\text{C}}{0.0429 \text{ m}} (11.01) = 6.696 \text{ W/m}^2\cdot^\circ\text{C}$$

Heat loss by both natural convection and radiation heat can be expressed as

$$\dot{Q} = hA_s(T_s - T_\infty) + \varepsilon A_s \sigma (T_s^4 - T_{surr}^4)$$

$$8 \text{ W} = (6.696 \text{ W/m}^2 \cdot ^\circ\text{C})(0.03 \text{ m}^2)(T_s - 20)^\circ\text{C} + (0.8)(0.03 \text{ m}^2)(5.67 \times 10^{-8})[(T_s + 273)^4 - (20 + 273 \text{ K})^4]$$

Its solution is

$$T_s = \mathbf{42.6^\circ\text{C}}$$

which is sufficiently close to the assumed value of  $45^\circ\text{C}$  in the evaluation of the properties and  $h$ .

(c) **Horizontal, hot surface facing down** Again we assume the surface temperature to be  $45^\circ\text{C}$  and use the properties evaluated above. The characteristic length in this case is, from part (b),  $L_c = 0.0429 \text{ m}$ . Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L_c^3}{\nu^2} Pr = \frac{(9.81 \text{ m/s}^2)(0.003273 \text{ K}^{-1})(45 - 20 \text{ K})(0.0429 \text{ m})^3}{(1.631 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7275) = 1.728 \times 10^5$$

$$Nu = 0.27 Ra^{1/4} = 0.27(1.728 \times 10^5)^{1/4} = 5.505$$

$$h = \frac{k}{L_c} Nu = \frac{0.02607 \text{ W/m} \cdot ^\circ\text{C}}{0.0429 \text{ m}} (5.505) = 3.3345 \text{ W/m}^2 \cdot ^\circ\text{C}$$

Considering both natural convection and radiation heat losses

$$\dot{Q} = hA_s(T_s - T_\infty) + \varepsilon A_s \sigma (T_s^4 - T_{surr}^4)$$

$$8 \text{ W} = (3.345 \text{ W/m}^2 \cdot ^\circ\text{C})(0.03 \text{ m}^2)(T_s - 20)^\circ\text{C} + (0.8)(0.03 \text{ m}^2)(5.67 \times 10^{-8})[(T_s + 273)^4 - (20 + 273 \text{ K})^4]$$

Its solution is

$$T_s = \mathbf{50.7^\circ\text{C}}$$

which is sufficiently close to the assumed value. Therefore, there is no need to repeat the calculations.

**Discussion** The assumed film temperature of  $T_f = 32.5^\circ\text{C}$  is an appropriate assumption, since the determined  $T_s = 46.6^\circ\text{C}$ ,  $T_s = 42.6^\circ\text{C}$ , and  $T_s = 50.7^\circ\text{C}$  in parts *a*, *b*, and *c*, respectively would give film temperatures of  $T_f = 33.3^\circ\text{C}$ ,  $T_f = 31.3^\circ\text{C}$ , and  $T_f = 35.4^\circ\text{C}$ , respectively. Otherwise,  $T_s$  would have to be solved iteratively.



**9-37** Prob. 9-36 is reconsidered. The effects of the room temperature and the emissivity of the board on the temperature of the hot surface of the board for different orientations of the board are to be investigated.

**Analysis** The problem is solved using EES, and the solution is given below.

**"GIVEN"**

L=0.2 [m]  
w=0.15 [m]  
T\_infinity=20 [C]  
Q\_dot=8 [W]  
epsilon=0.8  
T\_surr=T\_infinity

**"PROPERTIES"**

Fluid\$='air'  
k=Conductivity(Fluid\$, T=T\_film)  
Pr=Prandtl(Fluid\$, T=T\_film)  
rho=Density(Fluid\$, T=T\_film, P=101.3)  
mu=Viscosity(Fluid\$, T=T\_film)  
nu=mu/rho  
beta=1/(T\_film+273)  
T\_film=1/2\*(T\_s\_a+T\_infinity)  
sigma=5.67E-8 [W/m^2-K^4] "Stefan-Boltzmann constant"  
g=9.807 [m/s^2] "gravitational acceleration"

**"ANALYSIS"**

**"(a), plate is vertical"**

delta\_a=L  
Ra\_a=(g\*beta\*(T\_s\_a-T\_infinity)\*delta\_a^3)/nu^2\*Pr  
Nusselt\_a=0.59\*Ra\_a^0.25  
h\_a=k/delta\_a\*Nusselt\_a  
A=w\*L  
Q\_dot=h\_a\*A\*(T\_s\_a-T\_infinity)+epsilon\*A\*sigma\*((T\_s\_a+273)^4-(T\_surr+273)^4)

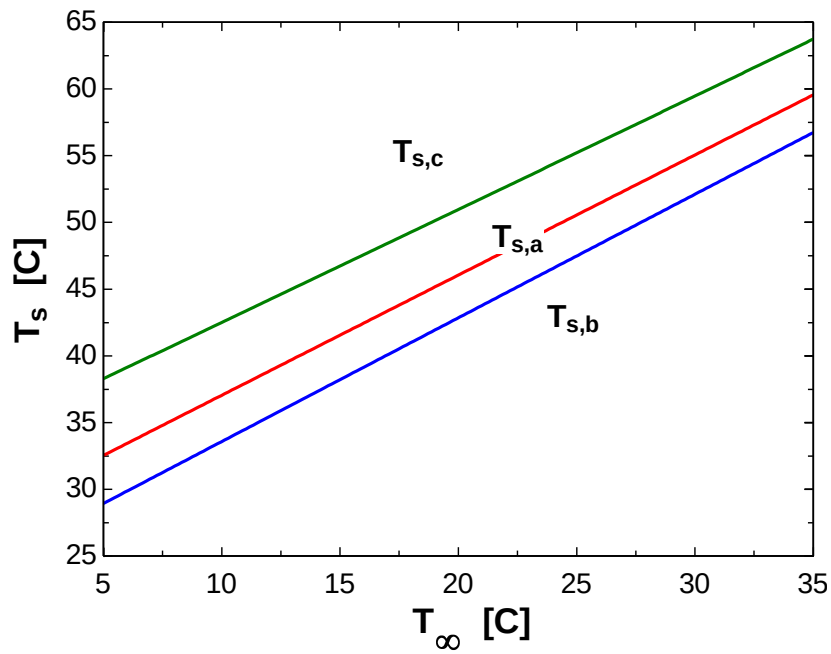
**"(b), plate is horizontal with hot surface facing up"**

delta\_b=A/p  
p=2\*(w+L)  
Ra\_b=(g\*beta\*(T\_s\_b-T\_infinity)\*delta\_b^3)/nu^2\*Pr  
Nusselt\_b=0.54\*Ra\_b^0.25  
h\_b=k/delta\_b\*Nusselt\_b  
Q\_dot=h\_b\*A\*(T\_s\_b-T\_infinity)+epsilon\*A\*sigma\*((T\_s\_b+273)^4-(T\_surr+273)^4)

**"(c), plate is horizontal with hot surface facing down"**

delta\_c=delta\_b  
Ra\_c=Ra\_b  
Nusselt\_c=0.27\*Ra\_c^0.25  
h\_c=k/delta\_c\*Nusselt\_c  
Q\_dot=h\_c\*A\*(T\_s\_c-T\_infinity)+epsilon\*A\*sigma\*((T\_s\_c+273)^4-(T\_surr+273)^4)

| $T_\infty$<br>[F] | $T_{s,a}$<br>[C] | $T_{s,b}$<br>[C] | $T_{s,c}$<br>[C] |
|-------------------|------------------|------------------|------------------|
| 5                 | 32.54            | 28.93            | 38.29            |
| 7                 | 34.34            | 30.79            | 39.97            |
| 9                 | 36.14            | 32.65            | 41.66            |
| 11                | 37.95            | 34.51            | 43.35            |
| 13                | 39.75            | 36.36            | 45.04            |
| 15                | 41.55            | 38.22            | 46.73            |
| 17                | 43.35            | 40.07            | 48.42            |
| 19                | 45.15            | 41.92            | 50.12            |
| 21                | 46.95            | 43.78            | 51.81            |
| 23                | 48.75            | 45.63            | 53.51            |
| 25                | 50.55            | 47.48            | 55.21            |
| 27                | 52.35            | 49.33            | 56.91            |
| 29                | 54.16            | 51.19            | 58.62            |
| 31                | 55.96            | 53.04            | 60.32            |
| 33                | 57.76            | 54.89            | 62.03            |
| 35                | 59.56            | 56.74            | 63.74            |



**9-38** C&S A horizontal ASTM A240 410S stainless steel plate has its upper surface subjected to natural convection with cold air. Heat is added to the plate at 70 W, and the lowest temperature that the cold air can reach without causing the plate surface temperature to cool below  $-30^{\circ}\text{C}$  is to be determined.

**Assumptions** 1 Steady state conditions. 2 Air is an ideal gas. 3 The local atmospheric pressure is 1 atm. 4 Thermal radiation is negligible.

**Properties** The properties of air at the film temperature of  $T_f = -50^{\circ}\text{C}$  are (Table A-15)  $\text{Pr} = 0.7440$ ,  $k = 0.01979 \text{ W/m}\cdot\text{K}$ ,  $\nu = 9.319 \times 10^{-6} \text{ m}^2/\text{s}$ ,  $\beta = 1/T_f = 1/(-50 + 273 \text{ K}) = 0.004484 \text{ K}^{-1}$

**Analysis** The characteristic length of the plate is

$$L_c = \frac{A_s}{p} = \frac{L^2}{4L} = \frac{L}{4} = \frac{0.6 \text{ m}}{4} = 0.15 \text{ m}$$

The Rayleigh number is

$$\text{Ra}_L = \frac{g\beta(T_s - T_\infty)L_c^3}{\nu^2} \text{Pr}$$

The Nusselt number for natural convection with horizontal hot surface facing up is determined from

$$\text{Nu} = \frac{hL_c}{k} = 0.15 \text{ Ra}_L^{1/3}$$

(Note: The Rayleigh number is assumed to be  $10^7 < \text{Ra}_L < 10^{11}$ . We will verify the Rayleigh number after finding the  $T_\infty$ . If  $10^4 < \text{Ra}_L < 10^7$ , then  $\text{Nu} = 0.54 \text{ Ra}_L^{1/4}$  would be used.)

So, the natural convection heat transfer coefficient is

$$h = 0.15 \frac{k}{L_c} \left[ \frac{g\beta(T_s - T_\infty)L_c^3}{\nu^2} \text{Pr} \right]^{1/3}$$

From energy balance, the heat rate added to the plate is equal to the heat transfer rate from the plate by natural convection to the cold air:

$$\dot{Q} = hA_s(T_s - T_\infty) = hL^2(T_s - T_\infty) = 70 \text{ W}$$

By substituting the equation for  $\text{Ra}_L$  into the equation for  $h$  and then substituting the resulting equation in the above Newton's law of cooling equation yields

$$0.15 \frac{k}{L_c} \left[ \frac{g\beta(T_s - T_\infty)L_c^3}{\nu^2} \text{Pr} \right]^{1/3} L^2(T_s - T_\infty) = 70 \text{ W}$$

or

$$0.15 \frac{(0.01979 \text{ W/m}\cdot\text{K})}{0.15 \text{ m}} \left[ \frac{(9.81 \text{ m/s}^2)(0.004484 \text{ K}^{-1})(T_s - T_\infty)(0.15 \text{ m})^3}{(9.319 \times 10^{-6} \text{ m}^2/\text{s})^2} (0.7440) \right]^{1/3} (0.6 \text{ m})^2 (T_s - T_\infty) = 70 \text{ W}$$

The temperature of the cold air can be solved implicitly by trial-and-error with  $T_s = -30^{\circ}\text{C}$ , which yields

$$T_\infty = -59.4^{\circ}\text{C}$$

The Rayleigh number with  $T_\infty = -59.4^{\circ}\text{C}$  is

$$\text{Ra}_L = \frac{g\beta(T_s - T_\infty)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.004484 \text{ K}^{-1})(-30 + 59.4 \text{ K})(0.15 \text{ m})^3}{(9.319 \times 10^{-6} \text{ m}^2/\text{s})^2} (0.7440) = 3.74 \times 10^7$$

So, the correlation for the Nusselt number is appropriate. (If  $\text{Ra}_L < 10^7$ , then the calculation for  $T_\infty$  would need to be repeated with a suitable correlation.)

**Discussion** With heat being added to the stainless steel plate at 70 W, the plate surface temperature would reach  $-30^{\circ}\text{C}$  by natural convection with the cold air at  $-59.4^{\circ}\text{C}$ . Thus, the lowest temperature that the air can reach without causing the plate surface temperature to cool below its minimum suitable temperature is  $-59.4^{\circ}\text{C}$ .

With  $T_\infty = -59.4^{\circ}\text{C}$  and  $T_s = -30^{\circ}\text{C}$ , the film temperature becomes  $-44.7^{\circ}\text{C}$ . Thus,  $-50^{\circ}\text{C}$  is a reasonable temperature to evaluate the properties of air and there is no need to repeat the calculations.

**9-39** Heat is generated in a horizontal plate while heat is lost from it by convection and radiation. The temperature of the plate when steady operating conditions are reached is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm.

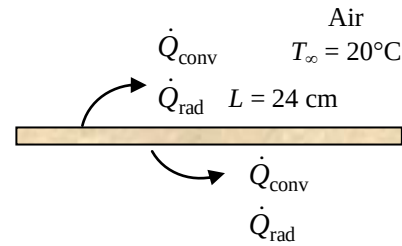
**Properties** We assume the surface temperature to be 50°C. Then the properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (50 + 20)/2 = 35^\circ\text{C}$  are (Table A-15)

$$k = 0.02625 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.655 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7268$$

$$\beta = \frac{1}{T_f} = \frac{1}{(35 + 273)\text{K}} = 0.003247 \text{ K}^{-1}$$



**Analysis** The characteristic length in this case is

$$L_c = \frac{A_s}{p} = \frac{(0.24 \text{ m})(0.20 \text{ m})}{2[(0.24 \text{ m}) + (0.20 \text{ m})]} = 0.05455 \text{ m}$$

The Rayleigh number is

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003247 \text{ K}^{-1})(50 - 20 \text{ K})(0.05455 \text{ m})^3}{(1.655 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7268) = 411,500$$

The Nusselt number relation for the top surface of the plate (horizontal hot surface, facing up) is

$$\text{Nu} = 0.54 \text{Ra}^{0.25} = 0.54(411,500)^{0.25} = 13.68$$

Then

$$h = \frac{k}{L_c} \text{Nu} = \frac{0.02625 \text{ W/m}\cdot^\circ\text{C}}{0.05455 \text{ m}} (13.68) = 6.581 \text{ W/m}^2\cdot^\circ\text{C}$$

and

$$\dot{Q}_{\text{top}} = hA(T_s - T_\infty) = (6.581 \text{ W/m}^2\cdot^\circ\text{C})(0.24 \times 0.20 \text{ m}^2)(T_s - 20)^\circ\text{C} = 0.3159(T_s - 20)$$

The Nusselt number relation for the bottom surface of the plate (horizontal hot surface, facing down) is

$$\text{Nu} = 0.27 \text{Ra}^{0.25} = 0.27(411,500)^{0.25} = 6.838$$

Then

$$h = \frac{k}{L_c} \text{Nu} = \frac{0.02625 \text{ W/m}\cdot^\circ\text{C}}{0.05455 \text{ m}} (6.838) = 3.291 \text{ W/m}^2\cdot^\circ\text{C}$$

$$\dot{Q}_{\text{bottom}} = hA(T_s - T_\infty) = (3.291 \text{ W/m}^2\cdot^\circ\text{C})(0.24 \times 0.20 \text{ m}^2)(T_s - 20)^\circ\text{C} = 0.1580(T_s - 20)$$

Considering that radiation heat loss to surroundings occur both from top and bottom surfaces, it may be expressed as

$$\begin{aligned} \dot{Q}_{\text{rad}} &= 2\varepsilon A \sigma (T_s^4 - T_{\text{surr}}^4) \\ &= (0.9)(2)(0.24 \times 0.20 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4) [(T_s + 273 \text{ K})^4 - (17 + 273 \text{ K})^4] \\ &= 4.899 \times 10^{-9} [(T_s + 273 \text{ K})^4 - (17 + 273 \text{ K})^4] \end{aligned}$$

When the heat lost from the plate equals to the heat generated, the steady operating conditions are reached. The surface temperature in this case can be determined by trial-error or using EES to be

$$\begin{aligned} \dot{Q}_{\text{total}} &= \dot{Q}_{\text{top}} + \dot{Q}_{\text{bottom}} + \dot{Q}_{\text{rad}} \\ 20 \text{ W} &= 0.3159(T_s - 20) + 0.1580(T_s - 20) + 4.899 \times 10^{-9} [(T_s + 273 \text{ K})^4 - (17 + 273 \text{ K})^4] \\ T_s &= \mathbf{38.3^\circ\text{C}} \end{aligned}$$

The surface temperature is about 12°C below the assumed surface temperature of 50°C. For more accuracy, it is recommended to repeat the calculations with a lower value of surface temperature.

**9-40** Absorber plates whose back side is heavily insulated is placed horizontally outdoors. Solar radiation is incident on the plate. The equilibrium temperature of the plate is to be determined for two cases.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm.

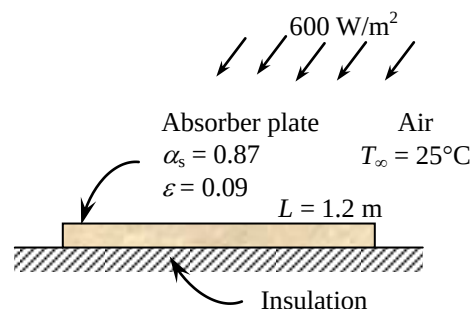
**Properties** The properties of air at 1 atm and the anticipated film temperature of  $(T_s + T_\infty)/2 = (115 + 25)/2 = 70^\circ\text{C}$  based on the problem statement are (Table A-15)

$$k = 0.02881 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.995 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7177$$

$$\beta = \frac{1}{T_f} = \frac{1}{(70 + 273)\text{K}} = 0.002915 \text{ K}^{-1}$$



**Analysis** The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown. We start the solution process by “guessing” the surface temperature to be  $115^\circ\text{C}$  for the evaluation of the properties and  $h$ . We will check the accuracy of this guess later and repeat the calculations if necessary. The characteristic length in this case is

$$L_c = \frac{A_s}{p} = \frac{(1.2 \text{ m})(0.8 \text{ m})}{2(1.2 \text{ m} + 0.8 \text{ m})} = 0.24 \text{ m}$$

Then,

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.002915 \text{ K}^{-1})(115 - 25 \text{ K})(0.24 \text{ m})^3}{(1.995 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7177) = 6.414 \times 10^7$$

The Nusselt number relation for the horizontal hot surface, facing up is

$$\text{Nu} = 0.15 \text{Ra}^{1/3} = 0.15(6.414 \times 10^7)^{1/3} = 60.04$$

$$h = \frac{k}{L_c} \text{Nu} = \frac{0.02881 \text{ W/m}\cdot^\circ\text{C}}{0.24 \text{ m}} (60.04) = 7.208 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = (0.8 \text{ m})(1.2 \text{ m}) = 0.96 \text{ m}^2$$

In steady operation, the heat gain by the plate by absorption of solar radiation must be equal to the heat loss by natural convection and radiation. Therefore,

$$\dot{Q} = \alpha \dot{q} A_s = (0.87)(600 \text{ W/m}^2)(0.96 \text{ m}^2) = 501.1 \text{ W}$$

$$\dot{Q} = h A_s (T_s - T_\infty) + \varepsilon A_s \sigma (T_s^4 - T_{\text{sky}}^4)$$

$$501.1 \text{ W} = (7.208 \text{ W/m}^2\cdot^\circ\text{C})(0.96 \text{ m}^2)(T_s - 25)^\circ\text{C} + (0.09)(0.96 \text{ m}^2)(5.67 \times 10^{-8})[(T_s + 273)^4 - (10 + 273 \text{ K})^4]$$

Its solution is  $T_s = 89.7^\circ\text{C}$

which is not very close to the assumed value of  $115^\circ\text{C}$ . We repeat the calculations at a new anticipated surface temperature of  $95^\circ\text{C}$ . The properties are to be evaluated at the film temperature of  $(95 + 25)/2 = 60^\circ\text{C}$ .

$$k = 0.02808 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.896 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7202$$

$$\beta = \frac{1}{T_f} = \frac{1}{(60 + 273)\text{K}} = 0.003003 \text{ K}^{-1}$$

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003003 \text{ K}^{-1})(95 - 25 \text{ K})(0.24 \text{ m})^3}{(1.896 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7202) = 5.711 \times 10^7$$

$$\text{Nu} = 0.15 \text{Ra}^{1/3} = 0.15(5.711 \times 10^7)^{1/3} = 57.77$$

$$h = \frac{k}{L_c} \text{Nu} = \frac{0.02808 \text{ W/m}\cdot^\circ\text{C}}{0.24 \text{ m}} (57.77) = 6.759 \text{ W/m}^2\cdot^\circ\text{C}$$

$$\begin{aligned}\dot{Q} &= hA_s(T_s - T_\infty) + \varepsilon A_s \sigma (T_s^4 - T_{sky}^4) \\ 501.1 \text{ W} &= (6.759 \text{ W/m}^2 \cdot ^\circ\text{C})(0.96 \text{ m}^2)(T_s - 25)^\circ\text{C} \\ &\quad + (0.09)(0.96 \text{ m}^2)(5.67 \times 10^{-8})[(T_s + 273)^4 - (10 + 273 \text{ K})^4] \\ T_s &= \mathbf{93.5^\circ\text{C}}\end{aligned}$$

This is close to the assumed surface temperature of  $95^\circ\text{C}$ . Therefore, there is no need to repeat the calculations.

If the absorber plate is made of ordinary aluminum which has a solar absorptivity of 0.28 and an emissivity of 0.07, the rate of solar gain becomes

$$\dot{Q} = \alpha \dot{q} A_s = (0.28)(600 \text{ W/m}^2)(0.96 \text{ m}^2) = 161.3 \text{ W}$$

Again noting that in steady operation the heat gain by the plate by absorption of solar radiation must be equal to the heat loss by natural convection and radiation, and using the convection coefficient determined above for convenience,

$$\begin{aligned}\dot{Q} &= hA_s(T_s - T_\infty) + \varepsilon A_s \sigma (T_s^4 - T_{sky}^4) \\ 161.3 \text{ W} &= (6.629 \text{ W/m}^2 \cdot ^\circ\text{C})(0.96 \text{ m}^2)(T_s - 25)^\circ\text{C} + (0.07)(0.96 \text{ m}^2)(5.67 \times 10^{-8})[(T_s + 273)^4 - (10 + 273 \text{ K})^4]\end{aligned}$$

Its solution is  $T_s = 47.8^\circ\text{C}$

Repeating the calculations at the new anticipated surface temperature of  $55^\circ\text{C}$  and the film temperature of  $(55+25)/2=40^\circ\text{C}$ , we obtain

$$h = 5.312 \text{ W/m}^2 \cdot ^\circ\text{C} \text{ and } T_s = \mathbf{53.0^\circ\text{C}}$$

**9-41** An absorber plate whose back side is heavily insulated is placed horizontally outdoors. Solar radiation is incident on the plate. The equilibrium temperature of the plate is to be determined for two cases.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm.

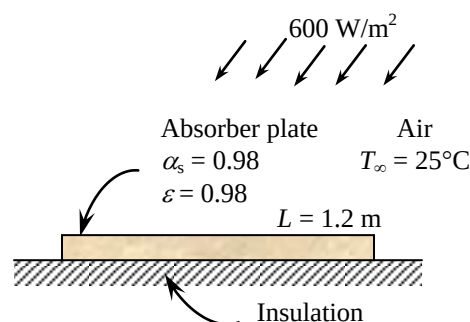
**Properties** The properties of air at 1 atm and the anticipated film temperature of  $(T_s + T_\infty)/2 = (65 + 25)/2 = 45^\circ\text{C}$  based on the problem statement are (Table A-15)

$$k = 0.02699 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.750 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7241$$

$$\beta = \frac{1}{T_f} = \frac{1}{(45 + 273)\text{K}} = 0.003145 \text{ K}^{-1}$$



**Analysis** The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown. We start the solution process by “guessing” the surface temperature to be  $65^\circ\text{C}$  for the evaluation of the properties and  $h$ . We will check the accuracy of this guess later and repeat the calculations if necessary. The characteristic length in this case is

$$L_c = \frac{A_s}{p} = \frac{(1.2 \text{ m})(0.8 \text{ m})}{2(1.2 \text{ m} + 0.8 \text{ m})} = 0.24 \text{ m}$$

Then,

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.00312 \text{ K}^{-1})(65 - 25 \text{ K})(0.24 \text{ m})^3}{(1.750 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7241) = 4.033 \times 10^7$$

The Nusselt number relation for the horizontal hot surface, facing up is

$$\text{Nu} = 0.15 \text{Ra}^{1/3} = 0.15(4.033 \times 10^7)^{1/3} = 51.44$$

$$h = \frac{k}{L_c} \text{Nu} = \frac{0.02699 \text{ W/m}\cdot^\circ\text{C}}{0.24 \text{ m}} (51.44) = 5.785 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = (0.8 \text{ m})(1.2 \text{ m}) = 0.96 \text{ m}^2$$

In steady operation, the heat gain by the plate by absorption of solar radiation must be equal to the heat loss by natural convection and radiation. Therefore,

$$\dot{Q} = \alpha \dot{q} A_s = (0.98)(600 \text{ W/m}^2)(0.96 \text{ m}^2) = 564.5 \text{ W}$$

$$\dot{Q} = h A_s (T_s - T_\infty) + \varepsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4)$$

$$564.5 \text{ W} = (5.785 \text{ W/m}^2\cdot^\circ\text{C})(0.96 \text{ m}^2)(T_s - 25)^\circ\text{C} + (0.98)(0.96 \text{ m}^2)(5.67 \times 10^{-8})[(T_s + 273)^4 - (10 + 273 \text{ K})^4]$$

Its solution is  $T_s = 64.2^\circ\text{C}$

which is very close to the assumed value. Therefore there is no need to repeat calculations.

For a white painted absorber plate, the solar absorptivity is 0.26 and the emissivity is 0.90. Then the rate of solar gain becomes

$$\dot{Q} = \alpha \dot{q} A_s = (0.26)(600 \text{ W/m}^2)(0.96 \text{ m}^2) = 149.8 \text{ W}$$

Again noting that in steady operation the heat gain by the plate by absorption of solar radiation must be equal to the heat loss by natural convection and radiation, and using the convection coefficient determined above for convenience (actually, we should calculate the new  $h$  using data at a lower temperature, and iterating if necessary for better accuracy),

$$\dot{Q} = h A_s (T_s - T_\infty) + \varepsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4)$$

$$149.8 \text{ W} = (5.785 \text{ W/m}^2\cdot^\circ\text{C})(0.96 \text{ m}^2)(T_s - 25)^\circ\text{C} + (0.90)(0.96 \text{ m}^2)(5.67 \times 10^{-8})[(T_s + 273)^4 - (10 + 273 \text{ K})^4]$$

Its solution is  $T_s = 32.1^\circ\text{C}$

Repeating the calculations at the new anticipated surface temperature of  $35^\circ\text{C}$  and the film temperature of  $(35 + 25)/2 = 30^\circ\text{C}$ , we obtain

$$h = 3.764 \text{ W/m}^2\cdot^\circ\text{C} \text{ and } T_s = 33.6^\circ\text{C}$$

**9-42** C&S A horizontal ASTM A203 B steel plate has its lower surface subjected to natural convection with cold hydrogen gas at  $-70^{\circ}\text{C}$ . Thermal radiation exchange occurs between the plate surface and the surrounding. The rate of heat addition to the plate needed to keep the plate surface from cooling below  $-30^{\circ}\text{C}$  is to be determined.

**Assumptions** 1 Steady state conditions. 2  $\text{H}_2$  is an ideal gas. 3 The local atmospheric pressure is 1 atm.

**Properties** The properties of  $\text{H}_2$  gas at the film temperature of  $T_f = (T_{\infty} + T_s)/2 = (-70 - 30)/2 = -50^{\circ}\text{C}$  are (Table A-16)  $\text{Pr} = 0.6562$ ,  $k = 0.1404 \text{ W/m}\cdot\text{K}$ ,  $\nu = 6.624 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\beta = 1/T_f = 1/(-50 + 273 \text{ K}) = 0.004484 \text{ K}^{-1}$

**Analysis** The characteristic length of the plate is

$$L_c = \frac{A_s}{p} = \frac{L^2}{4L} = \frac{L}{4} = \frac{0.6 \text{ m}}{4} = 0.15 \text{ m}$$

The Rayleigh number is

$$\text{Ra}_L = \frac{g\beta(T_s - T_{\infty})L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.004484 \text{ K}^{-1})(-30 + 70 \text{ K})(0.15 \text{ m})^3}{(6.624 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.6562) = 8.881 \times 10^5$$

The Nusselt number for natural convection with horizontal hot surface facing down is determined from

$$\text{Nu} = \frac{hL_c}{k} = 0.27 \text{ Ra}_L^{1/4}$$

So, the natural convection heat transfer coefficient is

$$h = 0.27 \frac{k}{L_c} \text{ Ra}_L^{1/4} = 0.27 \frac{(0.1404 \text{ W/m}\cdot\text{K})}{0.15 \text{ m}} (8.881 \times 10^5)^{1/4} = 7.758 \text{ W/m}^2 \cdot \text{K}$$

The heat transfer rate required to keep the plate surface at  $-30^{\circ}\text{C}$  can be determined from the total heat loss from the lower plate surface by natural convection and thermal radiation:

$$\begin{aligned} \dot{Q} &= \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}} = hA_s(T_s - T_{\infty}) + \varepsilon\sigma A_s(T_s^4 - T_{\text{surr}}^4) \\ &= (7.758 \text{ W/m}^2 \cdot \text{K})(0.6 \text{ m})^2(-30 + 70 \text{ K}) + (0.3)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(0.6 \text{ m})^2(243^4 - 200^4 \text{ K}^4) \\ &= 111.72 \text{ W} + 10.95 \text{ W} \\ &= \mathbf{122.7 \text{ W}} \end{aligned}$$

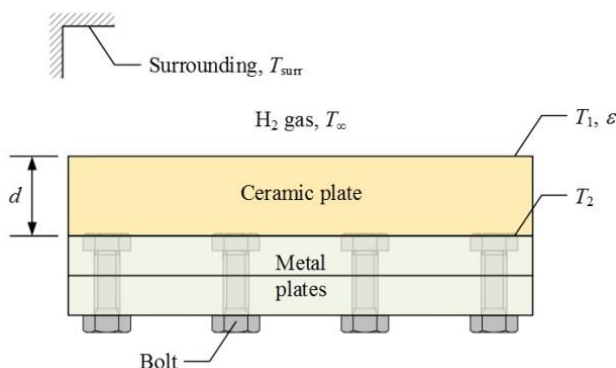
**Discussion** With a heat addition rate of 123 W or higher, the plate surface exposed to natural convection with the cold  $\text{H}_2$  gas and thermal radiation exchange with the surrounding can be maintained from being cooled below  $-30^{\circ}\text{C}$ .

Note that with an emissivity of 0.3, the heat loss from the plate surface by thermal radiation is almost 9% of the total heat loss. Due to the generally low natural convection heat transfer coefficient, thermal radiation can contribute significantly in the overall heat transfer.

**9-43** C&S Stainless steel bolts (ASTM A437 B4B) are used to secure two horizontal metal plates together. A ceramic plate is used for preventing the bolts in the metal plates from cooling below the minimum suitable temperature of  $-30^{\circ}\text{C}$ . The upper surface of the ceramic plate is subjected to natural convection with cold hydrogen gas at  $-60^{\circ}\text{C}$  and thermal radiation exchange with the surrounding. The minimum thickness of the plate is to be determined.

**Assumptions** 1 Steady state conditions. 2  $\text{H}_2$  is an ideal gas. 3 The local atmospheric pressure is 1 atm. 4 Conduction through the plate is one-dimensional. 5 Contact resistance at the interface is negligible.

**Properties** The properties of  $\text{H}_2$  gas at the film temperature of  $T_f = (T_{\infty} + T_s)/2 = (-60 - 40)/2 = -50^{\circ}\text{C}$  are (Table A-16)  $\text{Pr} = 0.6562$ ,  $k = 0.1404 \text{ W/m}\cdot\text{K}$ ,  $\nu = 6.624 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\beta = 1/T_f = 1/(-50 + 273 \text{ K}) = 0.004484 \text{ K}^{-1}$



**Analysis** The characteristic length of the plate is

$$L_c = \frac{A_s}{p} = \frac{L^2}{4L} = \frac{L}{4} = \frac{0.6 \text{ m}}{4} = 0.15 \text{ m}$$

The Rayleigh number is

$$\text{Ra}_L = \frac{g\beta(T_1 - T_{\infty})L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.004484 \text{ K}^{-1})(-40 + 60 \text{ K})(0.15 \text{ m})^3}{(6.624 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.6562) = 4.4405 \times 10^5$$

The Nusselt number for natural convection with horizontal hot surface facing up is determined from

$$\text{Nu} = \frac{hL_c}{k} = 0.54 \text{ Ra}_L^{1/4}$$

So, the natural convection heat transfer coefficient is

$$h = 0.54 \frac{k}{L_c} \text{ Ra}_L^{1/4} = 0.54 \frac{(0.1404 \text{ W/m}\cdot\text{K})}{0.15 \text{ m}} (4.4405 \times 10^5)^{1/4} = 13.05 \text{ W/m}^2 \cdot \text{K}$$

Applying energy balance, with natural convection and thermal radiation at the ceramic plate surface and conduction through the ceramic plate,

$$\dot{q}_{\text{cond}} = \dot{q}_{\text{conv}} + \dot{q}_{\text{rad}}$$

$$k_{\text{plate}} \frac{T_2 - T_1}{d} = h(T_1 - T_{\infty}) + \varepsilon\sigma(T_1^4 - T_{\text{surr}}^4)$$

Solving for the thickness  $d$  of the ceramic plate, with  $T_2 = -30^{\circ}\text{C}$ , yields

$$\begin{aligned} d &= \frac{k_{\text{plate}}(T_2 - T_1)}{h(T_1 - T_{\infty}) + \varepsilon\sigma(T_1^4 - T_{\text{surr}}^4)} \\ &= \frac{(1.4 \text{ W/m}\cdot\text{K})(-30 + 40 \text{ K})}{(13.05 \text{ W/m}^2 \cdot \text{K})(-40 + 60 \text{ K}) + (0.9)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(233^4 - 213^4 \text{ K}^4)} = \mathbf{0.0457 \text{ m}} \end{aligned}$$

**Discussion** The minimum thickness required by the ceramic plate to prevent the surface in contact with the stainless steel bolts from cooling below  $-30^{\circ}\text{C}$  is 4.57 cm. With  $d > 4.57 \text{ cm}$ , the ceramic plate surface that is in contact with the bolts would be above the minimum suitable temperature for the ASTM A437 B4B stainless steel bolts.

**9-44** The required electrical power to maintain a specified surface temperature of a grill is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas. 3 Thermal properties are constant.

**Properties** The properties of air at  $T_f = (T_s + T_\infty)/2 = 90^\circ\text{C}$  are  $k = 0.03024 \text{ W/m}\cdot\text{K}$ ,  $\nu = 2.201 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 0.7132$  (from Table A-15). Also,  $\beta = 1/T_f = 2.755 \times 10^{-3} \text{ K}^{-1}$ .

**Analysis** Treating the grill as a horizontal circular plate, the characteristic length is

$$L_c = \frac{A_s}{p} = \frac{\pi D^2 / 4}{\pi D} = \frac{D}{4} = 0.0625 \text{ m}$$

The Rayleigh number ( $L_c = D/4$ ) is

$$\begin{aligned} \text{Ra}_L &= \frac{g\beta(T_s - T_\infty)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.002755 \text{ K}^{-1})(150 - 30) \text{ K}(0.0625 \text{ m})^3}{(2.201 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7132) \\ &= 1.166 \times 10^6 \end{aligned}$$

Since the grill has a hot upper surface, we use

$$\text{Nu} = 0.54 \text{Ra}_L^{1/4} = 0.54(1.166 \times 10^6)^{1/4} = 17.74$$

$$h = \text{Nu} \frac{k}{L_c} = (17.74) \frac{0.03024 \text{ W/m}\cdot\text{K}}{0.0625 \text{ m}} = 8.583 \text{ W/m}^2 \cdot \text{K}$$

The total rate of heat transfer on the grill surface is

$$\begin{aligned} \dot{Q}_{\text{total}} &= \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}} = hA_s(T_s - T_\infty) + \varepsilon\sigma A_s(T_s^4 - T_{\text{surr}}^4) \\ &= \frac{\pi}{4} (0.25 \text{ m})^2 [(8.583 \text{ W/m}^2 \cdot \text{K})(150 - 30) \text{ K} + 0.8(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(423^4 - 303^4) \text{ K}^4] \\ &= \mathbf{103 \text{ W}} \end{aligned}$$

**Discussion** To maintain a surface temperature of  $150^\circ\text{C}$ , the grill needs at least 103 W of electrical power.

**9-45** A can of engine oil placed vertically in the trunk of a car and the heat transfer from the ends of the can are negligible, determine the heat transfer rate from the can surface.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas. 3 Thermal properties are constant. 4 Radiation heat transfer is negligible.

**Properties** The properties of air at  $T_f = (T_s + T_\infty)/2 = 30^\circ\text{C}$  are  $k = 0.02588 \text{ W/m}\cdot\text{K}$ ,  $\nu = 1.608 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 0.7282$  (from Table A-15). Also,  $\beta = 1/T_f = 0.0033 \text{ K}^{-1}$ .

**Analysis** The Rayleigh number ( $L_c = L$ ) is

$$\begin{aligned}\text{Ra}_L &= \frac{g\beta(T_\infty - T_s)L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.0033 \text{ K}^{-1})(43 - 17)\text{K}(0.15 \text{ m})^3}{(1.608 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7282) \\ &= 8.00 \times 10^6\end{aligned}$$

Then

$$\frac{35L}{Gr_L^{1/4}} = \frac{35(0.15 \text{ m})}{(1.099 \times 10^7)^{1/4}} = 0.0912 \text{ m} < D$$

Since  $D \geq 35L / Gr_L^{1/4}$  is satisfied, we can treat this vertical cylinder as a vertical plate, and the Nusselt may be calculated with

$$\text{Nu} = 0.59\text{Ra}_L^{1/4} = 0.59(8.00 \times 10^6)^{1/4} = 31.38$$

Then, the heat transfer coefficient is

$$h = \frac{k}{L} \text{Nu} = \frac{0.02588 \text{ W/m}\cdot\text{K}}{0.15 \text{ m}} (31.38) = 5.414 \text{ W/m}^2 \cdot \text{K}$$

Hence, the rate of heat transfer is

$$\begin{aligned}\dot{Q} &= hA_s(T_\infty - T_s) = h\pi DL(T_\infty - T_s) \\ &= (5.414 \text{ W/m}^2 \cdot \text{K})\pi(0.1 \text{ m})(0.15 \text{ m})(43 - 17) \text{ K} \\ &= \mathbf{6.63 \text{ W}}\end{aligned}$$

**Discussion** For vertical cylinder, the characteristic length is its length.

**9-46** Flue gases are released to atmosphere using a cylindrical stack. The rates of heat transfer from the stack with and without wind cases are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm.

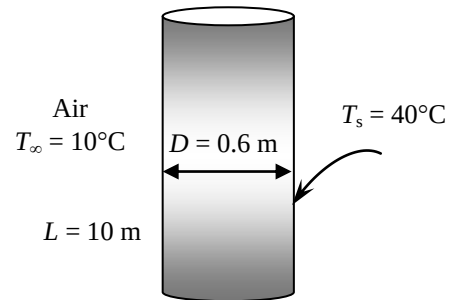
**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (40 + 10)/2 = 25^\circ\text{C}$  are (Table A-15)

$$k = 0.02551 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.562 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7296$$

$$\beta = \frac{1}{T_f} = \frac{1}{(25 + 273)\text{K}} = 0.003356 \text{ K}^{-1}$$



**Analysis** (a) When there is no wind heat transfer is by natural convection. The characteristic length in this case is the height of the stack,  $L_c = L = 10 \text{ m}$ . Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003356 \text{ K}^{-1})(40 - 10 \text{ K})(10 \text{ m})^3}{(1.562 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7296) = 2.953 \times 10^{12}$$

We can treat this vertical cylinder as a vertical plate since

$$\frac{35L}{Gr^{1/4}} = \frac{35(10)}{(2.953 \times 10^{12} / 0.7296)^{1/4}} = 0.246 < 0.6 \quad \text{and thus } D \geq \frac{35L}{Gr^{1/4}} \quad \text{The}$$

Nusselt number is determined from

$$Nu = 0.1Ra^{1/3} = 0.1(2.953 \times 10^{12})^{1/3} = 1435 \quad (\text{from Table 9-1})$$

Then

$$h = \frac{k}{L_c} Nu = \frac{0.02551 \text{ W/m}\cdot^\circ\text{C}}{10 \text{ m}} (1435) = 3.660 \text{ W/m}^2\cdot^\circ\text{C}$$

and

$$\dot{Q} = hA(T_s - T_\infty) = (3.660 \text{ W/m}^2\cdot^\circ\text{C})(\pi \times 0.6 \times 10 \text{ m}^2)(40 - 10)^\circ\text{C} = \mathbf{2070 \text{ W}}$$

(b) When the stack is exposed to 20 km/h winds, the heat transfer will be by forced convection. We have flow of air over a cylinder and the heat transfer rate is determined as follows:

$$\text{Re} = \frac{VD}{\nu} = \frac{(20 \times 1000 / 3600 \text{ m/s})(0.6 \text{ m})}{1.562 \times 10^{-5} \text{ m}^2/\text{s}} = 213,400$$

$$Nu = 0.027 \text{Re}^{0.805} \text{Pr}^{1/3} = 0.027(213,400)^{0.805} (0.7296)^{1/3} = 473.9 \quad (\text{from Table 7-1})$$

$$h = \frac{k}{D} Nu = \frac{0.02551 \text{ W/m}\cdot^\circ\text{C}}{0.6 \text{ m}} (473.9) = 20.15 \text{ W/m}^2\cdot^\circ\text{C}$$

$$\dot{Q} = hA(T_s - T_\infty) = (20.15 \text{ W/m}^2\cdot^\circ\text{C})(\pi \times 0.6 \times 10 \text{ m}^2)(40 - 10)^\circ\text{C} = \mathbf{11,390 \text{ W}}$$

**Discussion** There is more than five-fold increase in heat transfer due to winds.

**9-47** Water is boiling in a pan that is placed on top of a stove. The rate of heat loss from the cylindrical side surface of the pan by natural convection and radiation and the ratio of heat lost from the side surfaces of the pan to that by the evaporation of water are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm.

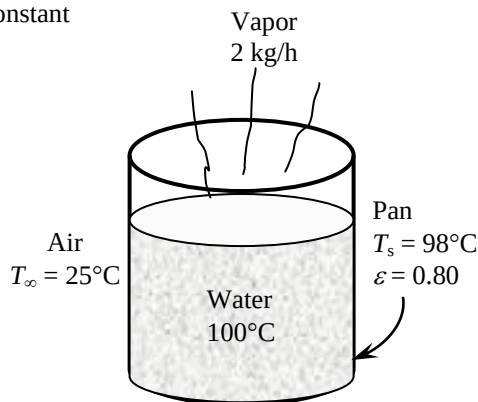
**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (98 + 25)/2 = 61.5^\circ\text{C}$  are (Table A-15)

$$k = 0.02819 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.910 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7198$$

$$\beta = \frac{1}{T_f} = \frac{1}{(61.5 + 273)\text{K}} = 0.00299 \text{ K}^{-1}$$



**Analysis** (a) The characteristic length in this case is the height of the pan,

$$L_c = L = 0.12 \text{ m. Then}$$

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.00299 \text{ K}^{-1})(98 - 25 \text{ K})(0.12 \text{ m})^3}{(1.910 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7198) = 7.299 \times 10^6$$

We can treat this vertical cylinder as a vertical plate since

$$\frac{35L}{\text{Gr}^{1/4}} = \frac{35(0.12)}{(7.299 \times 10^6 / 0.7198)^{1/4}} = 0.07443 < 0.25 \quad \text{and thus } D \geq \frac{35L}{\text{Gr}^{1/4}}$$

Therefore,

$$\text{Nu} = \left\{ 0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.492}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(7.299 \times 10^6)^{1/6}}{\left[ 1 + \left( \frac{0.492}{0.7198} \right)^{9/16} \right]^{8/27}} \right\}^2 = 28.60$$

$$h = \frac{k}{L} \text{Nu} = \frac{0.02819 \text{ W/m}\cdot^\circ\text{C}}{0.12 \text{ m}} (28.60) = 6.720 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = \pi DL = \pi(0.25 \text{ m})(0.12 \text{ m}) = 0.09425 \text{ m}^2$$

and

$$\dot{Q} = hA_s(T_s - T_\infty) = (6.720 \text{ W/m}^2\cdot^\circ\text{C})(0.09425 \text{ m}^2)(98 - 25)^\circ\text{C} = \mathbf{46.2 \text{ W}}$$

(b) The radiation heat loss from the pan is

$$\begin{aligned} \dot{Q}_{\text{rad}} &= \varepsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4) \\ &= (0.80)(0.09425 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4) [(98 + 273 \text{ K})^4 - (25 + 273 \text{ K})^4] = \mathbf{47.3 \text{ W}} \end{aligned}$$

(c) The heat loss by the evaporation of water is

$$\dot{Q} = \dot{m} h_{fg} = (1.5 / 3600 \text{ kg/s})(2257 \text{ kJ/kg}) = 0.9404 \text{ kW} = 940 \text{ W}$$

Then the ratio of the heat lost from the side surfaces of the pan to that by the evaporation of water then becomes

$$f = \frac{46.2 + 47.3}{940} = 0.099 = \mathbf{9.9\%}$$

**9-48** Water is boiling in a pan that is placed on top of a stove. The rate of heat loss from the cylindrical side surface of the pan by natural convection and radiation and the ratio of heat lost from the side surfaces of the pan to that by the evaporation of water are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm.

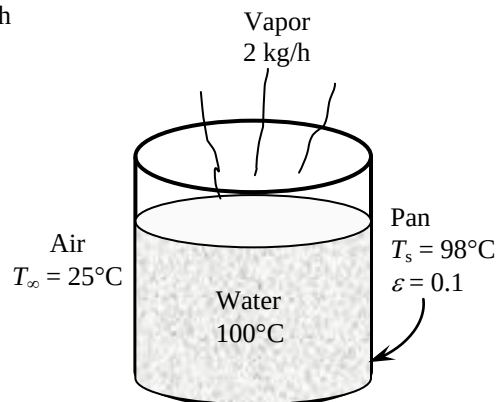
**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (98 + 25)/2 = 61.5^\circ\text{C}$  are (Table A-15)

$$k = 0.02819 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.910 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7198$$

$$\beta = \frac{1}{T_f} = \frac{1}{(61.5 + 273)\text{K}} = 0.00299 \text{ K}^{-1}$$



**Analysis** (a) The characteristic length in this case is the height of the pan,  $L_c = L = 0.12 \text{ m}$ . Then,

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.00299 \text{ K}^{-1})(98 - 25 \text{ K})(0.12 \text{ m})^3}{(1.910 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7198) = 7.299 \times 10^6$$

We can treat this vertical cylinder as a vertical plate since

$$\frac{35L}{\text{Gr}^{1/4}} = \frac{35(0.12)}{(7.299 \times 10^6 / 0.7198)^{1/4}} = 0.07443 < 0.25 \quad \text{and thus } D \geq \frac{35L}{\text{Gr}^{1/4}}$$

Therefore,

$$\text{Nu} = \left\{ 0.825 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.492}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(7.299 \times 10^6)^{1/6}}{\left[ 1 + \left( \frac{0.492}{0.7198} \right)^{9/16} \right]^{8/27}} \right\}^2 = 28.60$$

$$h = \frac{k}{L} \text{Nu} = \frac{0.02819 \text{ W/m}\cdot^\circ\text{C}}{0.12 \text{ m}} (28.60) = 6.720 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = \pi DL = \pi(0.25 \text{ m})(0.12 \text{ m}) = 0.09425 \text{ m}^2$$

and

$$\dot{Q} = hA_s(T_s - T_\infty) = (6.720 \text{ W/m}^2\cdot^\circ\text{C})(0.09425 \text{ m}^2)(98 - 25)^\circ\text{C} = \mathbf{46.2 \text{ W}}$$

(b) The radiation heat loss from the pan is

$$\begin{aligned} \dot{Q}_{rad} &= \varepsilon A_s \sigma (T_s^4 - T_{surr}^4) \\ &= (0.10)(0.09425 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4) [(98 + 273 \text{ K})^4 - (25 + 273 \text{ K})^4] = \mathbf{5.9 \text{ W}} \end{aligned}$$

(c) The heat loss by the evaporation of water is

$$\dot{Q} = \dot{m} h_{fg} = (1.5 / 3600 \text{ kg/s})(2257 \text{ kJ/kg}) = 0.9404 \text{ kW} = 940 \text{ W}$$

Then the ratio of the heat lost from the side surfaces of the pan to that by the evaporation of water then becomes

$$f = \frac{46.2 + 5.9}{940} = 0.055 = \mathbf{5.5\%}$$

**9-49** Some cans move slowly in a hot water container made of sheet metal. The rate of heat loss from the four side surfaces of the container and the annual cost of those heat losses are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm. 3 Heat loss from the top surface is disregarded.

**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (60 + 20)/2 = 40^\circ\text{C}$  are (Table A-15)

$$k = 0.02662 \text{ W/m} \cdot \text{K}$$

$$\nu = 1.702 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7255$$

$$\beta = \frac{1}{T_f} = \frac{1}{(40 + 273)\text{K}} = 0.003195 \text{ K}^{-1}$$

**Analysis** The characteristic length in this case is the height of the bath,  $L_c = L = 0.5 \text{ m}$ . Then,

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003195 \text{ K}^{-1})(60 - 20 \text{ K})(0.5 \text{ m})^3}{(1.702 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7255) = 3.925 \times 10^8$$

$$\text{Nu} = \left\{ 0.825 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.492}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(3.925 \times 10^8)^{1/6}}{\left[ 1 + \left( \frac{0.492}{0.7255} \right)^{9/16} \right]^{8/27}} \right\}^2 = 92.50$$

$$h = \frac{k}{L} \text{Nu} = \frac{0.02662 \text{ W/m} \cdot ^\circ\text{C}}{0.5 \text{ m}} (92.50) = 4.925 \text{ W/m}^2 \cdot \text{K}$$

$$A_s = 2[(0.5 \text{ m})(1 \text{ m}) + (0.5 \text{ m})(3.5 \text{ m})] = 4.5 \text{ m}^2$$

and

$$\dot{Q} = hA_s(T_s - T_\infty) = (4.925 \text{ W/m}^2 \cdot \text{K})(4.5 \text{ m}^2)(60 - 20)^\circ\text{C} = 886.5 \text{ W}$$

The radiation heat loss is

$$\begin{aligned} \dot{Q}_{\text{rad}} &= \varepsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4) \\ &= (0.7)(4.5 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4) [(60 + 273 \text{ K})^4 - (20 + 273 \text{ K})^4] \\ &= 879.9 \text{ W} \end{aligned}$$

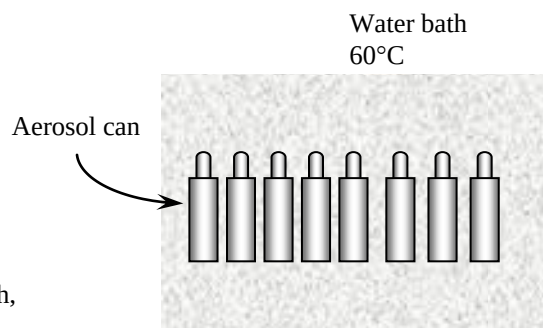
Then the total rate of heat loss becomes

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{natural convection}} + \dot{Q}_{\text{rad}} = 886.5 + 879.9 = \mathbf{1766 \text{ W}}$$

The amount and cost of the heat loss during one year is

$$Q_{\text{total}} = \dot{Q}_{\text{total}} \Delta t = (1.766 \text{ kW})(8760 \text{ h}) = 15,470 \text{ kWh}$$

$$\text{Cost} = (15,470 \text{ kWh})(\$0.085/\text{kWh}) = \mathbf{\$1315}$$



**9-50** Some cans move slowly in a hot water container made of sheet metal. It is proposed to insulate the side and bottom surfaces of the container for \$350. The simple payback period of the insulation to pay for itself from the energy it saves is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm. 3 Heat loss from the top surface is disregarded.

**Properties** Insulation will drop the outer surface temperature to a value close to the ambient temperature. The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature, which is unknown.

We evaluate air properties at a film temperature of  $(T_s + T_\infty)/2 = 23^\circ\text{C}$  and 1 atm based on the problem statement. Then, for an air temperature of  $T_\infty = 20^\circ\text{C}$ , the corresponding surface temperature is  $T_s = 26^\circ\text{C}$ . The properties of air at 1 atm and  $23^\circ\text{C}$  are (Table A-15)

$$k = 0.02536 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.543 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7301$$

$$\beta = \frac{1}{T_f} = \frac{1}{(23 + 273)\text{K}} = 0.00338 \text{ K}^{-1}$$

**Analysis** We start the solution process by “guessing” the outer surface temperature to be  $26^\circ\text{C}$ . We will check the accuracy of this guess later and repeat the calculations if necessary with a better guess based on the results obtained. The characteristic length in this case is the height of the tank,  $L_c = L = 0.5 \text{ m}$ . Then,

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.00338 \text{ K}^{-1})(26 - 20 \text{ K})(0.5 \text{ m})^3}{(1.543 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7301) = 7.622 \times 10^7$$

$$\text{Nu} = \left\{ 0.825 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.492}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(7.622 \times 10^7)^{1/6}}{\left[ 1 + \left( \frac{0.492}{0.7301} \right)^{9/16} \right]^{8/27}} \right\}^2 = 56.53$$

$$h = \frac{k}{L} \text{Nu} = \frac{0.02536 \text{ W/m}\cdot^\circ\text{C}}{0.5 \text{ m}} (56.53) = 2.868 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = 2[(0.5 \text{ m})(1.10 \text{ m}) + (0.5 \text{ m})(3.60 \text{ m})] = 4.7 \text{ m}^2$$

Then the total rate of heat loss from the outer surface of the insulated tank by convection and radiation becomes

$$\begin{aligned} \dot{Q} &= \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}} = hA_s(T_s - T_\infty) + \varepsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4) \\ &= (2.868 \text{ W/m}^2\cdot^\circ\text{C})(4.7 \text{ m}^2)(26 - 20)^\circ\text{C} + (0.1)(4.7 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[(26 + 273 \text{ K})^4 - (20 + 273 \text{ K})^4] \\ &= 97.5 \text{ W} \end{aligned}$$

In steady operation, the heat lost by the side surfaces of the tank must be equal to the heat lost from the exposed surface of the insulation by convection and radiation, which must be equal to the heat conducted through the insulation. The second condition requires the surface temperature to be

$$\dot{Q} = \dot{Q}_{\text{insulation}} = kA_s \frac{T_{\text{tank}} - T_s}{L} \rightarrow 97.5 \text{ W} = (0.035 \text{ W/m}\cdot^\circ\text{C})(4.7 \text{ m}^2) \frac{(60 - T_s)^\circ\text{C}}{0.05 \text{ m}}$$

It gives  $T_s = 30.4^\circ\text{C}$ , which is sufficiently close to the assumed temperature,  $26^\circ\text{C}$ . Therefore, there is no need to repeat the calculations. The total amount of heat loss and its cost during one year are

$$Q_{\text{total}} = \dot{Q}_{\text{total}} \Delta t = (97.5 \text{ W})(8760 \text{ h}) = 853.7 \text{ kWh}$$

$$\text{Cost} = (853.7 \text{ kWh})(\$0.085/\text{kWh}) = \$72.6$$

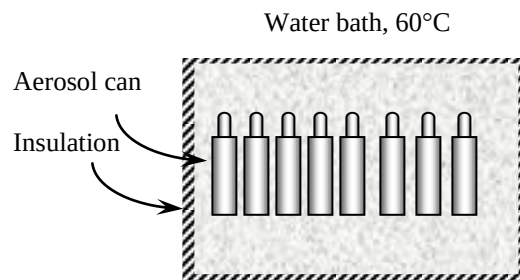
Then money saved during a one-year period due to insulation becomes

$$\text{Money saved} = \text{Cost}_{\text{without insulation}} - \text{Cost}_{\text{with insulation}} = \$1315 - \$72.6 = \$1242$$

where \$1116 is obtained from the solution of Problem 9-29. The insulation will pay for itself in

$$\text{Payback period} = \frac{\text{Cost}}{\text{Money saved}} = \frac{\$350}{\$1242/\text{yr}} = \mathbf{0.282 \text{ yr} = 103 \text{ days}}$$

**Discussion** We would definitely recommend the installation of insulation in this case.



**9-51** A room is to be heated by a cylindrical coal-burning stove. The surface temperature of the stove and the amount of coal burned during a 14-h period are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm. 4 The temperature of the outer surface of the stove is constant. 5 The heat transfer from the bottom surface is negligible. 6 The heat transfer coefficient at the top surface is the same as that on the side surface.

**Properties** The properties of air at 1 atm and the anticipated film temperature of  $(T_s + T_\infty)/2 = (130 + 24)/2 = 77^\circ\text{C}$  are (Table A-15)

$$k = 0.02931 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 2.066 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7161$$

$$\beta = \frac{1}{T_f} = \frac{1}{(77 + 273)\text{K}} = 0.002857 \text{ K}^{-1}$$

**Analysis** The characteristic length in this case is the height of the cylinder,  $L_c = L = 1.2 \text{ m}$ . Then,

$$\text{Gr} = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} = \frac{(9.81 \text{ m/s}^2)(0.002857 \text{ K}^{-1})(130 - 24 \text{ K})(1.2 \text{ m})^3}{(2.066 \times 10^{-5} \text{ m}^2/\text{s})^2} = 1.203 \times 10^{10}$$

A vertical cylinder can be treated as a vertical plate when

$$D (= 0.50 \text{ m}) \geq \frac{35L}{\text{Gr}^{1/4}} = \frac{35(1.2 \text{ m})}{(1.203 \times 10^{10})^{1/4}} = 0.1268 \text{ m}$$

which is satisfied. That is, the Nusselt number relation for a vertical plate can be used for side surfaces.

$$\text{Ra} = \text{GrPr} = (1.203 \times 10^{10})(0.7161) = 8.615 \times 10^9$$

$$\text{Nu} = \left\{ 0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.492}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(8.615 \times 10^9)^{1/6}}{\left[ 1 + \left( \frac{0.492}{0.7161} \right)^{9/16} \right]^{8/27}} \right\}^2 = 241.0$$

$$h = \frac{k}{L} \text{Nu} = \frac{0.02931 \text{ W/m}\cdot^\circ\text{C}}{1.2 \text{ m}} (241.0) = 5.886 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = \pi DL + \pi D^2 / 4 = \pi(0.5 \text{ m})(1.2 \text{ m}) + \pi(0.5 \text{ m})^2 / 4 = 2.081 \text{ m}^2$$

Then the surface temperature of the stove is determined from

$$\begin{aligned} \dot{Q} &= \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}} = hA_s(T_s - T_\infty) + \varepsilon\sigma A_s(T_s^4 - T_{\text{surr}}^4) \\ 1500 \text{ W} &= (5.886 \text{ W/m}^2\cdot^\circ\text{C})(2.081 \text{ m}^2)(T_s - 297) \\ &\quad + (0.85)(2.081 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)(T_s^4 - 287^4) \\ T_s &= 350.9 \text{ K} = 77.9^\circ\text{C} \end{aligned}$$

This surface temperature is not close to the value assumed for the evaluation of properties and  $h$ . We repeat calculations at the anticipated surface temperature of  $86^\circ\text{C}$  and the film temperature of  $(86 + 24)/2 = 55^\circ\text{C}$  as follows:

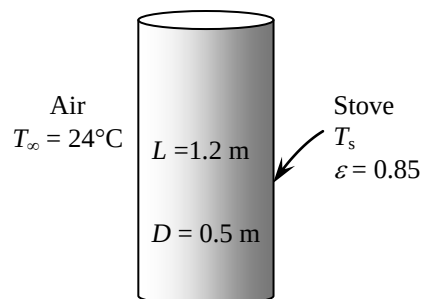
$$k = 0.02772 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.847 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7215$$

$$\beta = \frac{1}{T_f} = \frac{1}{(55 + 273)\text{K}} = 0.003049 \text{ K}^{-1}$$

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003049 \text{ K}^{-1})(86 - 24 \text{ K})(1.2 \text{ m})^3}{(1.847 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7215) = 6.777 \times 10^9$$



$$Nu = \left\{ 0.825 + \frac{0.387Ra^{1/6}}{\left[ 1 + \left( \frac{0.492}{Pr} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(6.777 \times 10^9)^{1/6}}{\left[ 1 + \left( \frac{0.492}{0.7215} \right)^{9/16} \right]^{8/27}} \right\}^2 = 223.6$$

$$h = \frac{k}{L} Nu = \frac{0.02772 \text{ W/m} \cdot ^\circ\text{C}}{1.2 \text{ m}} (223.6) = 5.165 \text{ W/m}^2 \cdot ^\circ\text{C}$$

$$\dot{Q} = \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}} = hA_s(T_s - T_\infty) + \varepsilon \sigma A_s(T_s^4 - T_{\text{surr}}^4)$$

$$1500 \text{ W} = (5.165 \text{ W/m}^2 \cdot ^\circ\text{C})(2.081 \text{ m}^2)(T_s - 297) \\ + (0.85)(2.081 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(T_s^4 - 287^4)$$

$$T_s = 353.7 \text{ K} = \mathbf{80.7^\circ\text{C}}$$

This is close to the assumed value, and thus there is no need to repeat the calculations. The amount of coal used is determined from

$$Q = \dot{Q} \Delta t = (1.5 \text{ kJ/s})(14 \text{ h/day} \times 3600 \text{ s/h}) = 75,600 \text{ kJ}$$

$$m_{\text{coal}} = \frac{Q / \eta}{HV} = \frac{(75,600 \text{ kJ}) / 0.65}{30,000 \text{ kJ/kg}} = \mathbf{3.88 \text{ kg}}$$

**9-52** A cylinder with specified length and diameter, the orientation of the cylinder that would achieve higher heat transfer rate is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas. 3 Thermal properties are constant. 4 Radiation heat transfer is negligible.

**Properties** The properties of air at  $T_f = (T_s + T_\infty)/2 = 30^\circ\text{C}$  are  $k = 0.02588 \text{ W/m}\cdot\text{K}$ ,  $\nu = 1.608 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 0.7282$  (from Table A-15). Also,  $\beta = 1/T_f = 0.0033 \text{ K}^{-1}$ .

**Analysis** For vertical orientation, the Rayleigh number ( $L_c = L$ ) is

$$\begin{aligned}\text{Ra}_L &= \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.0033 \text{ K}^{-1})(43 - 17)\text{K}(0.15 \text{ m})^3}{(1.608 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7282) \\ &= 8.00 \times 10^6\end{aligned}$$

Then

$$\frac{35L}{Gr_L^{1/4}} = \frac{35(0.15 \text{ m})}{(1.099 \times 10^7)^{1/4}} = 0.0912 \text{ m} < D$$

Since  $D \geq 35L / Gr_L^{1/4}$  is satisfied, we can treat this vertical cylinder as a vertical plate, and the Nusselt may be calculated with

$$\begin{aligned}\text{Nu}_{\text{vert}} &= \left\{ 0.825 + \frac{0.387\text{Ra}_L^{1/6}}{[1 + (0.492/\text{Pr})^{9/16}]^{8/27}} \right\}^2 \\ &= \left\{ 0.825 + \frac{0.387(8.00 \times 10^6)^{1/6}}{[1 + (0.492/0.7282)^{9/16}]^{8/27}} \right\}^2 = 29.39\end{aligned}$$

For horizontal orientation, the Rayleigh number ( $L_c = D$ ) is

$$\begin{aligned}\text{Ra}_D &= \frac{g\beta(T_s - T_\infty)D^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.0033 \text{ K}^{-1})(43 - 17)\text{K}(0.1 \text{ m})^3}{(1.608 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7282) \\ &= 2.37 \times 10^6\end{aligned}$$

The Nusselt number for horizontal cylinder is

$$\text{Nu}_{\text{horiz}} = \left\{ 0.6 + \frac{0.387\text{Ra}_D^{1/6}}{[1 + (0.559/\text{Pr})^{9/16}]^{8/27}} \right\}^2 = \left\{ 0.6 + \frac{0.387(2.37 \times 10^6)^{1/6}}{[1 + (0.559/0.7282)^{9/16}]^{8/27}} \right\}^2 = 18.64$$

Hence, the ratio of heat transfer rate is

$$\frac{\dot{Q}_{\text{vert}}}{\dot{Q}_{\text{horiz}}} = \frac{h_{\text{vert}}(k/L)A_s\Delta T}{h_{\text{horiz}}(k/L)A_s\Delta T} = \frac{\text{Nu}_{\text{vert}}(k/L)A_s\Delta T}{\text{Nu}_{\text{horiz}}(k/L)A_s\Delta T} = \frac{\text{Nu}_{\text{vert}}}{\text{Nu}_{\text{horiz}}} = \frac{29.39}{18.64} = \mathbf{1.58}$$

**Discussion** For the same  $\Delta T$ , the rate of heat transfer for the vertical orientation is 58% larger than that for the horizontal orientation.

**9-53** A soda can placed horizontally in a refrigerator compartment and the heat transfer from the ends of the can are negligible, determine the heat transfer rate from the can surface.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas. 3 Thermal properties are constant. 4 Radiation heat transfer is negligible.

**Properties** The properties of air at  $T_f = (T_s + T_\infty)/2 = 20^\circ\text{C}$  are  $k = 0.02514 \text{ W/m}\cdot\text{K}$ ,  $\nu = 1.516 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 0.7309$  (from Table A-15). Also,  $\beta = 1/T_f = 0.003413 \text{ K}^{-1}$ .

**Analysis** The Rayleigh number ( $L_c = D$ ) is

$$\text{Ra}_D = \frac{g\beta(T_s - T_\infty)D^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003413 \text{ K}^{-1})(36 - 4) \text{ K}(0.06 \text{ m})^3}{(1.516 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7309)$$

$$= 7.36 \times 10^5$$

The Nusselt number for horizontal cylinder is

$$\text{Nu} = \left\{ 0.6 + \frac{0.387 \text{Ra}_D^{1/6}}{[1 + (0.559/\text{Pr})^{9/16}]^{8/27}} \right\}^2 = \left\{ 0.6 + \frac{0.387(7.36 \times 10^5)^{1/6}}{[1 + (0.559/0.7309)^{9/16}]^{8/27}} \right\}^2 = 13.39$$

Then, the heat transfer coefficient is

$$h = \frac{k}{D} \text{Nu} = \frac{0.02514 \text{ W/m}\cdot\text{K}}{0.06 \text{ m}} (13.39) = 5.61 \text{ W/m}^2 \cdot \text{K}$$

Hence, the rate of heat transfer is

$$\begin{aligned} \dot{Q} &= hA_s(T_s - T_\infty) = h\pi DL(T_s - T_\infty) \\ &= (5.61 \text{ W/m}^2 \cdot \text{K})\pi(0.06 \text{ m})(0.15 \text{ m})(36 - 4) \text{ K} \\ &= \mathbf{5.08 \text{ W}} \end{aligned}$$

**Discussion** For horizontal cylinder, the characteristic length is its diameter.

**9-54** C&S A boiler supplies hot water to a dishwasher through a pipe at 10 g/s. The pipe dimensions are given. The water exits the boiler at 85°C. The pipe section between the boiler and the dishwasher is exposed to natural convection. The water temperature entering the dishwasher is to be determined whether it meets the ANSI/NSF 3 standard.

**Assumptions** 1 Steady state conditions. 2 Air is an ideal gas. 3 The local atmospheric pressure is 1 atm. 4 Thermal radiation is negligible. 5 Surface temperature of the pipe is constant.

**Properties** The properties of air at the film temperature of  $T_f = (T_\infty + T_s)/2 = (20 + 50)/2 = 35^\circ\text{C}$  are (Table A-15)  $\text{Pr} = 0.7268$ ,  $k = 0.02625 \text{ W/m}\cdot\text{K}$ ,  $\nu = 1.655 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\beta = 1/T_f = 1/(35 + 273 \text{ K}) = 0.003247 \text{ K}^{-1}$ . For water  $c_p = 4.20 \text{ kJ/kg}\cdot\text{K}$

**Analysis** The characteristic length of the horizontal pipe is  $D = 0.02 \text{ m}$ , and the Rayleigh number is

$$\text{Ra}_D = \frac{g\beta(T_s - T_\infty)D^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003247 \text{ K}^{-1})(50 - 20 \text{ K})(0.02 \text{ m})^3}{(1.655 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7268) = 20285$$

The Nusselt number for natural convection in this case is determined from

$$\text{Nu} = \frac{hD}{k} = \left\{ 0.6 + \frac{0.387 \text{ Ra}_D^{1/6}}{[1 + (0.559/\text{Pr})^{9/16}]^{8/27}} \right\}^2$$

So, the natural convection heat transfer coefficient is

$$h = \frac{k}{D} \left\{ 0.6 + \frac{0.387 \text{ Ra}_D^{1/6}}{[1 + (0.559/\text{Pr})^{9/16}]^{8/27}} \right\}^2 = \frac{0.02625 \text{ W/m}\cdot\text{K}}{0.02 \text{ m}} \left\{ 0.6 + \frac{0.387 (20285)^{1/6}}{[1 + (0.559/0.7268)^{9/16}]^{8/27}} \right\}^2$$

$$h = 6.828 \text{ W/m}^2 \cdot \text{K}$$

The rate of heat loss from the hot water flowing in the pipe is equal to the heat transfer rate by natural convection between the pipe surface at the ambient air:

$$\dot{Q} = \dot{m}c_p(T_{\text{in}} - T_{\text{out}}) = hA_s(T_s - T_\infty)$$

where  $T_{\text{in}}$  is the water temperature exiting the boiler (entering the pipe), and  $T_{\text{out}}$  is the water temperature entering the dishwasher (exiting the pipe). Solving for  $T_{\text{out}}$  yields,

$$T_{\text{out}} = T_{\text{in}} - \frac{h(\pi DL)}{\dot{m}c_p}(T_s - T_\infty) = 85^\circ\text{C} - \frac{(6.828 \text{ W/m}^2 \cdot \text{K})\pi(0.02 \text{ m})(20 \text{ m})}{(0.01 \text{ kg/s})(4200 \text{ J/kg}\cdot\text{K})}(50 - 20^\circ\text{C}) = \mathbf{78.9^\circ\text{C}} < 85^\circ\text{C}$$

**Discussion** The hot water entering the dishwasher is about 6°C lower than the temperature required by the ANSI/NSF 3 standard. To increase the water temperature entering the dishwasher, we can add insulation on the pipe to reduce the heat loss from the pipe surface. We can increase the water mass flow rate. We can reduce the pipe distance between the boiler and the dishwasher. We can increase the water temperature coming out from the boiler.

Note that this analysis does not account for the heat loss from the water due to thermal radiation exchange at the pipe surface with the surrounding. If the effect of thermal radiation is included, the water temperature entering the dishwasher would be lower than 78.9°C.

**9-55** Heat generated by the electrical resistance of a bare cable is dissipated to the surrounding air. The surface temperature of the cable is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm. 4 The temperature of the surface of the cable is constant.

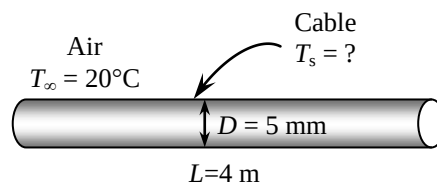
**Properties** We evaluate air properties at a film temperature of  $(T_s + T_\infty)/2 = 60^\circ\text{C}$  and 1 atm based on the problem statement. Then, for an air temperature of  $T_\infty = 20^\circ\text{C}$ , the corresponding surface temperature is  $T_s = 100^\circ\text{C}$ . The properties of air at 1 atm and  $60^\circ\text{C}$  are (Table A-15)

$$k = 0.02808 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.896 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7202$$

$$\beta = \frac{1}{T_f} = \frac{1}{(60 + 273)\text{K}} = 0.003003 \text{ K}^{-1}$$



**Analysis** The characteristic length in this case is the outer diameter of the pipe,  $L_c = D = 0.005 \text{ m}$ . Then,

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)D^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003003 \text{ K}^{-1})(100 - 20 \text{ K})(0.005 \text{ m})^3}{(1.896 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7202) = 590.2$$

$$\text{Nu} = \left\{ 0.6 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + (0.559/\text{Pr})^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.6 + \frac{0.387(590.2)^{1/6}}{\left[ 1 + (0.559/0.7202)^{9/16} \right]^{8/27}} \right\}^2 = 2.346$$

$$h = \frac{k}{D} \text{Nu} = \frac{0.02808 \text{ W/m}\cdot^\circ\text{C}}{0.005 \text{ m}} (2.346) = 13.17 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = \pi DL = \pi(0.005 \text{ m})(4 \text{ m}) = 0.06283 \text{ m}^2$$

$$\dot{Q} = hA_s(T_s - T_\infty)$$

$$(60 \text{ V})(1.5 \text{ A}) = (13.17 \text{ W/m}^2\cdot^\circ\text{C})(0.06283 \text{ m}^2)(T_s - 20)^\circ\text{C}$$

$$T_s = 128.8^\circ\text{C}$$

which is not close to the assumed value of  $100^\circ\text{C}$ . Repeating calculations for an assumed surface temperature of  $120^\circ\text{C}$ ,  $[T_f = (T_s + T_\infty)/2 = (120 + 20)/2 = 70^\circ\text{C}]$

$$k = 0.02881 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.995 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7177$$

$$\beta = \frac{1}{T_f} = \frac{1}{(70 + 273)\text{K}} = 0.002915 \text{ K}^{-1}$$

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)D^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.002915 \text{ K}^{-1})(120 - 20 \text{ K})(0.005 \text{ m})^3}{(1.995 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7177) = 644.6$$

$$\text{Nu} = \left\{ 0.6 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + (0.559/\text{Pr})^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.6 + \frac{0.387(644.6)^{1/6}}{\left[ 1 + (0.559/0.7177)^{9/16} \right]^{8/27}} \right\}^2 = 2.387$$

$$h = \frac{k}{D} \text{Nu} = \frac{0.02881 \text{ W/m}\cdot^\circ\text{C}}{0.005 \text{ m}} (2.387) = 13.76 \text{ W/m}^2\cdot^\circ\text{C}$$

$$\dot{Q} = hA_s(T_s - T_\infty)$$

$$(60 \text{ V})(1.5 \text{ A}) = (13.76 \text{ W/m}^2\cdot^\circ\text{C})(0.06283 \text{ m}^2)(T_s - 20)^\circ\text{C}$$

$$T_s = 124.1^\circ\text{C}$$

which is sufficiently close to the assumed value of  $120^\circ\text{C}$ .

**9-56** A cylindrical resistance heater is placed horizontally in a fluid. The outer surface temperature of the resistance wire is to be determined for two different fluids.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm. 4 Any heat transfer by radiation is ignored. 5 Properties are evaluated at 500°C for air and 40°C for water.

**Properties** The properties of air at 1 atm and 500°C are (Table A-15)

$$k = 0.05572 \text{ W/m} \cdot ^\circ\text{C}$$

$$\nu = 7.804 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.6986$$

$$\beta = \frac{1}{T_f} = \frac{1}{(500 + 273)\text{K}} = 0.001294 \text{ K}^{-1}$$

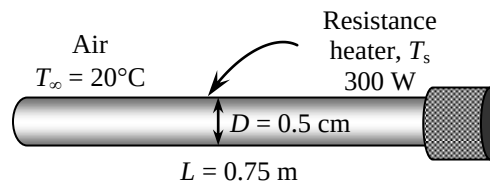
The properties of water at 40°C are (Table A-9)

$$k = 0.631 \text{ W/m} \cdot ^\circ\text{C}$$

$$\nu = \mu / \rho = 0.6582 \times 10^{-6} \text{ m}^2/\text{s}$$

$$\text{Pr} = 4.32$$

$$\beta = 0.000377 \text{ K}^{-1}$$



**Analysis** (a) The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown. We start the solution process by “guessing” the surface temperature to be 1200°C for the calculation of  $h$ . We will check the accuracy of this guess later and repeat the calculations if necessary. The characteristic length in this case is the outer diameter of the wire,  $L_c = D = 0.005 \text{ m}$ . Then,

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)D^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.001294 \text{ K}^{-1})(1200 - 20)^\circ\text{C}(0.005 \text{ m})^3}{(7.804 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.6986) = 214.7$$

$$\text{Nu} = \left\{ 0.6 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + (0.559 / \text{Pr})^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.6 + \frac{0.387(214.7)^{1/6}}{\left[ 1 + (0.559 / 0.6986)^{9/16} \right]^{8/27}} \right\}^2 = 1.919$$

$$h = \frac{k}{D} \text{Nu} = \frac{0.05572 \text{ W/m} \cdot ^\circ\text{C}}{0.005 \text{ m}} (1.919) = 21.38 \text{ W/m}^2 \cdot ^\circ\text{C}$$

$$A_s = \pi DL = \pi(0.005 \text{ m})(0.75 \text{ m}) = 0.01178 \text{ m}^2$$

and

$$\dot{Q} = hA_s(T_s - T_\infty) \rightarrow 300 \text{ W} = (21.38 \text{ W/m}^2 \cdot ^\circ\text{C})(0.01178 \text{ m}^2)(T_s - 20)^\circ\text{C} \rightarrow T_s = \mathbf{1211^\circ\text{C}}$$

which is close to the assumed value of 1200°C used in the evaluation of  $h$ . Therefore, there is no need to repeat calculations.

(b) For the case of water, we “guess” the surface temperature to be 40°C. The characteristic length in this case is the outer diameter of the wire,  $L_c = D = 0.005 \text{ m}$ . Then,

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)D^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.000377 \text{ K}^{-1})(40 - 20 \text{ K})(0.005 \text{ m})^3}{(0.6582 \times 10^{-6} \text{ m}^2/\text{s})^2} (4.32) = 92,197$$

$$\text{Nu} = \left\{ 0.6 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + (0.559 / \text{Pr})^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.6 + \frac{0.387(92,197)^{1/6}}{\left[ 1 + (0.559 / 4.32)^{9/16} \right]^{8/27}} \right\}^2 = 8.986$$

$$h = \frac{k}{D} \text{Nu} = \frac{0.631 \text{ W/m} \cdot ^\circ\text{C}}{0.005 \text{ m}} (8.986) = 1134 \text{ W/m}^2 \cdot ^\circ\text{C}$$

and

$$\dot{Q} = hA_s(T_s - T_\infty) \rightarrow 300 \text{ W} = (1134 \text{ W/m}^2 \cdot ^\circ\text{C})(0.01178 \text{ m}^2)(T_s - 20)^\circ\text{C} \rightarrow T_s = \mathbf{42.5^\circ\text{C}}$$

which is sufficiently close to the assumed value of 40°C in the evaluation of the properties and  $h$ . Therefore, there is no need to repeat calculations.

**9-57** A horizontal hot water pipe passes through a large room. The rate of heat loss from the pipe by natural convection and radiation is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm. 4 The temperature of the outer surface of the pipe is constant.

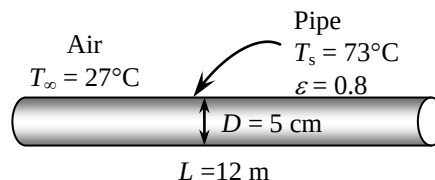
**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (73 + 27)/2 = 50^\circ\text{C}$  are (Table A-15)

$$k = 0.02735 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.798 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7228$$

$$\beta = \frac{1}{T_f} = \frac{1}{(50 + 273)\text{K}} = 0.003096 \text{ K}^{-1}$$



**Analysis** (a) The characteristic length in this case is the outer diameter of the pipe,  $L_c = D = 0.05 \text{ m}$ . Then,

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)D^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003096 \text{ K}^{-1})(73 - 27 \text{ K})(0.05 \text{ m})^3}{(1.798 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7228) = 3.905 \times 10^5$$

$$\text{Nu} = \left\{ 0.6 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + (0.559/\text{Pr})^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.6 + \frac{0.387(3.905 \times 10^5)^{1/6}}{\left[ 1 + (0.559/0.7228)^{9/16} \right]^{8/27}} \right\}^2 = 11.23$$

$$h = \frac{k}{D} \text{Nu} = \frac{0.02735 \text{ W/m}\cdot^\circ\text{C}}{0.05 \text{ m}} (11.23) = 6.143 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = \pi DL = \pi(0.05 \text{ m})(12 \text{ m}) = 1.885 \text{ m}^2$$

$$\dot{Q} = hA_s(T_s - T_\infty) = (6.143 \text{ W/m}^2\cdot^\circ\text{C})(1.885 \text{ m}^2)(73 - 27)^\circ\text{C} = \mathbf{533 \text{ W}}$$

(b) The radiation heat loss from the pipe is

$$\begin{aligned} \dot{Q}_{\text{rad}} &= \varepsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4) \\ &= (0.8)(1.885 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4) [(73 + 273 \text{ K})^4 - (27 + 273 \text{ K})^4] \\ &= \mathbf{533 \text{ W}} \end{aligned}$$

**9-58** A thick fluid flows through a pipe in calm ambient air. The pipe is heated electrically. The power rating of the electric resistance heater and the cost of electricity during a 15-h period are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm.

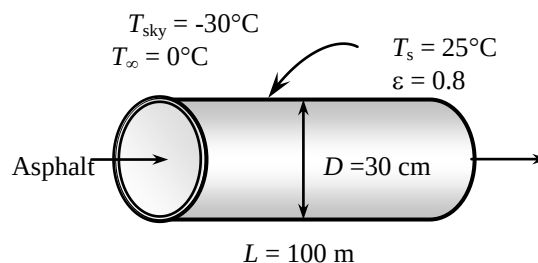
**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (25 + 0)/2 = 12.5^\circ\text{C}$  are (Table A-15)

$$k = 0.02458 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.448 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7330$$

$$\beta = \frac{1}{T_f} = \frac{1}{(12.5 + 273)\text{K}} = 0.003503 \text{ K}^{-1}$$



**Analysis** The characteristic length in this case is the outer diameter of the pipe,  $L_c = D = 0.3 \text{ m}$ . Then,

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003503 \text{ K}^{-1})(25 - 0 \text{ K})(0.3 \text{ m})^3}{(1.448 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7330) = 8.106 \times 10^7$$

$$\text{Nu} = \left\{ 0.6 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + (0.559/\text{Pr})^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.6 + \frac{0.387(8.106 \times 10^7)^{1/6}}{\left[ 1 + (0.559/0.7330)^{9/16} \right]^{8/27}} \right\}^2 = 53.29$$

$$h = \frac{k}{L_c} \text{Nu} = \frac{0.02458 \text{ W/m}\cdot^\circ\text{C}}{0.3 \text{ m}} (53.29) = 4.366 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = \pi DL = \pi(0.3 \text{ m})(100 \text{ m}) = 94.25 \text{ m}^2$$

and

$$\dot{Q} = hA_s(T_s - T_\infty) = (4.366 \text{ W/m}^2\cdot^\circ\text{C})(94.25 \text{ m}^2)(25 - 0)^\circ\text{C} = 10,287 \text{ W}$$

The radiation heat loss from the cylinder is

$$\begin{aligned} \dot{Q}_{\text{rad}} &= \epsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4) \\ &= (0.8)(94.25 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[(25 + 273 \text{ K})^4 - (-30 + 273 \text{ K})^4] = 18,808 \text{ W} \end{aligned}$$

Then,

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{natural convection}} + \dot{Q}_{\text{radiation}} = 10,287 + 18,808 = 29,094 \text{ W} = \mathbf{29.1 \text{ kW}}$$

The total amount and cost of heat loss during a 15 hour period is

$$Q = \dot{Q} \Delta t = (29.1 \text{ kW})(15 \text{ h}) = 436.5 \text{ kWh}$$

$$\text{Cost} = (436.5 \text{ kWh})(\$0.09/\text{kWh}) = \mathbf{\$39.3}$$

**9-59** A fluid flows through a pipe in calm ambient air. The pipe is heated electrically. The thickness of the insulation needed to reduce the losses by 85% and the money saved during 15-h are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm.

**Properties** Insulation will drop the outer surface temperature to a value close to the ambient temperature, and possibly below it because of the very low sky temperature for radiation heat loss. For convenience, we use the properties of air at 1 atm and 5°C (the anticipated film temperature) (Table A-15),

$$k = 0.02401 \text{ W/m} \cdot ^\circ\text{C}$$

$$\nu = 1.382 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7350$$

$$\beta = \frac{1}{T_f} = \frac{1}{(5 + 273) \text{ K}} = 0.003597 \text{ K}^{-1}$$

**Analysis** The rate of heat loss in the previous problem was obtained to be 29,094 W. Noting that insulation will cut down the heat losses by 85%, the rate of heat loss will be

$$\dot{Q} = (1 - 0.85) \dot{Q}_{\text{no insulation}} = 0.15 \times 29,094 \text{ W} = 4364 \text{ W}$$

The amount of energy and money insulation will save during a 15-h period is simply determined from

$$Q_{\text{saved, total}} = \dot{Q}_{\text{saved}} \Delta t = (0.85 \times 29,094 \text{ kW})(15 \text{ h}) = 370.9 \text{ kWh}$$

$$\text{Money saved} = (\text{Energy saved})(\text{Unit cost of energy}) = (370.9 \text{ kWh})(\$0.09/\text{kWh}) = \mathbf{\$33.4}$$

The characteristic length in this case is the outer diameter of the insulated pipe,  $L_c = D + 2t_{\text{insul}} = 0.3 + 2t_{\text{insul}}$  where  $t_{\text{insul}}$  is the thickness of insulation in m. Then the problem can be formulated for  $T_s$  and  $t_{\text{insul}}$  as follows:

$$Ra = \frac{g\beta(T_s - T_\infty)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003597 \text{ K}^{-1})[(T_s - 273) \text{ K}](0.3 + 2t_{\text{insul}})^3}{(1.382 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7350)$$

$$Nu = \left\{ 0.6 + \frac{0.387Ra^{1/6}}{\left[ 1 + (0.559/\text{Pr})^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.6 + \frac{0.387Ra^{1/6}}{\left[ 1 + (0.559/0.7350)^{9/16} \right]^{8/27}} \right\}^2$$

$$h = \frac{k}{L_c} Nu = \frac{0.02401 \text{ W/m} \cdot ^\circ\text{C}}{L_c} Nu$$

$$A_s = \pi D_o L = \pi(0.3 + 2t_{\text{insul}})(100 \text{ m})$$

The total rate of heat loss from the outer surface of the insulated pipe by convection and radiation becomes

$$\begin{aligned} \dot{Q} &= \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}} = hA_s(T_s - T_\infty) + \varepsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4) \\ 4364 &= hA_s(T_s - 273) + (0.1)A_s(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)[T_s^4 - (-30 + 273 \text{ K})^4] \end{aligned}$$

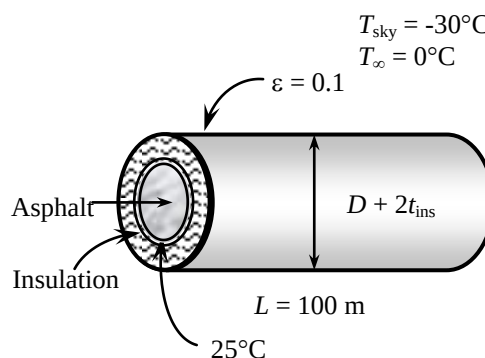
In steady operation, the heat lost by the side surfaces of the pipe must be equal to the heat lost from the exposed surface of the insulation by convection and radiation, which must be equal to the heat conducted through the insulation. Therefore,

$$\dot{Q} = \dot{Q}_{\text{insulation}} = \frac{2\pi k L (T_{\text{tank}} - T_s)}{\ln(D_o / D)} \rightarrow 4364 \text{ W} = \frac{2\pi(0.035 \text{ W/m} \cdot ^\circ\text{C})(100 \text{ m})(298 - T_s) \text{ K}}{\ln[(0.3 + 2t_{\text{insul}}) / 0.3]}$$

The solution of all of the equations above simultaneously using an equation solver gives

$$T_s = 281.5 \text{ K} = 8.5^\circ\text{C} \text{ and } t_{\text{insul}} = \mathbf{0.013 \text{ m} = 1.3 \text{ cm}}.$$

Note that the film temperature is  $(8.5 + 0)/2 = 4.25^\circ\text{C}$  which is very close to the assumed value of  $5^\circ\text{C}$ . Therefore, there is no need to repeat the calculations using properties at this new film temperature.



**9-60** C&S A boiler supplies hot water to an equipment through a pipe at 10 g/s. The pipe dimensions are given. The water enters the equipment at 98°C. The pipe section that is exposed to natural convection is 30 m long. The water temperature exiting the boiler is to be determined whether it meets the ASME Boiler and Pressure Vessel Code.

**Assumptions** **1** Steady state conditions. **2** Air is an ideal gas. **3** The local atmospheric pressure is 1 atm. **4** Surface temperature of the pipe is constant.

**Properties** The properties of air at the film temperature of  $T_f = (T_\infty + T_s)/2 = (20 + 80)/2 = 50^\circ\text{C}$  are (Table A-15)  $\text{Pr} = 0.7228$ ,  $k = 0.02735 \text{ W/m}\cdot\text{K}$ ,  $\nu = 1.798 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\beta = 1/T_f = 1/(50 + 273 \text{ K}) = 0.003096 \text{ K}^{-1}$ . For water  $c_p = 4.20 \text{ kJ/kg}\cdot\text{K}$

**Analysis** The characteristic length of the horizontal pipe is  $D = 0.02 \text{ m}$ , and the Rayleigh number is

$$\text{Ra}_D = \frac{g\beta(T_s - T_\infty)D^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003096 \text{ K}^{-1})(80 - 20 \text{ K})(0.02 \text{ m})^3}{(1.798 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7228) = 32595$$

The Nusselt number for natural convection in this case is determined from

$$\text{Nu} = \frac{hD}{k} = \left\{ 0.6 + \frac{0.387 \text{ Ra}_D^{1/6}}{[1 + (0.559/\text{Pr})^{9/16}]^{8/27}} \right\}^2$$

So, the natural convection heat transfer coefficient is

$$h = \frac{k}{D} \left\{ 0.6 + \frac{0.387 \text{ Ra}_D^{1/6}}{[1 + (0.559/\text{Pr})^{9/16}]^{8/27}} \right\}^2 = \frac{0.02735 \text{ W/m}\cdot\text{K}}{0.02 \text{ m}} \left\{ 0.6 + \frac{0.387 (32595)^{1/6}}{[1 + (0.559/0.7228)^{9/16}]^{8/27}} \right\}^2$$

$$h = 7.998 \text{ W/m}^2 \cdot \text{K}$$

The rate of heat loss from the hot water flowing in the pipe is equal to the heat transfer rate by natural convection and thermal radiation between the pipe surface at the ambient air:

$$\dot{Q} = \dot{m}c_p(T_{\text{in}} - T_{\text{out}}) = hA_s(T_s - T_\infty) + \varepsilon\sigma A_s(T_s^4 - T_{\text{surr}}^4)$$

where  $T_{\text{in}}$  is the water temperature exiting the boiler (entering the pipe), and  $T_{\text{out}}$  is the water temperature entering the equipment (exiting the pipe). Solving for  $T_{\text{in}}$  yields,

$$T_{\text{in}} = \frac{\pi DL}{\dot{m}c_p} [h(T_s - T_\infty) + \varepsilon\sigma(T_s^4 - T_{\text{surr}}^4)] + T_{\text{out}}$$

$$T_{\text{in}} = \frac{\pi(0.02 \text{ m})(30 \text{ m})}{(0.01 \text{ kg/s})(4200 \text{ J/kg}\cdot\text{K})} \left[ \left( 7.998 \frac{\text{W}}{\text{m}^2} \cdot \text{K} \right) (80 - 20 \text{ K}) + (0.6)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(353^4 - 293^4 \text{ K}^4) \right] = 132^\circ\text{C} > 120^\circ\text{C}$$

**Discussion** The hot water at the boiler outlet is at 132°C, which is 12°C higher than the service restriction of the ASME Boiler and Pressure Vessel Code. Thus, this operation is not in compliance with the ASME Boiler and Pressure Vessel Code (ASME BPVC.IV-2015, HG-101).

**9-61** An insulated electric wire is exposed to calm air. The temperature at the interface of the wire and the plastic insulation is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm.

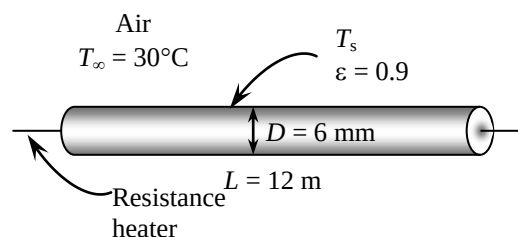
**Properties** The properties of air at 1 atm and the anticipated film temperature of  $(T_s + T_\infty)/2 = (50 + 30)/2 = 40^\circ\text{C}$  based on the problem statement are (Table A-15)

$$k = 0.02662 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.702 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7255$$

$$\beta = \frac{1}{T_f} = \frac{1}{(40 + 273)\text{K}} = 0.003195 \text{ K}^{-1}$$



**Analysis** The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown. We start the solution process by “guessing” the surface temperature to be  $50^\circ\text{C}$  for the evaluation of the properties and  $h$ . We will check the accuracy of this guess later and repeat the calculations if necessary. The characteristic length in this case is the outer diameter of the insulated wire  $L_c = D = 0.006 \text{ m}$ . Then,

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)D^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003195 \text{ K}^{-1})(50 - 30 \text{ K})(0.006 \text{ m})^3}{(1.702 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7255) = 339.3$$

$$\text{Nu} = \left\{ 0.6 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + (0.559/\text{Pr})^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.6 + \frac{0.387(339.3)^{1/6}}{\left[ 1 + (0.559/0.7255)^{9/16} \right]^{8/27}} \right\}^2 = 2.101$$

$$h = \frac{k}{D} \text{Nu} = \frac{0.02662 \text{ W/m}\cdot^\circ\text{C}}{0.006 \text{ m}} (2.101) = 9.327 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = \pi DL = \pi(0.006 \text{ m})(12 \text{ m}) = 0.2262 \text{ m}^2$$

The rate of heat generation, and thus the rate of heat transfer is

$$\dot{Q} = VI = (7 \text{ V})(10 \text{ A}) = 70 \text{ W}$$

Considering both natural convection and radiation, the total rate of heat loss can be expressed as

$$\begin{aligned} \dot{Q} &= hA_s(T_s - T_\infty) + \varepsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4) \\ 70 \text{ W} &= (9.327 \text{ W/m}^2\cdot^\circ\text{C})(0.226 \text{ m}^2)(T_s - 30)^\circ\text{C} \\ &\quad + (0.9)(0.2262 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[(T_s + 273)^4 - (30 + 273 \text{ K})^4] \end{aligned}$$

Its solution is

$$T_s = 49.9^\circ\text{C}$$

which is very close to the assumed value of  $50^\circ\text{C}$ . Then the temperature at the interface of the wire and the plastic cover in steady operation becomes

$$\dot{Q} = \frac{2\pi kL}{\ln(D_2/D_1)}(T_i - T_s) \longrightarrow T_i = T_s + \frac{\dot{Q} \ln(D_2/D_1)}{2\pi kL} = 49.9^\circ\text{C} + \frac{(70 \text{ W}) \ln(6/3)}{2\pi(0.20 \text{ W/m}\cdot^\circ\text{C})(12 \text{ m})} = \mathbf{53.1^\circ\text{C}}$$

**Discussion** The assumed film temperature of  $T_f = 40^\circ\text{C}$  is an appropriate assumption, since the determined  $T_s = 49.9^\circ\text{C}$  would give a film temperature of  $T_f = 39.95^\circ\text{C}$ . Otherwise,  $T_s$  would have to be solved iteratively.

**9-62** A steam pipe extended from one end of a plant to the other with no insulation on it. The rate of heat loss from the steam pipe and the annual cost of those heat losses are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm.

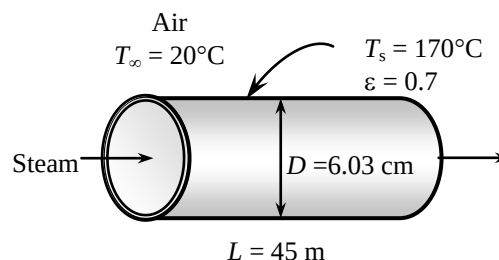
**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (170 + 20)/2 = 95^\circ\text{C}$  are (Table A-15)

$$k = 0.0306 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 2.254 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7122$$

$$\beta = \frac{1}{T_f} = \frac{1}{(95 + 273)\text{K}} = 0.002717 \text{ K}^{-1}$$



**Analysis** The characteristic length in this case is the outer diameter of the pipe,  $L_c = D = 0.0603 \text{ m}$ . Then,

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)D^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.002717 \text{ K}^{-1})(170 - 20 \text{ K})(0.0603 \text{ m})^3}{(2.254 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7122) = 1.229 \times 10^6$$

$$\text{Nu} = \left\{ 0.6 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + (0.559/\text{Pr})^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.6 + \frac{0.387(1.229 \times 10^6)^{1/6}}{\left[ 1 + (0.559/0.7122)^{9/16} \right]^{8/27}} \right\}^2 = 15.41$$

$$h = \frac{k}{D} \text{Nu} = \frac{0.0306 \text{ W/m}\cdot^\circ\text{C}}{0.0603 \text{ m}} (15.41) = 7.821 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = \pi DL = \pi(0.0603 \text{ m})(45 \text{ m}) = 8.525 \text{ m}^2$$

Then the total rate of heat transfer by natural convection and radiation becomes

$$\begin{aligned} \dot{Q} &= hA_s(T_s - T_\infty) + \varepsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4) \\ &= (8.525 \text{ W/m}^2\cdot^\circ\text{C})(11.37 \text{ m}^2)(170 - 20)^\circ\text{C} \\ &\quad + (0.7)(8.525 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[(170 + 273 \text{ K})^4 - (20 + 273 \text{ K})^4] \\ &= 20,539 \text{ W} = \mathbf{20.5 \text{ kW}} \end{aligned}$$

The total amount of gas consumption and its cost during a one-year period is

$$Q_{\text{gas}} = \frac{\dot{Q}\Delta t}{\eta} = \frac{20.539 \text{ kJ/s}}{0.84} \left( \frac{1 \text{ therm}}{105,500 \text{ kJ}} \right) (8760 \text{ h/yr} \times 3600 \text{ s/h}) = 7308 \text{ therms/yr}$$

$$\text{Cost} = (7308 \text{ therms/yr})(\$1.10/\text{therm}) = \mathbf{\$8039/\text{yr}}$$



**9-63** Prob. 9-62 is reconsidered. The effect of the surface temperature of the steam pipe on the rate of heat loss from the pipe and the annual cost of this heat loss is to be investigated.

**Analysis** The problem is solved using EES, and the solution is given below.

#### "GIVEN"

L=45 [m]  
D=0.0603 [m]  
T<sub>s</sub>=170 [C]  
T<sub>infinity</sub>=20 [C]  
epsilon=0.7  
T<sub>surr</sub>=T<sub>infinity</sub>  
eta<sub>furnace</sub>=0.84  
UnitCost=1.10 [\$/therm]  
time=24\*365 [h]

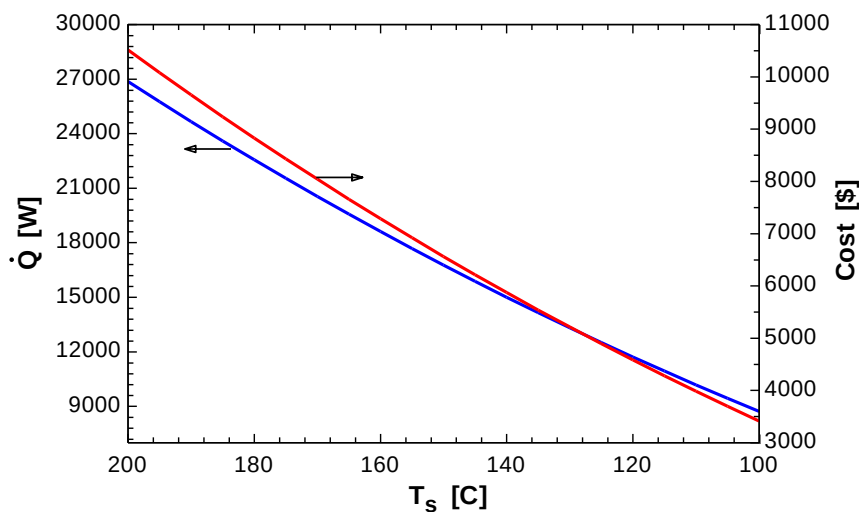
#### "PROPERTIES"

Fluid\$='air'  
k=Conductivity(Fluid\$, T=T<sub>film</sub>)  
Pr=Prandtl(Fluid\$, T=T<sub>film</sub>)  
rho=Density(Fluid\$, T=T<sub>film</sub>, P=101.3)  
mu=Viscosity(Fluid\$, T=T<sub>film</sub>)  
nu=mu/rho  
beta=1/(T<sub>film</sub>+273)  
T<sub>film</sub>=1/2\*(T<sub>s</sub>+T<sub>infinity</sub>)  
sigma=5.67E-8 [W/m^2-K^4] "Stefan-Boltzmann constant"  
g=9.807 [m/s^2] "gravitational acceleration"

#### "ANALYSIS"

delta=D  
Ra=(g\*beta\*(T<sub>s</sub>-T<sub>infinity</sub>)\*delta^3)/nu^2\*Pr  
Nusselt=(0.6+(0.387\*Ra^(1/6))/(1+(0.559/Pr)^(9/16)))^(8/27)\*2  
h=k/delta\*Nusselt  
A=pi\*D\*L  
Q<sub>dot</sub>=h\*A\*(T<sub>s</sub>-T<sub>infinity</sub>)+epsilon\*A\*sigma\*((T<sub>s</sub>+273)^4-(T<sub>surr</sub>+273)^4)  
Q<sub>gas</sub>=(Q<sub>dot</sub>\*time)/eta<sub>furnace</sub>\*Convert(h, s)\*Convert(J, kJ)\*Convert(kJ, therm)  
Cost=Q<sub>gas</sub>\*UnitCost

| T <sub>s</sub> [C] | Q̇ [W] | Cost [\$] |
|--------------------|--------|-----------|
| 100                | 8726   | 3416      |
| 105                | 9445   | 3697      |
| 110                | 10183  | 3986      |
| 115                | 10939  | 4282      |
| 120                | 11714  | 4585      |
| 125                | 12507  | 4896      |
| 130                | 13320  | 5214      |
| 135                | 14153  | 5540      |
| 140                | 15005  | 5873      |
| 145                | 15876  | 6214      |
| 150                | 16768  | 6563      |
| 155                | 17680  | 6920      |
| 160                | 18613  | 7286      |
| 165                | 19567  | 7659      |
| 170                | 20542  | 8041      |
| 175                | 21539  | 8431      |
| 180                | 22558  | 8830      |
| 185                | 23600  | 9237      |
| 190                | 24664  | 9654      |
| 195                | 25751  | 10080     |
| 200                | 26862  | 10514     |



**9-64** A cylindrical propane tank is exposed to calm ambient air. The propane is slowly vaporized due to a crack developed at the top of the tank. The time it will take for the tank to empty is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm. 4 Radiation heat transfer is negligible.

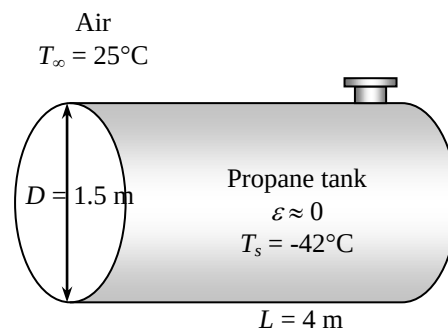
**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (-42 + 25)/2 = -8.5^\circ\text{C}$  are (Table A-15)

$$k = 0.02299 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.265 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7383$$

$$\beta = \frac{1}{T_f} = \frac{1}{(-8.5 + 273)\text{K}} = 0.003781 \text{ K}^{-1}$$



**Analysis** The tank gains heat through its cylindrical surface as well as its circular end surfaces. For convenience, we take the heat transfer coefficient at the end surfaces of the tank to be the same as that of its side surface. (The alternative is to treat the end surfaces as a vertical plate, but this will double the amount of calculations without providing much improvement in accuracy since the area of the end surfaces is much smaller and it is circular in shape rather than being rectangular). The characteristic length in this case is the outer diameter of the tank,  $L_c = D = 1.5 \text{ m}$ . Then,

$$Ra = \frac{g\beta(T_\infty - T_s)D^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003781 \text{ K}^{-1})[(25 - (-42))\text{K}](1.5 \text{ m})^3}{(1.265 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7383) = 3.869 \times 10^{10}$$

$$Nu = \left\{ 0.6 + \frac{0.387 Ra^{1/6}}{\left[ 1 + (0.559/\text{Pr})^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.6 + \frac{0.387 (3.869 \times 10^{10})^{1/6}}{\left[ 1 + (0.559/0.7383)^{9/16} \right]^{8/27}} \right\}^2 = 374.1$$

$$h = \frac{k}{D} Nu = \frac{0.02299 \text{ W/m}\cdot^\circ\text{C}}{1.5 \text{ m}} (374.1) = 5.733 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = \pi DL + 2\pi D^2/4 = \pi(1.5 \text{ m})(4 \text{ m}) + 2\pi(1.5 \text{ m})^2/4 = 22.38 \text{ m}^2$$

and

$$\dot{Q} = hA_s(T_\infty - T_s) = (5.733 \text{ W/m}^2\cdot^\circ\text{C})(22.38 \text{ m}^2)[(25 - (-42))^\circ\text{C}] = 8598 \text{ W}$$

The total mass and the rate of evaporation of propane are

$$m = \rho V = \rho \frac{\pi D^2}{4} L = (581 \text{ kg/m}^3) \frac{\pi(1.5 \text{ m})^2}{4} (4 \text{ m}) = 4107 \text{ kg}$$

$$\dot{m} = \frac{\dot{Q}}{h_{fg}} = \frac{8.598 \text{ kJ/s}}{425 \text{ kJ/kg}} = 0.02023 \text{ kg/s}$$

and it will take

$$\Delta t = \frac{m}{\dot{m}} = \frac{4107 \text{ kg}}{0.02023 \text{ kg/s}} = 202,996 \text{ s} = \mathbf{56.4 \text{ hours}}$$

for the propane tank to empty.

**9-65** Hot water flows in a horizontal pipe with a known inner surface temperature. The pipe outer surface is exposed to cool air. The outer surface temperature of the pipe is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Surface temperatures are constant. 3 The pipe thermal conductivity is constant. 4 Radiation heat transfer is negligible. 5 Local atmospheric pressure is 1 atm. 6 The film temperature is 40°C (this assumption will be verified).

**Properties** The properties of air at the assumed  $T_f = 40^\circ\text{C}$  are  $k = 0.02662 \text{ W/m}\cdot\text{K}$ ,  $\nu = 1.702 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 0.7255$  (Table A-15), and  $\beta = 1/T_f = 1/313 \text{ K}$ .

**Analysis** With the assumption that  $T_f = 40^\circ\text{C}$ , the pipe outer surface temperature is estimated as

$$T_{s,o} = 2T_f - T_\infty = 68^\circ\text{C}$$

The Rayleigh number is

$$\text{Ra} = \frac{g\beta(T_{s,o} - T_\infty)D_o^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(40 + 273 \text{ K})^{-1}(68 - 12) \text{ K} (0.045 \text{ m})^3}{(1.702 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7255) = 400,561$$

The Nusselt number relation for horizontal cylinder is

$$\text{Nu} = \left\{ 0.6 + \frac{0.387\text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.559}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.6 + \frac{0.387(400,561)^{1/6}}{\left[ 1 + \left( \frac{0.559}{0.7255} \right)^{9/16} \right]^{8/27}} \right\}^2 = 11.31$$

$$h = \frac{k}{D_o} \text{Nu} = \frac{0.02662 \text{ W/m}\cdot\text{K}}{0.045 \text{ m}} (11.31) = 6.6905 \text{ W/m}^2 \cdot \text{K}$$

Thus, the rate of heat transfer balance for conduction through the pipe wall and natural convection on the outer surface is

$$\dot{Q}_{\text{cyl}} = \dot{Q}_{\text{conv}}$$

$$\frac{2\pi L k_{\text{cyl}}}{\ln(D_o / D_i)} (T_{s,i} - T_{s,o}) = h A_s (T_{s,o} - T_\infty) \quad \rightarrow \quad T_{s,o} = 67.3^\circ\text{C}$$

where  $A_s = \pi D_o L$

**Discussion** The assumed film temperature of  $T_f = 40^\circ\text{C}$  is an appropriate assumption, since the determined  $T_{s,o} = 67.3^\circ\text{C}$  would give a film temperature of  $T_f = 39.7^\circ\text{C}$ . Otherwise,  $T_{s,o}$  would have to be solved iteratively.

**9-66** C&S A horizontal CPVC pipe with water flowing inside, while the outer surface is exposed to hot quiescence air at 107°C. The pipe dimensions are given. The convection heat transfer coefficient for the internal flow is given. The temperature of the water is to be determined such that the CPVC pipe temperature does not go above 93°C, which is the recommended maximum temperature by the ASME Code for Process Piping.

**Assumptions** 1 Steady state conditions. 2 Air is an ideal gas. 3 The local atmospheric pressure is 1 atm. 4 Surface temperature of the pipe is constant. 5 Conduction through the pipe wall is one-dimensional. 6 Thermal radiation is negligible.

**Properties** The properties of air at the film temperature of  $T_f = (T_\infty + T_s)/2 = (107 + 93)/2 = 100^\circ\text{C}$  are (Table A-15)  $\text{Pr} = 0.7111$ ,  $k = 0.03095 \text{ W/m}\cdot\text{K}$ ,  $\nu = 2.306 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\beta = 1/T_f = 1/(100 + 273 \text{ K}) = 0.002681 \text{ K}^{-1}$

**Analysis** The characteristic length of the horizontal pipe is  $D = 0.02 \text{ m}$ , and the Rayleigh number is

$$\text{Ra}_D = \frac{g\beta(T_\infty - T_s)D^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.002681 \text{ K}^{-1})(107 - 93 \text{ K})(0.02 \text{ m})^3}{(2.306 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7111) = 3939$$

The Nusselt number for natural convection in this case is determined from

$$\text{Nu} = \frac{hD}{k} = \left\{ 0.6 + \frac{0.387 \text{Ra}_D^{1/6}}{[1 + (0.559/\text{Pr})^{9/16}]^{8/27}} \right\}^2$$

So, the natural convection heat transfer coefficient is

$$h = \frac{k}{D} \left\{ 0.6 + \frac{0.387 \text{Ra}_D^{1/6}}{[1 + (0.559/\text{Pr})^{9/16}]^{8/27}} \right\}^2 = \frac{0.03095 \text{ W/m}\cdot\text{K}}{0.02 \text{ m}} \left\{ 0.6 + \frac{0.387 (3939)^{1/6}}{[1 + (0.559/0.7111)^{9/16}]^{8/27}} \right\}^2$$

$$h = 5.452 \text{ W/m}^2 \cdot \text{K}$$

The heat transfer rate from the hot air to the CPVC pipe outer surface is

$$\dot{Q} = hA_{s,o}(T_\infty - T_s) = h(\pi D_o L)(T_\infty - T_s)$$

Also, the heat transfer rate from the hot air to the water flowing inside the pipe is

$$\dot{Q} = \frac{T_\infty - T_{\text{water}}}{\frac{1}{hA_{s,o}} + \frac{\ln(D_o/D_i)}{2\pi Lk} + \frac{1}{h_{\text{water}}A_{s,i}}} = \frac{T_\infty - T_{\text{water}}}{\frac{1}{h(\pi D_o L)} + \frac{\ln(D_o/D_i)}{2\pi Lk} + \frac{1}{h_{\text{water}}(\pi D_i L)}}$$

Thus,

$$h(\pi D_o L)(T_\infty - T_s) = \frac{T_\infty - T_{\text{water}}}{\frac{1}{h(\pi D_o L)} + \frac{\ln(D_o/D_i)}{2\pi Lk} + \frac{1}{h_{\text{water}}(\pi D_i L)}}$$

Solving for the water temperature, yields

$$\begin{aligned} T_{\text{water}} &= T_\infty - hD_o(T_\infty - T_s) \left[ \frac{1}{hD_o} + \frac{\ln(D_o/D_i)}{2k} + \frac{1}{h_{\text{water}}D_i} \right] \\ T_{\text{water}} &= 107^\circ\text{C} \\ &\quad - (5.452 \text{ W/m}^2 \cdot \text{K})(0.02 \text{ m})(107 - 93 \text{ K}) \left[ \frac{1}{(5.452 \text{ W/m}^2 \cdot \text{K})(0.02 \text{ m})} + \frac{\ln(20/15)}{2(0.136 \text{ W/m}\cdot\text{K})} \right. \\ &\quad \left. + \frac{1}{(50 \text{ W/m}^2 \cdot \text{K})(0.015 \text{ m})} \right] = \mathbf{89.35^\circ\text{C}} \end{aligned}$$

**Discussion** To keep the CPVC pipe from heating above the recommended maximum temperature of 93°C by the hot air, the water flowing inside the pipe needs to be at about 89°C or lower. The lower temperature of the water keeps the pipe cool. The highest temperature in the pipe is at the outer surface, since it is exposed to the hot air.

**9-67** Hot engine flows in a horizontal pipe with a known inner surface temperature. The pipe outer surface is covered with a layer of insulation. The outer surface temperature is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Surface temperatures are constant. 3 Thermal conductivities of pipe and insulation are constant. 4 Contact resistance is negligible. 5 Radiation heat transfer is negligible. 6 Local atmospheric pressure is 1 atm.

**Properties** We first assume the film temperature is  $T_f = 50^\circ\text{C}$ . Then, the properties of air at  $T_f = 50^\circ\text{C}$  are  $k = 0.02735 \text{ W/m}\cdot\text{K}$ ,  $\nu = 1.798 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 0.7228$  (Table A-15), and  $\beta = 1/T_f = 1/323 \text{ K}$ .

The thermal conductivities of the pipe and the insulation are  $k_{\text{pipe}} = 15 \text{ W/m}\cdot\text{K}$  and  $k_{\text{ins}} = 0.15 \text{ W/m}\cdot\text{K}$ , respectively.

**Analysis** With the assumption that  $T_f = 50^\circ\text{C}$ , the outer surface temperature is estimated as

$$T_{s,o} = 2T_f - T_\infty = 90^\circ\text{C}$$

The Rayleigh number is

$$\text{Ra} = \frac{g\beta(T_{s,o} - T_\infty)D_o^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(50 + 273 \text{ K})^{-1}(90 - 10) \text{ K} (0.07 \text{ m})^3}{(1.798 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7228) = 1.863 \times 10^6$$

The Nusselt number relation for horizontal cylinder is

$$\text{Nu} = \left\{ 0.6 + \frac{0.387\text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.559}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.6 + \frac{0.387(1.863 \times 10^6)^{1/6}}{\left[ 1 + \left( \frac{0.559}{0.7228} \right)^{9/16} \right]^{8/27}} \right\}^2 = 17.38$$

$$h = \frac{k}{D_o} \text{Nu} = \frac{0.02735 \text{ W/m}\cdot\text{K}}{0.07 \text{ m}} (17.38) = 6.791 \text{ W/m}^2 \cdot \text{K}$$

From Chapter 3, the total thermal resistance for the pipe wall and the insulation is

$$R_{\text{total}} = \frac{\ln(D_{\text{interface}} / D_i)}{2\pi k_{\text{pipe}} L} + \frac{\ln(D_o / D_{\text{interface}})}{2\pi k_{\text{ins}} L}$$

$$= \frac{\ln(0.06 / 0.05)}{2\pi (15 \text{ W/m}\cdot\text{K})(2 \text{ m})} + \frac{\ln(0.07 / 0.06)}{2\pi (0.15 \text{ W/m}\cdot\text{K})(2 \text{ m})} = 0.08275 \text{ K/W}$$

Thus, the rate of heat transfer balance for conduction through the cylindrical layers and natural convection on the outer surface is

$$\dot{Q}_{\text{cyl}} = \dot{Q}_{\text{conv}}$$

$$\frac{(T_{s,i} - T_{s,o})}{R_{\text{total}}} = h(\pi D_o L)(T_{s,o} - T_\infty) \rightarrow T_{s,o} = 74.15^\circ\text{C} \quad (\text{first iteration})$$

The above solution is repeated iteratively until  $T_{s,o}$  converges to  $T_{s,o} = 74.8^\circ\text{C}$ .

**Discussion** The results from the iterations are as follows:

| Iter | $T_{s,o} [^\circ\text{C}]$ | Ra                  | Nu    | $h [\text{W/m}^2 \cdot \text{K}]$ |
|------|----------------------------|---------------------|-------|-----------------------------------|
| 1    | 90                         | $1.863 \times 10^6$ | 17.38 | 6.791                             |
| 2    | 74.15                      | $1.672 \times 10^6$ | 16.86 | 6.447                             |
| 3    | 74.80                      | $1.681 \times 10^6$ | 16.88 | 6.462                             |
| 4    | <b>74.77</b>               | $1.681 \times 10^6$ | 16.88 | 6.462                             |

As  $T_{s,o}$  changes through the iterations, so does the film temperature used for evaluating the properties.



**9-68** Reconsider Prob. 9-67. Hot engine flows in a horizontal pipe with a known inner surface temperature. The pipe outer surface is covered with a layer of insulation. The effect of the insulation layer thickness on the outer surface temperature is to be evaluated.

**Analysis** The problem is solved using EES, and the solution is given below.

"GIVEN"

L=2 [m] "Length exposed to natural convection"  
D\_i=0.05 [m] "Inner pipe diameter"  
t\_pipe=5e-3 [m] "Pipe wall thickness"  
T\_infinity=10 [C] "Cool air temperature"  
T\_s\_i=90 [C] "Inner surface temperature"

"PROPERTIES"

g=9.81 [m/s^2]  
Fluid\$='air'  
k=Conductivity(Fluid\$, T=T\_film)  
Pr=Prandtl(Fluid\$, T=T\_film)  
rho=Density(Fluid\$, T=T\_film, P=101.3)  
mu=Viscosity(Fluid\$, T=T\_film)  
nu=mu/rho  
beta=Volexpcoef(Fluid\$, T=T\_film)  
T\_film=1/2\*(T\_s\_o+T\_infinity)

"Pipe and insulation"

k\_ins=0.15 [W/m-K]  
k\_pipe=15 [W/m-K]

"ANALYSIS"

"Natural convection"

Gr=(g\*beta\*(T\_s\_o-T\_infinity)\*D\_o^3)/nu^2  
Ra=(g\*beta\*(T\_s\_o-T\_infinity)\*D\_o^3)/nu^2\*Pr  
Nusselt=(0.6+0.387\*Ra^(1/6))/((1+(0.559/Pr)^(9/16))^(8/27))^2  
h=k/D\_o\*Nusselt

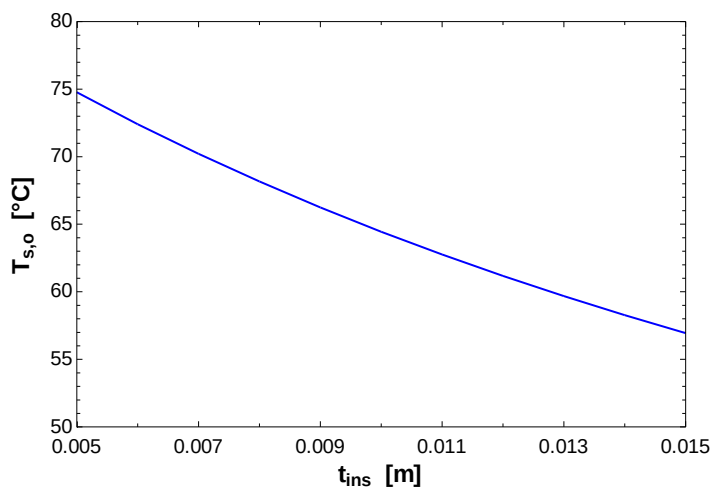
"Heat conduction through cylindrical layers"

D\_interface=D\_i+t\_pipe\*2  
D\_o=D\_interface+t\_ins\*2  
R\_pipe=ln(D\_interface/D\_i)/(2\*pi\*L\*k\_pipe)  
R\_ins=ln(D\_o/D\_interface)/(2\*pi\*L\*k\_ins)  
R\_total=R\_pipe+R\_ins

"Heat balance"

Q\_dot=(T\_s\_i-T\_s\_o)/R\_total "Heat conduction through cylindrical layers"  
Q\_dot=h\*A\_s\*(T\_s\_o-T\_infinity) "Heat loss by natural convection"  
A\_s=pi\*D\_o\*L "Surface area exposed to natural convection"

| t <sub>ins</sub> [m] | T <sub>s,o</sub> [°C] |
|----------------------|-----------------------|
| 0.005                | 74.77                 |
| 0.006                | 72.41                 |
| 0.007                | 70.21                 |
| 0.008                | 68.16                 |
| 0.009                | 66.24                 |
| 0.010                | 64.45                 |
| 0.011                | 62.76                 |
| 0.012                | 61.18                 |
| 0.013                | 59.68                 |
| 0.014                | 58.27                 |
| 0.015                | 56.94                 |



**Discussion** As the insulation layer thickness increases, the heat loss through the pipe is reduced. Therefore the outer surface temperature decreases with increasing thickness of insulation.

**9-69** A hot fluid flowing as a fully-developed laminar flow inside a horizontal pipe with constant surface temperature. The pipe outer surface temperature is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Surface temperatures are constant. 3 Radiation heat transfer is negligible. 4 Local atmospheric pressure is 1 atm. 5 The film temperature is 50°C (this assumption will be verified).

**Properties** We first assume the film temperature is  $T_f = 50^\circ\text{C}$ . Then, the properties of air at  $T_f = 50^\circ\text{C}$  and 1 atm are  $k = 0.02735 \text{ W/m}\cdot\text{K}$ ,  $\nu = 1.798 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 0.7228$  (Table A-15), and  $\beta = 1/T_f = 1/323 \text{ K}$ .

**Analysis** With the assumption that  $T_f = 50^\circ\text{C}$ , the outer surface temperature is estimated as

$$T_{s,o} = 2T_f - T_\infty = 90^\circ\text{C}$$

The Rayleigh number is

$$\text{Ra} = \frac{g\beta(T_{s,o} - T_\infty)D_o^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(50 + 273 \text{ K})^{-1}(90 - 10) \text{ K} (0.045 \text{ m})^3}{(1.798 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7228) = 495,031$$

At the outer surface, the natural convection Nusselt number relation for horizontal cylinder is

$$\text{Nu} = \left\{ 0.6 + \frac{0.387\text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.559}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.6 + \frac{0.387(495,031)^{1/6}}{\left[ 1 + \left( \frac{0.559}{0.7228} \right)^{9/16} \right]^{8/27}} \right\}^2 = 11.98$$

$$h_o = \frac{k}{D_o} \text{Nu} = \frac{0.02735 \text{ W/m}\cdot\text{K}}{0.045 \text{ m}} (11.98) = 7.2812 \text{ W/m}^2 \cdot \text{K}$$

At the inner surface, the forced convection heat transfer coefficient for laminar fully-developed flow with constant surface temperature is (see Chapter 8)

$$h_i = 3.66 \frac{k_{\text{fluid}}}{D_i} = 3.66 \frac{0.72 \text{ W/m}\cdot\text{K}}{0.035 \text{ m}} = 75.29 \text{ W/m}^2 \cdot \text{K}$$

Thus, the rate of heat transfer balance for internal forced convection at the pipe inner surface and the natural convection at the outer surface is

$$\dot{Q}_{\text{conv},i} = \dot{Q}_{\text{conv},o}$$

$$h_i(\pi D_i L) \Delta T_{\text{avg}} = h_o(\pi D_o L)(T_{s,o} - T_\infty) \rightarrow T_{s,o} = \frac{h_i D_i}{h_o D_o} \Delta T_{\text{avg}} + T_\infty$$

$$T_{s,o} = \frac{(75.29 \text{ W/m}^2 \cdot \text{K})(0.035 \text{ m})}{(7.2812 \text{ W/m}^2 \cdot \text{K})(0.045 \text{ m})} (10^\circ\text{C}) + 10^\circ\text{C} = \mathbf{90.4^\circ\text{C}}$$

**Discussion** The assumed film temperature  $T_f = 50^\circ\text{C}$  is an appropriate assumption, since the determined  $T_{s,o} = 90.4^\circ\text{C}$  would give a film temperature of  $T_f = 50.2^\circ\text{C}$ . Otherwise,  $T_{s,o}$  would have to be solved iteratively.

**9-70** A hot liquid flowing inside a horizontal pipe with a known mass flow rate and temperature difference of the pipe inlet and outlet. The pipe outer surface temperature is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Surface temperatures are constant. 3 Local atmospheric pressure is 1 atm. 4 The film temperature is 35°C (this assumption will be verified). 5 The  $T_{\text{surr}}$  is the same as the air temperature.

**Properties** We first assume the film temperature is  $T_f = 35^\circ\text{C}$ . Then, the properties of air at  $T_f = 35^\circ\text{C}$  are  $k = 0.02625 \text{ W/m}\cdot\text{K}$ ,  $\nu = 1.655 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 0.7268$  (Table A-15), and  $\beta = 1/T_f = 1/308 \text{ K}$ .

The emissivity of the black oxidized copper pipe outer surface is  $\varepsilon = 0.78$  (Table A-18)

**Analysis** With the assumption that  $T_f = 35^\circ\text{C}$ , the outer surface temperature is estimated as

$$T_{s,o} = 2T_f - T_\infty = 60^\circ\text{C}$$

The Rayleigh number is

$$\text{Ra} = \frac{g\beta(T_{s,o} - T_\infty)D_o^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(35 + 273 \text{ K})^{-1}(60 - 10) \text{ K}(0.055 \text{ m})^3}{(1.655 \times 10^{-5} \text{ m}^2/\text{s})^2}(0.7268) = 703,065$$

At the outer surface, the natural convection Nusselt number relation for horizontal cylinder is

$$\text{Nu} = \left\{ 0.6 + \frac{0.387\text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.559}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.6 + \frac{0.387(703,065)^{1/6}}{\left[ 1 + \left( \frac{0.559}{0.7268} \right)^{9/16} \right]^{8/27}} \right\}^2 = 13.21$$

$$h = \frac{k}{D_o} \text{Nu} = \frac{0.02625 \text{ W/m}\cdot\text{K}}{0.055 \text{ m}}(13.21) = 6.3048 \text{ W/m}^2 \cdot \text{K}$$

Thus, the outer surface temperature can be solved using

$$\dot{Q} = h(\pi D_o L)(T_{s,o} - T_\infty) + \varepsilon \sigma (\pi D_o L)(T_{s,o}^4 - T_{\text{surr}}^4)$$

$$\dot{m} c_p \Delta T = h(\pi D_o L)(T_{s,o} - T_\infty) + \varepsilon \sigma (\pi D_o L)(T_{s,o}^4 - T_{\text{surr}}^4) \rightarrow T_{s,o} = 60.3^\circ\text{C}$$

where  $\dot{m} = 0.05 \text{ kg/s}$ ,  $c_p = 1000 \text{ J/kg}\cdot\text{K}$ ,  $\Delta T = 10^\circ\text{C}$

**Discussion** The assumed film temperature  $T_f = 35^\circ\text{C}$  is an appropriate assumption, since the determined  $T_{s,o} = 60.3^\circ\text{C}$  would give a film temperature of  $T_f = 35.2^\circ\text{C}$ . Otherwise,  $T_{s,o}$  would have to be solved iteratively.

**9-71**  A hot liquid flowing inside a horizontal pipe with a known mass flow rate and temperature difference of the pipe inlet and outlet. The pipe outer surface temperature is to be determined whether or not it is safe from thermal burn hazards.

**Assumptions** 1 Steady operating conditions exist. 2 Surface temperatures are constant. 3 Local atmospheric pressure is 1 atm. 4 The film temperature is 35°C (this assumption will be verified). 5 The  $T_{\text{surr}}$  is the same as the air temperature.

**Properties** The properties of air at  $T_f = 25^\circ\text{C}$  and 1 atm pressure are  $k = 0.02551 \text{ W/m}\cdot\text{K}$ ,  $\nu = 1.562 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 0.7296$  (Table A-15), and  $\beta = 1/T_f = 1/298 \text{ K}$ . The emissivity of the black painted surface is given as  $\varepsilon = 0.88$ .

**Analysis** With the assumption that  $T_f = 25^\circ\text{C}$ , the outer surface temperature is estimated as

$$T_{s,o} = 2T_f - T_\infty = 40^\circ\text{C}$$

The Rayleigh number is

$$\text{Ra} = \frac{g\beta(T_{s,o} - T_\infty)D_o^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(25 + 273 \text{ K})^{-1}(40 - 10) \text{ K}(0.055 \text{ m})^3}{(1.562 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7296) = 491,343$$

At the outer surface, the natural convection Nusselt number relation for horizontal cylinder is

$$\text{Nu} = \left\{ 0.6 + \frac{0.387\text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.559}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.6 + \frac{0.387(491,343)^{1/6}}{\left[ 1 + \left( \frac{0.559}{0.7296} \right)^{9/16} \right]^{8/27}} \right\}^2 = 11.97$$

$$h = \frac{k}{D_o} \text{Nu} = \frac{0.02551 \text{ W/m}\cdot\text{K}}{0.055 \text{ m}} (11.97) = 5.5519 \text{ W/m}^2 \cdot \text{K}$$

Thus, the outer surface temperature can be solved using

$$\dot{Q} = h(\pi D_o L)(T_{s,o} - T_\infty) + \varepsilon \sigma (\pi D_o L)(T_{s,o}^4 - T_{\text{surr}}^4)$$

$$\dot{m} c_p \Delta T = h(\pi D_o L)(T_{s,o} - T_\infty) + \varepsilon \sigma (\pi D_o L)(T_{s,o}^4 - T_{\text{surr}}^4) \rightarrow T_{s,o} = 39.9^\circ\text{C}$$

where

$$\dot{m} = 0.028 \text{ kg/s}, \quad c_p = 1000 \text{ J/kg}\cdot\text{K}, \quad \Delta T = 10^\circ\text{C}$$

**Discussion** The black painted surface is sufficient to keep the outer surface temperature below  $45^\circ\text{C}$  to prevent thermal burn hazards.

The assumed film temperature  $T_f = 25^\circ\text{C}$  is appropriate for evaluating the properties of air, since the determined  $T_{s,o} = 39.9^\circ\text{C}$  would give a film temperature of  $T_f = 24.95^\circ\text{C}$ .

The emissivity of black paint is also listed in Table A-18 as  $\varepsilon = 0.88$ .

**9-72** The average surface temperature of a human head is to be determined when it is not covered.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm. 4 The head can be approximated as a 30-cm-diameter sphere.

**Properties** The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown. We start the solution process by “guessing” the surface temperature to be 30°C for the evaluation of the properties and  $h$ . We will check the accuracy of this guess later and repeat the calculations if necessary. The properties of air at 1 atm and the anticipated film temperature of  $(T_s + T_\infty)/2 = (30 + 20)/2 = 25^\circ\text{C}$  are (Table A-15)

$$k = 0.02551 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.562 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7296$$

$$\beta = \frac{1}{T_f} = \frac{1}{(25 + 273)\text{K}} = 0.003356 \text{ K}^{-1}$$

**Analysis** The characteristic length for a spherical object is  $L_c = D = 0.3 \text{ m}$ . Then,

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)D^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003356 \text{ K}^{-1})(30 - 20 \text{ K})(0.3 \text{ m})^3}{(1.562 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7296) = 2.658 \times 10^7$$

$$\text{Nu} = 2 + \frac{0.589 \text{Ra}^{1/4}}{\left[1 + \left(\frac{0.469}{\text{Pr}}\right)^{9/16}\right]^{4/9}} = 2 + \frac{0.589(2.658 \times 10^7)^{1/4}}{\left[1 + \left(\frac{0.469}{0.7296}\right)^{9/16}\right]^{4/9}} = 34.73$$

$$h = \frac{k}{D} \text{Nu} = \frac{0.02551 \text{ W/m}\cdot^\circ\text{C}}{0.3 \text{ m}} (34.73) = 2.953 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = \pi D^2 = \pi (0.3 \text{ m})^2 = 0.2827 \text{ m}^2$$

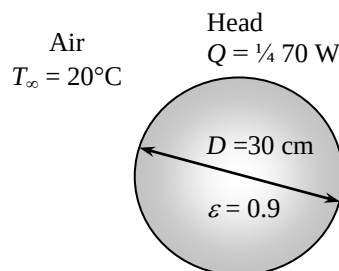
Considering both natural convection and radiation, the total rate of heat loss can be written as

$$\begin{aligned} \dot{Q} &= hA_s(T_s - T_\infty) + \varepsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4) \\ (70/4 \text{ W}) &= (2.953 \text{ W/m}^2\cdot^\circ\text{C})(0.2827 \text{ m}^2)(T_s - 20)^\circ\text{C} \\ &\quad + (0.9)(0.2827 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[(T_s + 273 \text{ K})^4 - (20 + 273 \text{ K})^4] \end{aligned}$$

Its solution is

$$T_s = 27.5^\circ\text{C}$$

which is sufficiently close to the assumed value in the evaluation of the properties and  $h$ . Therefore, there is no need to repeat calculations.



**9-73** The equilibrium temperature of a light glass bulb in a room is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm. 4 The light bulb is approximated as an 8-cm-diameter sphere.

**Properties** The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown. We start the solution process by “guessing” the surface temperature to be 170°C for the evaluation of the properties and  $h$ . We will check the accuracy of this guess later and repeat the calculations if necessary. The properties of air at 1 atm and the anticipated film temperature of  $(T_s + T_\infty)/2 = (170 + 25)/2 = 97.5^\circ\text{C}$  are (Table A-15)

$$k = 0.03077 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 2.279 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7116$$

$$\beta = \frac{1}{T_f} = \frac{1}{(97.5 + 273)\text{K}} = 0.002699 \text{ K}^{-1}$$

**Analysis** The characteristic length in this case is  $L_c = D = 0.08 \text{ m}$ .

Then,

$$\begin{aligned} Ra &= \frac{g\beta(T_s - T_\infty)D^3}{\nu^2} \text{Pr} \\ &= \frac{(9.81 \text{ m/s}^2)(0.002699 \text{ K}^{-1})(170 - 25 \text{ K})(0.08 \text{ m})^3}{(2.279 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7116) = 2.694 \times 10^6 \end{aligned}$$

$$Nu = 2 + \frac{0.589 Ra^{1/4}}{\left[1 + (0.469/\text{Pr})^{9/16}\right]^{4/9}} = 2 + \frac{0.589(2.694 \times 10^6)^{1/4}}{\left[1 + (0.469/0.7116)^{9/16}\right]^{4/9}} = 20.42$$

Then

$$h = \frac{k}{D} Nu = \frac{0.03077 \text{ W/m}\cdot^\circ\text{C}}{0.08 \text{ m}} (20.42) = 7.854 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = \pi D^2 = \pi (0.08 \text{ m})^2 = 0.02011 \text{ m}^2$$

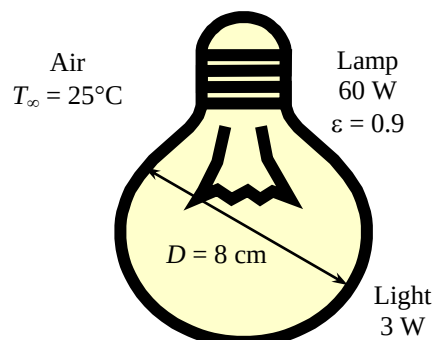
Considering both natural convection and radiation, the total rate of heat loss can be written as

$$\begin{aligned} \dot{Q} &= hA_s(T_s - T_\infty) + \varepsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4) \\ (0.95 \times 60) \text{ W} &= (7.854 \text{ W/m}^2\cdot^\circ\text{C})(0.02011 \text{ m}^2)(T_s - 25)^\circ\text{C} \\ &\quad + (0.9)(0.02011 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[(T_s + 273)^4 - (25 + 273 \text{ K})^4] \end{aligned}$$

Its solution is

$$T_s = 175^\circ\text{C}$$

which is sufficiently close to the value assumed in the evaluation of properties and  $h$ . Therefore, there is no need to repeat calculations.



**9-74** Water in a tank is to be heated by a spherical heater. The heating time is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 The temperature of the outer surface of the sphere is constant.

**Properties** Using the average temperature for water  $(15+45)/2=30^\circ\text{C}$  as the fluid temperature, the properties of water at the film temperature of  $(T_s+T_\infty)/2 = (85+30)/2 = 57.5^\circ\text{C}$  are (Table A-9)

$$k = 0.6515 \text{ W/m}\cdot^\circ\text{C}$$

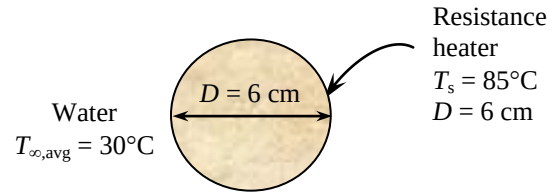
$$\nu = 0.493 \times 10^{-6} \text{ m}^2/\text{s}$$

$$\text{Pr} = 3.12$$

$$\beta = 0.501 \times 10^{-3} \text{ K}^{-1}$$

Also, the properties of water at  $30^\circ\text{C}$  are (Table A-15)

$$\rho = 996 \text{ kg/m}^3 \text{ and } c_p = 4178 \text{ J/kg}\cdot^\circ\text{C}$$



**Analysis** The characteristic length in this case is  $L_c = D = 0.06 \text{ m}$ . Then,

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)D^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.501 \times 10^{-3} \text{ K}^{-1})(85 - 30 \text{ K})(0.06 \text{ m})^3}{(0.493 \times 10^{-6} \text{ m}^2/\text{s})^2} (3.12) = 7.495 \times 10^8$$

$$\text{Nu} = 2 + \frac{0.589 \text{Ra}^{1/4}}{\left[1 + (0.469/\text{Pr})^{9/16}\right]^{4/9}} = 2 + \frac{0.589(7.495 \times 10^8)^{1/4}}{\left[1 + (0.469/3.12)^{9/16}\right]^{4/9}} = 87.44$$

$$h = \frac{k}{D} \text{Nu} = \frac{0.6515 \text{ W/m}\cdot^\circ\text{C}}{0.06 \text{ m}} (87.44) = 949.5 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = \pi D^2 = \pi (0.06 \text{ m})^2 = 0.01131 \text{ m}^2$$

The rate of heat transfer by convection is

$$\dot{Q}_{\text{conv}} = hA_s(T_s - T_\infty) = (949.5 \text{ W/m}^2\cdot^\circ\text{C})(0.01131 \text{ m}^2)(85 - 30) = 590.6 \text{ W}$$

The mass of water in the container is

$$m = \rho V = (996 \text{ kg/m}^3)(0.040 \text{ m}^3) = 39.84 \text{ kg}$$

The amount of heat transfer to the water is

$$Q = mc_p(T_2 - T_1) = (39.84 \text{ kg})(4178 \text{ J/kg}\cdot^\circ\text{C})(45 - 15)^\circ\text{C} = 4.994 \times 10^6 \text{ J}$$

Then the time the heater should be on becomes

$$\Delta t = \frac{Q}{\dot{Q}} = \frac{4.994 \times 10^6 \text{ J}}{590.6 \text{ J/s}} = 8456 \text{ s} = \mathbf{2.35 \text{ hours}}$$

**9-75** Chemical reaction in a spherical tank is releasing heat and the surface temperature is known. The heat rate from the reaction is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Surface temperature is constant. 3 Thermal resistance on the tank wall is negligible. 4 Local atmospheric pressure is 1 atm. 5 The  $T_{\text{surr}}$  is the same as the air temperature.

**Properties** The properties of air at  $T_f = (T_s + T_\infty)/2 = (50 + 20)/2 = 35^\circ\text{C}$  are  $k = 0.02625 \text{ W/m}\cdot\text{K}$ ,  $\nu = 1.655 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 0.7268$  (Table A-15), and  $\beta = 1/T_f = 1/308 \text{ K}$ .

The emissivity of the tank is given as  $\varepsilon = 0.35$ .

**Analysis** The Rayleigh number is

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)D^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(35 + 273 \text{ K})^{-1}(50 - 20) \text{ K} (2 \text{ m})^3}{(1.655 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7268) = 2.0284 \times 10^{10}$$

The Nusselt number relation for sphere is

$$\text{Nu} = 2 + \frac{0.589 \text{Ra}^{1/4}}{\left[1 + \left(\frac{0.469}{\text{Pr}}\right)^{9/16}\right]^{4/9}} = 2 + \frac{0.589(2.0284 \times 10^{10})^{1/4}}{\left[1 + \left(\frac{0.469}{0.7268}\right)^{9/16}\right]^{4/9}} = 173.96$$

$$h = \frac{k}{D} \text{Nu} = \frac{0.02625 \text{ W/m}\cdot\text{K}}{2 \text{ m}} (173.96) = 2.2832 \text{ W/m}^2 \cdot \text{K}$$

Thus, the heat rate from the reaction is

$$\begin{aligned} \dot{Q} &= h(\pi D^2)(T_s - T_\infty) + \varepsilon \sigma (\pi D^2)(T_s^4 - T_{\text{surr}}^4) \\ \dot{Q} &= \pi (2 \text{ m})^2 [(2.2832 \text{ W/m}^2 \cdot \text{K})(50 - 20) \text{ K} + (5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(0.35)(323^4 - 293^4) \text{ K}^4] \\ &= \mathbf{1737 \text{ W}} \end{aligned}$$

**Discussion** At the tank surface, natural convection contributed to about 50% of the 1737 W heat loss. From Table A-18, the emissivity value of 0.35 is a suitable value for lightly oxidized stainless steel surface.

**9-76** A spherical tank is filled with a hot liquid and the inner surface temperature is known. The outer surface temperature is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Surface temperatures are constant. 3 The tank wall thermal conductivity is constant. 4 Local atmospheric pressure is 1 atm. 5 The  $T_{\text{surr}}$  is the same as the air temperature. 6 The film temperature is  $40^\circ\text{C}$  (this assumption will be verified).

**Properties** The properties of air at the assumed  $T_f = 40^\circ\text{C}$  are  $k = 0.02662 \text{ W/m}\cdot\text{K}$ ,  $\nu = 1.702 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 0.7255$  (Table A-15), and  $\beta = 1/T_f = 1/313 \text{ K}$ .

The thermal conductivity and the emissivity of the tank are given as  $k_{\text{sph}} = 0.15 \text{ W/m}\cdot\text{K}$  and  $\varepsilon = 0.35$ , respectively.

**Analysis** With the assumption that  $T_f = 40^\circ\text{C}$ , the tank outer surface temperature is estimated as

$$T_{s,o} = 2T_f - T_\infty = 60^\circ\text{C}$$

The Rayleigh number is

$$\text{Ra} = \frac{g\beta(T_{s,o} - T_\infty)D_o^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(40 + 273 \text{ K})^{-1}(60 - 20) \text{ K}(3.06 \text{ m})^3}{(1.702 \times 10^{-5} \text{ m}^2/\text{s})^2}(0.7255) = 8.9964 \times 10^{10}$$

where

$$D_o = D_i + 2t = 3.06 \text{ m}$$

The Nusselt number relation for sphere is

$$\text{Nu} = 2 + \frac{0.589\text{Ra}^{1/4}}{\left[1 + \left(\frac{0.469}{\text{Pr}}\right)^{9/16}\right]^{4/9}} = 2 + \frac{0.589(8.9964 \times 10^{10})^{1/4}}{\left[1 + \left(\frac{0.469}{0.7255}\right)^{9/16}\right]^{4/9}} = 251.5$$

$$h = \frac{k}{D_o} \text{Nu} = \frac{0.02662 \text{ W/m}\cdot\text{K}}{3.06 \text{ m}}(251.5) = 2.1879 \text{ W/m}^2\cdot\text{K}$$

Thus, the outer surface temperature can be solved using

$$\dot{Q}_{\text{cond}} = \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}}$$

$$\frac{2\pi k_{\text{sph}} D_o D_i}{D_o - D_i} (T_{s,i} - T_{s,o}) = h(\pi D_o^2)(T_{s,o} - T_\infty) + \varepsilon \sigma (\pi D_o^2)(T_{s,o}^4 - T_{\text{surr}}^4) \rightarrow T_{s,o} = 61.1^\circ\text{C}$$

**Discussion** The assumed film temperature  $T_f = 40^\circ\text{C}$  is an appropriate assumption, since the determined  $T_{s,o} = 61.1^\circ\text{C}$  would give a film temperature of  $T_f = 40.6^\circ\text{C}$ . Otherwise,  $T_{s,o}$  would have to be solved iteratively.

**9-77**  A metal spherical tank is filled with chemicals in exothermic reaction and the heat generation is known. The type of surface to be used so that the outer surface temperature is safe from thermal burn is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Surface temperature is constant. 3 Local atmospheric pressure is 1 atm. 4 The  $T_{\text{surr}}$  is the same as the air temperature.

**Properties** The properties of air at the assumed  $T_f = 30^\circ\text{C}$  and 1 atm pressure are  $k = 0.02588 \text{ W/m}\cdot\text{K}$ ,  $\nu = 1.608 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 0.7282$  (Table A-15), and  $\beta = 1/T_f = 1/303 \text{ K}$ .

The emissivities for the unpainted and painted outer surface are  $\varepsilon_{\text{unpaint}} = 0.20$  and  $\varepsilon_{\text{paint}} = 0.88$ , respectively.

**Analysis** With the assumption that  $T_f = 30^\circ\text{C}$ , the outer surface temperature is estimated as

$$T_{s,o} = 2T_f - T_\infty = 45^\circ\text{C}$$

The Rayleigh number is

$$\text{Ra} = \frac{g\beta(T_{s,o} - T_\infty)D_o^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(30 + 273 \text{ K})^{-1}(45 - 15) \text{ K}(3.02 \text{ m})^3}{(1.608 \times 10^{-5} \text{ m}^2/\text{s})^2}(0.7282) = 7.5344 \times 10^{10}$$

where  $D_o = D_i + 2t_{\text{tank}} = 3.02 \text{ m}$

The Nusselt number relation for sphere is

$$\text{Nu} = 2 + \frac{0.589\text{Ra}^{1/4}}{\left[1 + \left(\frac{0.469}{\text{Pr}}\right)^{9/16}\right]^{4/9}} = 2 + \frac{0.589(7.5344 \times 10^{10})^{1/4}}{\left[1 + \left(\frac{0.469}{0.7282}\right)^{9/16}\right]^{4/9}} = 240.78$$

$$h = \frac{k}{D_o} \text{Nu} = \frac{0.02588 \text{ W/m}\cdot\text{K}}{3.02 \text{ m}}(240.78) = 2.0634 \text{ W/m}^2 \cdot \text{K}$$

Thus, the outer surface temperature can be solved using

$$\dot{Q}_{\text{reaction}} = \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}}$$

$$\dot{e}_{\text{gen}}(\pi D_i^3/6) = h(\pi D_o^2)(T_{s,o} - T_\infty) + \varepsilon\sigma(\pi D_o^2)(T_{s,o}^4 - T_{\text{surr}}^4)$$

where  $\dot{e}_{\text{gen}} = 450 \text{ W/m}^3$

For an unpainted outer surface ( $\varepsilon = \varepsilon_{\text{unpaint}} = 0.20$ ), we have  $\rightarrow T_{s,o} = 77.5^\circ\text{C}$  (above  $45^\circ\text{C}$ )

For a painted outer surface ( $\varepsilon = \varepsilon_{\text{paint}} = 0.88$ ), we have  $\rightarrow T_{s,o} = 44.2^\circ\text{C}$  (below  $45^\circ\text{C}$ )

**Discussion** To prevent thermal burn hazards, the outer surface should be painted with black paint. The black paint increases the radiation heat transfer from the outer surface, thus reducing the surface temperature.

Note that  $30^\circ\text{C}$  is an appropriate temperature to evaluate the air properties for the painted surface, since  $T_{s,o} = 44.2^\circ\text{C}$  would give a film temperature of  $T_f = 29.6^\circ\text{C}$ .

For the unpainted surface, a more appropriate temperature for evaluating the air properties would be  $45^\circ\text{C}$ . By performing an iterative solution, the outer surface temperature for the unpainted surface would converge to about  $73^\circ\text{C}$ .

**9-78**  A metal spherical tank is filled with hot liquid and the inner surface temperature is known. The tank is covered with a layer of insulation and outer surface temperature is to be determined whether it is safe for thermal burn prevention.

**Assumptions** 1 Steady operating conditions exist. 2 Surface temperatures are constant. 3 The tank wall thermal conductivity is constant. 4 Local atmospheric pressure is 1 atm. 5 The  $T_{\text{surr}}$  is the same as the air temperature. 6 The film temperature is 30°C (this assumption will be verified).

**Properties** The properties of air at the assumed  $T_f = 30^\circ\text{C}$  and 1 atm pressure are  $k = 0.02588 \text{ W/m}\cdot\text{K}$ ,  $\nu = 1.608 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 0.7282$  (Table A-15), and  $\beta = 1/T_f = 1/303 \text{ K}$ . The thermal conductivity of the tank is given as  $k_{\text{tank}} = 15 \text{ W/m}\cdot\text{K}$ .

The thermal conductivity and the emissivity of the insulation are given as  $k_{\text{ins}} = 0.15 \text{ W/m}\cdot\text{K}$  and  $\varepsilon = 0.35$ , respectively.

**Analysis** With the assumption that  $T_f = 30^\circ\text{C}$ , the outer surface temperature is estimated as

$$T_{s,o} = 2T_f - T_\infty = 44^\circ\text{C}$$

The Rayleigh number is

$$\text{Ra} = \frac{g\beta(T_{s,o} - T_\infty)D_o^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(30 + 273 \text{ K})^{-1}(44 - 16) \text{ K}(3.16 \text{ m})^3}{(1.608 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7282) = 8.0561 \times 10^{10}$$

where  $D_o = D_i + 2(t_{\text{tank}} + t_{\text{ins}}) = 3.16 \text{ m}$

The Nusselt number relation for sphere is

$$\text{Nu} = 2 + \frac{0.589\text{Ra}^{1/4}}{\left[1 + \left(\frac{0.469}{\text{Pr}}\right)^{9/16}\right]^{4/9}} = 2 + \frac{0.589(8.0561 \times 10^{10})^{1/4}}{\left[1 + \left(\frac{0.469}{0.7282}\right)^{9/16}\right]^{4/9}} = 244.8$$

$$h = \frac{k}{D_o} \text{Nu} = \frac{0.02588 \text{ W/m}\cdot\text{K}}{3.16 \text{ m}} (244.8) = 2.0049 \text{ W/m}^2 \cdot \text{K}$$

Thus, the outer surface temperature can be solved using

$$\dot{Q}_{\text{cond}} = \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}}$$

$$\frac{(T_{s,i} - T_{s,o})}{R_{\text{total}}} = h(\pi D_o^2)(T_{s,o} - T_\infty) + \varepsilon \sigma (\pi D_o^2)(T_{s,o}^4 - T_{\text{surr}}^4) \rightarrow T_{s,o} = 43.5^\circ\text{C}$$

where the total thermal resistance for the spherical layers is

$$R_{\text{total}} = \frac{D_{\text{interface}} - D_i}{2\pi k_{\text{tank}} D_{\text{interface}} D_i} + \frac{D_o - D_{\text{interface}}}{2\pi k_{\text{ins}} D_o D_{\text{interface}}} = 0.01559 \text{ K/W}$$

**Discussion** The insulation layer is able to keep the outer surface temperature 1.5°C below the safe temperature of 45°C. For additional safety precaution, the outer surface temperature can be further reduced by increasing insulation thickness or paint the outer surface to achieve higher emissivity.

The assumed film temperature  $T_f = 30^\circ\text{C}$  is an appropriate assumption, since the determined  $T_{s,o} = 43.5^\circ\text{C}$  would give a film temperature of  $T_f = 29.8^\circ\text{C}$ . Otherwise,  $T_{s,o}$  would have to be solved iteratively.

## Natural Convection from Finned Surfaces and PCBs

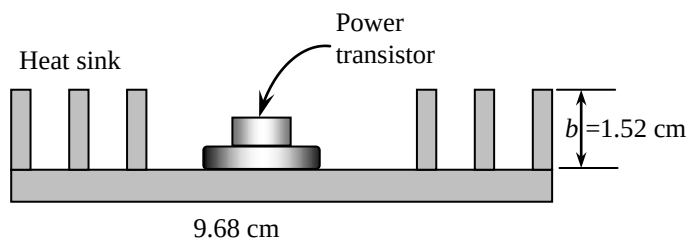
**9-79C** Finned surfaces are frequently used in practice to enhance heat transfer by providing a larger heat transfer surface area. Finned surfaces are referred to as heat sinks in the electronics industry since they provide a medium to which the waste heat generated in the electronic components can be transferred effectively.

**9-80C** A heat sink with closely packed fins will have greater surface area for heat transfer, but smaller heat transfer coefficient because of the extra resistance the additional fins introduce to fluid flow through the interfin passages.

**9-81C** Removing some of the fins on the heat sink will decrease heat transfer surface area, but will increase heat transfer coefficient. The decrease on heat transfer surface area more than offsets the increase in heat transfer coefficient, and thus heat transfer rate will decrease. In the second case, the decrease on heat transfer coefficient more than offsets the increase in heat transfer surface area, and thus heat transfer rate will again decrease.

**9-82** An aluminum heat sink of rectangular profile oriented vertically is used to cool a power transistor. The average natural convection heat transfer coefficient is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 Radiation heat transfer from the sink is negligible. 4 The entire sink is at the base temperature.



**Analysis** The total surface area of the heat sink is

$$A_{\text{fins}} = 2nLb = (2)(6)(0.0762 \text{ m})(0.0152 \text{ m}) + (2)(0.0048 \text{ m})(0.0762 \text{ m}) = 0.01463 \text{ m}^2$$

$$A_{\text{unfinned}} = (4)(0.0145 \text{ m})(0.0762 \text{ m}) + (0.0317 \text{ m})(0.0762 \text{ m}) = 0.006835 \text{ m}^2$$

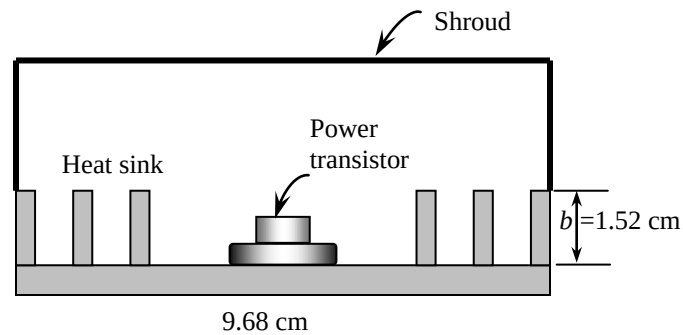
$$A_{\text{total}} = A_{\text{fins}} + A_{\text{unfinned}} = 0.01463 + 0.006835 = 0.021465 \text{ m}^2$$

Then the average natural convection heat transfer coefficient becomes

$$\dot{Q} = hA_{\text{total}}(T_s - T_\infty) \longrightarrow h = \frac{\dot{Q}}{A_{\text{total}}(T_s - T_\infty)} = \frac{18 \text{ W}}{(0.021465 \text{ m}^2)(120 - 22)^\circ\text{C}} = 8.56 \text{ W/m}^2 \cdot ^\circ\text{C}$$

**9-83** Aluminum heat sinks of rectangular profile oriented vertically are used to cool a power transistor. A shroud is placed very close to the tips of fins. The average natural convection heat transfer coefficient is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 Radiation heat transfer from the sink is negligible. 4 The entire sink is at the base temperature.



**Analysis** The total surface area of the shrouded heat sink is

$$A_{\text{fins}} = 2nLb = (2)(6)(0.0762 \text{ m})(0.0152 \text{ m}) = 0.013898 \text{ m}^2$$

$$A_{\text{unfinned}} = (4)(0.0145 \text{ m})(0.0762 \text{ m}) + (0.0317 \text{ m})(0.0762 \text{ m}) = 0.006835 \text{ m}^2$$

$$A_{\text{shroud}} = (2)(0.0968 \text{ m})(0.0762 \text{ m}) = 0.014752 \text{ m}^2$$

$$A_{\text{total}} = A_{\text{fins}} + A_{\text{unfinned}} + A_{\text{shroud}} = 0.013898 + 0.006835 + 0.014752 = 0.035486 \text{ m}^2$$

Then the average natural convection heat transfer coefficient becomes

$$\dot{Q} = hA_{\text{total}}(T_s - T_\infty) \longrightarrow h = \frac{\dot{Q}}{A_{\text{total}}(T_s - T_\infty)} = \frac{18 \text{ W}}{(0.035486 \text{ m}^2)(108 - 22)^\circ\text{C}} = \mathbf{5.90 \text{ W/m}^2 \cdot ^\circ\text{C}}$$

**9-84** A heat sink with equally spaced rectangular fins is to be used to cool a hot surface. The optimum fin spacing and the rate of heat transfer from the heat sink are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The atmospheric pressure at that location is 1 atm.

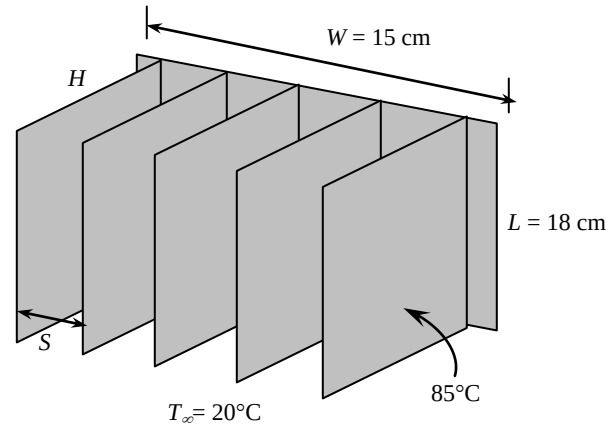
**Properties** The properties of air at 1 atm and 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (85 + 20)/2 = 52.5^\circ\text{C}$  are (Table A-15)

$$k = 0.02753 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.823 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7222$$

$$\beta = \frac{1}{T_f} = \frac{1}{(52.5 + 273)\text{K}} = 0.003072 \text{ K}^{-1}$$



**Analysis** The characteristic length in this case is the height of the surface  $L_c = L = 0.18 \text{ m}$ . Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003072 \text{ K}^{-1})(85 - 20 \text{ K})(0.18 \text{ m})^3}{(1.823 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7222) = 2.483 \times 10^7$$

The optimum fin spacing is

$$S = 2.714 \frac{L}{Ra^{1/4}} = 2.714 \frac{0.18 \text{ m}}{(2.483 \times 10^7)^{1/4}} = 0.006921 \text{ m} = \mathbf{6.921 \text{ mm}}$$

The heat transfer coefficient for this optimum fin spacing case is

$$h = 1.307 \frac{k}{S} = 1.307 \frac{0.02753 \text{ W/m}\cdot^\circ\text{C}}{0.006921 \text{ m}} = 5.199 \text{ W/m}^2\cdot^\circ\text{C}$$

The number of fins and the total heat transfer surface area is

$$n = \frac{w}{S + t} \cong \frac{w}{s} = \frac{0.15}{0.006921} \cong 22 \text{ fins}$$

$$A_s = 2nLH = 2 \times 22 \times (0.18 \text{ m})(0.04 \text{ m}) = 0.3168 \text{ m}^2$$

Then the rate of natural convection heat transfer becomes

$$\dot{Q} = hA_s(T_s - T_\infty) = (5.199 \text{ W/m}^2\cdot^\circ\text{C})(0.3168 \text{ m}^2)(85 - 20)^\circ\text{C} = \mathbf{107.1 \text{ W}}$$

## Natural Convection inside Enclosures

**9-85C** We would recommend putting the hot fluid into the upper compartment of the container. In this case no convection currents will develop in the enclosure since the lighter (hot) fluid will always be on top of the heavier (cold) fluid.

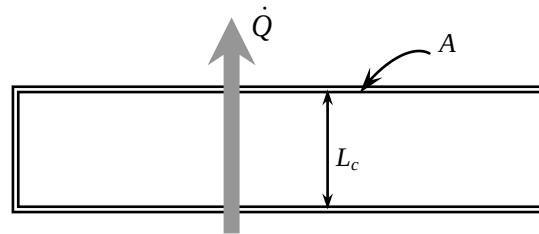
**9-86C** We would disagree with this recommendation since the air space introduces some thermal resistance to heat transfer. The thermal resistance of air space will be zero only when the convection coefficient approaches infinity, which is never the case. However, when the air space is eliminated, so is its thermal resistance.

**9-87C** Yes, dividing the air space into two compartments will retard air motion in the air space, and thus slow down heat transfer by natural convection. The vinyl sheet will also act as a radiation shield and reduce heat transfer by radiation.

**9-88C** The effective thermal conductivity of an enclosure represents the enhancement on heat transfer as result of convection currents relative to conduction. The ratio of the effective thermal conductivity to the ordinary thermal conductivity yields Nusselt number  $Nu = k_{eff} / k$ .

**9-89** Conduction thermal resistance of a medium is expressed as  $R = L / (kA)$ . Thermal resistance of a rectangular enclosure can be expressed by replacing  $L$  with characteristic length of enclosure  $L_c$ , and thermal conductivity  $k$  with effective thermal conductivity  $k_{eff}$  to give

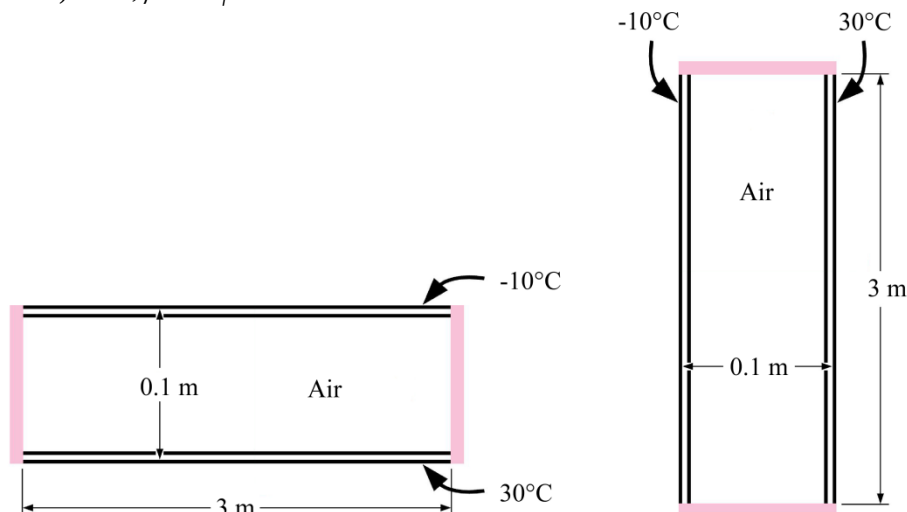
$$R = L_c / (k_{eff} A) = L_c / (kNuA)$$



**9-90** A rectangular enclosure consists of two surfaces separated by an air gap, and the ratio of the heat transfer rate for the horizontal orientation (with hotter surface at the bottom) to that of vertical orientation is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas. 3 Thermal properties are constant.

**Properties** The properties of air at  $T_f = (T_s + T_\infty)/2 = 10^\circ\text{C}$  are  $k = 0.02439 \text{ W/m}\cdot\text{K}$ ,  $\nu = 1.426 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 0.7336$  (from Table A-15). Also,  $\beta = 1/T_f = 3.534 \times 10^{-3} \text{ K}^{-1}$ .



**Analysis** The characteristic length for both cases is  $L_c = 0.1 \text{ m}$ . The Rayleigh number is

$$\text{Ra}_L = \frac{g\beta(T_s - T_\infty)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003534 \text{ K}^{-1})(30 + 10) \text{ K}(0.1 \text{ m})^3}{(1.426 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7336) = 5.003 \times 10^6$$

The aspect ratio for this rectangular enclosure is  $H/L = 30 \text{ m}$ . The Nusselt numbers for horizontal and vertical orientations are

$$\text{Nu}_{\text{horiz}} = 1 + 1.44 \left[ 1 - \frac{1708}{\text{Ra}_L} \right]^+ + \left[ \frac{\text{Ra}_L^{1/3}}{18} - 1 \right]^+ = 10.94$$

$$\text{Nu}_{\text{vert}} = 0.42 \text{Ra}^{1/4} \text{Pr}^{0.012} \left( \frac{H}{L} \right)^{-0.3} = 7.134$$

Hence, the ratio of heat transfer rate is

$$\frac{\dot{Q}_{\text{horiz}}}{\dot{Q}_{\text{vert}}} = \frac{h_{\text{horiz}}(k/L)A_s\Delta T}{h_{\text{vert}}(k/L)A_s\Delta T} = \frac{\text{Nu}_{\text{horiz}}(k/L)A_s\Delta T}{\text{Nu}_{\text{vert}}(k/L)A_s\Delta T} = \frac{\text{Nu}_{\text{horiz}}}{\text{Nu}_{\text{vert}}} = \frac{10.94}{7.134} = \mathbf{1.53}$$

**Discussion** For the same  $\Delta T$ , the rate of heat transfer for the horizontal orientation is 53% larger than that for the vertical orientation.

**9-91** The absorber plate and the glass cover of a flat-plate solar collector are maintained at specified temperatures. The rate of heat loss from the absorber plate by natural convection is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 Heat loss by radiation is negligible. 4 The air pressure in the enclosure is 1 atm.

**Properties** The properties of air at 1 atm and the average temperature of  $(T_1 + T_2)/2 = (80 + 40)/2 = 60^\circ\text{C}$  are (Table A-15)

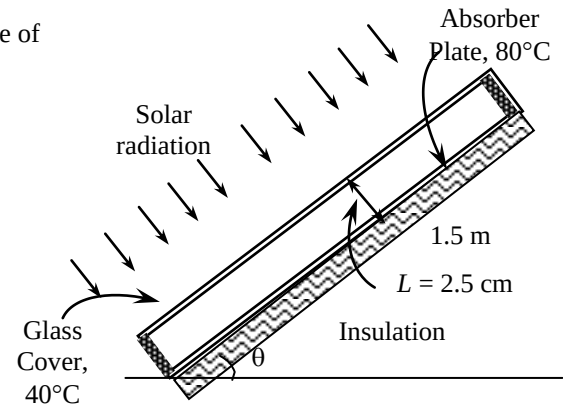
$$k = 0.02808 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.896 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7202$$

$$\beta = \frac{1}{T_f} = \frac{1}{(60 + 273)\text{K}} = 0.003003 \text{ K}^{-1}$$

**Analysis** For  $\theta = 0^\circ$ , we have horizontal rectangular enclosure. The characteristic length in this case is the distance between the two glasses  $L_c = L = 0.025 \text{ m}$ . Then,



$$\text{Ra} = \frac{g\beta(T_1 - T_2)L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003003 \text{ K}^{-1})(80 - 40 \text{ K})(0.025 \text{ m})^3}{(1.896 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7202) = 3.689 \times 10^4$$

$$\begin{aligned} \text{Nu} &= 1 + 1.44 \left[ 1 - \frac{1708}{\text{Ra}} \right]^+ + \left[ \frac{\text{Ra}^{1/3}}{18} - 1 \right]^+ \\ &= 1 + 1.44 \left[ 1 - \frac{1708}{3.689 \times 10^4} \right]^+ + \left[ \frac{(3.689 \times 10^4)^{1/3}}{18} - 1 \right]^+ = 3.223 \end{aligned}$$

Then

$$A_s = H \times W = (1.5 \text{ m})(3 \text{ m}) = 4.5 \text{ m}^2$$

$$\dot{Q} = k \text{Nu}_s \frac{T_1 - T_2}{L} = (0.02808 \text{ W/m}\cdot^\circ\text{C})(3.223)(4.5 \text{ m}^2) \frac{(80 - 40)^\circ\text{C}}{0.025 \text{ m}} = \mathbf{652 \text{ W}}$$

For  $\theta = 30^\circ$ , we obtain

$$\begin{aligned} \text{Nu} &= 1 + 1.44 \left[ 1 - \frac{1708}{\text{Ra} \cos \theta} \right]^+ \left[ 1 - \frac{1708(\sin 1.8\theta)^{1.6}}{\text{Ra} \cos \theta} \right] + \left[ \frac{(\text{Ra} \cos \theta)^{1/3}}{18} - 1 \right]^+ \\ &= 1 + 1.44 \left[ 1 - \frac{1708}{(3.689 \times 10^4) \cos(30)} \right]^+ \left[ 1 - \frac{1708[\sin(1.8 \times 30)]^{1.6}}{(3.689 \times 10^4) \cos(30)} \right] + \left[ \frac{[(3.689 \times 10^4) \cos(30)]^{1/3}}{18} - 1 \right]^+ \\ &= 3.074 \end{aligned}$$

$$\dot{Q} = k \text{Nu}_s \frac{T_1 - T_2}{L} = (0.02808 \text{ W/m}\cdot^\circ\text{C})(3.074)(4.5 \text{ m}^2) \frac{(80 - 40)^\circ\text{C}}{0.025 \text{ m}} = \mathbf{621 \text{ W}}$$

For  $\theta = 90^\circ$ , we have vertical rectangular enclosure. The Nusselt number for this geometry and orientation can be determined from  $(\text{Ra} = 3.689 \times 10^4)$  - same as that for horizontal case)

$$\text{Nu} = 0.42 \text{Ra}^{1/4} \text{Pr}^{0.012} \left( \frac{H}{L} \right)^{-0.3} = 0.42(3.689 \times 10^4)^{1/4} (0.7202)^{0.012} \left( \frac{1.5 \text{ m}}{0.025 \text{ m}} \right)^{-0.3} = 1.69$$

$$\dot{Q} = k \text{Nu}_s \frac{T_1 - T_2}{L} = (0.02808 \text{ W/m}\cdot^\circ\text{C})(1.69)(4.5 \text{ m}^2) \frac{(80 - 40)^\circ\text{C}}{0.025 \text{ m}} = \mathbf{344 \text{ W}}$$

**Discussion** Caution is advised for the vertical case since the condition  $H/L < 40$  is not satisfied.

**9-92** Two glasses of a double pane window are maintained at specified temperatures. The fraction of heat transferred through the enclosure by radiation is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The air pressure in the enclosure is 1 atm.

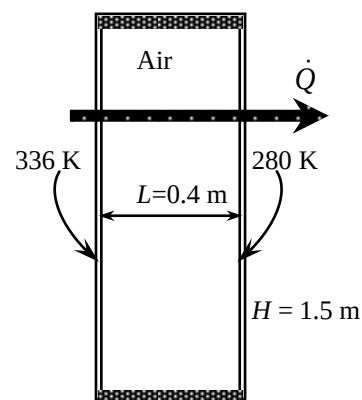
**Properties** The properties of air at 1 atm and the average temperature of  $(T_1 + T_2)/2 = (280 + 336)/2 = 308 \text{ K} = 35^\circ\text{C}$  are (Table A-15E)

$$k = 0.02625 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.655 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7268$$

$$\beta = \frac{1}{T_f} = \frac{1}{308 \text{ K}} = 0.003247 \text{ K}^{-1}$$



**Analysis** The characteristic length in this case is the distance between the two glasses,  $L_c = L = 0.4 \text{ m}$ . Then,

$$Ra_L = \frac{g\beta(T_1 - T_2)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003247 \text{ K}^{-1})(336 - 280 \text{ K})(0.4 \text{ m})^3}{(1.655 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7268) = 3.029 \times 10^8$$

The aspect ratio of the geometry is  $H/L = 1.5/0.4 = 3.75$ . For this value of  $H/L$  the Nusselt number can be determined from

$$Nu = 0.22 \left( \frac{\text{Pr}}{0.2 + \text{Pr}} Ra \right)^{0.28} \left( \frac{H}{L} \right)^{-1/4} = 0.22 \left( \frac{0.7268}{0.2 + 0.7268} (3.029 \times 10^8) \right)^{0.28} \left( \frac{1.5}{0.4} \right)^{-1/4} = 35.00$$

Then,

$$A_s = H \times W = (1.5 \text{ m})(3 \text{ m}) = 4.5 \text{ m}^2$$

$$\dot{Q}_{\text{conv}} = kNu_s \frac{T_1 - T_2}{L} = (0.02625 \text{ W/m}\cdot^\circ\text{C})(35.00)(4.5 \text{ m}^2) \frac{(336 - 280) \text{ K}}{0.4 \text{ m}} = 578.9 \text{ W}$$

The effective emissivity is

$$\frac{1}{\varepsilon_{\text{eff}}} = \frac{1}{\varepsilon_1} + \frac{1}{\varepsilon_2} - 1 = \frac{1}{0.15} + \frac{1}{0.90} - 1 = 6.778 \longrightarrow \varepsilon_{\text{eff}} = 0.1475$$

The rate of heat transfer by radiation is

$$\begin{aligned} \dot{Q}_{\text{rad}} &= \varepsilon_{\text{eff}} A_s \sigma (T_1^4 - T_2^4) \\ &= (0.1475)(4.5 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)[(336 \text{ K})^4 - (280 \text{ K})^4] = 248.4 \text{ W} \end{aligned}$$

Then the fraction of heat transferred through the enclosure by radiation becomes

$$f_{\text{rad}} = \frac{\dot{Q}_{\text{rad}}}{\dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}}} = \frac{248.4}{578.9 + 248.4} = \mathbf{0.30}$$

**9-93** A double pane window with an air gap is considered. The rate of heat transfer through the window by natural convection the temperature of the outer glass layer are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The air pressure in the enclosure is 1 atm. 4 Radiation heat transfer is neglected.

**Properties** For natural convection between the inner surface of the window and the room air, the properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (18 + 26)/2 = 22^\circ\text{C}$  are (Table A-15)

$$k = 0.02529 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.534 \times 10^{-5} \text{ m}^2/\text{s}$$

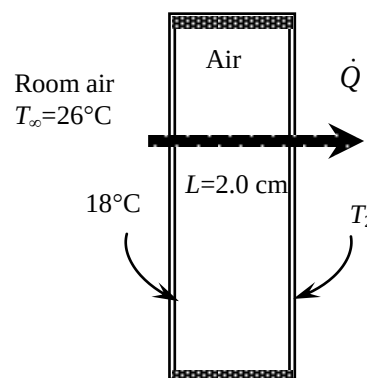
$$\text{Pr} = 0.7304$$

$$\beta = \frac{1}{T_f} = \frac{1}{(22 + 273)\text{K}} = 0.00339 \text{ K}^{-1}$$

For natural convection between the two glass sheets separated by an air gap, the properties of air at 1 atm and the anticipated average temperature of  $(T_1 + T_2)/2 = (18 + 0)/2 = 9^\circ\text{C}$  are (Table A-15)

$$k = 0.02431 \text{ W/m}\cdot^\circ\text{C}, \quad \nu = 1.417 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7339, \quad \beta = \frac{1}{T_f} = \frac{1}{(9 + 273)\text{K}} = 0.003546 \text{ K}^{-1}$$



**Analysis** We first calculate the natural convection heat transfer between the room air and the inner surface of the window.

$$L_c = H = 1.5 \text{ m}$$

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)H^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.00339 \text{ K}^{-1})(26 - 18)\text{K}(1.5 \text{ m})^3}{(1.534 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7304) = 2.787 \times 10^9$$

$$\text{Nu} = \left\{ 0.825 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.492}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(2.787 \times 10^9)^{1/6}}{\left[ 1 + \left( \frac{0.492}{0.7304} \right)^{9/16} \right]^{8/27}} \right\}^2 = 169.5$$

$$h = \frac{k}{H} \text{Nu} = \frac{0.02529 \text{ W/m}\cdot^\circ\text{C}}{1.5 \text{ m}} (169.5) = 2.858 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = H \times W = (1.5 \text{ m})(2.8 \text{ m}) = 4.2 \text{ m}^2$$

$$\dot{Q}_{\text{conv}} = hA_s(T_\infty - T_s) = (2.858 \text{ W/m}^2\cdot^\circ\text{C})(4.2 \text{ m}^2)(26 - 18)^\circ\text{C} = \mathbf{96.0 \text{ W}}$$

Next, we consider the natural convection between the two glass sheets separated by an air gap.

$$L_c = L = 2.0 \text{ cm}$$

$$\text{Ra} = \frac{g\beta(T_1 - T_2)L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003546 \text{ K}^{-1})(18 - 0)\text{K}(0.020 \text{ m})^3}{(1.417 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7339) = 18,309$$

$$\text{Nu} = 0.42 \text{Ra}^{1/4} \text{Pr}^{0.012} \left( \frac{H}{L} \right)^{-0.3} = 0.42(18,309)^{1/4} (0.7339)^{0.012} \left( \frac{1.5 \text{ m}}{0.020 \text{ m}} \right)^{-0.3} = 1.333$$

Under steady operation, the rate of heat transfer between the room air and the inner surface of the window is equal to the heat transfer through the air gap. Setting these two equal to each other we obtain the temperature of the outer glass sheet

$$\dot{Q} = k \text{Nu} A_s \frac{T_1 - T_2}{L} \longrightarrow 96.0 \text{ W} = (0.02431 \text{ W/m}\cdot^\circ\text{C})(1.333)(4.2 \text{ m}^2) \frac{(18 - T_2)^\circ\text{C}}{0.020 \text{ m}} \longrightarrow T_2 = \mathbf{3.9^\circ\text{C}}$$

which is sufficiently close to the assumed temperature  $0^\circ\text{C}$ . Therefore, there is no need to repeat the calculations.

**Discussion** The aspect ratio for this geometry is  $H/L = (1.5 \text{ m})/(0.020 \text{ m}) = 75$ . There is no Nusselt number relation given in the text covering this value of aspect ratio. One of the closest aspect ratio ranges is given by Eq. 9-54 as  $10 < H/L < 40$ . We used Eq. 9-54 knowing that there will be some error in the calculation of Nusselt number.

**9-94** Two glasses of a double pane window are maintained at specified temperatures. The rate of heat transfer through the window by natural convection and radiation, and the R-value of insulation are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The air pressure in the enclosure is 1 atm.

**Properties** The properties of air at 1 atm and the average temperature of  $(T_1 + T_2)/2 = (18 + 4)/2 = 11^\circ\text{C}$  are (Table A-15)

$$k = 0.02446 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.435 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7333$$

$$\beta = \frac{1}{T_f} = \frac{1}{(11 + 273) \text{ K}} = 0.003521 \text{ K}^{-1}$$

**Analysis** (a) The characteristic length in this case is the distance between the two glasses,  $L_c = L = 2 \text{ cm}$ . Then,

$$\text{Ra}_L = \frac{g\beta(T_1 - T_2)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003521 \text{ K}^{-1})(18 - 4 \text{ K})(0.02 \text{ m})^3}{(1.435 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7333) = 13,776$$

The aspect ratio of the geometry is  $H/L = 150/2 = 75$  (which is somewhat over 40, but still close enough for an approximate analysis). For these values of  $H/L$  and  $\text{Ra}_L$ , the Nusselt number can be determined from

$$\text{Nu} = 0.42 \text{Ra}^{1/4} \text{Pr}^{0.012} \left(\frac{H}{L}\right)^{-0.3} = 0.42(13,776)^{1/4} (0.7333)^{0.012} \left(\frac{1.5 \text{ m}}{0.02 \text{ m}}\right)^{-0.3} = 1.241$$

Then,

$$A_s = H \times W = (1.5 \text{ m})(1.8 \text{ m}) = 2.7 \text{ m}^2$$

$$\dot{Q} = k \text{Nu} A_s \frac{T_1 - T_2}{L} = (0.02446 \text{ W/m}\cdot^\circ\text{C})(1.241)(2.7 \text{ m}^2) \frac{(18 - 4)^\circ\text{C}}{0.02 \text{ m}} = \mathbf{57.4 \text{ W}}$$

(b) The rate of heat transfer by radiation is

$$\begin{aligned} \dot{Q}_{\text{rad}} &= \varepsilon A_s \sigma (T_1^4 - T_2^4) \\ &= (0.82)(2.16 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)[(18 + 273 \text{ K})^4 - (4 + 273 \text{ K})^4] = \mathbf{161.1 \text{ W}} \end{aligned}$$

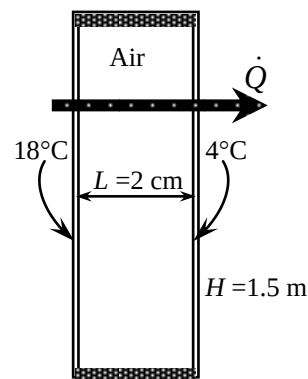
Then the total rate of heat transfer is

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{convection}} + \dot{Q}_{\text{rad}} = 57.4 + 161.1 = 218.5 \text{ W}$$

Then the effective thermal conductivity of the air, which also accounts for the radiation effect and the R-value become

$$\dot{Q}_{\text{total}} = k_{\text{eff}} A_s \frac{T_1 - T_2}{L} \longrightarrow k_{\text{eff}} = \frac{\dot{Q} L}{A_s (T_1 - T_2)} = \frac{(218.5 \text{ W})(0.02 \text{ m})}{(2.7 \text{ m}^2)(18 - 4)^\circ\text{C}} = 0.1156 \text{ W/m}\cdot^\circ\text{C}$$

$$R_{\text{value}} = \frac{L}{k_{\text{eff}}} = \frac{0.02 \text{ m}}{0.1156 \text{ W/m}\cdot^\circ\text{C}} = 0.173 \text{ m}^2 \cdot ^\circ\text{C/W} = \mathbf{R - 0.17}$$





**9-95** Prob. 9-94 is reconsidered. The effect of the air gap thickness on the rates of heat transfer by natural convection and radiation, and the  $R$ -value of insulation is to be investigated.

**Analysis** The problem is solved using EES, and the solution is given below.

**"GIVEN"**

H=1.5 [m]  
W=1.8 [m]  
L=0.02 [m]  
T\_1=18 [C]  
T\_2=4 [C]  
epsilon\_eff=0.82

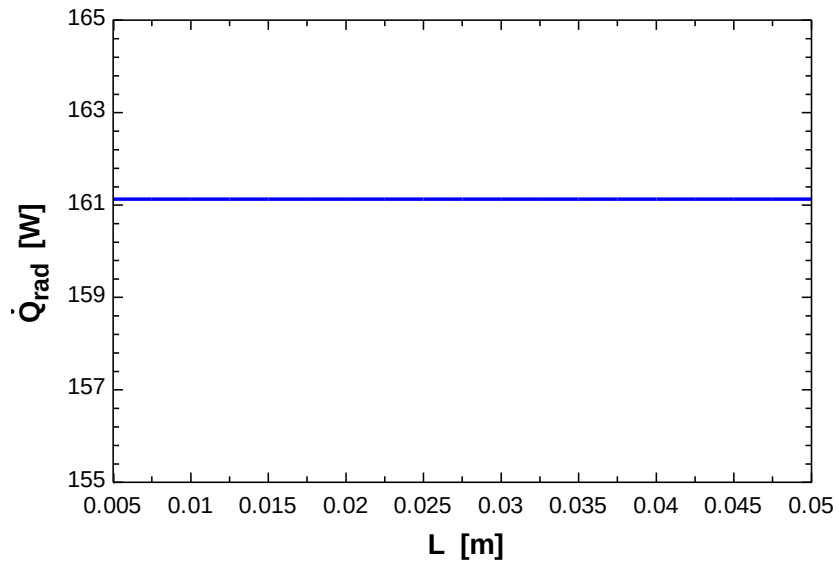
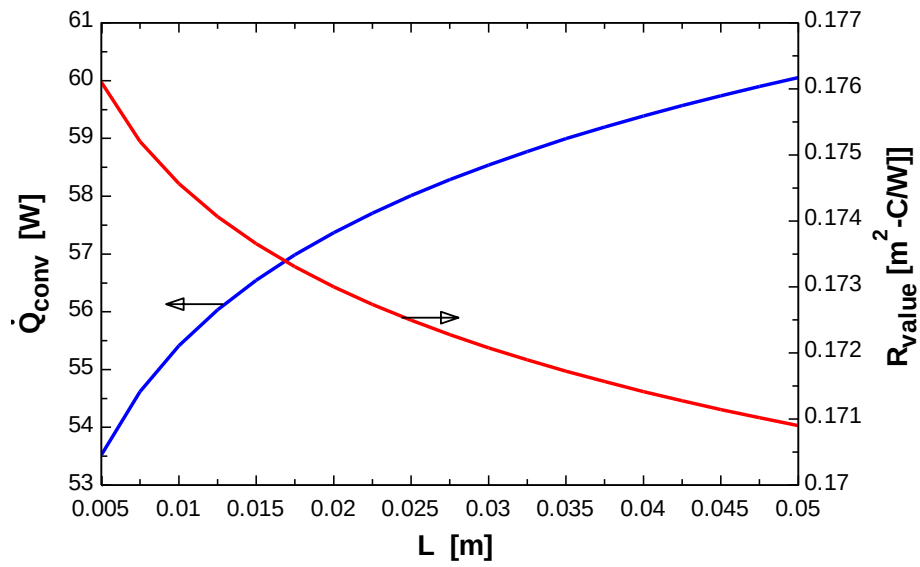
**"PROPERTIES"**

Fluid\$='air'  
k=Conductivity(Fluid\$, T=T\_ave)  
Pr=Prandtl(Fluid\$, T=T\_ave)  
rho=Density(Fluid\$, T=T\_ave, P=101.325)  
mu=Viscosity(Fluid\$, T=T\_ave)  
nu=mu/rho  
beta=1/(T\_ave+273)  
T\_ave=1/2\*(T\_1+T\_2)  
g=9.81 [m/s^2] "gravitational acceleration"  
sigma=5.67E-8 [W/m^2-K^4] "Stefan-Boltzmann constant"

**"ANALYSIS"**

Ra=(g\*beta\*(T\_1-T\_2)\*L^3)/nu^2\*Pr  
Ratio=H/L  
Nusselt=0.42\*Ra^0.25\*Pr^0.012\*(H/L)^(-0.3)  
A=H\*W  
Q\_dot\_conv=k\*Nusselt\*A\*(T\_1-T\_2)/L  
Q\_dot\_rad=epsilon\_eff\*A\*sigma\*((T\_1+273)^4-(T\_2+273)^4)  
Q\_dot\_total=Q\_dot\_conv+Q\_dot\_rad  
Q\_dot\_total=k\_eff\*A\*(T\_1-T\_2)/L  
R\_value=L/k\_eff

| L<br>[m] | Q <sub>conv</sub><br>[W] | Q <sub>rad</sub><br>[W] | R-value<br>[m <sup>2</sup> -C/W] |
|----------|--------------------------|-------------------------|----------------------------------|
| 0.005    | 53.52                    | 161.1                   | 0.1761                           |
| 0.0075   | 54.62                    | 161.1                   | 0.1752                           |
| 0.01     | 55.41                    | 161.1                   | 0.1746                           |
| 0.0125   | 56.03                    | 161.1                   | 0.1741                           |
| 0.015    | 56.55                    | 161.1                   | 0.1737                           |
| 0.0175   | 56.98                    | 161.1                   | 0.1733                           |
| 0.02     | 57.37                    | 161.1                   | 0.173                            |
| 0.0225   | 57.7                     | 161.1                   | 0.1727                           |
| 0.025    | 58.01                    | 161.1                   | 0.1725                           |
| 0.0275   | 58.29                    | 161.1                   | 0.1723                           |
| 0.03     | 58.54                    | 161.1                   | 0.1721                           |
| 0.0325   | 58.78                    | 161.1                   | 0.1719                           |
| 0.035    | 58.99                    | 161.1                   | 0.1717                           |
| 0.0375   | 59.2                     | 161.1                   | 0.1716                           |
| 0.04     | 59.39                    | 161.1                   | 0.1714                           |
| 0.0425   | 59.57                    | 161.1                   | 0.1713                           |
| 0.045    | 59.74                    | 161.1                   | 0.1711                           |
| 0.0475   | 59.9                     | 161.1                   | 0.171                            |
| 0.05     | 60.05                    | 161.1                   | 0.1709                           |



**9-96** A simple solar collector is built by placing a clear plastic tube around a garden hose. The rate of heat loss from the water in the hose per meter of its length by natural convection is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 Heat loss by radiation is negligible. 3 The air pressure in the enclosure is 1 atm.

**Properties** Based on the problem statement, the properties of air at 1 atm and the anticipated average temperature of  $(T_i + T_o)/2 = (65 + 35)/2 = 50^\circ\text{C}$  are (Table A-15)

$$k = 0.02735 \text{ W/m}\cdot^\circ\text{C}, \quad \nu = 1.798 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7228, \quad \beta = \frac{1}{T_f} = \frac{1}{(50 + 273)\text{K}} = 0.003096 \text{ K}^{-1}$$

**Analysis** We assume the plastic tube temperature to be  $35^\circ\text{C}$ . We will check this assumption later, and repeat calculations, if necessary. The characteristic length in this case is

$$L_c = \frac{D_o - D_i}{2} = \frac{5 - 1.6}{2} = 1.7 \text{ cm}$$

Then,

$$\text{Ra} = \frac{g\beta(T_i - T_o)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003096 \text{ K}^{-1})(65 - 35 \text{ K})(0.017 \text{ m})^3}{(1.798 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7228) = 10,000$$

The effective thermal conductivity is

$$F_{\text{cyl}} = \frac{[\ln(D_o/D_i)]^4}{L_c^3(D_i^{-3/5} + D_o^{-3/5})^5} = \frac{[\ln(0.05/0.016)]^4}{(0.017 \text{ m})^3[(0.016 \text{ m})^{-3/5} + (0.05 \text{ m})^{-3/5}]^5} = 0.1821$$

$$k_{\text{eff}} = 0.386k \left( \frac{\text{Pr}}{0.861 + \text{Pr}} \right)^{1/4} (F_{\text{cyl}} \text{Ra})^{1/4}$$

$$= 0.386(0.02735 \text{ W/m}\cdot^\circ\text{C}) \left( \frac{0.7228}{0.861 + 0.7228} \right)^{1/4} [(0.1821)(10,000)]^{1/4} = 0.05670 \text{ W/m}\cdot^\circ\text{C}$$

Then the rate of heat transfer between the cylinders becomes

$$\dot{Q} = \frac{2\pi k_{\text{eff}}}{\ln(D_o/D_i)} (T_i - T_o) = \frac{2\pi(0.05670 \text{ W/m}\cdot^\circ\text{C})}{\ln(0.05/0.016)} (65 - T_o) \quad (\text{Eq. 1})$$

Now we will calculate heat transfer from plastic tube to the ambient air by natural convection. Note that we should find a result close to the value we have already calculated since in steady operation they must be equal to each other. Also note that we neglect radiation heat transfer. We will use the same assumption for the plastic tube temperature (i.e.,  $35^\circ\text{C}$ ). The properties of air at 1 atm and the film temperature of  $T_{\text{avg}} = (T_s + T_\infty)/2 = (35 + 26)/2 = 30.5^\circ\text{C}$  are

$$k = 0.02592 \text{ W/m}\cdot^\circ\text{C}, \quad \nu = 1.613 \times 10^{-5} \text{ m}^2/\text{s},$$

$$\text{Pr} = 0.7281, \quad \text{and } \beta = 1/T_f = 1/(30.5 + 273)\text{K} = 0.003295 \text{ K}^{-1}$$

The characteristic length in this case is the outer diameter of the solar collector  $L_c = D_o = 0.05 \text{ m}$ . Then,

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)D_o^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003295 \text{ K}^{-1})(35 - 26 \text{ K})(0.05 \text{ m})^3}{(1.613 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7281) = 1.018 \times 10^5$$

$$\text{Nu} = \left\{ 0.6 + \frac{0.387 \text{Ra}^{1/6}}{[1 + (0.559/\text{Pr})^{9/16}]^{8/27}} \right\}^2 = \left\{ 0.6 + \frac{0.387(1.018 \times 10^5)^{1/6}}{[1 + (0.559/0.7281)^{9/16}]^{8/27}} \right\}^2 = 7.838$$

$$A_o = \pi D_o L = \pi(0.05 \text{ m})(1 \text{ m}) = 0.1571 \text{ m}^2 \quad h_o = \frac{k}{D_o} \text{Nu} = \frac{0.02592 \text{ W/m}\cdot^\circ\text{C}}{0.05 \text{ m}} (7.838) = 4.063 \text{ W/m}^2\cdot^\circ\text{C}$$

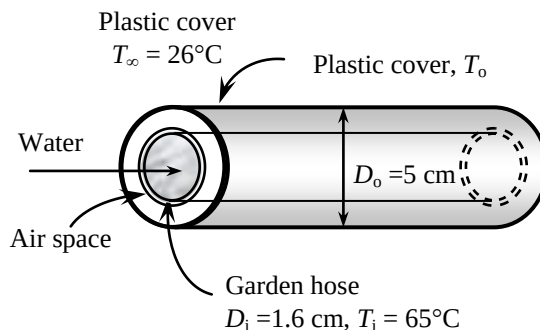
$$\dot{Q} = hA_o(T_o - T_\infty) = (4.063 \text{ W/m}^2\cdot^\circ\text{C})(0.1571 \text{ m}^2)(T_o - 26)^\circ\text{C} \quad (\text{Eq. 2})$$

Solving Eq. 1 and Eq. 2 simultaneously, we find

$$T_o = 38.8^\circ\text{C}, \quad \dot{Q} = 8.18 \text{ W}$$

Repeating the calculations at the new average temperature for enclosure analysis and at the new film temperature for convection at the outer surface analysis using the new calculated temperature  $38.8^\circ\text{C}$ , we find

$$T_o = 39.0^\circ\text{C}, \quad \dot{Q} = \mathbf{8.22 \text{ W}}$$





**9-97** Prob. 9-96 is reconsidered. The rate of heat loss from the water by natural convection as a function of the ambient air temperature is to be plotted.

**Analysis** The problem is solved using EES, and the solution is given below.

**"GIVEN"**

D\_1=0.016 [m]  
 D\_2=0.05 [m]  
 T\_1=65 "[C]"  
 T\_infinity=26 [C]  
 Length=1 [m] "unit length of the tube is considered"

**"PROPERTIES for enclosure"**

Fluid\$='air'  
 k\_1=Conductivity(Fluid\$, T=T\_ave)  
 Pr\_1=Prandtl(Fluid\$, T=T\_ave)  
 rho\_1=Density(Fluid\$, T=T\_ave, P=101.3)  
 mu\_1=Viscosity(Fluid\$, T=T\_ave)  
 nu\_1=mu\_1/rho\_1  
 beta\_1=1/(T\_ave+273)  
 T\_ave=1/2\*(T\_1+T\_2)  
 g=9.807 [m/s^2]

**"ANALYSIS for enclosure"**

L=(D\_2-D\_1)/2  
 Ra\_1=(g\*beta\_1\*(T\_1-T\_2)\*L^3)/nu\_1^2\*Pr\_1  
 F\_cyl=(ln(D\_2/D\_1))^4/(L^3\*(D\_1^(-3/5)+D\_2^(-3/5))^5)  
 k\_eff=0.386\*k\_1\*(Pr\_1/(0.861+Pr\_1))^0.25\*(F\_cyl\*Ra\_1)^0.25  
 Q\_dot=(2\*pi\*k\_eff)/ln(D\_2/D\_1)\*(T\_1-T\_2)

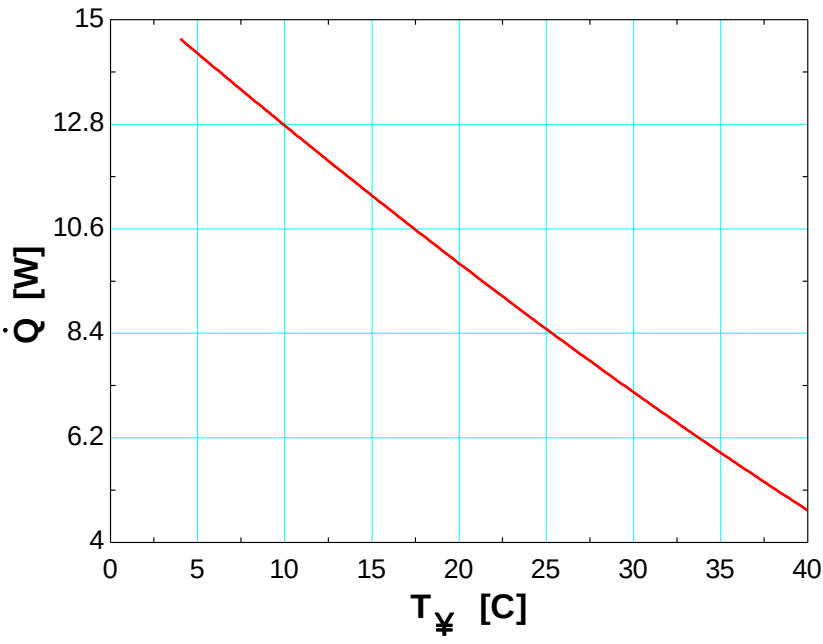
**"PROPERTIES for convection on the outer surface"**

k\_2=Conductivity(Fluid\$, T=T\_film)  
 Pr\_2=Prandtl(Fluid\$, T=T\_film)  
 rho\_2=Density(Fluid\$, T=T\_film, P=101.3)  
 mu\_2=Viscosity(Fluid\$, T=T\_film)  
 nu\_2=mu\_2/rho\_2  
 beta\_2=1/(T\_film+273)  
 T\_film=1/2\*(T\_2+T\_infinity)

**"ANALYSIS for convection on the outer surface"**

delta=D\_2  
 Ra\_2=(g\*beta\_2\*(T\_2-T\_infinity)\*delta^3)/nu\_2^2\*Pr\_2  
 Nusselt=(0.6+(0.387\*Ra\_2^(1/6))/(1+(0.559/Pr\_2)^(9/16)))^(8/27))^2  
 h=k\_2/delta\*Nusselt  
 A=pi\*D\_2\*Length  
 Q\_dot=h\*A\*(T\_2-T\_infinity)

| $T_{\infty}$ [W] | $\dot{Q}$ [W] |
|------------------|---------------|
| 4                | 14.6          |
| 6                | 13.98         |
| 8                | 13.37         |
| 10               | 12.77         |
| 12               | 12.18         |
| 14               | 11.59         |
| 16               | 11.01         |
| 18               | 10.44         |
| 20               | 9.871         |
| 22               | 9.314         |
| 24               | 8.764         |
| 26               | 8.222         |
| 28               | 7.688         |
| 30               | 7.163         |
| 32               | 6.647         |
| 34               | 6.139         |
| 36               | 5.641         |
| 38               | 5.153         |
| 40               | 4.675         |



**9-98** The space between the two concentric cylinders is filled with water or air. The rate of heat transfer from the outer cylinder to the inner cylinder by natural convection is to be determined for both cases.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The air pressure in the enclosure is 1 atm. 4 Heat transfer by radiation is negligible.

**Properties** The properties of water air at the average temperature of  $(T_i + T_o)/2 = (54 + 106)/2 = 80^\circ\text{C}$  are (Table A-9)

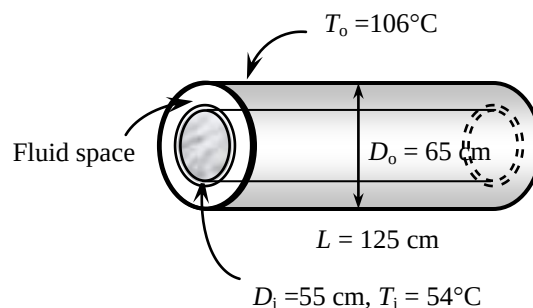
$$k = 0.670 \text{ W/m}\cdot^\circ\text{C}, \quad \nu = 3.653 \times 10^{-7} \text{ m}^2/\text{s}$$

$$\text{Pr} = 2.22, \quad \beta = 0.653 \times 10^{-3} \text{ K}^{-1}$$

The properties of air at 1 atm and the average temperature of  $(T_i + T_o)/2 = (54 + 106)/2 = 80^\circ\text{C}$  are (Table A-15)

$$k = 0.02953 \text{ W/m}\cdot^\circ\text{C}, \quad \nu = 2.097 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7154, \quad \beta = \frac{1}{T_f} = \frac{1}{(80 + 273)\text{K}} = 0.002833 \text{ K}^{-1}$$



**Analysis** (a) The fluid is water:

$$L_c = \frac{D_o - D_i}{2} = \frac{65 - 55}{2} = 5 \text{ cm}.$$

$$\text{Ra} = \frac{g\beta(T_o - T_i)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.653 \times 10^{-3} \text{ K}^{-1})(106 - 54)\text{K}(0.05 \text{ m})^3}{(3.653 \times 10^{-7} \text{ m}^2/\text{s})^2} (2.22) = 6.927 \times 10^8$$

The effective thermal conductivity is

$$F_{\text{cyl}} = \frac{\left[ \ln \frac{D_o}{D_i} \right]^4}{L_c^3 (D_i^{-3/5} + D_o^{-3/5})^5} = \frac{\left[ \ln \frac{0.65 \text{ m}}{0.55 \text{ m}} \right]^4}{(0.05 \text{ m})^3 [(0.55 \text{ m})^{-7/5} + (0.65 \text{ m})^{-7/5}]^5} = 0.04136$$

$$k_{\text{eff}} = 0.386k \left( \frac{\text{Pr}}{0.861 + \text{Pr}} \right)^{1/4} (F_{\text{cyl}} \text{Ra})^{1/4}$$

$$= 0.386(0.670 \text{ W/m}\cdot^\circ\text{C}) \left( \frac{2.22}{0.861 + 2.22} \right)^{1/4} [(0.04136)(6.927 \times 10^8)]^{1/4} = 17.43 \text{ W/m}\cdot^\circ\text{C}$$

Then the rate of heat transfer between the cylinders becomes

$$\dot{Q} = \frac{2\pi k_{\text{eff}}}{\ln \left( \frac{D_o}{D_i} \right)} (T_o - T_i) = \frac{2\pi(17.43 \text{ W/m}\cdot^\circ\text{C})}{\ln \left( \frac{0.65 \text{ m}}{0.55 \text{ m}} \right)} (106 - 54) = 34,090 \text{ W} = \mathbf{34.1 \text{ kW}}$$

(b) The fluid is air:

$$\text{Ra} = \frac{g\beta(T_o - T_i)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.002833 \text{ K}^{-1})(106 - 54)\text{K}(0.05 \text{ m})^3}{(2.097 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7154) = 2.939 \times 10^5$$

The effective thermal conductivity is

$$F_{\text{cyl}} = \frac{\left[ \ln \frac{D_o}{D_i} \right]^4}{L_c^3 (D_i^{-3/5} + D_o^{-3/5})^5} = \frac{\left[ \ln \frac{0.65 \text{ m}}{0.55 \text{ m}} \right]^4}{(0.05 \text{ m})^3 [(0.55 \text{ m})^{-7/5} + (0.65 \text{ m})^{-7/5}]^5} = 0.04136$$

$$k_{\text{eff}} = 0.386k \left( \frac{\text{Pr}}{0.861 + \text{Pr}} \right)^{1/4} (F_{\text{cyl}} \text{Ra})^{1/4}$$

$$= 0.386(0.02953 \text{ W/m}\cdot^\circ\text{C}) \left( \frac{0.7154}{0.861 + 0.7154} \right)^{1/4} [(0.04136)(2.939 \times 10^5)]^{1/4} = 0.09824 \text{ W/m}\cdot^\circ\text{C}$$

Then the rate of heat transfer between the cylinders becomes

$$\dot{Q} = \frac{2\pi k_{\text{eff}}}{\ln \left( \frac{D_o}{D_i} \right)} (T_o - T_i) = \frac{2\pi(0.09824 \text{ W/m}\cdot^\circ\text{C})}{\ln \left( \frac{0.65 \text{ m}}{0.55 \text{ m}} \right)} (106 - 54) = \mathbf{192 \text{ W}}$$

**9-99**  A hot fluid flowing inside a horizontal tube with a known mass flow rate and temperature difference between the tube inlet and outlet. The tube is enclosed in a concentric cylindrical thin cover. The concentric outer cover temperature is to be determined whether it is safe from thermal burn hazards.

**Assumptions** 1 Steady operating conditions exist. 2 Surface temperatures are constant. 3 Air is an ideal gas with constant properties. 4 Heat loss by radiation is negligible. 5 The air pressure in the enclosure is 1 atm.

**Properties** The properties of air at the assumed  $T_{\text{avg}} = 80^\circ\text{C}$  and 1 atm pressure are  $k = 0.02953 \text{ W/m}\cdot\text{K}$ ,  $\nu = 2.097 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 0.7154$  (Table A-15), and  $\beta = 1/T_{\text{avg}} = 1/353 \text{ K}$ .

**Analysis** With the assumption that  $T_{\text{avg}} = 80^\circ\text{C}$ , the outer surface temperature is estimated as

$$T_o = 2T_{\text{avg}} - T_i = 40^\circ\text{C}$$

The Rayleigh number is

$$\text{Ra} = \frac{g\beta(T_i - T_o)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(80 + 273 \text{ K})^{-1}(120 - 40) \text{ K}(0.0125 \text{ m})^3}{(2.097 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7154) = 7064$$

where

$$L_c = (D_o - D_i) / 2 = 0.0125 \text{ m}$$

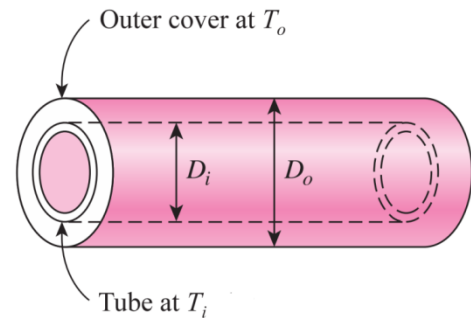
The effective thermal conductivity is

$$F_{\text{cyl}} = \frac{[\ln(D_o / D_i)]^4}{L_c^3 (D_i^{-3/5} + D_o^{-3/5})^5}$$

$$= \frac{\ln(0.05 / 0.025)^4}{(0.0125 \text{ m})^3 [(0.025 \text{ m})^{-3/5} + (0.05 \text{ m})^{-3/5}]^5} = 0.1466$$

$$k_{\text{eff}} = 0.386k \left( \frac{\text{Pr}}{0.861 + \text{Pr}} \right)^{1/4} (F_{\text{cyl}} \text{Ra})^{1/4}$$

$$= 0.386(0.02953 \text{ W/m}\cdot\text{K}) \left( \frac{0.7154}{0.861 + 0.7154} \right)^{1/4} [(0.1466)(7064)]^{1/4} = 0.05307 \text{ W/m}\cdot\text{K}$$



Thus, the temperature of the outer cylinder can be determined using

$$\dot{Q} = \frac{2\pi k_{\text{eff}}}{\ln(D_o / D_i)} (T_i - T_o)$$

$$\dot{m} c_p \Delta T = \frac{2\pi k_{\text{eff}}}{\ln(D_o / D_i)} (T_i - T_o) \rightarrow T_o = 41^\circ\text{C}$$

where

$$\dot{m} = 0.005 \text{ kg/s}, \quad c_p = 950 \text{ J/kg}\cdot\text{K}, \quad \Delta T = 8^\circ\text{C}, \quad T_i = 120^\circ\text{C}$$

**Discussion** The air gap between the concentric cylinders is sufficient to keep the outer cover temperature below  $45^\circ\text{C}$  to prevent thermal burn hazards.

The assumed average temperature  $T_{\text{avg}} = 80^\circ\text{C}$  is appropriate for evaluating the air properties, since the determined  $T_o = 41^\circ\text{C}$  would give an average temperature of  $T_{\text{avg}} = 80.5^\circ\text{C}$ .

**9-100** Two surfaces of a spherical enclosure are maintained at specified temperatures. The rate of heat transfer through the enclosure is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Nitrogen is an ideal gas with constant properties. 3 Radiation heat transfer is negligible

**Properties** The properties of nitrogen at  $T_f = (T_s + T_\infty)/2 = 150^\circ\text{C}$  are  $k = 0.03416 \text{ W/m}\cdot\text{K}$ ,  $\nu = 2.851 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 0.7025$  (from Table A-16). Also,  $\beta = 1/T_f = 0.002364 \text{ K}^{-1}$ .

**Analysis** The characteristic length in this case is determined from

$$L_c = \frac{D_o - D_i}{2} = \frac{10 - 5}{2} \text{ cm} = 2.5 \text{ cm}$$

The Rayleigh number is

$$\begin{aligned} \text{Ra}_L &= \frac{g\beta(T_o - T_i)L_c^3}{\nu^2} \text{Pr} \\ &= \frac{(9.81 \text{ m/s}^2)(2.364 \times 10^{-3} \text{ K}^{-1})(200 - 100) \text{ K}(0.025 \text{ m})^3}{(2.851 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7025) \\ &= 3.132 \times 10^4 \end{aligned}$$

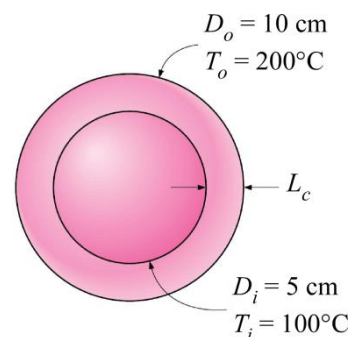
The effective thermal conductivity is

$$\begin{aligned} F_{\text{sph}} &= \frac{L_c}{(D_i D_o)^4 (D_i^{-7/5} + D_o^{-7/5})^5} = \frac{(0.025 \text{ m})}{[(0.1 \text{ m})(0.05 \text{ m})]^4 [(0.05 \text{ m})^{-7/5} + (0.1 \text{ m})^{-7/5}]^5} \\ &= 0.006268 \\ k_{\text{eff}} &= 0.74 \left( \frac{\text{Pr}}{0.861 + \text{Pr}} \right)^{1/4} (F_{\text{sph}} \text{Ra}_L)^{1/4} k \\ &= 0.74 \left( \frac{0.7025}{0.861 + 0.7025} \right)^{1/4} [(0.006268)(3.132 \times 10^4)]^{1/4} (0.03416 \text{ W/m}\cdot\text{K}) \\ &= 0.07747 \text{ W/m}\cdot\text{K} \end{aligned}$$

Then the rate of heat transfer between the spheres becomes

$$\dot{Q} = k_{\text{eff}} \frac{\pi D_i D_o}{L_c} (T_o - T_i) = (0.07747 \text{ W/m}\cdot\text{K}) \frac{\pi(0.1 \text{ m})(0.05 \text{ m})}{(0.025 \text{ m})} (200 - 100) \text{ K} = \mathbf{4.87 \text{ W}}$$

**Discussion** Note that if  $k_{\text{eff}} < k$ , then we use  $k = k_{\text{eff}}$ .



**9-101** Two surfaces of a spherical enclosure are maintained at specified temperatures. The rate of heat transfer through the enclosure is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The air pressure in the enclosure is 1 atm.

**Properties** The properties of air at 1 atm and the average temperature of  $(T_1 + T_2)/2 = (350 + 275)/2 = 312.5 \text{ K} = 39.5^\circ\text{C}$  are (Table A-15)

$$k = 0.02658 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.697 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7256$$

$$\beta = \frac{1}{T_f} = \frac{1}{312.5 \text{ K}} = 0.003200 \text{ K}^{-1}$$

**Analysis** The characteristic length in this case is determined from

$$L_c = \frac{D_2 - D_1}{2} = \frac{25 - 15}{2} = 5 \text{ cm.}$$

Then,

$$Ra = \frac{g\beta(T_1 - T_2)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003200 \text{ K}^{-1})(350 - 275 \text{ K})(0.05 \text{ m})^3}{(1.697 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7256) = 7.415 \times 10^5$$

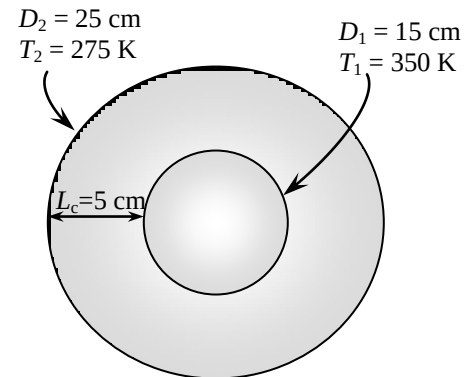
The effective thermal conductivity is

$$F_{\text{sph}} = \frac{L_c}{(D_i D_o)^4 (D_i^{-7/5} + D_o^{-7/5})^5} = \frac{0.05 \text{ m}}{[(0.15 \text{ m})(0.25 \text{ m})]^4 [(0.15 \text{ m})^{-7/5} + (0.25 \text{ m})^{-7/5}]^5} = 0.005900$$

$$\begin{aligned} k_{\text{eff}} &= 0.74k \left( \frac{\text{Pr}}{0.861 + \text{Pr}} \right)^{1/4} (F_{\text{sph}} Ra)^{1/4} \\ &= 0.74(0.02658 \text{ W/m}\cdot^\circ\text{C}) \left( \frac{0.7256}{0.861 + 0.7256} \right)^{1/4} [(0.005900)(7.415 \times 10^5)]^{1/4} \\ &= 0.1315 \text{ W/m}\cdot^\circ\text{C} \end{aligned}$$

Then the rate of heat transfer between the spheres becomes

$$\dot{Q} = k_{\text{eff}} \pi \left( \frac{D_i D_o}{L_c} \right) (T_i - T_o) = (0.1315 \text{ W/m}\cdot^\circ\text{C}) \pi \left[ \frac{(0.15 \text{ m})(0.25 \text{ m})}{(0.05 \text{ m})} \right] (350 - 275) \text{ K} = \mathbf{23.3 \text{ W}}$$





**9-102** Prob. 9-101 is reconsidered. The rate of natural convection heat transfer as a function of the hot surface temperature of the sphere is to be plotted.

**Analysis** The problem is solved using EES, and the solution is given below.

**"GIVEN"**

D\_1=0.15 [m]  
D\_2=0.25 [m]  
T\_1=350 [K]  
T\_2=275 [K]

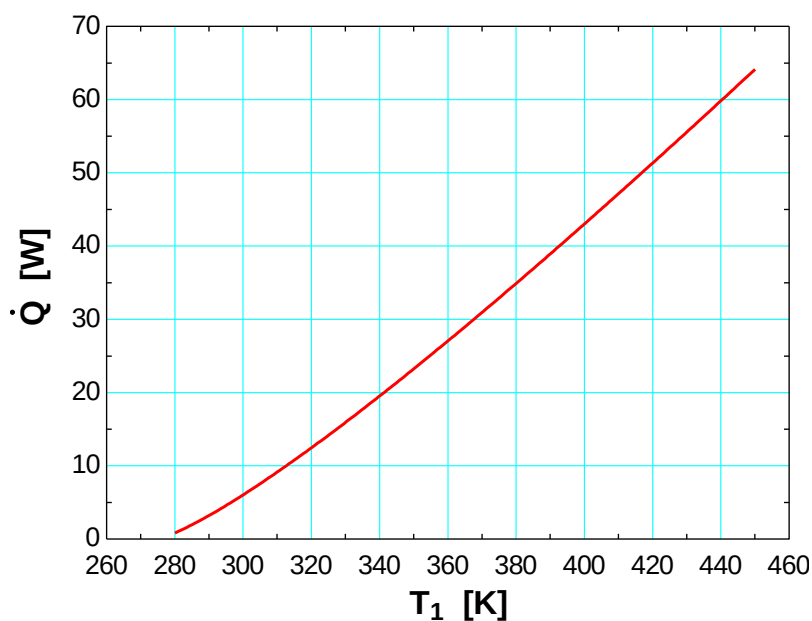
**"PROPERTIES"**

Fluid\$='air'  
k=Conductivity(Fluid\$, T=T\_ave)  
Pr=Prandtl(Fluid\$, T=T\_ave)  
rho=Density(Fluid\$, T=T\_ave, P=101.3)  
mu=Viscosity(Fluid\$, T=T\_ave)  
nu=mu/rho  
beta=1/T\_ave  
T\_ave=1/2\*(T\_1+T\_2)  
g=9.807 [m/s^2]

**"ANALYSIS"**

L=(D\_2-D\_1)/2  
Ra=(g\*beta\*(T\_1-T\_2)\*L^3)/nu^2\*Pr  
F\_sph=L/((D\_1\*D\_2)^4\*(D\_1^(-7/5)+D\_2^(-7/5))^5)  
k\_eff=0.74\*k\*(Pr/(0.861+Pr))^0.25\*(F\_sph\*Ra)^0.25  
Q\_dot=k\_eff\*pi\*(D\_1\*D\_2)/L\*(T\_1-T\_2)

| T <sub>1</sub><br>[K] | Q̇<br>[W] |
|-----------------------|-----------|
| 250                   | -         |
| 260                   | -         |
| 270                   | -         |
| 280                   | 0.8153    |
| 290                   | 3.202     |
| 300                   | 6.032     |
| 310                   | 9.139     |
| 320                   | 12.45     |
| 330                   | 15.92     |
| 340                   | 19.52     |
| 350                   | 23.23     |
| 360                   | 27.04     |
| 370                   | 30.93     |
| 380                   | 34.89     |
| 390                   | 38.92     |
| 400                   | 43.01     |
| 410                   | 47.15     |
| 420                   | 51.33     |
| 430                   | 55.56     |
| 440                   | 59.83     |
| 450                   | 64.13     |



**9-103** Two surfaces of a spherical enclosure are maintained at specified temperatures. The rate of heat transfer through the enclosure is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas. 3 Radiation heat transfer is not considered.

**Properties** The properties of air at the average temperature of  $T_{\text{avg}} = (T_i + T_o)/2 = (320 + 280)/2 = 300 \text{ K} = 27^\circ\text{C}$  and 1 atm pressure are:  $k = 0.02566 \text{ W/m}\cdot\text{K}$ ,  $\text{Pr} = 0.7290$ ,  $\nu = 1.58 \times 10^{-5} \text{ m}^2/\text{s}$  (Table A-15), and  $\beta = 1/T_{\text{avg}} = 1/300 \text{ K}$ .

**Analysis** We have a spherical enclosure filled with air. The characteristic length in this case is the distance between the two spheres,

$$L_c = (D_o - D_i)/2 = (0.3 - 0.2)/2 = 0.05 \text{ m}$$

The Rayleigh number is

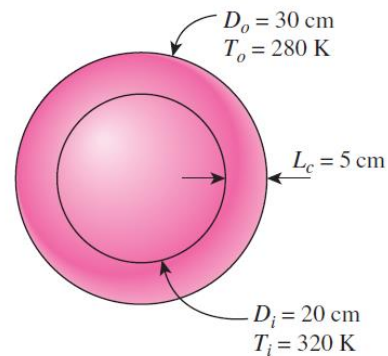
$$\begin{aligned} \text{Ra} &= \frac{g\beta(T_i - T_o)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(300 \text{ K})^{-1}(320 - 280) \text{ K} (0.05 \text{ m})^3}{(1.58 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.729) \\ &= 4.775 \times 10^5 \end{aligned}$$

The effective thermal conductivity is

$$\begin{aligned} F_{\text{sph}} &= \frac{L_c}{(D_i D_o)^4 (D_i^{-7/5} + D_o^{-7/5})^5} = \frac{0.05 \text{ m}}{[(0.2 \text{ m})(0.3 \text{ m})]^4 [(0.2 \text{ m})^{-7/5} + (0.3 \text{ m})^{-7/5}]^5} = 0.0005229 \\ k_{\text{eff}} &= 0.74k \left( \frac{\text{Pr}}{0.861 + \text{Pr}} \right)^{1/4} (F_{\text{sph}} \text{Ra})^{1/4} \\ &= 0.74(0.02566 \text{ W/m}\cdot\text{K}) \left( \frac{0.729}{0.861 + 0.729} \right)^{1/4} [(0.0005229)(4.775 \times 10^5)]^{1/4} \\ &= 0.1105 \text{ W/m}\cdot\text{K} \end{aligned}$$

Then, the rate of heat transfer between spheres becomes

$$\begin{aligned} \dot{Q} &= k_{\text{eff}} \pi \left( \frac{D_i D_o}{L_c} \right) (T_i - T_o) \\ &= (0.1105) \pi \left( \frac{0.2 \text{ m} \times 0.3 \text{ m}}{0.05 \text{ m}} \right) (320 - 280) \text{ K} = \mathbf{16.7 \text{ W}} \end{aligned}$$



**Discussion** Note that the air in the spherical enclosure acts like a stationary fluid whose thermal conductivity is  $k_{\text{eff}}/k = 0.1105/0.02566 = 4.3$  times that of air as a result of natural convection currents. Also, radiation heat transfer between spheres is usually significant, and should be considered in a complete analysis.

**9-104**  A metal spherical tank is filled with a solution undergoing an exothermic reaction and the heat generation is known. The tank is enclosed by a concentric outer cover to prevent thermal burn hazards. The temperature of the outer cover is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Surface temperatures are constant. 3 Air is an ideal gas with constant properties. 4 Heat loss by radiation is negligible. 5 The air pressure in the enclosure is 1 atm.

**Properties** The properties of air at the assumed  $T_{\text{avg}} = 80^\circ\text{C}$  and 1 atm pressure are  $k = 0.02953 \text{ W/m}\cdot\text{K}$ ,  $\nu = 2.097 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 0.7154$  (Table A-15), and  $\beta = 1/T_f = 1/353 \text{ K}$ .

**Analysis** With the assumption that  $T_{\text{avg}} = 80^\circ\text{C}$ , the outer surface temperature is estimated as

$$T_o = 2T_{\text{avg}} - T_i = 40^\circ\text{C}$$

The Rayleigh number is

$$\begin{aligned} \text{Ra} &= \frac{g\beta(T_i - T_o)L_c^3}{\nu^2} \text{Pr} \\ &= \frac{(9.81 \text{ m/s}^2)(80 + 273 \text{ K})^{-1}(120 - 40) \text{ K} (0.05 \text{ m})^3}{(2.097 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7154) = 452,112 \end{aligned}$$

where

$$L_c = (D_o - D_i) / 2 = 0.05 \text{ m}$$

The effective thermal conductivity is

$$\begin{aligned} F_{\text{sph}} &= \frac{L_c}{(D_i D_o)^4 (D_i^{-7/5} + D_o^{-7/5})^5} = \frac{0.05 \text{ m}}{[(3.0 \text{ m})(3.1 \text{ m})]^4 [(3.0 \text{ m})^{-7/5} + (3.1 \text{ m})^{-7/5}]^5} = 0.0005117 \\ k_{\text{eff}} &= 0.74k \left( \frac{\text{Pr}}{0.861 + \text{Pr}} \right)^{1/4} (F_{\text{sph}} \text{Ra})^{1/4} \\ &= 0.74(0.02953 \text{ W/m}\cdot\text{K}) \left( \frac{0.7154}{0.861 + 0.7154} \right)^{1/4} [(0.0005117)(452,112)]^{1/4} = 0.06995 \text{ W/m}\cdot\text{K} \end{aligned}$$

Thus, the temperature of the outer cover can be determined using

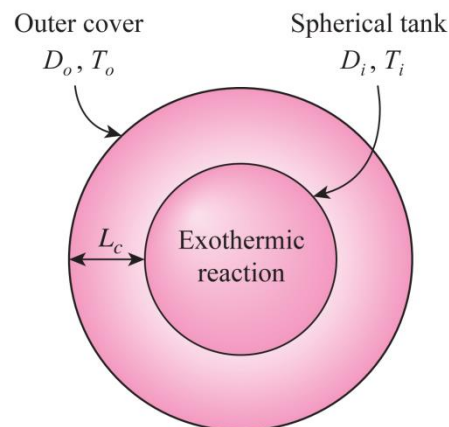
$$\begin{aligned} \dot{Q} &= k_{\text{eff}} \pi \left( \frac{D_i D_o}{L_c} \right) (T_i - T_o) \\ \dot{e}_{\text{gen}} (\pi D_i^3 / 6) &= k_{\text{eff}} \pi \left( \frac{D_i D_o}{L_c} \right) (T_i - T_o) \rightarrow T_o = 39.4^\circ\text{C} \end{aligned}$$

where

$$\dot{e}_{\text{gen}} = 233 \text{ W/m}^3$$

**Discussion** The air gap between the concentric spheres is sufficient to keep the outer cover temperature below  $45^\circ\text{C}$  to prevent thermal burn hazards.

The assumed average temperature  $T_{\text{avg}} = 80^\circ\text{C}$  is appropriate for evaluating the air properties, since the determined  $T_o = 39.4^\circ\text{C}$  would give an average temperature of  $T_{\text{avg}} = 79.7^\circ\text{C}$ .



## Combined Natural and Forced Convection

**9-105C** In combined natural and forced convection, the natural convection is negligible when  $Gr / Re^2 < 0.1$ . Otherwise it is not.

**9-106C** In assisting or transverse flows, natural convection enhances forced convection heat transfer while in opposing flow it hurts forced convection.

**9-107C** When neither natural nor forced convection is negligible, it is not correct to calculate each separately and to add them to determine the total convection heat transfer. Instead, the correlation

$$Nu_{\text{combined}} = \left( Nu_{\text{forced}}^n + Nu_{\text{natural}}^n \right)^{1/n}$$

based on the experimental studies should be used.

**9-108** A vertical plate in air is considered. The forced motion velocity above which natural convection heat transfer from the plate is negligible is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The atmospheric pressure at that location is 1 atm.

**Properties** The properties of air at 1 atm and 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (85 + 30)/2 = 57.5^\circ\text{C}$  are (Table A-15)

$$\nu = 1.872 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\beta = \frac{1}{T_f} = \frac{1}{(57.5 + 273)\text{K}} = 0.003026 \text{ K}^{-1}$$

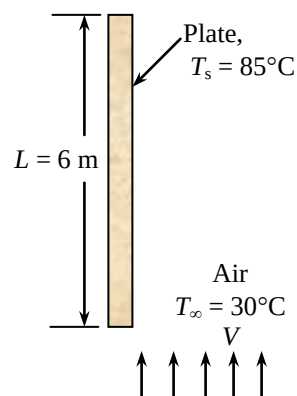
**Analysis** The characteristic length is the height of the plate,  $L_c = L = 6 \text{ m}$ . The Grashof and Reynolds numbers are

$$Gr = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} = \frac{(9.81 \text{ m/s}^2)(0.003026 \text{ K}^{-1})(85 - 30 \text{ K})(6 \text{ m})^3}{(1.872 \times 10^{-5} \text{ m}^2/\text{s})^2} = 1.006 \times 10^{12}$$

$$Re = \frac{VL}{\nu} = \frac{V_\infty(6 \text{ m})}{1.872 \times 10^{-5} \text{ m}^2/\text{s}} = 3.205 \times 10^5 V$$

and the forced motion velocity above which natural convection heat transfer from this plate is negligible is

$$\frac{Gr}{Re^2} = 0.1 \longrightarrow \frac{1.006 \times 10^{12}}{(3.205 \times 10^5 V)^2} = 0.1 \longrightarrow V = \mathbf{9.90 \text{ m/s}}$$



**9-109** Thin square plates coming out of the oven in a production facility are cooled by blowing ambient air horizontally parallel to their surfaces. The air velocity above which the natural convection effects on heat transfer are negligible is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The atmospheric pressure at that location is 1 atm.

**Properties** The properties of air at 1 atm and 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (270 + 18)/2 = 144^\circ\text{C}$  are (Table A-15)

$$\nu = 2.791 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\beta = \frac{1}{T_f} = \frac{1}{(144 + 273)\text{K}} = 0.002398 \text{ K}^{-1}$$

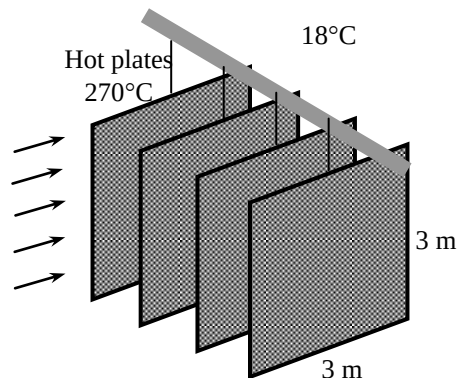
**Analysis** The characteristic length is the height of the plate  $L_c = L = 3 \text{ m}$ . The Grashof and Reynolds numbers are

$$Gr = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} = \frac{(9.81 \text{ m/s}^2)(0.002398 \text{ K}^{-1})(270 - 18 \text{ K})(3 \text{ m})^3}{(2.791 \times 10^{-5} \text{ m}^2/\text{s})^2} = 2.055 \times 10^{11}$$

$$Re = \frac{VL}{\nu} = \frac{V(3 \text{ m})}{2.791 \times 10^{-5} \text{ m}^2/\text{s}} = 1.075 \times 10^5 V$$

and the forced motion velocity above which natural convection heat transfer from this plate is negligible is

$$\frac{Gr}{Re^2} = 0.1 \longrightarrow \frac{2.055 \times 10^{11}}{(1.075 \times 10^5 V)^2} = 0.1 \longrightarrow V = \mathbf{13.3 \text{ m/s}}$$



**9-110** The significance of natural convection to the heat transfer process on a vertical rod with water flowing across its outer surface is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Properties are constant. 3 The rod is orientated such that the characteristic length is its length.

**Properties** The properties of water at  $T_f = (T_s + T_\infty)/2 = 80^\circ\text{C}$  are  $\rho = 971.8 \text{ kg/m}^3$ ,  $\mu = 0.355 \times 10^{-3} \text{ kg/m}\cdot\text{s}$ , and  $\beta = 0.653 \times 10^{-3} \text{ K}^{-1}$  (Table A-9).

**Analysis** The Reynolds number for the cross flow is

$$Re = \frac{\rho VD}{\mu} = \frac{(971.8 \text{ kg/m}^3)(0.5 \text{ m/s})(0.150 \text{ m})}{0.355 \times 10^{-3} \text{ kg/m}\cdot\text{s}} = 2.05 \times 10^5$$

For vertical cylinder, the Grashof number with  $L_c = L$  is

$$\begin{aligned} Gr_L &= \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} = \frac{g\beta(T_s - T_\infty)L^3}{(\mu/\rho)^2} \\ &= \frac{(9.81 \text{ m/s}^2)(0.653 \times 10^{-3} \text{ K}^{-1})(120 - 40) \text{ K}(1 \text{ m})^3}{(0.355 \times 10^{-3} / 971.8)^2 \text{ m}^4/\text{s}^2} = 3.84 \times 10^{12} \end{aligned}$$

Hence,

$$\frac{Gr_L}{Re^2} = \frac{3.84 \times 10^{12}}{(2.05 \times 10^5)^2} = \mathbf{91.4}$$

Since  $Gr_L/Re^2 \gg 1$ , therefore natural convection effects dominate forced convection effects. Natural convection effects are important to the heat transfer process.

**Discussion** If  $Gr_L/Re^2 \ll 1$ , then forced convection effects dominate natural convection effects.

**9-111** The significance of natural convection to the heat transfer process on a horizontal rod with water flowing across its outer surface is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Properties are constant. 3 The rod is orientated such that the characteristic length is its diameter.

**Properties** The properties of water at  $T_f = (T_s + T_\infty)/2 = 80^\circ\text{C}$  are  $\rho = 971.8 \text{ kg/m}^3$ ,  $\mu = 0.355 \times 10^{-3} \text{ kg/m}\cdot\text{s}$ , and  $\beta = 0.653 \times 10^{-3} \text{ K}^{-1}$  (Table A-9).

**Analysis** The Reynolds number for the cross flow is

$$\text{Re} = \frac{\rho V D}{\mu} = \frac{(971.8 \text{ kg/m}^3)(0.2 \text{ m/s})(0.150 \text{ m})}{0.355 \times 10^{-3} \text{ kg/m}\cdot\text{s}} = 8.21 \times 10^4$$

For horizontal cylinder, the Grashof number with  $L_c = D$  is

$$\begin{aligned} \text{Gr}_D &= \frac{g\beta(T_s - T_\infty)D^3}{\nu^2} = \frac{g\beta(T_s - T_\infty)D^3}{(\mu/\rho)^2} \\ &= \frac{(9.81 \text{ m/s}^2)(0.653 \times 10^{-3} \text{ K}^{-1})(120 - 40) \text{ K}(0.15 \text{ m})^3}{(0.355 \times 10^{-3} / 971.8)^2 \text{ m}^4/\text{s}^2} = 1.296 \times 10^{10} \end{aligned}$$

Hence,

$$\frac{\text{Gr}_D}{\text{Re}^2} = \frac{6.48 \times 10^8}{(8.21 \times 10^4)^2} = \mathbf{1.92}$$

Since  $\text{Gr}_D/\text{Re}^2 \approx 1$ , therefore natural convection and forced convection effects are both important to the heat transfer process.

**Discussion** When both natural convection and forced convection effects are important, the process is called combined natural and forced convection or mixed convection.

**9-112** A vertical plate in air is considered. The forced motion velocity above which natural convection heat transfer from the plate is negligible is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The atmospheric pressure at that location is 1 atm.

**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (85 + 30)/2 = 57.5^\circ\text{C}$  are (Table A-15)

$$\nu = 1.872 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\beta = \frac{1}{T_f} = \frac{1}{(57.5 + 273) \text{ K}} = 0.003026 \text{ K}^{-1}$$

**Analysis** The characteristic length is the height of the plate,  $L_c = L = 5 \text{ m}$ .

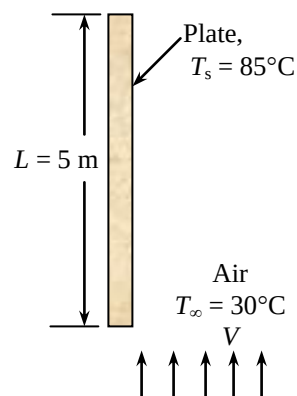
The Grashof and Reynolds numbers are

$$\text{Gr} = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} = \frac{(9.81 \text{ m/s}^2)(0.003026 \text{ K}^{-1})(85 - 30) \text{ K}(5 \text{ m})^3}{(1.872 \times 10^{-5} \text{ m}^2/\text{s})^2} = 5.824 \times 10^{11}$$

$$\text{Re} = \frac{VL}{\nu} = \frac{V_\infty(5 \text{ m})}{1.872 \times 10^{-5} \text{ m}^2/\text{s}} = 2.671 \times 10^5 V$$

and the forced motion velocity above which natural convection heat transfer from this plate is negligible is

$$\frac{\text{Gr}}{\text{Re}^2} = 0.1 \longrightarrow \frac{5.824 \times 10^{11}}{(2.671 \times 10^5 V)^2} = 0.1 \longrightarrow V = \mathbf{9.04 \text{ m/s}}$$





**9-113** Prob. 9-112 is reconsidered. The forced motion velocity above which natural convection heat transfer is negligible as a function of the plate temperature is to be plotted.

**Analysis** The problem is solved using EES, and the solution is given below.

**"GIVEN"**

L=5 [m]  
T<sub>s</sub>=85 [C]  
T<sub>infinity</sub>=30 [C]

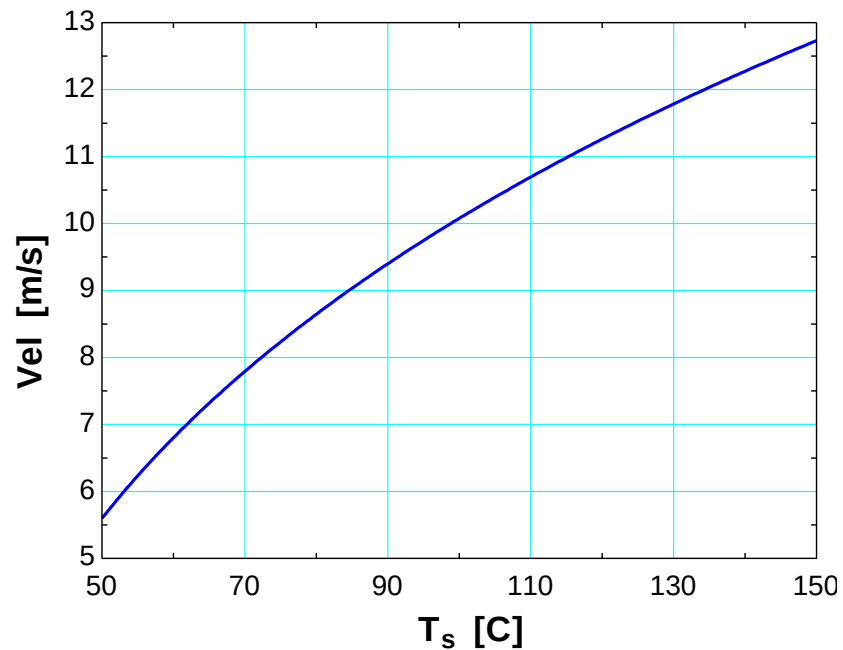
**"PROPERTIES"**

Fluid\$='air'  
rho=Density(Fluid\$, T=T<sub>film</sub>, P=101.3)  
mu=Viscosity(Fluid\$, T=T<sub>film</sub>)  
nu=mu/rho  
beta=1/(T<sub>film</sub>+273)  
T<sub>film</sub>=1/2\*(T<sub>s</sub>+T<sub>infinity</sub>)  
g=9.807 [m/s<sup>2</sup>]

**"ANALYSIS"**

Gr=(g\*beta\*(T<sub>s</sub>-T<sub>infinity</sub>)\*L<sup>3</sup>)/nu<sup>2</sup>  
Re=(Vel\*L)/nu  
Gr/Re<sup>2</sup>=0.1

| T <sub>s</sub><br>[C] | Vel<br>[m/s] |
|-----------------------|--------------|
| 50                    | 5.598        |
| 55                    | 6.233        |
| 60                    | 6.801        |
| 65                    | 7.318        |
| 70                    | 7.793        |
| 75                    | 8.233        |
| 80                    | 8.646        |
| 85                    | 9.033        |
| 90                    | 9.4          |
| 95                    | 9.747        |
| 100                   | 10.08        |
| 105                   | 10.39        |
| 110                   | 10.69        |
| 115                   | 10.98        |
| 120                   | 11.26        |
| 125                   | 11.53        |
| 130                   | 11.79        |
| 135                   | 12.03        |
| 140                   | 12.27        |
| 145                   | 12.51        |
| 150                   | 12.73        |



**9-114** A circuit board is cooled by a fan that blows air upwards. The average temperature on the surface of the circuit board is to be determined for two cases.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The atmospheric pressure at that location is 1 atm.

**Properties** Based on the problem statement, the properties of air at 1 atm and 1 atm and the anticipated film temperature of  $(T_s + T_\infty)/2 = (60 + 35)/2 = 47.5^\circ\text{C}$  are (Table A-15)

$$k = 0.02717 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.774 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7235$$

$$\beta = \frac{1}{T_f} = \frac{1}{(47.5 + 273)\text{K}} = 0.00312 \text{ K}^{-1}$$

**Analysis** We assume the surface temperature to be  $60^\circ\text{C}$ . We will check this assumption later on and repeat calculations with a better assumption, if necessary. The characteristic length in this case is the length of the board in the flow (vertical) direction,  $L_c = L = 0.12 \text{ m}$ . Then the Reynolds number becomes

$$\text{Re} = \frac{VL}{\nu} = \frac{(0.5 \text{ m/s})(0.12 \text{ m})}{1.774 \times 10^{-5} \text{ m}^2/\text{s}} = 3383$$

which is less than critical Reynolds number ( $5 \times 10^5$ ). Therefore the flow is laminar and the forced convection Nusselt number and  $h$  are determined from

$$\text{Nu} = \frac{hL}{k} = 0.664 \text{Re}^{0.5} \text{Pr}^{1/3} = 0.664(3383)^{0.5} (0.7235)^{1/3} = 34.67$$

$$h = \frac{k}{L} \text{Nu} = \frac{0.02717 \text{ W/m}\cdot^\circ\text{C}}{0.12 \text{ m}} (34.67) = 7.85 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = L \times W = (0.12 \text{ m})(0.2 \text{ m}) = 0.024 \text{ m}^2$$

Then,  $\dot{Q} = hA_s(T_s - T_\infty) \longrightarrow T_s = T_\infty + \frac{\dot{Q}}{hA_s} = 35^\circ\text{C} + \frac{(100)(0.05 \text{ W})}{(7.85 \text{ W/m}^2\cdot^\circ\text{C})(0.024 \text{ m}^2)} = \mathbf{61.5^\circ\text{C}}$

which is sufficiently close to the assumed value in the evaluation of properties. Therefore, there is no need to repeat calculations.

(b) The Rayleigh number is

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.00312 \text{ K}^{-1})(60 - 35 \text{ K})(0.12 \text{ m})^3}{(1.774 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7235) = 3.041 \times 10^6$$

$$\text{Nu} = \left\{ 0.825 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.492}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(3.041 \times 10^6)^{1/6}}{\left[ 1 + \left( \frac{0.492}{0.7235} \right)^{9/16} \right]^{8/27}} \right\}^2 = 22.42$$

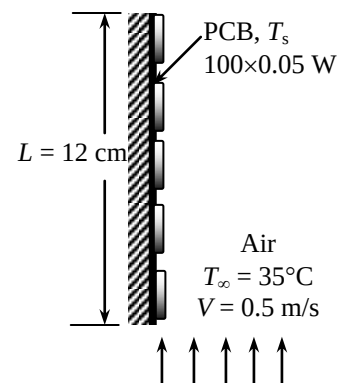
This is an assisting flow and the combined Nusselt number is determined from

$$\text{Nu}_{\text{combined}} = (\text{Nu}_{\text{forced}}^n + \text{Nu}_{\text{natural}}^n)^{1/n} = (34.67^3 + 22.42^3)^{1/3} = 37.55$$

Then,  $h = \frac{k}{L} \text{Nu}_{\text{combined}} = \frac{0.02717 \text{ W/m}\cdot^\circ\text{C}}{0.12 \text{ m}} (37.55) = 8.502 \text{ W/m}^2\cdot^\circ\text{C}$

and  $\dot{Q} = hA_s(T_s - T_\infty) \longrightarrow T_s = T_\infty + \frac{\dot{Q}}{hA_s} = 35^\circ\text{C} + \frac{(100)(0.05 \text{ W})}{(8.502 \text{ W/m}^2\cdot^\circ\text{C})(0.024 \text{ m}^2)} = \mathbf{59.5^\circ\text{C}}$

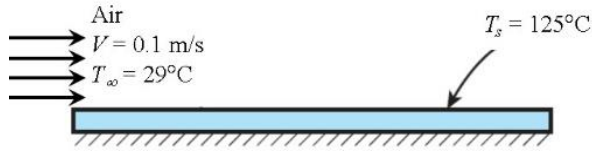
which is sufficiently close to the assumed surface temperature value of  $60^\circ\text{C}$  used in the evaluation of properties. Therefore, there is no need to repeat calculations. The natural convection lowers the surface temperature in this case by about  $2^\circ\text{C}$ .



**9-115** A horizontal square plate is being cooled by blowing air over the surface of the plate. Determine the Nusslet number for this case.

**Assumptions** 1 Steady state conditions. 2 Air is an ideal gas. 3 The local atmospheric pressure is 1 atm. 4 Thermal radiation is negligible. 5 Backside of the plate is insulated.

**Properties** The properties of air at the film temperature of  $T_f = (T_\infty + T_s)/2 = (29 + 125)/2 = 77^\circ\text{C}$  are (Table A-15)  $\text{Pr} = 0.7161$ ,  $k = 0.02931 \text{ W/m}\cdot\text{K}$ ,  $\nu = 2.066 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\beta = 1/T_f = 1/(77 + 273 \text{ K}) = 0.002857 \text{ K}^{-1}$



**Analysis** The characteristic length of the plate is

$$L_c = \frac{A_s}{p} = \frac{L^2}{4L} = \frac{L}{4} = \frac{0.25 \text{ m}}{4} = 0.0625 \text{ m}$$

The Grashof, Rayleigh, and Reynolds numbers are

$$\text{Gr}_L = \frac{g\beta(T_s - T_\infty)L_c^3}{\nu^2} = \frac{(9.81 \text{ m/s}^2)(0.002857 \text{ K}^{-1})(125 - 29 \text{ K})(0.0625 \text{ m})^3}{(2.066 \times 10^{-5} \text{ m}^2/\text{s})^2} = 1.539 \times 10^6$$

$$\text{Ra}_L = \text{Gr}_L \text{Pr} = (1.539 \times 10^6)(0.7161) = 1.102 \times 10^6$$

$$\text{Re} = \frac{VL}{\nu} = \frac{(0.1 \text{ m/s})(0.25 \text{ m})}{2.066 \times 10^{-5} \text{ m}^2/\text{s}} = 1.21 \times 10^3$$

Hence,

$$\frac{\text{Gr}_L}{\text{Re}^2} = \frac{1.539 \times 10^6}{(1.21 \times 10^3)^2} = 1.05$$

Noting that  $\frac{\text{Gr}_L}{\text{Re}^2} \approx 1$ , both natural convection and forced convection are significant in this case, and we have mixed flow.

The average Nusselt number for natural convection with horizontal hot surface facing up with  $\text{Ra}$  between  $10^4$  and  $10^7$  is determined from

$$\text{Nu}_{\text{natural}} = 0.54 \text{Ra}_L^{1/4} = 0.54 (1.102 \times 10^6)^{1/4} = 17.5$$

The average Nusselt number for laminar flow ( $\text{Re} < 5 \times 10^5$ ) forced convection over a flat plate is determined from

$$\text{Nu}_{\text{forced}} = 0.664 \text{Re}^{0.5} \text{Pr}^{1/3} = 0.664 (1.21 \times 10^3)^{0.5} (0.7161)^{1/3} = 20.7$$

Finally, the combined Nusselt numbers for the assisting flow is (use  $n=3.5$  for horizontal plate)

$$\text{Nu}_{\text{combined}} = (\text{Nu}_{\text{forced}}^{3.5} + \text{Nu}_{\text{natural}}^{3.5})^{1/3.5} = (20.7^{3.5} + 17.5^{3.5})^{1/3.5} = \mathbf{23.5}$$

**Discussion** Note that the combined Nusselt number is 34% higher for natural convection Nusselt number and about 14% higher than forced convection Nusselt number. Therefore, both natural and forced convection are important in this case.

## Special Topic: Heat Transfer through Windows

**9-116C** Windows are considered in three regions when analyzing heat transfer through them because the structure and properties of the frame are quite different than those of the glazing. As a result, heat transfer through the frame and the edge section of the glazing adjacent to the frame is two-dimensional. Even in the absence of solar radiation and air infiltration, heat transfer through the windows is more complicated than it appears to be. Therefore, it is customary to consider the windows in three regions when analyzing heat transfer through them: (1) the *center-of-glass*, (2) the *edge-of-glass*, and (3) the *frame* regions. When the heat transfer coefficient for all three regions are known, the overall U-value of the window is determined from

$$U_{\text{window}} = (U_{\text{center}} A_{\text{center}} + U_{\text{edge}} A_{\text{edge}} + U_{\text{frame}} A_{\text{frame}}) / A_{\text{window}}$$

where  $A_{\text{window}}$  is the window area, and  $A_{\text{center}}$ ,  $A_{\text{edge}}$ , and  $A_{\text{frame}}$  are the areas of the center, edge, and frame sections of the window, respectively, and  $U_{\text{center}}$ ,  $U_{\text{edge}}$ , and  $U_{\text{frame}}$  are the heat transfer coefficients for the center, edge, and frame sections of the window.

**9-117C** Of the three similar double pane windows with air gap widths of 5, 10, and 20 mm, the U-factor and thus the rate of heat transfer through the window will be a minimum for the window with 10-mm air gap, as can be seen from Fig. 9-37.

**9-118C** In an ordinary double pane window, about half of the heat transfer is by radiation. A practical way of reducing the radiation component of heat transfer is to reduce the emissivity of glass surfaces by coating them with low-emissivity (or “low-e”) material.

**9-119C** When a thin polyester film is used to divide the 20-mm wide air of a double pane window space into two 10-mm wide layers, both (a) convection and (b) radiation heat transfer through the window will be reduced.

**9-120C** When a double pane window whose air space is flashed and filled with argon gas, (a) convection heat transfer will be reduced but (b) radiation heat transfer through the window will remain the same.

**9-121C** The heat transfer rate through the glazing of a double pane window is higher at the edge section than it is at the center section because of the two-dimensional effects due to heat transfer through the frame.

**9-122C** The U-factors of windows with aluminum frames will be highest because of the higher conductivity of aluminum. The U-factors of wood and vinyl frames are comparable in magnitude.

**9-123** The U-factor for the center-of-glass section of a double pane window is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Heat transfer through the window is one-dimensional. 3 The thermal resistance of glass sheets is negligible.

**Properties** The emissivity of clear glass is given to be 0.84. The values of  $h_i$  and  $h_o$  for winter design conditions are  $h_i = 8.29 \text{ W/m}^2 \cdot ^\circ\text{C}$  and  $h_o = 34.0 \text{ W/m}^2 \cdot ^\circ\text{C}$  (from the text).

**Analysis** Disregarding the thermal resistance of glass sheets, which are small, the U-factor for the center region of a double pane window is determined from

$$\frac{1}{U_{\text{center}}} \cong \frac{1}{h_i} + \frac{1}{h_{\text{space}}} + \frac{1}{h_o}$$

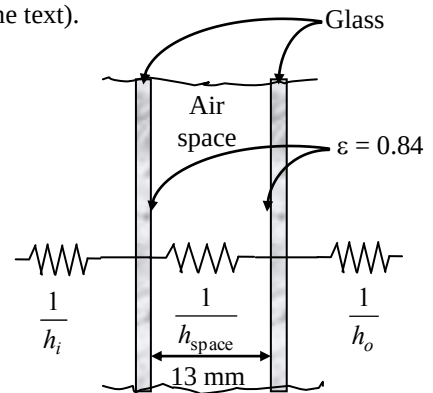
where  $h_i$ ,  $h_{\text{space}}$ , and  $h_o$  are the heat transfer coefficients at the inner surface of window, the air space between the glass layers, and the outer surface of the window, respectively. The effective emissivity of the air space of the double pane window is

$$\varepsilon_{\text{effective}} = \frac{1}{1/\varepsilon_1 + 1/\varepsilon_2 - 1} = \frac{1}{1/0.84 + 1/0.84 - 1} = 0.72$$

For this value of emissivity and an average air space temperature of  $10^\circ\text{C}$  with a temperature difference across the air space to be  $15^\circ\text{C}$ , we read  $h_{\text{space}} = 5.7 \text{ W/m}^2 \cdot ^\circ\text{C}$  from Table 9-3 for 13-mm thick air space. Therefore,

$$\frac{1}{U_{\text{center}}} = \frac{1}{8.29} + \frac{1}{5.7} + \frac{1}{34.0} \longrightarrow U_{\text{center}} = \mathbf{3.07 \text{ W/m}^2 \cdot ^\circ\text{C}}$$

**Discussion** The overall U-factor of the window will be higher because of the edge effects of the frame.



**9-124** The overall U-factor of a window is given to be  $U = 2.76 \text{ W/m}^2 \cdot ^\circ\text{C}$  for 12 km/h winds outside. The new U-factor when the wind velocity outside is doubled is to be determined.

**Assumptions** Thermal properties of the windows and the heat transfer coefficients are constant.

**Properties** The heat transfer coefficients at the outer surface of the window are  $h_o = 22.7 \text{ W/m}^2 \cdot ^\circ\text{C}$  for 12 km/h winds, and  $h_o = 34.0 \text{ W/m}^2 \cdot ^\circ\text{C}$  for 24 km/h winds (from the text).

**Analysis** The corresponding convection resistances for the outer surfaces of the window are

$$R_{o, 12 \text{ km/h}} = \frac{1}{h_{o, 12 \text{ km/h}}} = \frac{1}{22.7 \text{ W/m}^2 \cdot ^\circ\text{C}} = 0.044 \text{ m}^2 \cdot ^\circ\text{C/W}$$

$$R_{o, 24 \text{ km/h}} = \frac{1}{h_{o, 24 \text{ km/h}}} = \frac{1}{34.0 \text{ W/m}^2 \cdot ^\circ\text{C}} = 0.029 \text{ m}^2 \cdot ^\circ\text{C/W}$$

Also, the R-value of the window at 12 km/h winds is

$$R_{\text{window}, 12 \text{ km/h}} = \frac{1}{U_{\text{window}, 12 \text{ km/h}}} = \frac{1}{2.76 \text{ W/m}^2 \cdot ^\circ\text{C}} = 0.362 \text{ m}^2 \cdot ^\circ\text{C/W}$$

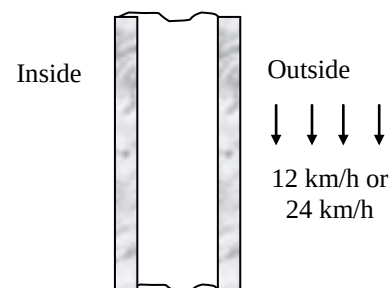
Noting that all thermal resistances are in series, the thermal resistance of the window for 24 km/h winds is determined by replacing the convection resistance for 12 km/h winds by the one for 24 km/h:

$$R_{\text{window}, 24 \text{ km/h}} = R_{\text{window}, 12 \text{ km/h}} - R_{o, 12 \text{ km/h}} + R_{o, 24 \text{ km/h}} = 0.362 - 0.044 + 0.029 = 0.347 \text{ m}^2 \cdot ^\circ\text{C/W}$$

Then the U-factor for the case of 24 km/h winds becomes

$$U_{\text{window}, 24 \text{ km/h}} = \frac{1}{R_{\text{window}, 24 \text{ km/h}}} = \frac{1}{0.347 \text{ m}^2 \cdot ^\circ\text{C/W}} = \mathbf{2.88 \text{ W/m}^2 \cdot ^\circ\text{C}}$$

**Discussion** Note that doubling of the wind velocity increases the U-factor only slightly ( about 4%) from 2.76 to 2.88  $\text{W/m}^2 \cdot ^\circ\text{C}$ .



**9-125** The existing wood framed single pane windows of an older house in Wichita are to be replaced by double-door type vinyl framed double pane windows with an air space of 6.4 mm. The amount of money the new windows will save the home owner per month is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Heat transfer through the window is one-dimensional. 3 Thermal properties of the windows and the heat transfer coefficients are constant. 4 Infiltration heat losses are not considered.

**Properties** The U-factors of the windows are  $5.57 \text{ W/m}^2 \cdot ^\circ\text{C}$  for the old single pane windows, and  $3.20 \text{ W/m}^2 \cdot ^\circ\text{C}$  for the new double pane windows (Table 9-6).

**Analysis** The rate of heat transfer through the window can be determined from

$$\dot{Q}_{\text{window}} = U_{\text{overall}} A_{\text{window}} (T_i - T_o)$$

where  $T_i$  and  $T_o$  are the indoor and outdoor air temperatures, respectively,  $U_{\text{overall}}$  is the U-factor (the overall heat transfer coefficient) of the window, and  $A_{\text{window}}$  is the window area. Noting that the heaters will turn on only when the outdoor temperature drops below  $18^\circ\text{C}$ , the rates of heat transfer due to electric heating for the old and new windows are determined to be

$$\dot{Q}_{\text{window, old}} = (5.57 \text{ W/m}^2 \cdot ^\circ\text{C})(17 \text{ m}^2)(18 - 7.1)^\circ\text{C} = 1032 \text{ W}$$

$$\dot{Q}_{\text{window, new}} = (3.20 \text{ W/m}^2 \cdot ^\circ\text{C})(17 \text{ m}^2)(18 - 7.1)^\circ\text{C} = 593 \text{ W}$$

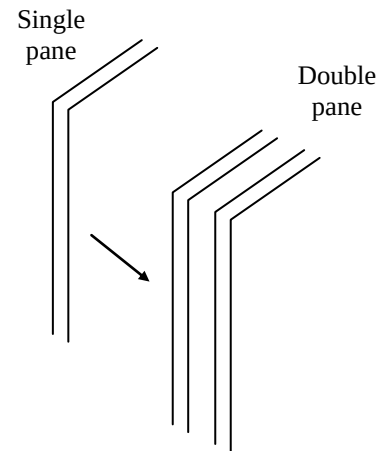
$$\dot{Q}_{\text{saved}} = \dot{Q}_{\text{window, old}} - \dot{Q}_{\text{window, new}} = 1032 - 593 = 439 \text{ W}$$

Then the electrical energy and cost savings per month becomes

$$\text{Energy savings} = \dot{Q}_{\text{saved}} \Delta t = (0.439 \text{ kW})(30 \times 24 \text{ h/month}) = 316 \text{ kWh/month}$$

$$\text{Cost savings} = (\text{Energy savings})(\text{Unit cost of energy}) = (316 \text{ kWh/month})(\$0.085/\text{kWh}) = \mathbf{\$26.9/\text{month}}$$

**Discussion** We would obtain the same result if we used the actual indoor temperature (probably  $22^\circ\text{C}$ ) for  $T_i$  instead of the balance point temperature of  $18^\circ\text{C}$ .



**9-126** The windows of a house in Atlanta are of double door type with wood frames and metal spacers. The average rate of heat loss through the windows in winter is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Heat transfer through the window is one-dimensional. 3 Thermal properties of the windows and the heat transfer coefficients are constant. 4 Infiltration heat losses are not considered.

**Properties** The U-factor of the window is given in Table 9-6 to be  $2.13 \text{ W/m}^2 \cdot ^\circ\text{C}$ .

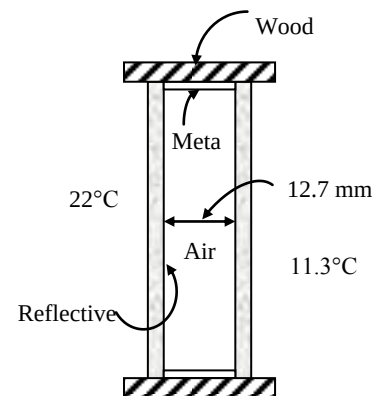
**Analysis** The rate of heat transfer through the window can be determined from

$$\dot{Q}_{\text{window, avg}} = U_{\text{overall}} A_{\text{window}} (T_i - T_{o, \text{avg}})$$

where  $T_i$  and  $T_o$  are the indoor and outdoor air temperatures, respectively,  $U_{\text{overall}}$  is the U-factor (the overall heat transfer coefficient) of the window, and  $A_{\text{window}}$  is the window area. Substituting,

$$\dot{Q}_{\text{window, avg}} = (2.13 \text{ W/m}^2 \cdot ^\circ\text{C})(14 \text{ m}^2)(22 - 11.3)^\circ\text{C} = \mathbf{319 \text{ W}}$$

**Discussion** This is the “average” rate of heat transfer through the window in winter in the absence of any infiltration.



**9-127** The  $R$ -value of the common double door windows that are double pane with 6.4-mm of air space and have aluminum frames is to be compared to the  $R$ -value of  $R$ -2.3 wall. It is also to be determined if more heat is transferred through the windows or the walls.

**Assumptions** 1 Steady operating conditions exist. 2 Heat transfer through the window is one-dimensional. 3 Thermal properties of the windows and the heat transfer coefficients are constant. 4 Infiltration heat losses are not considered.

**Properties** The  $U$ -factor of the window is given in Table 9-6 to be  $4.55 \text{ W/m}^2 \cdot ^\circ\text{C}$

**Analysis** The  $R$ -value of the windows is simply the inverse of its  $U$ -factor, and is determined to be

$$R_{\text{window}} = \frac{1}{U} = \frac{1}{4.55 \text{ W/m}^2 \cdot ^\circ\text{C}} = 0.220 \text{ m}^2 \cdot ^\circ\text{C/W}$$

which is less than 2.3. Therefore, the  $R$ -value of a double pane window is **much less** than the  $R$ -value of an  $R$ -2.3 wall.

Now consider a  $1\text{-m}^2$  section of a wall. The solid wall and the window areas of this section are  $A_{\text{wall}} = 0.8 \text{ m}^2$  and  $A_{\text{window}} = 0.2 \text{ m}^2$ . Then the rates of heat transfer through the two sections are determined to be

$$\dot{Q}_{\text{wall}} = U_{\text{wall}} A_{\text{wall}} (T_i - T_o) = A_{\text{wall}} \frac{T_i - T_o}{R - \text{value, wall}} = (0.8 \text{ m}^2) \frac{\Delta T (^\circ\text{C})}{2.3 \text{ m}^2 \cdot ^\circ\text{C/W}} = 0.348 \Delta T \text{ W}$$

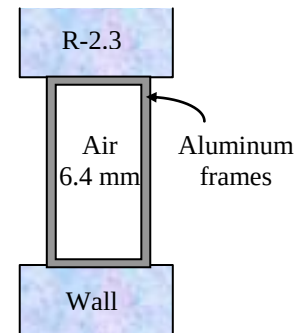
$$\dot{Q}_{\text{window}} = U_{\text{window}} A_{\text{window}} (T_i - T_o) = A_{\text{window}} \frac{T_i - T_o}{R - \text{value}} = (0.2 \text{ m}^2) \frac{\Delta T (^\circ\text{C})}{0.220 \text{ m}^2 \cdot ^\circ\text{C/W}} = 0.909 \Delta T \text{ W}$$

Therefore, the rate of heat transfer through a double pane window is **much more** than the rate of heat transfer through an  $R$ -2.3 wall.

**Discussion** The ratio of heat transfer through the wall and through the window is

$$\frac{\dot{Q}_{\text{window}}}{\dot{Q}_{\text{wall}}} = \frac{0.909 \text{ W}}{0.348 \text{ W}} = 2.61$$

Therefore, 2.61 times more heat is lost through the windows than through the walls although the windows occupy only 20% of the wall area.



**9-128** The rate of heat loss through a double-door wood framed window and the inner surface temperature are to be determined for the cases of single pane, double pane, and low-e triple pane windows.

**Assumptions** 1 Steady operating conditions exist. 2 Heat transfer through the window is one-dimensional. 3 Thermal properties of the windows and the heat transfer coefficients are constant. 4 Infiltration heat losses are not considered.

**Properties** The U-factors of the windows are given in Table 9-6.

**Analysis** The rate of heat transfer through the window can be determined from

$$\dot{Q}_{\text{window}} = U_{\text{overall}} A_{\text{window}} (T_i - T_o)$$

where  $T_i$  and  $T_o$  are the indoor and outdoor air temperatures, respectively,  $U_{\text{overall}}$  is the U-factor (the overall heat transfer coefficient) of the window, and  $A_{\text{window}}$  is the window area which is determined to be

$$A_{\text{window}} = \text{Height} \times \text{Width} = (1.2 \text{ m})(1.8 \text{ m}) = 2.16 \text{ m}^2$$

The U-factors for the three cases can be determined directly from Table 9-6 to be 5.57, 2.86, and 1.46  $\text{W/m}^2 \cdot ^\circ\text{C}$ , respectively. Also, the inner surface temperature of the window glass can be determined from Newton's law,

$$\dot{Q}_{\text{window}} = h_i A_{\text{window}} (T_i - T_{\text{glass}}) \longrightarrow T_{\text{glass}} = T_i - \frac{\dot{Q}_{\text{window}}}{h_i A_{\text{window}}}$$

where  $h_i$  is the heat transfer coefficient on the inner surface of the window which is determined from Table 9-5 to be  $h_i = 8.3 \text{ W/m}^2 \cdot ^\circ\text{C}$ . Then the rate of heat loss and the interior glass temperature for each case are determined as follows:

(a) Single glazing:

$$\dot{Q}_{\text{window}} = (5.57 \text{ W/m}^2 \cdot ^\circ\text{C})(2.16 \text{ m}^2)[20 - (-8)^\circ\text{C}] = \mathbf{337 \text{ W}}$$

$$T_{\text{glass}} = T_i - \frac{\dot{Q}_{\text{window}}}{h_i A_{\text{window}}} = 20^\circ\text{C} - \frac{337 \text{ W}}{(8.29 \text{ W/m}^2 \cdot ^\circ\text{C})(2.16 \text{ m}^2)} = \mathbf{1.2^\circ\text{C}}$$

(b) Double glazing (13 mm air space):

$$\dot{Q}_{\text{window}} = (2.86 \text{ W/m}^2 \cdot ^\circ\text{C})(2.16 \text{ m}^2)[20 - (-8)^\circ\text{C}] = \mathbf{173 \text{ W}}$$

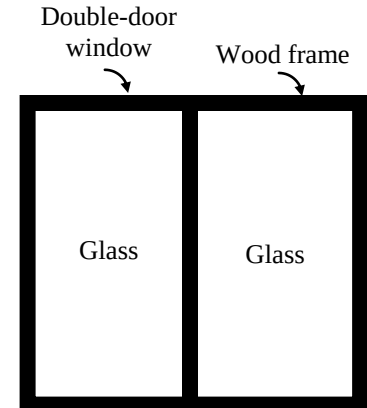
$$T_{\text{glass}} = T_i - \frac{\dot{Q}_{\text{window}}}{h_i A_{\text{window}}} = 20^\circ\text{C} - \frac{173 \text{ W}}{(8.29 \text{ W/m}^2 \cdot ^\circ\text{C})(2.16 \text{ m}^2)} = \mathbf{10.3^\circ\text{C}}$$

(c) Triple glazing (13 mm air space, low-e coated):

$$\dot{Q}_{\text{window}} = (1.46 \text{ W/m}^2 \cdot ^\circ\text{C})(2.16 \text{ m}^2)[20 - (-8)^\circ\text{C}] = \mathbf{88.3 \text{ W}}$$

$$T_{\text{glass}} = T_i - \frac{\dot{Q}_{\text{window}}}{h_i A_{\text{window}}} = 20 - \frac{88.3 \text{ W}}{(8.3 \text{ W/m}^2 \cdot ^\circ\text{C})(2.16 \text{ m}^2)} = \mathbf{15.1^\circ\text{C}}$$

**Discussion** Note that heat loss through the window will be reduced by 49 percent in the case of double glazing and by 74 percent in the case of triple glazing relative to the single glazing case. Also, in the case of single glazing, the low inner glass surface temperature will cause considerable discomfort in the occupants because of the excessive heat loss from the body by radiation. It is raised from  $1.2^\circ\text{C}$  to  $10.3^\circ\text{C}$  in the case of double glazing and to  $15.1^\circ\text{C}$  in the case of triple glazing.



**9-129** The overall U-factor for a double-door type window is to be determined, and the result is to be compared to the value listed in Table 9-6.

**Assumptions** 1 Steady operating conditions exist. 2 Heat transfer through the window is one-dimensional.

**Properties** The U-factors for the various sections of windows are given in Table 9-6.

**Analysis** The areas of the window, the glazing, and the frame are

$$A_{\text{window}} = \text{Height} \times \text{Width} = (2 \text{ m})(2.4 \text{ m}) = 4.80 \text{ m}^2$$

$$A_{\text{glazing}} = 2 \times \text{Height} \times \text{Width} = 2(1.92 \text{ m})(1.14 \text{ m}) = 4.38 \text{ m}^2$$

$$A_{\text{frame}} = A_{\text{window}} - A_{\text{glazing}} = 4.80 - 4.38 = 0.42 \text{ m}^2$$

The edge-of-glass region consists of a 6.5-cm wide band around the perimeter of the glazings, and the areas of the center and edge sections of the glazing are determined to be

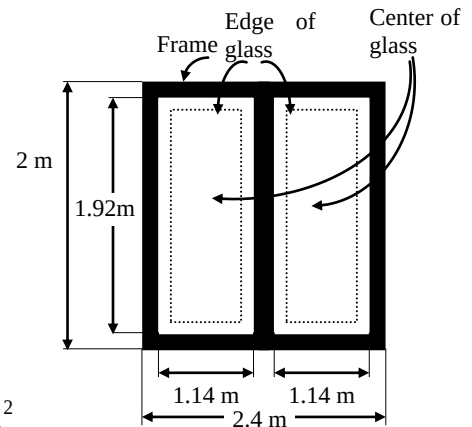
$$A_{\text{center}} = 2(\text{Height} \times \text{Width}) = 2(1.92 - 0.13 \text{ m})(1.14 - 0.13 \text{ m}) = 3.62 \text{ m}^2$$

$$A_{\text{edge}} = A_{\text{glazing}} - A_{\text{center}} = 4.38 - 3.62 = 0.76 \text{ m}^2$$

The U-factor for the frame section is determined from Table 9-4 to be  $U_{\text{frame}} = 2.8 \text{ W/m}^2 \cdot ^\circ\text{C}$ . The U-factor for the center and edge sections are determined from Table 9-6 to be  $U_{\text{center}} = 2.78 \text{ W/m}^2 \cdot ^\circ\text{C}$  and  $U_{\text{edge}} = 3.40 \text{ W/m}^2 \cdot ^\circ\text{C}$ . Then the overall U-factor of the entire window becomes

$$\begin{aligned} U_{\text{window}} &= (U_{\text{center}} A_{\text{center}} + U_{\text{edge}} A_{\text{edge}} + U_{\text{frame}} A_{\text{frame}}) / A_{\text{window}} \\ &= (2.78 \times 3.62 + 3.40 \times 0.76 + 2.8 \times 0.42) / 4.80 \\ &= \mathbf{2.88 \text{ W/m}^2 \cdot ^\circ\text{C}} \end{aligned}$$

**Discussion** The overall U-factor listed in Table 9-6 for the specified type of window is  $2.86 \text{ W/m}^2 \cdot ^\circ\text{C}$ , which is sufficiently close to the value obtained above.



## Review Problems

**9-130** An electric resistance space heater filled with oil is placed against a wall. The power rating of the heater and the time it will take for the heater to reach steady operation when it is first turned on are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 Heat transfer from the back, bottom, and top surfaces are disregarded. 4 The local atmospheric pressure is 1 atm.

**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (75 + 25)/2 = 50^\circ\text{C}$  are (Table A-15)

$$k = 0.02735 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.798 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7228$$

$$\beta = \frac{1}{T_f} = \frac{1}{(50 + 273)\text{K}} = 0.003096 \text{ K}^{-1}$$

**Analysis** Heat transfer from the top and bottom surfaces are said to be negligible, and thus the heat transfer area in this case consists of the three exposed side surfaces. The characteristic length is the height of the box,  $L_c = L = 0.5 \text{ m}$ . Then,

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003096 \text{ K}^{-1})(75 - 25 \text{ K})(0.5 \text{ m})^3}{(1.798 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7228) = 4.244 \times 10^8$$

$$\text{Nu} = \left\{ 0.825 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.492}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(4.244 \times 10^8)^{1/6}}{\left[ 1 + \left( \frac{0.492}{0.7228} \right)^{9/16} \right]^{8/27}} \right\}^2 = 94.68$$

$$h = \frac{k}{L} \text{Nu} = \frac{0.02735 \text{ W/m}\cdot^\circ\text{C}}{0.5 \text{ m}} (94.68) = 5.179 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = (0.5 \text{ m})(0.8 \text{ m}) + 2(0.15 \text{ m})(0.5 \text{ m}) = 0.55 \text{ m}^2$$

and

$$\dot{Q} = hA_s(T_s - T_\infty) = (5.179 \text{ W/m}^2\cdot^\circ\text{C})(0.55 \text{ m}^2)(75 - 25)^\circ\text{C} = 142.4 \text{ W}$$

The radiation heat loss is

$$\begin{aligned} \dot{Q}_{\text{rad}} &= \varepsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4) \\ &= (0.8)(0.55 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[(75 + 273 \text{ K})^4 - (25 + 273 \text{ K})^4] = 169.1 \text{ W} \end{aligned}$$

Then the total rate of heat transfer, thus the power rating of the heater becomes

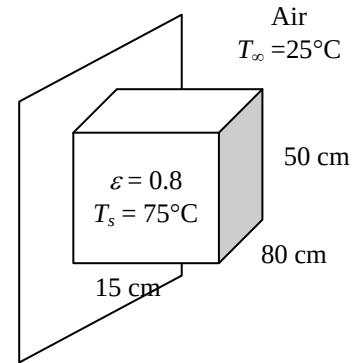
$$\dot{Q}_{\text{total}} = 142.4 + 169.1 = \mathbf{311.5 \text{ W}}$$

The specific heat of the oil at the average temperature of the oil is  $2006 \text{ J/kg}\cdot^\circ\text{C}$  (Table A-13). Then the amount of heat transfer needed to raise the temperature of the oil to the steady operating temperature and the time it takes become

$$Q = mc_p(T_2 - T_1) = (45 \text{ kg})(2006 \text{ J/kg}\cdot^\circ\text{C})(75 - 25)^\circ\text{C} = 4.514 \times 10^6 \text{ J}$$

$$Q = \dot{Q}\Delta t \longrightarrow \Delta t = \frac{Q}{\dot{Q}} = \frac{4.514 \times 10^6 \text{ kJ}}{311.5 \text{ J/s}} = 14,490 \text{ s} = \mathbf{4.03 \text{ h}}$$

which is not practical. Therefore, the surface temperature of the heater must be allowed to be higher than  $75^\circ\text{C}$ .



**9-131** A hot part of the vertical front section of a natural gas furnace in a plant is considered. The rate of heat loss from this section and the annual cost of this heat loss are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm. 4 Any heat transfer from other surfaces of the tank is disregarded.

**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (110 + 25)/2 = 67.5^\circ\text{C}$  are (Table A-15)

$$k = 0.02863 \text{ W/m} \cdot ^\circ\text{C}$$

$$\nu = 1.97 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7183$$

$$\beta = \frac{1}{T_f} = \frac{1}{(67.5 + 273)\text{K}} = 0.002937 \text{ K}^{-1}$$

**Analysis** The characteristic length in this case is the height of that section of furnace,  $L_c = L = 1.5 \text{ m}$ . Then,

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.002937 \text{ K}^{-1})(110 - 25 \text{ K})(1.5 \text{ m})^3}{(1.97 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7183) = 1.530 \times 10^{10}$$

$$\text{Nu} = \left\{ 0.825 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.492}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(1.530 \times 10^{10})^{1/6}}{\left[ 1 + \left( \frac{0.492}{0.7183} \right)^{9/16} \right]^{8/27}} \right\}^2 = 289.1$$

$$h = \frac{k}{L} \text{Nu} = \frac{0.02863 \text{ W/m} \cdot ^\circ\text{C}}{1.5 \text{ m}} (289.1) = 5.518 \text{ W/m}^2 \cdot ^\circ\text{C}$$

$$A_s = (1.5 \text{ m})(1 \text{ m}) = 1.5 \text{ m}^2$$

and

$$\dot{Q} = hA_s(T_s - T_\infty) = (5.518 \text{ W/m}^2 \cdot ^\circ\text{C})(1.5 \text{ m}^2)(110 - 25)^\circ\text{C} = 703.5 \text{ W}$$

The radiation heat loss is

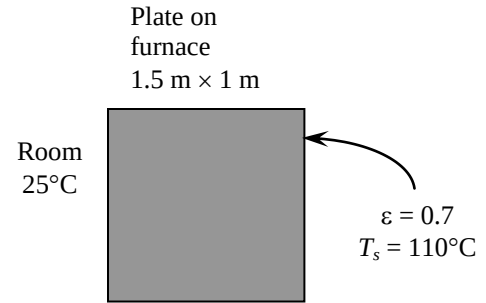
$$\begin{aligned} \dot{Q}_{rad} &= \varepsilon A_s \sigma (T_{surr}^4 - T_s^4) \\ &= (0.7)(1.5 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)[(110 + 273 \text{ K})^4 - (25 + 273 \text{ K})^4] \\ &= 811.6 \text{ W} \end{aligned}$$

$$\dot{Q}_{total} = 703.5 + 811.6 = \mathbf{1515 \text{ W}}$$

The amount and cost of natural gas used to overcome this heat loss per year is

$$Q_{gas} = \dot{Q}_{gas} \Delta t = \frac{\dot{Q}_{total}}{0.79} \Delta t = \frac{(1.515 \text{ kJ/s})}{0.79} (310 \text{ days/yr} \times 10 \text{ hr/day} \times 3600 \text{ s/hr}) = 2.140 \times 10^7 \text{ kJ}$$

$$\text{Cost} = (2.140 \times 10^7 / 105,500 \text{ therm})(\$1.20/\text{therm}) = \mathbf{\$243}$$



**9-132** An electric hot water heater is located in a small room. A hot water tank insulation kit is available for \$60. The payback period of this insulation to pay for itself from the energy it saves is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm. 4 Any heat transfer from the top and bottom surfaces of the tank is disregarded. 5 The thermal resistance of the metal sheet is negligible.

**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (40 + 20)/2 = 30^\circ\text{C}$  are (Table A-15)

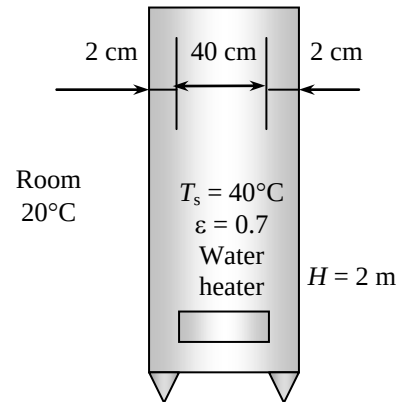
$$k = 0.02588 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.608 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7282$$

$$\beta = \frac{1}{T_f} = \frac{1}{(30 + 273)\text{K}} = 0.0033 \text{ K}^{-1}$$

**Analysis** The characteristic length in this case is the height of the heater,  $L_c = L = 2 \text{ m}$ . Then,



$$\text{Ra} = \frac{g\beta(T_\infty - T_s)L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.0033 \text{ K}^{-1})(40 - 20 \text{ K})(2 \text{ m})^3}{(1.608 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7282) = 1.459 \times 10^{10}$$

$$\text{Nu} = \left\{ 0.825 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.492}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(1.459 \times 10^{10})^{1/6}}{\left[ 1 + \left( \frac{0.492}{0.7282} \right)^{9/16} \right]^{8/27}} \right\}^2 = 285.4$$

$$h = \frac{k}{L} \text{Nu} = \frac{0.02588 \text{ W/m}\cdot^\circ\text{C}}{2 \text{ m}} (285.4) = 3.693 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = \pi DL = \pi(0.44 \text{ m})(2 \text{ m}) = 2.765 \text{ m}^2$$

and

$$\dot{Q} = hA_s(T_\infty - T_s) = (3.693 \text{ W/m}^2\cdot^\circ\text{C})(2.765 \text{ m}^2)(40 - 20)^\circ\text{C} = 204.2 \text{ W}$$

The radiation heat loss is

$$\begin{aligned} \dot{Q}_{\text{rad}} &= \varepsilon A_s \sigma (T_{\text{surr}}^4 - T_s^4) \\ &= (0.7)(2.765 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[(40 + 273 \text{ K})^4 - (20 + 273 \text{ K})^4] = 244.5 \text{ W} \end{aligned}$$

and

$$\dot{Q}_{\text{total}} = 204.2 + 244.5 = 448.7 \text{ W}$$

The reduction in heat loss after adding insulation is

$$\dot{Q} = (0.80)(448.7) = 359.0 \text{ W}$$

The amount of heat and money saved per hour is

$$Q_{\text{saved}} = \dot{Q}_{\text{saved}} \Delta t = (0.3590 \text{ kW})(1 \text{ h}) = 0.3590 \text{ kWh}$$

$$\text{Money saved} = (0.3590 \text{ kWh})(\$0.08/\text{kWh}) = \$0.02872$$

Then it will take

$$\Delta t = \frac{\$60}{\$0.02872} = 2089 \text{ h} = \mathbf{87.0 \text{ days}}$$

for the additional insulation to pay for itself from the energy it saves.

**9-133** A plate inclined at  $30^\circ$  with the top surface of the plate well insulated. The rate of heat loss from the plate is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas. 3 Thermal properties are constant. 4 Radiation heat transfer is negligible.

**Properties** The properties of air at  $T_f = (T_s + T_\infty)/2 = 30^\circ\text{C}$  are  $k = 0.02588 \text{ W/m}\cdot\text{K}$ ,  $\nu = 1.608 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 0.7282$  (from Table A-15). Also,  $\beta = 1/T_f = 0.0033 \text{ K}^{-1}$ .

**Analysis** The Rayleigh number ( $L_c = L$ ) is

$$\begin{aligned}\text{Ra}_L &= \frac{g \cos \theta \beta (T_s - T_\infty) L^3}{\nu^2} \text{Pr} \\ &= \frac{(9.81 \text{ m/s}^2)(\cos 30^\circ)(0.0033 \text{ K}^{-1})(60 - 0) \text{ K}(0.5 \text{ m})^3}{(1.608 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7282) \\ &= 5.922 \times 10^8\end{aligned}$$

The Nusselt number is calculated using the correlation for vertical plate,

$$\text{Nu} = \left\{ 0.825 + \frac{0.387 \text{Ra}_L^{1/6}}{[1 + (0.492/\text{Pr})^{9/16}]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(5.922 \times 10^8)^{1/6}}{[1 + (0.492/0.7282)^{9/16}]^{8/27}} \right\}^2 = 104.9$$

The heat transfer coefficient is

$$h = \text{Nu} \frac{k}{L} = (104.9) \frac{0.02588 \text{ W/m}\cdot\text{K}}{0.5 \text{ m}} = 5.43 \text{ W/m}^2 \cdot \text{K}$$

Hence, the rate of heat loss is

$$\dot{Q} = hA_s(T_s - T_\infty) = (5.43 \text{ W/m}^2 \cdot \text{K})(0.5 \text{ m})(0.5 \text{ m})(60 - 0) \text{ K} = \mathbf{81.5 \text{ W}}$$

**Discussion** The inclined plate with well insulated top surface can be treated as an inclined plate with hot surface down.

**9-134** A group of 25 transistors are cooled by attaching them to a square aluminum plate and mounting the plate on the wall of a room. The required size of the plate to limit the surface temperature to 50°C is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm. 4 Any heat transfer from the back side of the plate is negligible.

**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (50 + 30)/2 = 40^\circ\text{C}$  are (Table A-15)

$$k = 0.02662 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.702 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7255$$

$$\beta = \frac{1}{T_f} = \frac{1}{(40 + 273)\text{K}} = 0.003195 \text{ K}^{-1}$$

**Analysis** The Rayleigh number can be determined in terms of the characteristic length (length of the plate) to be

$$\text{Ra} = \frac{g\beta(T_\infty - T_s)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003195 \text{ K}^{-1})(50 - 30 \text{ K})(L)^3}{(1.702 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7255) = 1.571 \times 10^9 L^3$$

The Nusselt number relation is

$$\text{Nu} = \left\{ 0.825 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.492}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(1.571 \times 10^9 L^3)^{1/6}}{\left[ 1 + \left( \frac{0.492}{0.7255} \right)^{9/16} \right]^{8/27}} \right\}^2$$

The heat transfer coefficient is

$$h = \frac{k}{L} \text{Nu} = \frac{0.02662 \text{ W/m}\cdot^\circ\text{C}}{L} \text{Nu}$$

$$A_s = L^2$$

Noting that both the surface and surrounding temperatures are known, the rate of convection and radiation heat transfer are expressed as

$$\dot{Q}_{\text{conv}} = hA_s(T_s - T_\infty) = \frac{0.02662 \text{ W/m}\cdot^\circ\text{C}}{L} \text{Nu} L^2 (50 - 30)^\circ\text{C}$$

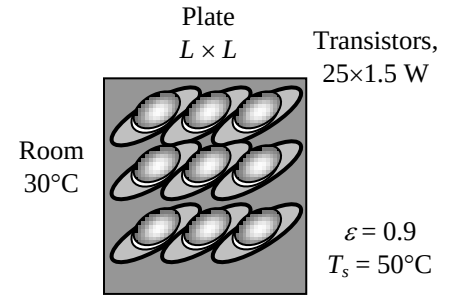
$$\dot{Q}_{\text{rad}} = \varepsilon A_s \sigma (T_s^4 - T_{\text{sky}}^4) = (0.9) L^2 (5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4) [(50 + 273)^4 - (30 + 273)^4] \text{K}^4 = 125.3 L^2$$

The rate of total heat transfer is expressed as

$$\begin{aligned} \dot{Q}_{\text{total}} &= \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}} \\ 25 \times (1.5 \text{ W}) &= \frac{0.02662 \text{ W/m}\cdot^\circ\text{C}}{L} \text{Nu} L^2 (50 - 30)^\circ\text{C} + 125.3 L^2 \end{aligned}$$

Substituting Nusselt number expression above into this equation and solving for  $L$ , the length of the plate is determined to be

$$L = \mathbf{0.426 \text{ m}}$$



**9-135** A group of 25 transistors are cooled by attaching them to a square aluminum plate and positioning the plate horizontally in a room. The required size of the plate to limit the surface temperature to 50°C is to be determined for two cases.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm. 4 Any heat transfer from the back side of the plate is negligible.

**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (50 + 30)/2 = 40^\circ\text{C}$  are (Table A-15)

$$k = 0.02662 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.702 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7255$$

$$\beta = \frac{1}{T_f} = \frac{1}{(40 + 273)\text{K}} = 0.003195 \text{ K}^{-1}$$

**Analysis** The characteristic length and the Rayleigh number for the horizontal case are determined to be

$$L_c = \frac{A_s}{p} = \frac{L^2}{4L} = \frac{L}{4}$$

$$\text{Ra} = \frac{g\beta(T_\infty - T_s)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003195 \text{ K}^{-1})(50 - 30 \text{ K})(L/4)^3}{(1.702 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7255) = 2.454 \times 10^7 L^3$$

Noting that both the surface and surrounding temperatures are known, the rate of radiation heat transfer is determined to be

$$\dot{Q}_{\text{rad}} = \varepsilon A_s \sigma (T_s^4 - T_{\text{sky}}^4) = (0.9)L^2 (5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4) [(50 + 273)^4 - (30 + 273)^4] \text{ K}^4 = 125.3 L^2$$

(a) **Hot surface facing up:** We assume  $\text{Ra} < 10^7$  and thus  $L < 0.74 \text{ m}$  so that we can determine the Nu number from Eq. 9-22. Then the Nusselt number and the convection heat transfer coefficient become

$$\text{Nu} = 0.54 \text{Ra}^{1/4} = 0.54(2.454 \times 10^7 L^3)^{1/4} = 38.0 L^{3/4}$$

Then,

$$h = \frac{k}{L} \text{Nu} = \frac{0.02662 \text{ W/m}\cdot^\circ\text{C}}{L/4} (38.0 L^{3/4}) = 4.047 L^{-1/4} \text{ W/m}^2 \cdot ^\circ\text{C}$$

$$A_s = L^2$$

The rate of convection heat transfer is

$$\dot{Q}_{\text{conv}} = h A_s (T_s - T_\infty) = (4.047 L^{-1/4}) L^2 (50 - 30) = 80.94 L^{7/4} \text{ W}$$

Then,

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}} \longrightarrow 25 \times (1.5 \text{ W}) = 80.94 L^{7/4} + 125.3 L^2 \text{ W}$$

Solving for  $L$ , the length of the plate is determined to be

$$L = \mathbf{0.407 \text{ m}}$$

Note that  $L < 0.75 \text{ m}$ , and therefore the assumption of  $\text{Ra} < 10^7$  is verified. That is,

(b) **Hot surface facing down:** The Nusselt number in this case is determined from

$$\text{Nu} = 0.27 \text{Ra}^{1/4} = 0.27(2.454 \times 10^7 L^3)^{1/4} = 19.0 L^{3/4}$$

$$\text{Then, } h = \frac{k}{L_c} \text{Nu} = \frac{0.02662 \text{ W/m}\cdot^\circ\text{C}}{L/4} (19.0 L^{3/4}) = 2.023 L^{-1/4}$$

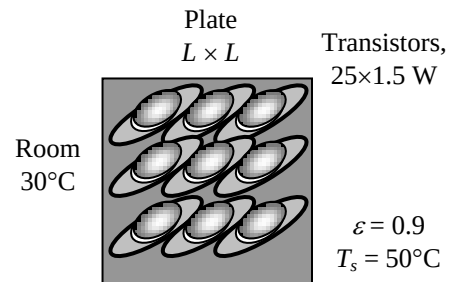
The rate of convection heat transfer is

$$\dot{Q}_{\text{conv}} = h A_s (T_s - T_\infty) = (2.023 L^{-1/4}) L^2 (50 - 30) = 40.47 L^{7/4} \text{ W}$$

$$\text{Then, } \dot{Q}_{\text{total}} = \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}} \longrightarrow 25 \times (1.5 \text{ W}) = 40.47 L^{7/4} + 125.3 L^2 \text{ W}$$

Solving for  $L$ , the length of the plate is determined to be

$$L = \mathbf{0.464 \text{ m}}$$



**9-136** An  $L \times L$  horizontal plate is placed in a quiescent air with the hot surface facing up, and the expressions, having the form  $Nu = C Ra_L^n$ , for the average heat transfer coefficient are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties.

**Properties** The properties of air at  $T_f = 20^\circ\text{C}$  are  $k = 0.02514 \text{ W/m}\cdot\text{K}$ ,  $\nu = 1.516 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $Pr = 0.7309$  (from Table A-15). Also,  $\beta = 1/T_f = 0.003413 \text{ K}^{-1}$ .

**Analysis** For horizontal plate, the characteristic length is

$$L_c = \frac{A_s}{p} = \frac{L^2}{4L} = \frac{L}{4}$$

The Rayleigh number ( $L_c = L/4$ ) is

$$\begin{aligned} Ra_L &= \frac{g\beta(T_s - T_\infty)L_c^3}{\nu^2} Pr = \frac{(9.81 \text{ m/s}^2)(0.003413 \text{ K}^{-1})\Delta T(L/4)^3}{(1.516 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7309) \\ &= 1.664 \times 10^6 \Delta T L^3 \end{aligned}$$

For  $10^4 < Ra_L < 10^7$ , we have

$$Nu = \frac{hL_c}{k} = \frac{hL}{4k} = 0.54 Ra_L^{1/4} \quad \rightarrow \quad h = 2.16 \frac{k}{L} Ra_L^{1/4}$$

Substituting the  $Ra_L$  yields

$$h = 2.16 \left( \frac{0.02514}{L} \right) (1.664 \times 10^6 \Delta T L^3)^{1/4} = 1.95 (\Delta T / L)^{1/4} \quad 10^4 < Ra_L < 10^7$$

For  $10^7 < Ra_L < 10^{11}$ , we have

$$Nu = \frac{hL_c}{k} = \frac{hL}{4k} = 0.15 Ra_L^{1/3} \quad \rightarrow \quad h = 0.6 \frac{k}{L} Ra_L^{1/3}$$

Substituting the  $Ra_L$  yields

$$h = 0.6 \left( \frac{0.02514}{L} \right) (1.664 \times 10^6 \Delta T L^3)^{1/3} = 1.79 \Delta T^{1/3} \quad 10^7 < Ra_L < 10^{11}$$

**Discussion** The average heat transfer coefficient for  $10^4 < Ra_L < 10^7$  is dependent on  $\Delta T$  and  $L$ . For  $10^{10} < Ra_L < 10^{13}$ , the average heat transfer coefficient is not influenced by  $L$ .

**9-137** A flat-plate solar collector placed horizontally on the flat roof of a house is exposed to the calm ambient air. The rate of heat loss from the collector by natural convection and radiation are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm.

**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (42 + 8)/2 = 25^\circ\text{C}$  are (Table A-15)

$$k = 0.02551 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.562 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7296$$

$$\beta = \frac{1}{T_f} = \frac{1}{(25 + 273)\text{K}} = 0.003356 \text{ K}^{-1}$$

**Analysis** The characteristic length in this case is determined from

$$L_c = \frac{A_s}{p} = \frac{(1.5 \text{ m})(4.5 \text{ m})}{2(1.5 \text{ m} + 4.5 \text{ m})} = 0.5625 \text{ m}$$

Then,

$$\text{Ra} = \frac{g\beta(T_\infty - T_s)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003356 \text{ K}^{-1})(42 - 8 \text{ K})(0.5625 \text{ m})^3}{(1.562 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7296) = 5.957 \times 10^8$$

$$\text{Nu} = 0.15 \text{Ra}^{1/3} = 0.15(5.957 \times 10^8)^{1/3} = 126.2$$

$$h = \frac{k}{L_c} \text{Nu} = \frac{0.02551 \text{ W/m}\cdot^\circ\text{C}}{0.5625 \text{ m}} (126.2) = 5.724 \text{ W/m}^2\cdot^\circ\text{C}$$

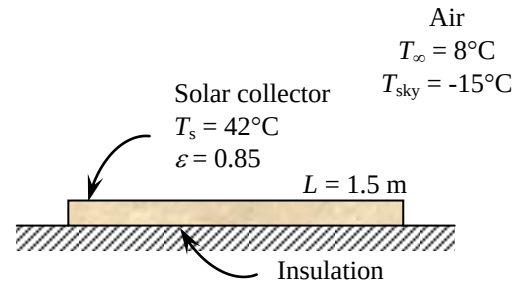
$$A_s = (1.5 \text{ m})(4.5 \text{ m}) = 6.75 \text{ m}^2$$

and

$$\dot{Q}_{\text{conv}} = hA_s(T_s - T_\infty) = (5.724 \text{ W/m}^2\cdot^\circ\text{C})(6.75 \text{ m}^2)(42 - 8)^\circ\text{C} = \mathbf{1314 \text{ W}}$$

Heat transfer rate by radiation is

$$\begin{aligned} \dot{Q}_{\text{rad}} &= \varepsilon A_s \sigma (T_{\text{surr}}^4 - T_s^4) \\ &= (0.85)(6.75 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[(42 + 273 \text{ K})^4 - (-15 + 273 \text{ K})^4] \\ &= \mathbf{1762 \text{ W}} \end{aligned}$$



**9-138** A horizontal skylight made of a single layer of glass on the roof of a house is considered. The rate of heat loss through the skylight is to be determined for two cases.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm.

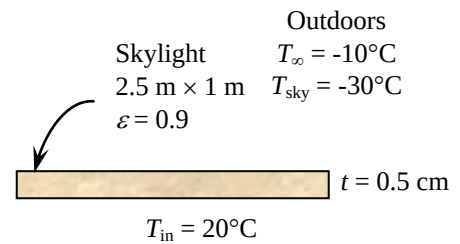
**Properties** Based on the problem statement, the properties of air at 1 atm and the anticipated film temperature of  $(T_s + T_\infty)/2 = (-4 - 10)/2 = -7^\circ\text{C}$  are (Table A-15)

$$k = 0.02311 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.278 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.738$$

$$\beta = \frac{1}{T_f} = \frac{1}{(-7 + 273)\text{K}} = 0.003759 \text{ K}^{-1}$$



**Analysis** We assume radiation heat transfer inside the house to be negligible. We start the calculations by “guessing” the glass temperature to be  $-4^\circ\text{C}$  for the evaluation of the properties and  $h$ . We will check the accuracy of this guess later and repeat the calculations if necessary. The characteristic length in this case is determined from

$$L_c = \frac{A_s}{p} = \frac{(1\text{ m})(2.5\text{ m})}{2(1\text{ m} + 2.5\text{ m})} = 0.357 \text{ m}.$$

Then,

$$\text{Ra} = \frac{g\beta(T_1 - T_2)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81\text{ m/s}^2)(0.003759\text{ K}^{-1})[-4 - (-10)\text{ K}](0.357\text{ m})^3}{(1.278 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.738) = 4.553 \times 10^7$$

$$\text{Nu} = 0.15\text{Ra}^{1/3} = 0.15(4.553 \times 10^7)^{1/3} = 53.56$$

$$h_o = \frac{k}{L_c} \text{Nu} = \frac{0.02311 \text{ W/m}\cdot^\circ\text{C}}{0.357 \text{ m}} (53.56) = 3.467 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = (1\text{ m})(2.5\text{ m}) = 2.5 \text{ m}^2$$

Using the assumed value of glass temperature, the radiation heat transfer coefficient is determined to be

$$\begin{aligned} h_{\text{rad}} &= \varepsilon\sigma(T_s + T_{\text{sky}})(T_s^2 + T_{\text{sky}}^2) \\ &= 0.9(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[(-4 + 273) + (-30 + 273)][(-4 + 273)^2 + (-30 + 273)^2] \text{ K}^3 \\ &= 3.433 \text{ W/m}^2\cdot\text{K} \end{aligned}$$

Then the combined convection and radiation heat transfer coefficient outside becomes

$$h_{o,\text{combined}} = h_o + h_{\text{rad}} = 3.467 + 3.433 = 6.90 \text{ W/m}^2$$

Again we take the glass temperature to be  $-4^\circ\text{C}$  for the evaluation of the properties and  $h$  for the inner surface of the skylight. The properties of air at 1 atm and the film temperature of  $T_f = (-4 + 20)/2 = 8^\circ\text{C}$  are (Table A-15)

$$k = 0.02424 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.408 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7342$$

$$\beta = \frac{1}{T_f} = \frac{1}{(8 + 273)\text{K}} = 0.003559 \text{ K}^{-1}$$

The characteristic length in this case is also 0.357 m. Then,

$$\text{Ra} = \frac{g\beta(T_1 - T_2)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81\text{ m/s}^2)(0.003559\text{ K}^{-1})[20 - (-4)\text{ K}](0.357\text{ m})^3}{(1.408 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7342) = 1.412 \times 10^8$$

$$\text{Nu} = 0.27\text{Ra}^{1/4} = 0.27(1.412 \times 10^8)^{1/4} = 29.43$$

$$h_i = \frac{k}{L_c} \text{Nu} = \frac{0.02424 \text{ W/m}\cdot^\circ\text{C}}{0.357 \text{ m}} (29.43) = 1.998 \text{ W/m}^2\cdot^\circ\text{C}$$

Using the thermal resistance network, the rate of heat loss through the skylight is determined to be

$$\begin{aligned}\dot{Q}_{\text{skylight}} &= \frac{T_{s,i} - T_{\infty,o}}{R_{\text{conv},i} + R_{\text{cond,glas}} + R_{\text{combined},o}} \\ &= \frac{A_s (T_{\text{room}} - T_{\text{out}})}{\frac{1}{h_i} + \frac{t_{\text{glass}}}{k_{\text{glass}}} + \frac{1}{h}} = \frac{(2.5 \text{ m}^2)[20 - (-10)]^\circ\text{C}}{\frac{1}{1.998 \text{ W/m}^2 \cdot ^\circ\text{C}} + \frac{0.005 \text{ m}}{0.78 \text{ W/m} \cdot ^\circ\text{C}} + \frac{1}{6.90 \text{ W/m}^2 \cdot ^\circ\text{C}}} = 115 \text{ W}\end{aligned}$$

Using the same heat transfer coefficients for simplicity, the rate of heat loss through the roof in the case of R-5.34 construction is determined to be

$$\begin{aligned}\dot{Q}_{\text{roof}} &= \frac{T_{s,i} - T_{\infty,o}}{R_{\text{conv},i} + R_{\text{cond}} + R_{\text{combined},o}} \\ &= \frac{A_s (T_{\text{room}} - T_{\text{out}})}{\frac{1}{h_i} + R_{\text{glass}} + \frac{1}{h}} = \frac{(2.5 \text{ m}^2)[20 - (-10)]^\circ\text{C}}{\frac{1}{1.998 \text{ W/m}^2 \cdot ^\circ\text{C}} + 5.34 \text{ m}^2 \cdot ^\circ\text{C/W} + \frac{1}{6.90 \text{ W/m}^2 \cdot ^\circ\text{C}}} = 12.5 \text{ W}\end{aligned}$$

Therefore, a house loses  $115/12.5 \cong 9$  times more heat through the skylights than it does through an insulated wall of the same size.

Using Newton's law of cooling, the glass temperature corresponding to a heat transfer rate of 115 W is calculated to be  $-3.3^\circ\text{C}$ , which is sufficiently close to the assumed value of  $-4^\circ\text{C}$ . Therefore, there is no need to repeat the calculations.

**Discussion** The assumed film temperature of  $T_f = -7^\circ\text{C}$  is an appropriate assumption, since the determined  $T_s = -3.3^\circ\text{C}$  would give a film temperature of  $T_f = -6.65^\circ\text{C}$ . Otherwise,  $T_s$  would have to be solved iteratively.

**9-139** An electronic box is cooled internally by a fan blowing air into the enclosure. The fraction of the heat lost from the outer surfaces of the electronic box is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 Heat transfer from the base surface is disregarded. 4 The pressure of air inside the enclosure is 1 atm.

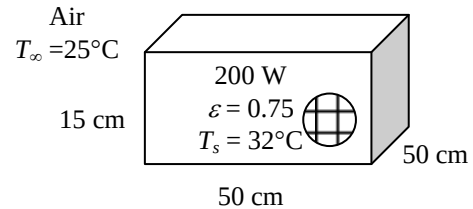
**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (32 + 15)/2 = 28.5^\circ\text{C}$  are (Table A-15)

$$k = 0.02577 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.594 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7286$$

$$\beta = \frac{1}{T_f} = \frac{1}{(28.5 + 273)\text{K}} = 0.003317 \text{ K}^{-1}$$



**Analysis** Heat loss from the horizontal top surface:

The characteristic length in this case is

$$L_c = \frac{A}{P} = \frac{(0.5 \text{ m})^2}{2[(0.5 \text{ m}) + (0.5 \text{ m})]} = 0.125 \text{ m}$$

Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003317 \text{ K}^{-1})(32 - 25 \text{ K})(0.125 \text{ m})^3}{(1.594 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7286) = 1.275 \times 10^6$$

$$Nu = 0.54 Ra^{1/4} = 0.54 (1.275 \times 10^6)^{1/4} = 18.15$$

$$h = \frac{k}{L_c} Nu = \frac{0.02577 \text{ W/m}\cdot^\circ\text{C}}{0.125 \text{ m}} (18.15) = 3.741 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_{top} = (0.5 \text{ m})^2 = 0.25 \text{ m}^2$$

and  $\dot{Q}_{top} = hA_{top}(T_s - T_\infty) = (3.741 \text{ W/m}^2\cdot^\circ\text{C})(0.25 \text{ m}^2)(32 - 25)^\circ\text{C} = 6.55 \text{ W}$

**Heat loss from vertical side surfaces:**

The characteristic length in this case is the height of the box  $L_c = L = 0.15 \text{ m}$ . Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003317 \text{ K}^{-1})(32 - 25 \text{ K})(0.15 \text{ m})^3}{(1.594 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7286) = 2.204 \times 10^6$$

$$Nu = \left\{ 0.825 + \frac{0.387 Ra^{1/6}}{\left[ 1 + \left( \frac{0.492}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387 (2.204 \times 10^6)^{1/6}}{\left[ 1 + \left( \frac{0.492}{0.7286} \right)^{9/16} \right]^{8/27}} \right\}^2 = 20.55$$

$$h = \frac{k}{L} Nu = \frac{0.02577 \text{ W/m}\cdot^\circ\text{C}}{0.15 \text{ m}} (20.55) = 3.530 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_{side} = 4(0.15 \text{ m})(0.5 \text{ m}) = 0.3 \text{ m}^2$$

and  $\dot{Q}_{side} = hA_{side}(T_s - T_\infty) = (3.530 \text{ W/m}^2\cdot^\circ\text{C})(0.3 \text{ m}^2)(32 - 25)^\circ\text{C} = 7.41 \text{ W}$

The radiation heat loss is

$$\begin{aligned} \dot{Q}_{rad} &= \varepsilon A_s \sigma (T_s^4 - T_{surr}^4) \\ &= (0.75)(0.25 + 0.3 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[(32 + 273 \text{ K})^4 - (25 + 273 \text{ K})^4] = 17.95 \text{ W} \end{aligned}$$

Then the fraction of the heat loss from the outer surfaces of the box is determined to be

$$f = \frac{(6.55 + 7.41 + 17.95) \text{ W}}{200 \text{ W}} = 0.160 = \mathbf{16.0\%}$$

**9-140** A 10-m tall exhaust stack discharging exhaust gases at a rate of 0.125 kg/s is subjected to solar radiation and natural convection at the outer surface. The outer surface temperature of the exhaust stack is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Properties are constant. 3 The surface temperature is constant. 4 Air is an ideal gas.

**Properties** The properties of air at 60°C are  $k = 0.02808 \text{ W/m}\cdot\text{K}$ ,  $\nu = 1.896 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 0.7202$  (from Table A-15). Also,  $\beta = 1/T_f = 0.003003 \text{ K}^{-1}$ .

**Analysis** Assume that the exhaust stack can be treated as a vertical plate, the Rayleigh number is

$$\begin{aligned} \text{Ra}_L &= \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} \text{Pr} \\ &= \frac{(9.81 \text{ m/s}^2)(0.003003 \text{ K}^{-1})(T_s - 306 \text{ K})(10 \text{ m})^3}{(1.896 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7202) \end{aligned} \quad (1)$$

Assuming the Rayleigh number is within  $10^{10} < \text{Ra}_L < 10^{13}$ , the Nusselt number for vertical plate is

$$\text{Nu} = 0.1 \text{Ra}_L^{1/3} \quad \text{or} \quad h = \left( \frac{0.02808 \text{ W/m}\cdot\text{K}}{10 \text{ m}} \right) 0.1 \text{Ra}_L^{1/3} \quad (2)$$

The outer surface area of the exhaust stack is

$$A_s = \pi DL = \pi(1 \text{ m})(10 \text{ m}) = 31.42 \text{ m}^2$$

The rate of heat loss from the exhaust gases in the exhaust stack can be determined from

$$\dot{Q}_{\text{loss}} = \dot{m}c_p(T_{\text{in}} - T_{\text{out}}) = (0.125 \text{ kg/s})(1600 \text{ J/kg}\cdot^\circ\text{C})(30)^\circ\text{C} = 6000 \text{ W}$$

The heat loss on the outer surface of the exhaust stack by radiation and convection can be expressed as

$$\frac{\dot{Q}_{\text{loss}}}{A_s} = h[T_s - T_\infty] + \varepsilon\sigma[T_s^4 - T_{\text{surr}}^4] - \alpha_s \dot{q}_{\text{solar}} \quad (3)$$

Equations (1), (2), and (3) can be solved simultaneously to get the surface temperature. Copy the following lines and paste on a blank EES screen to solve the above equation:

```
A_s=31.42
D=1
g=9.81
k=0.02808
L=10
Pr=0.7202
q_incident=500
Q_loss=6000
T_inf=33+273
T_surr=33+273
alpha=0.9
beta=1/(273+60)
epsilon=0.9
nu=1.896e-5
sigma=5.670e-8
Ra_L=g*beta*(T_s-T_inf)*L^3/nu^2*Pr
(h*L/(0.1*k))^3=Ra_L
Q_loss/A_s=h*(T_s-T_inf)+epsilon*sigma*(T_s^4-T_surr^4)-alpha*q_incident
```

Solving by EES software, we get

$$\text{Ra}_L = 3.219 \times 10^{12}, \quad h = 4.146 \text{ W/m}^2 \cdot \text{K}, \quad \text{and} \quad T_s = 360.5 \text{ K}$$

Therefore, the exhaust stack outer surface temperature is

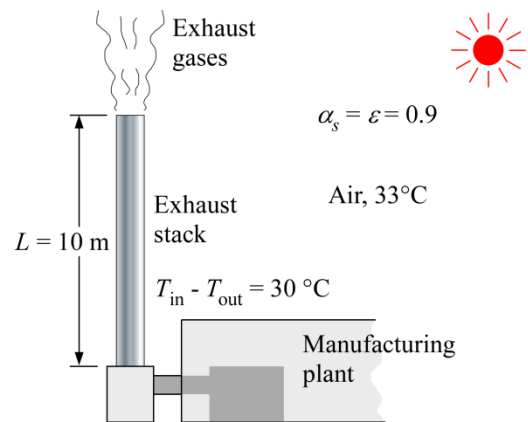
$$T_s = 87.5^\circ\text{C}$$

Now, we need to check if the assumption that the exhaust stack can be treated as a vertical plate is valid:

$$\frac{35L}{\text{Gr}_L^{1/4}} = \frac{35L}{(\text{Ra}_L / \text{Pr})^{1/4}} = \frac{35(10 \text{ m})}{(3.219 \times 10^{12} / 0.7202)^{1/4}} = 0.2407 < D$$

Since  $D \geq 35L / \text{Gr}_L^{1/4}$  is satisfied, we can treat the exhaust stack as a vertical plate.

**Discussion** The assumption that the Rayleigh number is within  $10^{10} < \text{Ra}_L < 10^{13}$  turned out to be appropriate. Also, the assumed film temperature of 60°C is valid, since  $(T_s + T_\infty)/2 = 60.3^\circ\text{C}$ .



**9-141** A vertically oriented cylindrical hot water tank is located in a bathroom. The rate of heat loss from the tank by natural convection and radiation is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm. 4 The temperature of the outer surface of the tank is constant.

**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (44 + 20)/2 = 32^\circ\text{C}$  are (Table A-15)

$$k = 0.02603 \text{ W/m} \cdot \text{K}$$

$$\nu = 1.627 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7276$$

$$\beta = \frac{1}{T_f} = \frac{1}{(32 + 273)\text{K}} = 0.003279 \text{ K}^{-1}$$

**Analysis** The characteristic length in this case is the height of the cylinder,  $L_c = L = 1.1 \text{ m}$ . Then,

$$\text{Gr} = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} = \frac{(9.81 \text{ m/s}^2)(0.003279 \text{ K}^{-1})(44 - 20 \text{ K})(1.1 \text{ m})^3}{(1.627 \times 10^{-5} \text{ m}^2/\text{s})^2} = 3.883 \times 10^9$$

A vertical cylinder can be treated as a vertical plate when

$$D (= 0.4 \text{ m}) \geq \frac{35L}{\text{Gr}^{1/4}} = \frac{35(1.1 \text{ m})}{(3.883 \times 10^9)^{1/4}} = 0.1542 \text{ m}$$

which is satisfied. That is, the Nusselt number relation for a vertical plate can be used for the side surfaces. For the top and bottom surfaces we use the relevant Nusselt number relations. First, for the side surfaces,

$$\text{Ra} = \text{GrPr} = (3.883 \times 10^9)(0.7276) = 2.825 \times 10^9$$

$$\text{Nu} = \left\{ 0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[ 1 + \left( \frac{0.492}{\text{Pr}} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(2.825 \times 10^9)^{1/6}}{\left[ 1 + \left( \frac{0.492}{0.7276} \right)^{9/16} \right]^{8/27}} \right\}^2 = 170.2$$

$$h = \frac{k}{L} \text{Nu} = \frac{0.02603 \text{ W/m} \cdot \text{K}}{1.1 \text{ m}} (170.2) = 4.027 \text{ W/m}^2 \cdot \text{K}$$

$$A_s = \pi DL = \pi(0.4 \text{ m})(1.1 \text{ m}) = 1.382 \text{ m}^2$$

$$\dot{Q}_{\text{side}} = hA_s(T_s - T_\infty) = (4.027 \text{ W/m}^2 \cdot \text{K})(1.382 \text{ m}^2)(44 - 20)^\circ\text{C} = 133.6 \text{ W}$$

For the top surface,

$$L_c = \frac{A_s}{p} = \frac{\pi D^2 / 4}{\pi D} = \frac{D}{4} = \frac{0.4 \text{ m}}{4} = 0.1 \text{ m}$$

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003279 \text{ K}^{-1})(44 - 20 \text{ K})(0.1 \text{ m})^3}{(1.627 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7276) = 2.123 \times 10^6$$

$$\text{Nu} = 0.54\text{Ra}^{1/4} = 0.54(2.123 \times 10^6)^{1/4} = 20.61$$

$$h = \frac{k}{L_c} \text{Nu} = \frac{0.02603 \text{ W/m} \cdot \text{K}}{0.1 \text{ m}} (20.61) = 5.365 \text{ W/m}^2 \cdot \text{K}$$

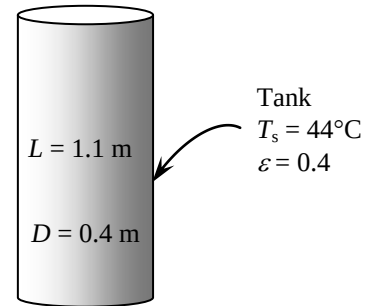
$$A_s = \pi D^2 / 4 = \pi(0.4 \text{ m})^2 / 4 = 0.1257 \text{ m}^2$$

$$\dot{Q}_{\text{top}} = hA_s(T_s - T_\infty) = (5.365 \text{ W/m}^2 \cdot \text{K})(0.1257 \text{ m}^2)(44 - 20)^\circ\text{C} = 16.2 \text{ W}$$

For the bottom surface,

$$\text{Nu} = 0.27\text{Ra}^{1/4} = 0.27(2.123 \times 10^6)^{1/4} = 10.31$$

$$h = \frac{k}{L_c} \text{Nu} = \frac{0.02603 \text{ W/m} \cdot \text{K}}{0.1 \text{ m}} (10.31) = 2.683 \text{ W/m}^2 \cdot \text{K}$$



$$\dot{Q}_{\text{bottom}} = hA_s(T_s - T_\infty) = (2.683 \text{ W/m}^2 \cdot \text{K})(0.1257 \text{ m}^2)(44 - 20)^\circ\text{C} = 8.1 \text{ W}$$

The total heat loss by natural convection is

$$\dot{Q}_{\text{conv}} = \dot{Q}_{\text{side}} + \dot{Q}_{\text{top}} + \dot{Q}_{\text{bottom}} = 133.6 + 16.2 + 8.1 = \mathbf{157.9 \text{ W}}$$

The radiation heat loss from the tank is

$$\begin{aligned}\dot{Q}_{\text{rad}} &= \varepsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4) \\ &= (0.4)(1.382 + 0.1257 + 0.1257 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4) \left[ (44 + 273 \text{ K})^4 - (20 + 273 \text{ K})^4 \right] \\ &= \mathbf{101.1 \text{ W}}\end{aligned}$$

**9-142** A cold cylinder is placed horizontally in hot air. The rates of heat transfer from the stack with and without wind cases are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm.

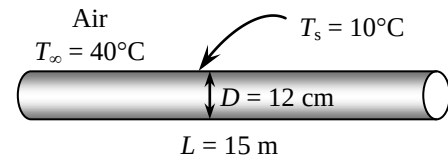
**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (40 + 10)/2 = 25^\circ\text{C}$  are (Table A-15)

$$k = 0.02551 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.562 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7296$$

$$\beta = \frac{1}{T_f} = \frac{1}{(25 + 273)\text{K}} = 0.003356 \text{ K}^{-1}$$



**Analysis** (a) When the stack is exposed to 10 m/s winds, the heat transfer will be by forced convection. We have flow of air over a cylinder and the heat transfer rate is determined as follows:

$$\text{Re} = \frac{VD}{\nu} = \frac{(10 \text{ m/s})(0.12 \text{ m})}{1.562 \times 10^{-5} \text{ m}^2/\text{s}} = 76,825$$

$$\text{Nu} = 0.027 \text{Re}^{0.805} \text{Pr}^{1/3} = 0.027(76,825)^{0.805} (0.7296)^{1/3} = 208.2 \quad (\text{from Table 7-1})$$

$$h = \frac{k}{D} \text{Nu} = \frac{0.02551 \text{ W/m}\cdot^\circ\text{C}}{0.12 \text{ m}} (208.2) = 44.27 \text{ W/m}^2\cdot^\circ\text{C}$$

$$\dot{Q}_{\text{forced conv}} = hA(T_\infty - T_s) = (44.27 \text{ W/m}^2\cdot^\circ\text{C})(\pi \times 0.12 \times 15 \text{ m}^2)(40 - 10)^\circ\text{C} = \mathbf{7510 \text{ W}}$$

(b) Without wind the heat transfer will be by natural convection. The characteristic length in this case is the outer diameter of the cylinder,  $L_c = D = 0.12 \text{ m}$ . Then,

$$\text{Ra} = \frac{g\beta(T_\infty - T_s)D^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003356 \text{ K}^{-1})(40 - 10 \text{ K})(0.12 \text{ m})^3}{(1.562 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7296) = 5.104 \times 10^6$$

$$\text{Nu} = \left\{ 0.6 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + (0.559/\text{Pr})^{9/16} \right]^{4/27}} \right\}^2 = \left\{ 0.6 + \frac{0.387(5.104 \times 10^6)^{1/6}}{\left[ 1 + (0.559/0.7296)^{9/16} \right]^{4/27}} \right\}^2 = 23.28$$

$$h = \frac{k}{D} \text{Nu} = \frac{0.02551 \text{ W/m}\cdot^\circ\text{C}}{0.12 \text{ m}} (23.28) = 4.949 \text{ W/m}^2\cdot^\circ\text{C}$$

$$\dot{Q}_{\text{nat. conv}} = hA(T_\infty - T_s) = (4.949 \text{ W/m}^2\cdot^\circ\text{C})(\pi \times 0.12 \times 15 \text{ m}^2)(40 - 10)^\circ\text{C} = \mathbf{840 \text{ W}}$$

**9-143** A hot water pipe passes through a basement. The temperature drop of water in the basement due to heat loss from the pipe is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm.

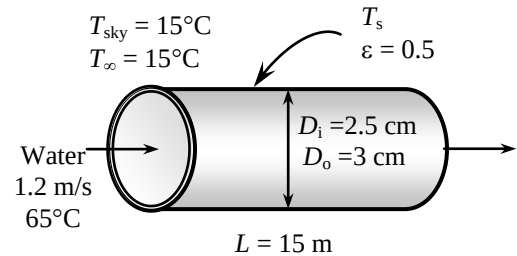
**Properties** The properties of air at 1 atm and the anticipated film temperature of  $(T_s + T_\infty)/2 = (65 + 15)/2 = 40^\circ\text{C}$  are (Table A-15)

$$k = 0.02662 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.702 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7255$$

$$\beta = \frac{1}{T_f} = \frac{1}{(40 + 273)\text{K}} = 0.003195 \text{ K}^{-1}$$



**Analysis** We expect the pipe temperature to be very close to the water temperature, and start the calculations by “guessing” the average outer surface temperature of the pipe to be  $65^\circ\text{C}$  for the evaluation of the properties and  $h$ . We will check the accuracy of this guess later and repeat the calculations if necessary. The characteristic length in this case is the outer diameter of the pipe,  $L_c = D_o = 0.03 \text{ m}$ . Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003195 \text{ K}^{-1})(65 - 15 \text{ K})(0.03 \text{ m})^3}{(1.702 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7255) = 1.060 \times 10^5$$

The natural convection Nusselt number can be determined from

$$Nu = \left\{ 0.6 + \frac{0.387 Ra^{1/6}}{\left[ 1 + (0.559/\text{Pr})^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.6 + \frac{0.387(1.060 \times 10^5)^{1/6}}{\left[ 1 + (0.559/0.7255)^{9/16} \right]^{8/27}} \right\}^2 = 7.918$$

$$h_o = \frac{k}{D_o} Nu = \frac{0.02662 \text{ W/m}\cdot^\circ\text{C}}{0.03 \text{ m}} (7.918) = 7.026 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_i = \pi D_i L = \pi(0.025 \text{ m})(15 \text{ m}) = 1.178 \text{ m}^2$$

$$A_o = \pi D_o L = \pi(0.03 \text{ m})(15 \text{ m}) = 1.414 \text{ m}^2$$

Using the assumed value of glass temperature, the radiation heat transfer coefficient is determined to be

$$\begin{aligned} h_{rad} &= \varepsilon \sigma (T_s + T_{surr})(T_s^2 + T_{surr}^2) \\ &= (0.5)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[(65 + 273) + (15 + 273)][(65 + 273)^2 + (15 + 273)^2] \text{ K}^3 \\ &= 3.500 \text{ W/m}^2\cdot\text{K} \end{aligned}$$

Then the combined convection and radiation heat transfer coefficient outside becomes

$$h_{o,combined} = h_o + h_{rad} = 7.026 + 3.500 = 10.526 \text{ W/m}^2\cdot\text{K}$$

and

$$\dot{Q} = \frac{T_{\text{water}} - T_\infty}{\frac{1}{h_i A_i} + \frac{\ln(D_o/D_i)}{4\pi k L} + \frac{1}{h_o A_o}} = \frac{65 - 15}{\frac{1}{(170)(1.178)} + \frac{\ln(3/2.5)}{4\pi(52)(15)} + \frac{1}{(10.526)(1.414)}} = 692.5 \text{ W}$$

The mass flow rate of water

$$\dot{m} = \rho A_c V = (980.4 \text{ kg/m}^3) \left[ \pi(0.025 \text{ m})^2 / 4 \right] (1.2 \text{ m/s}) = 0.5775 \text{ kg/s}$$

Then the temperature drop of water as it flows through the pipe becomes

$$\dot{Q} = \dot{m} c_p \Delta T \rightarrow \Delta T = \frac{\dot{Q}}{\dot{m} c_p} = \frac{692.5 \text{ W}}{(0.5775 \text{ kg/s})(4187 \text{ J/kg}\cdot^\circ\text{C})} = 0.29^\circ\text{C}$$

**Discussion** The outer surface temperature of the pipe can be determined from

$$\dot{Q} = h_o A_o (T_{s,o} - T_\infty) \rightarrow 692.5 \text{ W} = (10.526 \text{ W/m}^2\cdot\text{K})(1.414 \text{ m}^2)(T_{s,o} - 15) \rightarrow T_{s,o} = 61.5^\circ\text{C}$$

which is sufficiently close to the assumed value of  $65^\circ\text{C}$ . Therefore, there is no need to repeat the calculations.

**Discussion** The assumed film temperature of  $T_f = 40^\circ\text{C}$  is an appropriate assumption, since the determined  $T_s = 61.5^\circ\text{C}$  would give a film temperature of  $T_f = 38.3^\circ\text{C}$ . Otherwise,  $T_{s,o}$  would have to be solved iteratively.

**9-144** A small cylindrical resistor mounted on the lower part of a vertical circuit board. The approximate surface temperature of the resistor is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm. 4 Radiation effects are negligible. 5 Heat transfer through the connecting wires is negligible.

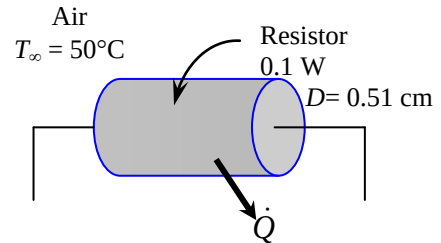
**Properties** Based on the problem statement, the properties of air at 1 atm and the anticipated film temperature of  $(T_s + T_\infty)/2 = (90 + 50)/2 = 70^\circ\text{C}$  are (Table A-15)

$$k = 0.02881 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.995 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7177$$

$$\beta = \frac{1}{T_f} = \frac{1}{(70 + 273)\text{K}} = 0.002915 \text{ K}^{-1}$$



**Analysis** The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown. We start the solution process by “guessing” the surface temperature to be  $90^\circ\text{C}$  for the evaluation of the properties and  $h$ . We will check the accuracy of this guess later and repeat the calculations if necessary. The characteristic length in this case is the diameter of resistor,  $L_c = D = 0.51 \text{ cm}$ . Then,

$$Ra = \frac{g\beta(T_s - T_\infty)D^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.002915 \text{ K}^{-1})(90 - 50 \text{ K})(0.0051 \text{ m})^3}{(1.995 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7177) = 273.6$$

$$Nu = \left\{ 0.6 + \frac{0.387 Ra^{1/6}}{\left[ 1 + (0.559/\text{Pr})^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.6 + \frac{0.387(273.6)^{1/6}}{\left[ 1 + (0.559/0.7177)^{9/16} \right]^{8/27}} \right\}^2 = 2.014$$

$$h = \frac{k}{D} Nu = \frac{0.02881 \text{ W/m}\cdot^\circ\text{C}}{0.0051 \text{ m}} (2.014) = 11.38 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = \pi DL + 2\pi D^2/4 = \pi(0.0051 \text{ m})(0.0076 \text{ m}) + 2\pi(0.0051 \text{ m})^2/4 = 0.0001626 \text{ m}^2$$

and

$$\dot{Q} = hA_s(T_s - T_\infty) \longrightarrow T_s = T_\infty + \frac{\dot{Q}}{hA_s} = 50^\circ\text{C} + \frac{0.1 \text{ W}}{(11.38 \text{ W/m}^2\cdot^\circ\text{C})(0.0001626 \text{ m}^2)} = \mathbf{104.0^\circ\text{C}}$$

which is sufficiently close to the assumed temperature for the evaluation of properties. Therefore, there is no need to repeat calculations.

**Discussion** The assumed film temperature of  $T_f = 90^\circ\text{C}$  is an appropriate assumption, since the determined  $T_s = 104^\circ\text{C}$  would give a film temperature of  $T_f = 77.0^\circ\text{C}$ . Otherwise,  $T_s$  would have to be solved iteratively.

**9-145** An industrial furnace that resembles a horizontal cylindrical enclosure whose end surfaces are well insulated. The highest allowable surface temperature of the furnace and the annual cost of this loss to the plant are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm.

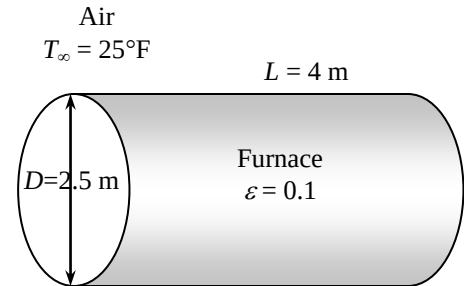
**Properties** The properties of air at 1 atm and the anticipated film temperature of  $(T_s + T_\infty)/2 = (65 + 25)/2 = 45^\circ\text{C}$  are (Table A-15)

$$k = 0.02699 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.750 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7241$$

$$\beta = \frac{1}{T_f} = \frac{1}{(45 + 273)\text{K}} = 0.003145 \text{ K}^{-1}$$



**Analysis** The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown. We start the solution process by “guessing” the surface temperature to be  $65^\circ\text{C}$  for the evaluation of the properties and  $h$ . We will check the accuracy of this guess later and repeat the calculations if necessary. The characteristic length in this case is the outer diameter of the furnace,  $L_c = D = 2.5 \text{ m}$ . Then,

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)D^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003145 \text{ K}^{-1})(65 - 25 \text{ K})(2.5 \text{ m})^3}{(1.750 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7241) = 4.559 \times 10^{10}$$

$$\text{Nu} = \left\{ 0.6 + \frac{0.387 \text{Ra}^{1/6}}{\left[ 1 + (0.559/\text{Pr})^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.6 + \frac{0.387(4.559 \times 10^{10})^{1/6}}{\left[ 1 + (0.559/0.7241)^{9/16} \right]^{8/27}} \right\}^2 = 393.3$$

$$h = \frac{k}{D} \text{Nu} = \frac{0.02699 \text{ W/m}\cdot^\circ\text{C}}{2.5 \text{ m}} (393.3) = 4.246 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = \pi DL = \pi(2.5 \text{ m})(4 \text{ m}) = 31.42 \text{ m}^2$$

The total rate of heat generated in the furnace is

$$\dot{Q}_{\text{gen}} = (0.82)(48 \text{ therms/h})(105,500 \text{ kJ/therm}) = 4.1525 \times 10^6 \text{ kJ/h}$$

Noting that 1% of the heat generated can be dissipated by natural convection and radiation,

$$\dot{Q} = (0.01)(4.1525 \times 10^6 \text{ kJ/h}) = 41,525 \text{ kJ/h}$$

The total rate of heat loss from the furnace by natural convection and radiation can be expressed as

$$\begin{aligned} \dot{Q} &= hA_s(T_s - T_\infty) + \varepsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4) \\ 41,525 \times (1000/3600) \text{ W} &= (4.246 \text{ W/m}^2\cdot^\circ\text{C})(31.42 \text{ m}^2)[T_s - (25 + 273 \text{ K})] \\ &\quad + (0.85)(31.42 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[T_s^4 - (25 + 273 \text{ K})^4] \end{aligned}$$

Its solution by the following EES equation is

$$41525 \times 1000/3600 = 4.246 \times 31.42 \times (T - 298) + 0.85 \times 31.42 \times 5.67 \times 10^{-8} \times (T^4 - 298^4)$$

$$T_s = 333.5 \text{ K} = \mathbf{60.5^\circ\text{C}}$$

which is close to the assumed value. Therefore, there is no need to repeat calculations. The total amount of heat loss and its cost during a 2800 hour period is

$$Q_{\text{total}} = \dot{Q}_{\text{total}} \Delta t = (41,525 \text{ kJ/h})(2800 \text{ h}) = 1.1627 \times 10^8 \text{ kJ}$$

$$\text{Cost} = (1.1627 \times 10^8 / 105,500 \text{ therm})(\$1.15 / \text{therm}) = \mathbf{\$1267}$$

**9-146** A spherical tank made of stainless steel is used to store iced water. The rate of heat transfer to the iced water and the amount of ice that melts during a 24-h period are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 Thermal resistance of the tank is negligible. 4 The local atmospheric pressure is 1 atm.

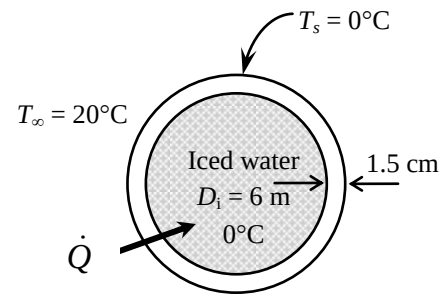
**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (0 + 20)/2 = 10^\circ\text{C}$  are (Table A-15)

$$k = 0.02439 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.426 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7336$$

$$\beta = \frac{1}{T_f} = \frac{1}{(10 + 273)\text{K}} = 0.003534 \text{ K}^{-1}$$



**Analysis** (a) The characteristic length in this case is  $L_c = D_o = 6.03 \text{ m}$ . Then,

$$\text{Ra} = \frac{g\beta(T_\infty - T_s)D_o^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003534 \text{ K}^{-1})(20 - 0 \text{ K})(6.03 \text{ m})^3}{(1.426 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7336) = 5.485 \times 10^{11}$$

$$\text{Nu} = 2 + \frac{0.589 \text{Ra}^{1/4}}{\left[1 + (0.469/\text{Pr})^{9/16}\right]^{4/9}} = 2 + \frac{0.589(5.485 \times 10^{11})^{1/4}}{\left[1 + (0.469/0.7336)^{9/16}\right]^{4/9}} = 394.5$$

$$h = \frac{k}{D_o} \text{Nu} = \frac{0.02439 \text{ W/m}\cdot^\circ\text{C}}{6.03 \text{ m}} (394.5) = 1.596 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = \pi D_o^2 = \pi (6.03 \text{ m})^2 = 114.2 \text{ m}^2$$

and

$$\dot{Q} = hA_s(T_\infty - T_s) = (1.596 \text{ W/m}^2\cdot^\circ\text{C})(114.2 \text{ m}^2)(20 - 0)^\circ\text{C} = 3646 \text{ W}$$

Heat transfer by radiation and the total rate of heat transfer are

$$\begin{aligned} \dot{Q}_{\text{rad}} &= \varepsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4) \\ &= (1)(114.2 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[(20 + 273 \text{ K})^4 - (0 + 273 \text{ K})^4] = 11,759 \text{ W} \end{aligned}$$

$$\dot{Q}_{\text{total}} = 3646 + 11,759 = 15,404 \text{ W} \approx \mathbf{15.4 \text{ kW}}$$

(b) The total amount of heat transfer during a 24-hour period is

$$Q = \dot{Q}\Delta t = (15.4 \text{ kJ/s})(24 \text{ h/day} \times 3600 \text{ s/h}) = 1.331 \times 10^6 \text{ kJ/day}$$

Then the amount of ice that melts during this period becomes

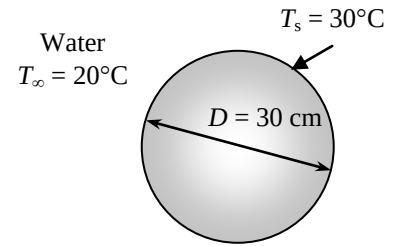
$$Q = mh_{if} \longrightarrow m = \frac{Q}{h_{if}} = \frac{1.331 \times 10^6 \text{ kJ}}{333.7 \text{ kJ/kg}} = \mathbf{3988 \text{ kg}}$$

**9-147** A spherical vessel is completely submerged in a large water-filled tank. The rates of heat transfer from the vessel by natural convection, conduction, and forced convection are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 The surface temperature is constant.

**Properties** The properties of water at the film temperature of  $(T_s + T_\infty)/2 = (30 + 20)/2 = 25^\circ\text{C}$  are (Table A-9)

$$\begin{aligned}\rho &= 997 \text{ kg/m}^3 & k &= 0.607 \text{ W/m}\cdot^\circ\text{C} \\ \mu &= 0.891 \times 10^{-3} \text{ kg/m}\cdot\text{s} & \nu &= \mu / \rho = 8.937 \times 10^{-7} \text{ m}^2/\text{s} \\ \text{Pr} &= 6.14 & \beta &= 0.247 \times 10^{-3} \text{ K}^{-1}\end{aligned}$$



**Analysis** (a) Heat transfer in this case will be by natural convection.

The characteristic length in this case is  $L_c = D = 0.3 \text{ m}$ . Then,

$$Ra = \frac{g\beta(T_s - T_\infty)D^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.247 \times 10^{-3} \text{ K}^{-1})(30 - 20 \text{ K})(0.3 \text{ m})^3}{(8.937 \times 10^{-7} \text{ m}^2/\text{s})^2} (6.14) = 5.029 \times 10^9$$

$$Nu = 2 + \frac{0.589 Ra^{1/4}}{\left[1 + (0.469 / \text{Pr})^{9/16}\right]^{4/9}} = 2 + \frac{0.589(5.029 \times 10^9)^{1/4}}{\left[1 + (0.469 / 6.14)^{9/16}\right]^{4/9}} = 144.8$$

Then

$$h = \frac{k}{D} Nu = \frac{0.607 \text{ W/m}\cdot^\circ\text{C}}{0.3 \text{ m}} (144.8) = 293.0 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = \pi D^2 = \pi (0.3 \text{ m})^2 = 0.2827 \text{ m}^2$$

The rate of heat transfer is

$$\dot{Q}_{\text{nat, conv}} = hA(T_s - T_\infty) = (293.0 \text{ W/m}^2\cdot^\circ\text{C})(0.2827 \text{ m}^2)(30 - 20)^\circ\text{C} = \mathbf{828 \text{ W}}$$

(b) When buoyancy force is neglected, there will be no convection currents (since  $\beta = 0$ ) and the heat transfer will be by conduction. Then Rayleigh number becomes zero ( $Ra = 0$ ). The Nusselt number in this case is

$$Nu = 2$$

Then

$$h = \frac{k}{D} Nu = \frac{0.607 \text{ W/m}\cdot^\circ\text{C}}{0.3 \text{ m}} (2) = 4.047 \text{ W/m}^2\cdot^\circ\text{C}$$

$$\dot{Q}_{\text{cond}} = hA(T_s - T_\infty) = (4.047 \text{ W/m}^2\cdot^\circ\text{C})(0.2827 \text{ m}^2)(30 - 20)^\circ\text{C} = \mathbf{11.4 \text{ W}}$$

(c) In this case, the heat transfer from the vessel is by forced convection. The properties of water at the free stream temperature of  $20^\circ\text{C}$  are (Table A-9)

$$\begin{aligned}\rho &= 998 \text{ kg/m}^3 & k &= 0.598 \text{ W/m}\cdot^\circ\text{C} \\ \mu_\infty &= 1.002 \times 10^{-3} \text{ kg/m}\cdot\text{s} & \nu &= \mu_\infty / \rho = 1.004 \times 10^{-6} \text{ m}^2/\text{s} \\ \mu_{s, @ 30^\circ\text{C}} &= 0.798 \times 10^{-3} \text{ kg/m}\cdot\text{s} & \text{Pr} &= 7.01\end{aligned}$$

The Reynolds number is

$$\text{Re} = \frac{VD}{\nu} = \frac{(0.2 \text{ m/s})(0.3 \text{ m})}{1.004 \times 10^{-6} \text{ m}^2/\text{s}} = 59,760$$

The Nusselt number is

$$\begin{aligned}Nu &= \frac{hD}{k} = 2 + \left[0.4 \text{Re}^{0.5} + 0.06 \text{Re}^{2/3}\right] \text{Pr}^{0.4} \left(\frac{\mu_\infty}{\mu_s}\right)^{1/4} \\ &= 2 + \left[0.4(59,760)^{0.5} + 0.06(59,760)^{2/3}\right] (7.01)^{0.4} \left(\frac{1.002 \times 10^{-3}}{0.798 \times 10^{-3}}\right)^{1/4} = 439.1\end{aligned}$$

The heat transfer coefficient is

$$h = \frac{k}{D} Nu = \frac{0.598 \text{ W/m}\cdot^\circ\text{C}}{0.3 \text{ m}} (439.1) = 875.3 \text{ W/m}^2\cdot^\circ\text{C}$$

The rate of heat transfer is

$$\dot{Q}_{\text{forced conv}} = hA(T_s - T_\infty) = (875.3 \text{ W/m}^2\cdot^\circ\text{C})(0.2827 \text{ m}^2)(30 - 20)^\circ\text{C} = \mathbf{2474 \text{ W}}$$

**9-148** A double-pane window consisting of two layers of glass separated by an air space is considered. The rate of heat transfer through the window and the temperature of its inner surface are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 Radiation effects are negligible. 4 The pressure of air inside the enclosure is 1 atm.

**Properties** We expect the average temperature of the air gap to be roughly the average of the indoor and outdoor temperatures, and evaluate the properties of air at 1 atm and the average temperature of  $(T_{\infty 1} + T_{\infty 2})/2 = (20 + 0)/2 = 10^\circ\text{C}$  are (Table A-15)

$$k = 0.02439 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.426 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7336$$

$$\beta = \frac{1}{T_f} = \frac{1}{(10 + 273)\text{K}} = 0.003534 \text{ K}^{-1}$$

**Analysis** We “guess” the temperature difference across the air gap to be  $15^\circ\text{C} = 15 \text{ K}$  for use in the Ra relation. The characteristic length in this case is the air gap thickness,  $L_c = L = 0.025 \text{ m}$ . Then,

$$\text{Ra} = \frac{g\beta(T_1 - T_2)L^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003534 \text{ K}^{-1})(15 \text{ K})(0.025 \text{ m})^3}{(1.426 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7336) = 2.931 \times 10^4$$

Then the Nusselt number and the heat transfer coefficient are determined to be

$$\text{Nu} = 0.42 \text{Ra}^{1/4} \text{Pr}^{0.012} \left(\frac{H}{L}\right)^{-0.3} = 0.42(2.931 \times 10^4)^{1/4} (0.7336)^{0.012} \left(\frac{1.2 \text{ m}}{0.025 \text{ m}}\right)^{-0.3} = 1.714$$

$$h_{\text{air}} = \frac{k}{L} \text{Nu} = \frac{0.02439 \text{ W/m}\cdot^\circ\text{C}}{0.025 \text{ m}} (1.714) = 1.672 \text{ W/m}^2\cdot^\circ\text{C}$$

Then the rate of heat transfer through this double pane window is determined to be

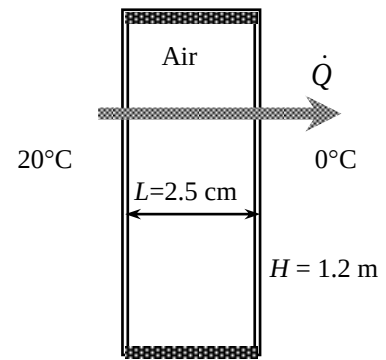
$$A_s = H \times W = (1.2 \text{ m})(2 \text{ m}) = 2.4 \text{ m}^2$$

$$\begin{aligned} \dot{Q} &= \frac{T_{\infty, i} - T_{\infty, o}}{R_{\text{conv}, i} + R_{\text{cond}, \text{glasses}} + R_{\text{conv}, \text{air}} + R_{\text{conv}, o}} = \frac{T_{\infty} - T_{s, i}}{\frac{1}{h_i A_s} + \frac{2t_{\text{glass}}}{k_{\text{glass}} A_s} + \frac{1}{h_{\text{air}} A_s} + \frac{1}{h_o A_s}} \\ &= \frac{20 - 0}{\frac{1}{(10)(2.4)} + \frac{2(0.003)}{(0.78)(2.4)} + \frac{1}{(1.672)(2.4)} + \frac{1}{(25)(2.4)}} = \mathbf{64.4 \text{ W}} \end{aligned}$$

**Check:** The temperature drop across the air gap is determined from

$$\dot{Q} = hA_s \Delta T \rightarrow \Delta T = \frac{\dot{Q}}{hA_s} = \frac{64.4 \text{ W}}{(1.672 \text{ W/m}^2\cdot^\circ\text{C})(2.4 \text{ m}^2)} = 16.0^\circ\text{C}$$

which is very close to the assumed value of  $15^\circ\text{C}$  used in the evaluation of the Ra number.



**9-149** A solar collector consists of a horizontal copper tube enclosed in a concentric thin glass tube. Water is heated in the tube, and the annular space between the copper and glass tube is filled with air. The rate of heat loss from the collector by natural convection is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 Radiation effects are negligible. 3 The pressure of air in the enclosure is 1 atm.

**Properties** The properties of air at 1 atm and the average temperature of  $(T_i + T_o)/2 = (60 + 32)/2 = 46^\circ\text{C}$  are (Table A-15)

$$k = 0.02706 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.760 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7238$$

$$\beta = \frac{1}{T_f} = \frac{1}{(46 + 273)\text{K}} = 0.003135 \text{ K}^{-1}$$

**Analysis** The characteristic length in this case is the distance between the two cylinders

$$L_c = \frac{D_o - D_i}{2} = \frac{(9 - 5) \text{ cm}}{2} = 2 \text{ cm}$$

and

$$\text{Ra} = \frac{g\beta(T_i - T_o)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003135 \text{ K}^{-1})(60 - 32 \text{ K})(0.02 \text{ m})^3}{(1.760 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7238) = 16,100$$

The effective thermal conductivity is

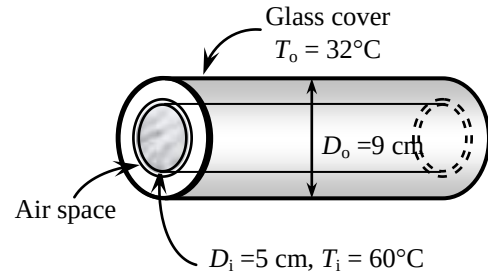
$$F_{\text{cyl}} = \frac{\left[ \ln \frac{D_o}{D_i} \right]^4}{L_c^3 (D_i^{-3/5} + D_o^{-3/5})^5} = \frac{\left[ \ln \frac{0.09 \text{ m}}{0.05 \text{ m}} \right]^4}{(0.02 \text{ m})^3 [(0.05 \text{ m})^{-7/5} + (0.09 \text{ m})^{-7/5}]^5} = 0.1303$$

$$k_{\text{eff}} = 0.386k \left( \frac{\text{Pr}}{0.861 + \text{Pr}} \right)^{1/4} (F_{\text{cyl}} \text{Ra})^{1/4}$$

$$= 0.386(0.02706 \text{ W/m}\cdot^\circ\text{C}) \left( \frac{0.7238}{0.861 + 0.7238} \right)^{1/4} [(0.1303)(16,100)]^{1/4} = 0.05811 \text{ W/m}\cdot^\circ\text{C}$$

Then the heat loss from the collector per meter length of the tube becomes

$$\dot{Q} = \frac{2\pi k_{\text{eff}}}{\ln \left( \frac{D_o}{D_i} \right)} (T_i - T_o) = \frac{2\pi(0.05811 \text{ W/m}\cdot^\circ\text{C})}{\ln \left( \frac{0.09 \text{ m}}{0.05 \text{ m}} \right)} (60 - 32)^\circ\text{C} = \mathbf{17.4 \text{ W}}$$



**9-150** A solar collector consists of a horizontal tube enclosed in a concentric thin glass tube is considered. The pump circulating the water fails. The temperature of the aluminum tube when equilibrium is established is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The local atmospheric pressure is 1 atm.

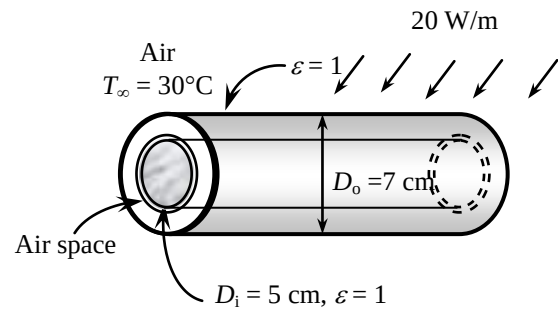
**Properties** We assume a surface temperature of 33°C for glass cover based on the problem statement. Then the properties of air at 1 atm and the anticipated film temperature of  $(T_s + T_\infty)/2 = (33 + 30)/2 = 31.5^\circ\text{C}$  are (Table A-15)

$$k = 0.02599 \text{ W/m} \cdot ^\circ\text{C}$$

$$\nu = 1.622 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7278$$

$$\beta = \frac{1}{T_f} = \frac{1}{(31.5 + 273)\text{K}} = 0.003284 \text{ K}^{-1}$$



**Analysis** This problem involves heat transfer from the aluminum tube to the glass cover, and from the outer surface of the glass cover to the surrounding ambient air. When steady operation is reached, these two heat transfers will be equal to the rate of heat gain. That is,

$$\dot{Q}_{\text{tube-glass}} = \dot{Q}_{\text{glass-ambient}} = \dot{Q}_{\text{solar gain}} = 20 \text{ W (per meter length)}$$

Now we assume the surface temperature of the glass cover to be 33°C. We will check this assumption later on, and repeat calculations with a better assumption, if necessary.

The characteristic length for the outer diameter of the glass cover  $L_c = D_o = 0.07 \text{ m}$ . Then,

$$Ra = \frac{g\beta(T_s - T_\infty)D_o^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003284 \text{ K}^{-1})(33 - 30 \text{ K})(0.07 \text{ m})^3}{(1.622 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7278) = 91,700$$

$$Nu = \left\{ 0.6 + \frac{0.387 Ra^{1/6}}{\left[ 1 + (0.559/\text{Pr})^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.6 + \frac{0.387(91,700)^{1/6}}{\left[ 1 + (0.559/0.7278)^{9/16} \right]^{8/27}} \right\}^2 = 7.626$$

$$A_s = \pi D_o L = \pi(0.07 \text{ m})(1 \text{ m}) = 0.2199 \text{ m}^2$$

$$h = \frac{k}{D_o} Nu = \frac{0.02599 \text{ W/m} \cdot ^\circ\text{C}}{0.07 \text{ m}} (7.626) = 2.832 \text{ W/m}^2 \cdot ^\circ\text{C}$$

and,

$$\dot{Q}_{\text{conv}} = hA_s(T_s - T_\infty) = (2.832 \text{ W/m}^2 \cdot ^\circ\text{C})(0.2199 \text{ m}^2)(T_{\text{glass}} - 30)^\circ\text{C}$$

The radiation heat loss is

$$\dot{Q}_{\text{rad}} = \epsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4) = (1)(0.2199 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4) [(T_{\text{glass}} + 273 \text{ K})^4 - (20 + 273 \text{ K})^4]$$

The expression for the total rate of heat transfer is

$$\begin{aligned} \dot{Q}_{\text{total}} &= \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}} \\ 20 \text{ W} &= (2.832 \text{ W/m}^2 \cdot ^\circ\text{C})(0.2199 \text{ m}^2)(T_{\text{glass}} - 30)^\circ\text{C} \\ &\quad + (1)(0.2199 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4) [(T_{\text{glass}} + 273 \text{ K})^4 - (20 + 273 \text{ K})^4] \end{aligned}$$

Its solution is

$$T_{\text{glass}} = 33.34^\circ\text{C}$$

which is sufficiently close to the assumed value of 33°C. Therefore, there is no need to repeat the calculations.

Now we will calculate heat transfer through the air layer between aluminum tube and glass cover. We will assume the aluminum tube temperature to be 45°C based on the problem statement and evaluate properties at the average temperature of  $(T_i + T_o)/2 = (45 + 33.34)/2 = 39.17^\circ\text{C}$  are (Table A-15)

$$k = 0.02656 \text{ W/m} \cdot ^\circ\text{C}$$

$$\nu = 1.694 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7257$$

$$\beta = \frac{1}{T_f} = \frac{1}{(39.17 + 273) \text{ K}} = 0.003203 \text{ K}^{-1}$$

The characteristic length in this case is the distance between the two cylinders,

$$L_c = (D_o - D_i) / 2 = (7 - 5) / 2 \text{ cm} = 1 \text{ cm}$$

Then,

$$\text{Ra} = \frac{g\beta(T_1 - T_2)L_c^3}{\nu^2} \text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003203 \text{ K}^{-1})(45 - 33.34 \text{ K})(0.01 \text{ m})^3}{(1.694 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7257) = 926.5$$

The effective thermal conductivity is

$$F_{\text{cyl}} = \frac{\left[ \ln \frac{D_o}{D_i} \right]^4}{L_c^3 (D_i^{-3/5} + D_o^{-3/5})^5} = \frac{\left[ \ln \frac{0.07 \text{ m}}{0.05 \text{ m}} \right]^4}{(0.01 \text{ m})^3 [(0.05 \text{ m})^{-3/5} + (0.07 \text{ m})^{-3/5}]^5} = 0.08085$$

$$k_{\text{eff}} = 0.386k \left( \frac{\text{Pr}}{0.861 + \text{Pr}} \right)^{1/4} (F_{\text{cyl}} \text{Ra})^{1/4}$$

$$= 0.386(0.02656 \text{ W/m} \cdot ^\circ\text{C}) \left( \frac{0.7257}{0.861 + 0.7257} \right)^{1/4} [(0.08085)(926.5)]^{1/4} = 0.02480 \text{ W/m} \cdot ^\circ\text{C}$$

The heat transfer expression is

$$\dot{Q} = \frac{2\pi k_{\text{eff}}}{\ln \left( \frac{D_o}{D_i} \right)} (T_1 - T_2) = \frac{2\pi(0.02480 \text{ W/m} \cdot ^\circ\text{C})}{\ln \left( \frac{0.07 \text{ m}}{0.05 \text{ m}} \right)} (T_{\text{tube}} - 33.34)^\circ\text{C}$$

The radiation heat loss is

$$A_s = \pi D_i L = \pi (0.05 \text{ m})^2 (1 \text{ m}) = 0.1571 \text{ m}^2$$

$$\dot{Q}_{\text{rad}} = \varepsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4)$$

$$= (1)(0.1571 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4) [(T_{\text{tube}} + 273 \text{ K})^4 - (33.34 + 273 \text{ K})^4]$$

The expression for the total rate of heat transfer is

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}}$$

$$20 \text{ W} = \frac{2\pi(0.02480 \text{ W/m} \cdot ^\circ\text{C})}{\ln \left( \frac{0.07 \text{ m}}{0.05 \text{ m}} \right)} (T_{\text{tube}} - 33.34)^\circ\text{C}$$

$$+ (1)(0.1571 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4) [(T_{\text{tube}} + 273 \text{ K})^4 - (33.34 + 273 \text{ K})^4]$$

Its solution is

$$T_{\text{tube}} = \mathbf{46.3^\circ\text{C}},$$

which is sufficiently close to the assumed value of  $45^\circ\text{C}$ . Therefore, there is no need to repeat the calculations.

**9-151** Two surfaces of a spherical enclosure are maintained at specified temperatures. Both inner and outer surfaces are black, and the rate of heat transfer on the inner surface is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Nitrogen is an ideal gas with constant properties. 3 The inner and outer surfaces are black.

**Properties** The properties of nitrogen at  $T_f = (T_s + T_\infty)/2 = 150^\circ\text{C}$  are  $k = 0.03416 \text{ W/m}\cdot\text{K}$ ,  $\nu = 2.851 \times 10^{-5} \text{ m}^2/\text{s}$ ,  $\text{Pr} = 0.7025$  (from Table A-16). Also,  $\beta = 1/T_f = 0.002364 \text{ K}^{-1}$ .

**Analysis** The characteristic length in this case is determined from

$$L_c = \frac{D_o - D_i}{2} = \frac{10 - 5}{2} \text{ cm} = 2.5 \text{ cm}$$

The Rayleigh number is

$$\begin{aligned} \text{Ra}_L &= \frac{g\beta(T_i - T_o)L_c^3}{\nu^2} \text{Pr} \\ &= \frac{(9.81 \text{ m/s}^2)(2.364 \times 10^{-3} \text{ K}^{-1})(200 - 100) \text{ K}(0.025 \text{ m})^3}{(2.851 \times 10^{-5} \text{ m}^2/\text{s})^2} (0.7025) \\ &= 3.132 \times 10^4 \end{aligned}$$

The effective thermal conductivity is

$$\begin{aligned} F_{\text{sph}} &= \frac{L_c}{(D_i D_o)^4 (D_i^{-7/5} + D_o^{-7/5})^5} = \frac{(0.025 \text{ m})}{[(0.1 \text{ m})(0.05 \text{ m})]^4 [(0.05 \text{ m})^{-7/5} + (0.1 \text{ m})^{-7/5}]^5} \\ &= 0.006268 \\ k_{\text{eff}} &= 0.74 \left( \frac{\text{Pr}}{0.861 + \text{Pr}} \right)^{1/4} (F_{\text{sph}} \text{Ra}_L)^{1/4} k \\ &= 0.74 \left( \frac{0.7025}{0.861 + 0.7025} \right)^{1/4} [(0.006268)(3.132 \times 10^4)]^{1/4} (0.03416 \text{ W/m}\cdot\text{K}) \\ &= 0.07747 \text{ W/m}\cdot\text{K} \end{aligned}$$

Then the rate of heat transfer by natural convection is

$$\dot{Q}_{\text{conv}} = k_{\text{eff}} \frac{\pi D_i D_o}{L_c} (T_i - T_o) = (0.07747 \text{ W/m}\cdot\text{K}) \frac{\pi(0.1 \text{ m})(0.05 \text{ m})}{(0.025 \text{ m})} (200 - 100) \text{ K} = 4.87 \text{ W}$$

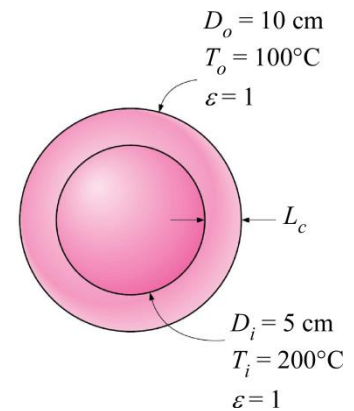
The rate of heat transfer by radiation on the inner surface is

$$\begin{aligned} \dot{Q}_{\text{rad}} &= \varepsilon \sigma A_{s,i} (T_i^4 - T_o^4) = \dot{Q}_{\text{rad}} = \sigma \pi D_i^2 (T_i - T_o) \\ &= (5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4) \pi (0.05 \text{ m})^2 (473^4 - 373^4) \text{ K}^4 = 13.67 \text{ W} \end{aligned}$$

Hence, the total rate of heat transfer on the inner surface is

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}} = 4.87 \text{ W} + 13.67 \text{ W} = \mathbf{18.5 \text{ W}}$$

**Discussion** In this problem, radiation heat transfer plays an important role, since the heat transfer rate by radiation is almost 3 times of that by natural convection.



## Fundamentals of Engineering (FE) Exam Problems

**9-152** Consider a hot boiled egg in a spacecraft that is filled with air at atmospheric pressure and temperature at all times. Disregarding any radiation effect, will the egg cool faster or slower when the spacecraft is in space instead of on the ground?

- (a) faster                      (b) no difference                      (c) slower                      (d) insufficient information

*Answer* (c) slower [there is no gravity and thus no natural convection currents in space]

**9-153** A hot object suspended by a string is to be cooled by natural convection in fluids whose volume changes differently with temperature at constant pressure. In which fluid will the rate of cooling be lowest?

With increasing temperature, a fluid whose volume

- (a) increases a lot    (b) increases slightly    (c) does not change    (d) decreases slightly    (e) decreases a lot

*Answer* (c) A fluid whose volume **does not change** [since there will be no natural convection currents in this case]

**9-154** A spherical block of dry ice at  $-79^{\circ}\text{C}$  is exposed to atmospheric air at  $30^{\circ}\text{C}$ . The general direction in which the air moves in this situation is

- (a) horizontal                      (b) up                      (c) down  
(d) recirculation around the sphere                      (e) no motion

*Answer* (c) down

**9-155** The primary driving force for natural convection is

- (a) shear stress forces                      (b) buoyancy forces                      (c) pressure forces  
(d) surface tension forces                      (e) None of them

*Answer* (b) buoyancy forces

**9-156** Consider a horizontal 0.7-m-wide and 0.85-m-long plate in a room at 30°C. Top side of the plate is insulated while the bottom side is maintained at 0°C. The rate of heat transfer from the room air to the plate by natural convection is

- (a) 36.8 W                      (b) 43.7 W                      (c) 128.5 W                      (d) 92.7 W                      (e) 69.7 W

(For air, use  $k = 0.02476 \text{ W/m}\cdot^\circ\text{C}$ ,  $\text{Pr} = 0.7323$ ,  $\nu = 1.470 \times 10^{-5} \text{ m}^2/\text{s}$ )

*Answer* (b) 43.7 W

**Solution** Solved by EES Software. Solutions can be verified by copying-and-pasting the following lines on a blank EES screen.

Width=0.7 [m]

Height=0.85 [m]

T\_infinity=30 [C]

T\_s=0 [C]

"The properties of air at  $(0+30)/2 = 15 \text{ C}$  are (Table A-15)"

$k=0.02476 \text{ [W/m}\cdot\text{C]}$

$\nu=1.470\text{E-}5 \text{ [m}^2/\text{s]}$

$\text{Pr}=0.7323$

$\beta=1/T_f$

$T_f=(T_s+T_{\text{infinity}})/2+273$

$g=9.81 \text{ [m/s}^2\text{]}$

$L=(\text{Width}\cdot\text{Height})/(2\cdot(\text{Width}+\text{Height}))$

$\text{Ra}=(g\cdot\beta\cdot(T_{\text{infinity}}-T_s)\cdot L^3)/\nu^2\cdot\text{Pr}$

$\text{Nus}=0.27\cdot\text{Ra}^{0.25}$

$h=k/L\cdot\text{Nus}$

$A_s=\text{Width}\cdot\text{Height}$

$\dot{Q}=h\cdot A_s\cdot(T_{\text{infinity}}-T_s)$

**9-157** A 4-m-long section of a 5-cm-diameter horizontal pipe in which a refrigerant flows passes through a room at 20°C. The pipe is not well insulated and the outer surface temperature of the pipe is observed to be -10°C. The emissivity of the pipe surface is 0.85 and the surrounding surfaces are at 15°C. The fraction of heat transferred to the pipe by radiation is

- (a) 0.24                      (b) 0.30                      (c) 0.37                      (d) 0.48                      (e) 0.58

(For air, use  $k = 0.02401 \text{ W/m}\cdot\text{°C}$ ,  $Pr = 0.735$ ,  $\nu = 1.382 \times 10^{-5} \text{ m}^2/\text{s}$ )

*Answer* (c) 0.37

**Solution** Solved by EES Software. Solutions can be verified by copying-and-pasting the following lines on a blank EES screen.

```
L=4 [m]
D=0.05 [m]
T_infinity=20 [C]
T_s=-10 [C]
T_surr=15 [C]
epsilon=0.85
```

"The properties of air at  $(20-10)/2 = 5 \text{ C}$  are (Table A-15)"

```
k=0.02401 [W/m-C]
nu=1.382E-5 [m^2/s]
Pr=0.7350
beta=1/T_f
T_f=(T_s+T_infinity)/2+273
g=9.81 [m/s^2]
sigma=5.67E-8 [W/m^2-K^4]
```

```
Ra=(g*beta*(T_infinity-T_s)*D^3)/nu^2*Pr
Nus=((0.6+(0.387*Ra^(1/6)))/(1+(0.559/Pr)^(9/16)))^(8/27))^2
h=k/D*Nus
A_s=pi*D*L
Q_dot_conv=h*A_s*(T_infinity-T_s)
Q_dot_rad=epsilon*A_s*sigma*((T_surr+273)^4-(T_s+273)^4)
f_rad=Q_dot_rad/(Q_dot_conv+Q_dot_rad)
```

"Some Wrong Solutions with Common Mistakes"

```
W_f_rad=Q_dot_rad/Q_dot_conv "Finding the ratio of radiation to convection"
```

**9-158** Consider a 0.2-m-diameter, 1.8-m-long horizontal cylinder in a room at 20°C. If the outer surface temperature of the cylinder is 40°C, the natural convection heat transfer coefficient is

- (a) 2.9 W/m<sup>2</sup>·°C      (b) 3.5 W/m<sup>2</sup>·°C      (c) 4.1 W/m<sup>2</sup>·°C      (d) 5.2 W/m<sup>2</sup>·°C      (e) 6.1 W/m<sup>2</sup>·°C  
 (For air, use  $k = 0.02588$  W/m·°C,  $Pr = 0.7282$ ,  $\nu = 1.608 \times 10^{-5}$  m<sup>2</sup>/s)

*Answer* (c) 4.1 W/m<sup>2</sup>·°C

**Solution** Solved by EES Software. Solutions can be verified by copying-and-pasting the following lines on a blank EES screen.

D=0.2 [m]

L=1.8 [m]

T\_infinity=20 [C]

T\_s=40 [C]

"The properties of air at (20+40)/2 = 30 C are (Table A-15)"

k=0.02588 [W/m-C]

nu=1.608E-5 [m^2/s]

Pr=0.7282

beta=1/T\_f

T\_f=(T\_s+T\_infinity)/2+273

g=9.81 [m/s^2]

Ra=(g\*beta\*(T\_s-T\_infinity)\*D^3)/nu^2\*Pr

Nus=((0.6+(0.387\*Ra^(1/6))/(1+(0.559/Pr)^(9/16)))^(8/27))^2

h=k/D\*Nus

**9-159** A 4-m-diameter spherical tank contains iced water at 0°C. The tank is thin-shelled and thus its outer surface temperature may be assumed to be same as the temperature of the iced water inside. Now the tank is placed in a large lake at 20°C. The rate at which the ice melts is

- (a) 0.42 kg/s      (b) 0.58 kg/s      (c) 0.70 kg/s      (d) 0.83 kg/s      (e) 0.98 kg/s

(For lake water, use  $k = 0.580$  W/m·°C,  $Pr = 9.45$ ,  $\nu = 0.1307 \times 10^{-5}$  m<sup>2</sup>/s,  $\beta = 0.138 \times 10^{-3}$  K<sup>-1</sup>)

*Answer* (a) 0.42 kg/s

**Solution** Solved by EES Software. Solutions can be verified by copying-and-pasting the following lines on a blank EES screen.

D=4 [m]

T\_infinity=20 [C]

T\_s=0 [C]

"The properties of water at (20+0)/2 = 10 C are (Table A-9)"

k=0.580 [W/m-C]

nu=0.1307E-5 [m^2/s]

Pr=9.45

beta=0.138E-3 [1/K]

g=9.81 [m/s^2]

Ra=(g\*beta\*(T\_infinity-T\_s)\*D^3)/nu^2\*Pr

Nus=2+(0.589\*Ra^(1/4))/(1+(0.469/Pr)^(9/16))^(4/9)

h=k/D\*Nus

A\_s=pi\*D^2

Q\_dot=h\*A\_s\*(T\_infinity-T\_s)

h\_if=333700 [J/kg]

m\_dot\_melt=Q\_dot/h\_if

**9-160** A vertical double-pane window consists of two sheets of glass separated by a 1.2-cm air gap at atmospheric pressure. The glass surface temperatures across the air gap are measured to be 278 K and 288 K. If it is estimated that the heat transfer by convection through the enclosure is 1.5 times that by pure conduction and that the rate of heat transfer by radiation through the enclosure is about the same magnitude as the convection, the effective emissivity of the two glass surfaces is

(a) 0.35                      (b) 0.48                      (c) 0.59                      (d) 0.84                      (e) 0.72

*Answer* (c) 0.59

**Solution** Solved by EES Software. Solutions can be verified by copying-and-pasting the following lines on a blank EES screen.

```
L=0.012 [m]
T1=288 [C]
T2=278 [C]
Nus=1.5
```

"The properties of air at  $(278+288)/2 = 283$  K =10 C are (Table A-15)"

```
k=0.02439 [W/m-K]
sigma=5.67E-8 [W/m^2-K^4]
```

```
q_dot_conv=k*Nus*(T1-T2)/L
q_dot_rad=q_dot_conv
q_dot_rad=epsilon_eff*sigma*(T1^4-T2^4)
```

**9-161** A horizontal 1.5-m-wide, 4.5-m-long double-pane window consists of two sheets of glass separated by a 3.5-cm gap filled with water. If the glass surface temperatures at the bottom and the top are measured to be 60°C and 40°C, respectively, the rate of heat transfer through the window is

- (a) 27.6 kW      (b) 39.4 kW      (c) 59.6 kW      (d) 66.4 kW      (e) 75.5 kW

(For water, use  $k = 0.644 \text{ W/m}\cdot^\circ\text{C}$ ,  $\text{Pr} = 3.55$ ,  $\nu = 0.554 \times 10^{-6} \text{ m}^2/\text{s}$ ,  $\beta = 0.451 \times 10^{-3} \text{ K}^{-1}$ . Also, the applicable correlation is  $\text{Nu} = 0.069\text{Ra}^{1/3} \text{Pr}^{0.074}$ ).

*Answer* (d) 66.4 kW

**Solution** Solved by EES Software. Solutions can be verified by copying-and-pasting the following lines on a blank EES screen.

```
Width=1.5 [m]
Length=4.5 [m]
L=0.035 [m]
T1=60 [C]
T2=40 [C]
```

"The properties of water at  $(60+40)/2 = 50 \text{ C}$  are (Table A-9)"

```
k=0.644 [W/m-C]
nu=0.554E-6 [m^2/s]
Pr=3.55
beta=0.451E-3 [1/K]
g=9.81 [m/s^2]
```

```
Ra=(g*beta*(T1-T2)*L^3)/nu^2*Pr
Nus=0.069*Ra^(1/3)*Pr^(0.074)
A_s=Width*Length
Q_dot=k*Nus*A_s*(T1-T2)/L
```

"Some Wrong Solutions with Common Mistakes"

```
W1_Nus=0.068*Ra^(1/3) "Relation for air gap, Eq. 9-45"
W1_Q_dot=k*W1_Nus*A_s*(T1-T2)/L
W2_Nus=0.195*Ra^(1/4) "Relation for air gap, Eq. 9-44"
W2_Q_dot=k*W2_Nus*A_s*(T1-T2)/L
```

**9-162** A vertical 0.9-m-high and 1.5-m-wide double-pane window consists of two sheets of glass separated by a 2.0-cm air gap at atmospheric pressure. If the glass surface temperatures across the air gap are measured to be 20°C and 30°C, the rate of heat transfer through the window is

- (a) 16.3 W                      (b) 21.7 W                      (c) 24.0 W                      (d) 31.3 W                      (e) 44.6 W

(For air, use  $k = 0.02551 \text{ W/m}\cdot^\circ\text{C}$ ,  $\text{Pr} = 0.7296$ ,  $\nu = 1.562 \times 10^{-5} \text{ m}^2/\text{s}$ . Also, the applicable correlation is  $\text{Nu} = 0.42\text{Ra}^{1/4} \text{Pr}^{0.012} (H/L)^{-0.3}$ )

*Answer* (b) 21.7 W

**Solution** Solved by EES Software. Solutions can be verified by copying-and-pasting the following lines on a blank EES screen.

```
H=0.9 [m]
W=1.5 [m]
L=0.020 [m]
T1=30 [C]
T2=20 [C]
```

"The properties of air at  $(20+30)/2 = 25 \text{ C}$  are (Table A-15)"

```
k=0.02551 [W/m-C]
nu=1.562E-5 [m^2/s]
Pr=0.7296
beta=1/T_ave
T_ave=(T1+T2)/2+273
g=9.81 [m/s^2]
```

```
Ra=(g*beta*(T1-T2)*L^3)/nu^2*Pr
Nus=0.42*Ra^(1/4)*Pr^0.012*(H/L)^(-0.3)
A_s=H*W
Q_dot=k*Nus*A_s*(T1-T2)/L
```

**9-163** Two concentric cylinders of diameters  $D_i = 30$  cm and  $D_o = 40$  cm and  $L = 5$  m are separated by air at 1 atm pressure. Heat is generated within the inner cylinder uniformly at a rate of  $1100 \text{ W/m}^3$  and the inner surface temperature of the outer cylinder is 300 K. The steady-state outer surface temperature of the inner cylinder is

- (a) 402 K                      (b) 415 K                      (c) 429 K                      (d) 442 K                      (e) 456 K

(For air, use  $k = 0.03095 \text{ W/m}\cdot^\circ\text{C}$ ,  $\text{Pr} = 0.7111$ ,  $\nu = 2.306 \times 10^{-5} \text{ m}^2/\text{s}$ .)

*Answer* (c) 429 K

**Solution** Solved by EES Software. Solutions can be verified by copying-and-pasting the following lines on a blank EES screen.

```
D_i=0.30 [m]
D_o=0.40 [m]
L=5 [m]
g_dot=1100 [W/m^3]
T_o=300 [K]
```

"The properties of air at 100 C are (Table A-15)"

```
k=0.03095 [W/m-K]
nu=2.306E-5 [m^2/s]
Pr=0.7111
beta=1/T_ave
T_ave=100 [C]+273 [K]
g=9.81 [m/s^2]
```

```
L_c=(D_o-D_i)/2
Ra=(g*beta*(T_i-T_o)*L_c^3)/nu^2*Pr
F_cyl=(ln(D_o/D_i))^4/(L_c^3*(D_i^(-3/5)+D_o^(-3/5))^5)
k_eff=k*0.386*(Pr/(0.861+Pr))^0.25*(F_cyl*Ra)^0.25
```

```
Vol=pi*D_i^2/4*L
Q_dot=g_dot*Vol
Q_dot=(2*pi*k_eff/Ln(D_o/D_i))*(T_i-T_o)
```

## 9-164 ..... 9-167 Design and Essay Problems

